

**RELAP5/MOD3.3 CODE MANUAL
VOLUME II:
APPENDIX A
INPUT REQUIREMENTS**

Nuclear Systems Analysis Operations

October 2010

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Rockville, Maryland
Idaho Falls, Idaho**

**Prepared for the
Division of Systems Research
Office of Nuclear Regulatory Research
U. S. Nuclear Regulatory Commission
Washington, DC 20555**

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A1 INTRODUCTION

This appendix completely describes data deck organization and data card requirements for all problem types allowed in RELAP5/MOD3.3Patch04. The card numbers are in numerical order.

A1.1 Control Format

Input is described in terms of input records or cards, where an input record or card is an 80-character record. Punched cards are nearly obsolete and one would be hard-pressed to find a key punch machine at most installations. Now, data are normally entered from interactive terminals, personal computers, or workstations, and the input usually exists only as disk files or is archived on tape. Data are usually viewed as lines on a CRT screen or lines of printed output. Nonetheless, card is used extensively in this input description to mean an input record.

RELAP5 attempts to read a 96-character record. If the actual input record is smaller, blank characters are added to the end of the input record to extend it to 96 characters. Each 96-character input record, preceded by a sequential card number starting at 1 and incrementing by 1, is printed as the first part of a problem output. Only the first 80 characters are used for RELAP5 input; the additional 16 columns are for use with editors or utility programs such as UPDATE.

Most interactive editors allow the input of at least 80 character records. With many terminals allowing only 80 characters per line, it is convenient to limit the data record to 72 characters so that the data and editor supplied line numbers fit on one line (eight columns for line number and separator, 72 columns of data). Some editors provide for the optional storing of editor line numbers following the data portion of the record. If the data field is 72 columns, the line numbers might be stored in columns 73 to 80. These line numbers will be processed by RELAP5 as input, since RELAP5 uses the first 80 characters. To avoid this, either request the editor to store line numbers starting at character position 81, put a terminating character before the line number, or don't store the line numbers. The line numbers, if saved, are listed in the output echo of the input data.

If the UPDATE program is used to maintain the input deck, the update command must be used to specify that the card data are 80 columns instead of the default of 72.

A1.2 Data Deck Organization

A RELAP5 problem input deck consists of at least one title card, optional comment cards, data cards, and a terminator card. A list of these input cards is printed at the beginning of each RELAP5 problem. The order of the title, data, and comment cards is not critical except that only the last title card and in the case of data cards having duplicate data card numbers, only the last data card is used. We recommend that for a base deck, the title card be first, followed by data cards in card number order. Comment cards should be used freely to document the input. For parameter studies and for temporary changes, a new title card with the inserted, modified, and deleted data cards and identifying comment cards should be placed just ahead of the terminating card. In this manner, a base deck is maintained, yet changes are easily made.

When card format punctuation errors, such as an alphanumeric character in numeric fields are detected, a line containing a caret (^) located under the character causing the error and a message giving

the card column of the error are printed. An error flag is set such that input processing continues, but the RELAP5 problem is terminated at the end of input processing. A standard RELAP5 error message (error message preceded by eight asterisks (*****) is printed if a card error is found. Usually a card error will cause additional error comments to be printed during further input processing when the program attempts to process the erroneous data. If the card number can not be interpreted correctly by RELAP5 because of alphabetic characters, punctuation characters, etc., in the number, the error message will indicate the sequential card number in the deck instead of using the caret to point to the character causing the error.

A1.3 Title Card

A title card must be entered for each RELAP5 problem. A title card is identified by an equal sign (=) as the first nonblank character. The title (remainder of the title card) is printed as the second line of the first page following the list of input data. If more than one title card is entered, the last one entered is used.

A1.4 Comment Cards

An asterisk (*) or a dollar sign (\$) appearing as the first nonblank character identifies the card as a comment card. Blank cards are treated as comment cards. The only processing of comment cards is the printing of their contents. Comment cards may be placed anywhere in the input deck except before continuation cards.

A1.5 Data Cards

Data cards may contain varying numbers of fields that may be integer, real (floating point), or alphanumeric. Blanks preceding and following fields are ignored.

The first field on a data card is a card identification number that must be an unsigned integer. The value for this number depends upon the data being entered and will be defined for each type. If the first field has an error or is not an integer, an error flag is set. Consequently, data on the card are not used, and the card will be identified by the card sequence number in the list of unused data cards. After each card number and the accompanying data are read, the card number is compared to previously entered card numbers. If a matching card number is found, the data entered on the previous card are replaced by data from the current card. If the card being processed contains only a card number, the card number and data from the last previous card with that card number are deleted. Deleting a nonexistent card is not considered an error. If a card causes replacement or deletion of data, a statement is printed indicating that the card is a replacement card.

NOTE: For new problems, the recommended approach for deleting a model feature in the calculation input is to remove the appropriate cards or “comment” them out. Use of the “card number only” method mentioned above is not recommended. For restart problems, the recommended approach for deleting a model feature is to use the “delete” word that is typically input on the first data card for a model feature. The “card number only” method works only if the input cards previously appear in the restart input deck, and if that is the case, it is recommended that the user either remove the appropriate cards or “comment” them out. In conclusion, the “card number only” method is not recommended for any problem.

Comment information may follow the data fields on any data card by beginning the comment with an asterisk (*) or dollar (\$) sign.

A numeric field must begin with either a digit (0 through 9), a sign (+ or -), or a decimal point (.). A comma or blank (with one exception, subsequently noted) terminates the numeric field. The numeric field has a number part and optionally an exponent part. A numeric field without a decimal point or an exponent is an integer field; a number with either a decimal point, an exponent, or both is a real field. A real number without a decimal point (i.e., with an exponent) is assumed to have a decimal point immediately in front of the first digit. The exponent part denotes the power of ten to be applied to the number part of the field. The exponent part has an E or e or D or d, a sign (+ or -), or both followed by a number giving the power of ten. These rules for real numbers are identical to those for entering data in FORTRAN E or F fields except that no blanks (with one exception) are allowed between characters to allow real data written by FORTRAN programs to be read. The exception is that a blank following an E or e or D or d denoting an exponent is treated as a plus sign. Acceptable ways of entering real numbers, all corresponding to the quantity 12.45, are illustrated by the following eight fields:

12.45, +12.45, 0.1245+2, 1.245+1, 1.245E 1, 1.245D+1, 1.245e 1, 1.245e+1

When entering a decimal zero for either an integer or floating point quantity, the 0 can be written in either form. Thus a floating point zero can be entered simply as 0 without a decimal point.

It is recommended that users enter real numbers that do not have a fractional part, i.e., six thousand, with a trailing 0 after the decimal point, e.g., 6.0E3, and not as 6.E3. It is also recommended that users enter real numbers that are less than 1.0 with a leading 0 before the decimal point. That way the decimal point is set apart from any trailing exponential part or any leading white space, and it will be more likely that the user will notice the lack of the decimal point. As stated above, a real number without a decimal point is interpreted as if the decimal point is front of the number. Thus, 6E3 is interpreted as 0.6E3 or 600.0 and not as 6000.0, which the user might hope it would be.

Alphanumeric fields have three forms. The most common alphanumeric form is a field that begins with a letter and terminates with a blank, a comma, or the end of the card. After the first alphabetic character, any characters except commas and blanks are allowed. The second form is a series of characters delimited by quotes (") or apostrophes ('). Either a quote or an apostrophe initiates the field, and the same character terminates the field. The delimiters are not part of the alphanumeric word. If the delimiter character is also a desired character within the field, two adjacent delimiting characters are treated as a character in the field. The third alphanumeric form is entered as nHz, where n is the number of characters in the field, and the field starts at the first column to the right of H and extends for n columns. With the exception of the delimiters (even these can be entered if entered in pairs), the last two alphanumeric forms can include any desired characters. In all cases, the maximum number of alphanumeric characters that can be stored in a word is eight. If the number of characters is less than eight, the word is left justified and padded to the right with blanks. If more than eight characters are entered, the field generates as many words as needed to store the field, eight characters per word, and the last word is padded with blanks as needed. Regardless of the alphanumeric type, at least one blank or comma must separate the field from the next field.

Most computers (e.g., workstations, CRAY, and IBM) hold only eight characters per word. All alphanumeric words required by RELAP5, such as components types, system names, or processing options, have thus been limited to eight characters. We highly recommend that the user limit all other one-word alphanumeric quantities to eight characters so that input decks can be easily used on all computer versions. Examples of such input are alphanumeric names entered to aid identification of components in output edits.

A1.6 Continuation Cards

A continuation card, indicated by a plus sign (+) as the first nonblank character on a card, may follow a data card or another continuation card. Fields on each card must be complete, that is, a field may not start on one card and be continued on the next card. The data card and each continuation card may have a comment field starting with an asterisk (*) or dollar (\$) sign. No card number field is entered on the continuation card, since continuation cards merely extend the amount of information that can be entered under one card number. Deleting a card deletes the data card and any associated continuation cards.

A1.7 Terminator Cards

The input data are terminated by a slash or a period card. The slash and period cards have a slash (/) and a period (.), respectively, as the first nonblank character. Comments may follow the slash and period on these cards.

When a slash card is used as the problem terminator, the list of card numbers and associated data used in a problem is passed to the next problem. Cards entered for the next problem are added to the passed list or act as replacement cards, depending on the card number. The resulting input is the same as if all previous slash cards were removed from the input data up to the last period card or the beginning of the input data.

When a period card is used as the problem terminator, all previous input is erased before the input to the next problem is processed.

A1.8 Sequential Expansion Format

Several different types of input are specified in sequential expansion format. This format consists of sets of data, each set containing one or more data items followed by an integer. The data items are the parameters to be expanded, and the integer is the termination point for the expansion. The expansion begins at 1 more than the termination point of the previous set and continues to the termination point of the current set. For the first set, the expansion begins at 1. The termination points are generally volume, junction, or mesh point numbers, and always form a strictly increasing sequence. The input description will indicate the number of words per set (always at least two) and the last terminating point. The terminating point of the last expansion set must equal the last terminating point. Two examples are given.

The first example is for the volume flow areas in a pipe component; the format is two words per set in sequential expansion format for nv sets. Using the number of volumes in the pipe (nv) as 10, the volume flow areas could be entered as

0010101 0.01,10 .

In this case, the volume flow areas for volumes 1 through 10 have the value 0.01.

The second example shows how the pipe volume friction data could be input. The input consists of three words per set for *nv* sets. The three words designate the wall roughness, hydraulic diameter (input of 0 causes the code to calculate it), and volume number. Possible data might be

0010801 1.0-6,0,8 1.0-3,0,9

0010802 1.0-6,0,10 .

Here, volumes 1 through 8 and 10 have the same values, and volume 9 has a different value.

A1.9 Upper/Lower Case Sensitivity

Historically, computer systems allowed only upper case alphabetic characters. Accordingly, the following input descriptions use upper case for required input, e.g., SNGLVOL, 1.25E5. Now, many systems have upper and lower case alphabetic characters, and some applications are case sensitive, others not. At the NRC, required input must be in lower case, and the user should check the requirements at other installations. At installations with both upper and lower case capability, there are utilities (tr) and editors that can easily switch alphabetic characters to the desired case.

A1.10 Data Card Requirements

In the following description of the data cards, the card number is given with a descriptive title of the data contained on the card. Next, an explanation is given of any variable data that are included in the card number. Then, the order of the data, the type, and the description of the data item are given. The type is indicated by A for alphanumeric, I for integer, and R for real.

A2 MISCELLANEOUS CONTROL

A2.1 Developmental Model Control, 1

This card has been added to the code for the convenience of developers in testing model improvements or new models. This card is not a standard input feature of the code. The description of this card has been added to the input requirements because several laboratories are receiving test versions to assist in the development and testing of the code. Anyone using this card must realize that they are selecting experimental options still under development. Furthermore, these options may change more frequently than the revision of this input manual. Thus, before using the options, users should obtain the brief listing of current options from the code (described below) and verify those descriptions against this manual.

The purpose of this card is to allow developers and analysts to quickly test new models by activating or deactivating a model through simple input instead of program modification, compilation, and loading. Ninety logical variables having only false or true values are provided and defined at the start of program execution as false. This input sets the logical variables to true or resets them to false at the beginning of a new problem or at any restart. Fortran IF statements added as part of the experimental coding activate or deactivate models based on the values of the logical variables.

As described above, up to 190 options can be defined and the options are identified with a number from 1 through 190. Which options are defined and what they control are very much version dependent. The usual practice is to enter the option capability using a currently unused option number as the new model or improvement is first coded. During further development and testing, the model may change and the effect of the option can change in a manner ranging from large to subtle. When the model has been completed or even abandoned, the production version of the model is coded and the option capability is removed. The option number is then available for reuse with a completely different model. Thus, the options are version dependent as to what option numbers are in use, what models they control, and the particular features of the models. Accordingly, these options should be used only by those in direct contact with the developers.

Each current option is described below. In light of the discussion above, the user should verify that the code version being used corresponds to this description. Programmers using this option feature are asked to include coding that issues error messages when unused options are selected and to issue a brief statement of the purpose of selected options. Remember, however, that all coding associated with these options is experimental and these output conventions may not be thoroughly checked.

Up to 191 numbers consisting of 0 or any of the currently available option numbers may be entered on this card. A positive nonzero number, *n*, activates option *n* by setting the logical variable *n* to true; a negative nonzero number, *-n*, deactivates option *n* by setting the logical variable *n* to false. Attempting to activate an unused option is an error, and attempting to deactivate an already inactive option or an unused option is also an error. Adding options or deactivating options is allowed at a restart; the previously defined options will remain. The status of the options is printed in any NEW or RESTART problem containing this card or a RESTART problem in which the restart point had an option selected. The printout includes a listing of the 190 option numbers and a false (option not selected) or a true (option selected) value plus the brief description of each selected option.

The number 0 is not an option number but may be entered to force the brief descriptions of all available options to be printed regardless of whether they are active. The 0 input should be used only once to observe the available options and then removed so that the list better emphasizes the selected options.

W1-191. Option number as described above.

Option 0. This option lists out all the available options.

Option 1. This option adds an explicit viscous-like term to the momentum equations in the nearly-implicit scheme. This term is documented in the discussion of the semi-implicit scheme.

Option 2. This option sets the interfacial mass transfer to 0 and the direct heating heat transfer coefficient to 10^5 . This option simulates the no mass transfer case in which the two phases are in thermal equilibrium.

Option 3. This option sets molten pool natural circulation for SCDAP.

Option 4. This option scales back the Courant limit to 75% of its calculated value so that the RELAP5 time step will always be less than or equal to 75% of the Courant limit.

Option 5. This option smooths the heat transfer coefficients using an exponential-type smoothing function $(1 - e^{-t/\tau})$. The time constant is set equal to Δt with the limitation that the time constant can not be greater than 0.5 second.

Option 6. This option outputs information to the screen and to the screen file every time Δt_{step} is called. It is useful for determining why the code is cutting the time step. The volume number for the volume that has the limiting Courant time step and the maximum mass error for each time step is printed on one line. In addition, the pressure and void fraction for each of these volumes are printed. If the time step is cut, the reason for the cut is given in the right-hand column.

Option 7. This option outputs information to the Reasonfile on the volume that caused a reason to be printed in the last column of the screen file. It is used in conjunction with option 6.

Option 8. This option provides time step control based on the change in void fraction and is designed to limit the time step when the void fraction in any cell is decreasing rapidly such as during periods of condensation. The time step will be repeated if the decrease in void fraction in any hydrodynamic volume is considered too large. The time step will be repeated if the new void fraction is less than 0.001 and the old void fraction was greater than 0.01. The time step will also be repeated if the new void fraction is between 0.01 and 0.10 and the change in void fraction during the time step exceeds 20% of the new void fraction. If the new void fraction exceeds 0.10, the time step will be repeated if the change exceeds 50% of the new void fraction. The time steps are repeated only when the void

fraction is decreasing. This option is being tested in conjunction with options 10 and 12 for improved performance at low pressures.

- Option 9. This option provides a user controlled (on/off) transition to plug flow when the vapor velocity exceeds the criterion for transition from stratified flow in horizontal geometries. The need for this change occurs during the reflood phase in the horizontal core of the Hanford N-Reactor. The process tubes in the core region are long and have a small diameter. As subcooled coolant enters the process tubes, high condensation occurs that results in local depressurization and high steam velocities towards the condensation site. Instabilities occur that eventually cause the code to fail. This change introduces a plug flow model that limits condensation to a value that is just large enough to condense all the vapor that can flow at the critical velocity, utilizing the full channel cross-sectional area. The critical velocity is the velocity large enough to cause transition from a stratified flow condition. The concept behind this model is that condensation lowers the local pressure and draws in steam which pushes the water into a plug. The area of the plug limits condensation and reduces steam flow which causes a return to stratified flow.
- Option 10. This option provides time step control based on change in pressure within a hydrodynamic volume. This change causes the code to repeat a time step if the change in pressure during a time step exceeds the old time value, the new time value, or 50,000 Pa. This time step control allows the pressure to change by no more than a factor of two during a time step. This change generally causes the code to run slower but more reliably. With this change activated, the code will more accurately track pressure waves and oscillations but may cause certain problems to run unacceptably slowly. This option has no effect if option 8 is not selected.
- Option 11. This option increases the interfacial liquid (HIF) and vapor (HIG) heat transfer coefficients when the pressure is near the critical pressure in order to force vaporization and/or condensation. The values computed from the RELAP5 correlations are too small, which results in the liquid becoming superheated and/or the vapor becoming subcooled. When this happens, the steam tables are not able to compute the thermodynamic properties and the code fails. This option is useful for supercritical pressure blowdowns. It essentially forces an equilibrium model near the critical pressure.
- Option 12. This option provides a user controlled (on/off) water packer developed for horizontal reactors such as the Hanford N-Reactor. The interfacial friction coefficient term for the momentum equation, c , is adjusted as a function of vapor void fraction. For void fractions of 0.001 or less, F_{ij} is forced to an arbitrarily large value of 10^{10} . For void fractions greater than 0.01, the regularly calculated F_{ij} is used. For void fractions between 0.001 and 0.01, a cubic interpolation scheme is used to adjust F_{ij} between the calculated value and 10^{10} . As a cell is calculated to fill with liquid and the calculated cell pressure rises, the lower inertia vapor is the phase first to respond, either moving on to the next cell or moving back to an

upsteam cell. As the void fraction decreases, the interfacial drag is increased, thus allowing the moving vapor to either drag liquid into the next cell or impede the incoming liquid from an upstream cell. In either case, the model eases the overfilling and overpressurization of the cell.

- Option 13. This option activates vertical stratification changes. It uses donor/acceptor donoring of void fraction, not density, however, if the volume is in the vertical stratification flow regime.
- Option 14. This option turns off constitutive relations and should only be used for testing advancement of the basic advancement scheme. This option is not operational now.
- Option 15. This option activates the older formulations for the light water transport properties (surface tension, viscosity, and thermal conductivity) for the new steam tables. The newer versions are dated September 1994 for surface tension, August 2003 for viscosity, and September 1998 for thermal conductivity. It is not needed if the old steam tables are being used.
- Option 16. This option activates the velocity flip/flop logic for the nearly-implicit advancement scheme.
- Option 17. This option allows the code to run at the Courant limit with a multiplication factor of four instead of two and a reduction factor of 0.25 instead of 0.5. These limits are used in the mass error check for the time step control.
- Option 18. This option adds the sharp interface and reverse void profile logic from MOD2.5.
- Option 19. This option uses the Bestion correlation (in MOD2.5) rather than the EPRI correlation (in MOD3) for bundles (when volume bundle flag $b = 1$).
- Option 20. This option implements a smooth bubbly-slug transition region.
- Option 21. This option interpolates for the density and temperature derivatives that are stored in steam tables instead of recomputing them from interpolated values of beta, kappa, and csubp. It is useful for calculations near the critical point and may also help other problems as well. It should be used when Option 26 is used for calculations that have volumes with supercritical pressures.
- Option 22. This option deactivates the Simplex root finding method that is used in the nwithair subroutine. The Simplex method converges the solution for the steam partial pressure and internal energy when the Newton method starts oscillating about the true solution.

- Option 23. This option selects a boron transport algorithm (Godunov) that greatly reduces the numerical diffusion of boron compared to the standard algorithm.
- Option 24. This option activates the original RELAP5 subcooled boiling model. The default subcooled boiling model is the SRL model, which is better at low pressures.
- Option 25. This option linearizes the interfacial heat transfer for the nearly implicit scheme.
- Option 26. This option uses the bilinear fits to the thermodynamic properties and the density and temperature partial derivatives with respect to pressure and internal energy in the **supercritical pressure region**. It is useful for problems that have thermodynamic states above the critical pressure. If it is used, then Option 21 should also be used to maintain continuity of the partial derivatives across the critical pressure line.
- Option 27. This option removes most of the timing and common block location from the output file so that it is easier to compare runs using two different versions of the code. This option can also be implemented by adding a new card, one with 27 for the card number and any integer for the first field.
- Option 28. This option turns on the modified dynamic gap conductance model which more accurately computes the fuel-cladding gap width for fuel slumping based on geometric analysis.
- Option 29. This option turns off the checking of the offtake junctions when the junction v-flag is 1, 2, 3, or 4.
- Option 30. This option introduces implicit coupling between SCDAP radiation and hydrodynamic advancement.
- Option 32. This option activates the water stretch logic.
- Option 36. This option limits metastable extrapolation to 50 K. Normally, the code fails when the metastable extrapolation exceeds 50 K, so this option allows the code to continue running by resetting the metastable extrapolation to 50 K whenever it exceeds 50 K.
- Option 37. This option turns off the umbrella model. When the umbrella model is on, an upper limit is placed on the liquid interfacial heat transfer coefficient (H_{if}) when the liquid is subcooled. The limit is umbrella shaped so as to force the coefficient to small values as the void fraction approaches 0.0 or 1.0.
- Option 38. This option increases hif when the liquid temperature is above the saturation temperature and approaches the liquid spinodal temperature. It is useful in supercritical pressure blowdown problems to convert the liquid to vapor before liquid temperature reaches the

spinodal temperature. Option 11 should also be considered if Option 38 is not sufficient to prevent superheated liquid states.

- Option 39. This option increases h_{ig} when the vapor temperature is below the saturation temperature and approaches the vapor spinodal temperature. It is useful in supercritical pressure blowdown problems to convert the vapor to liquid before the vapor temperature reaches the vapor spinodal temperature. Option 11 should also be considered if Option 39 is not sufficient to prevent subcooled vapor states.
- Option 41. This option includes energy dissipation due to form loss.
- Option 42. This option applies a stronger unchoking test for junctions with an abrupt area change.
- Option 43. This option uses iteration to calculate the hydrodynamic conditions at the throat for critical flow of vapor.
- Option 45. This option selects the newly developed model for condensing interphase heat transfer.
- Option 48. This option deactivates the linear implicit interfacial drag, wall friction, and K-loss logic.
- Option 50. This option activates at all junctions where choking option is activated the original RELAP5 critical flow model instead of the default Henry-Fauske critical flow model. The code does not allow a user to switch between the two critical flow models in the middle of a problem, e.g. switching from the Henry-Fauske to the original critical flow model after a steady-state run at the beginning of a transient run.
- Option 51. Normally, water packing is activated in all volumes unless specifically disabled by an input volume flag. This option disables water packing for all volumes.
- Option 52. Normally, the choking model is activated for all junctions unless specifically disabled by an input junction flag. This option disables the choking model for all junctions and there is not way to turn it on for a specific junction.
- Option 54. This option activates the Moody critical flow model. It is part of the CANDU package. The Moody model is a subset of the Henry-Fauske two-phase choking model in that it uses the Henry-Fauske model for values of x_{zero} less than 0.01, where $x_{zero} = \dot{\alpha}_{gj}\dot{\rho}_{gj} / (\dot{\alpha}_{gj}\dot{\rho}_{gj} + \dot{\alpha}_{fj}\dot{\rho}_{fj})$, and the dot indicates a junction donored quantity. Henry-Fauske is used for values of x_{zero} greater than $1.0e-6$ and less than 0.01. Thus, the Moody model will use the same single discharge coefficient entered for the Henry-Fauske model for all three regions, subcooled, two-phase, and vapor, just like the Henry-Fauske model does now. A separate discharge coefficient should not be entered for the three regions when the Moody model is used. Instead, enter a single discharge coefficient and the

thermal nonequilibrium constant as if you were using the Henry-Fauske model for the junction, valve, or branch input cards.

- Option 56. This option enforces $v_g = v_f$ at $\alpha_g = 1$ in FIDIS2.
- Option 60. This option modifies the time step control for Courant limit. It is turned on automatically now.
- Option 62. This option uses newly developed changes to the Chen F factor in PREDNB to minimize oscillations.
- Option 64. This option uses the old halving and doubling of the time step but the new Courant limit.
- Option 65. This option changes the subcooled boiling model by modifying the fraction of nucleate boiling heat flux that generates vapor when the bulk liquid is subcooled. The modification minimizes the “on/off” behavior associated with low-pressure/low-flow conditions.
- Option 66. This option implements donor/acceptor differencing in vertical stratification volumes.
- Option 67. This option implements velocity squared instead of (velocity)*(velocity-donored) for momentum flux.
- Option 68. This option implements (velocity-j)*(velocity-L - velocity-K) instead of (velocity)*(velocity-donored) for momentum flux.
- Option 70. This option uses explicit coupling between the continuity and momentum equations. It is useful for tracking sonic waves.
- Option 71. This option does not set the time-dependent volume length to zero. It keeps it the same as the user input. It is useful for tracking sonic waves.
- Option 74. This option turns on the new interphase mass transfer model.
- Option 75. This option relaxes the mass transfer at the wall. It is part of the Candu package.
- Option 77. This option introduces the new geometric restrictions on the hse model.
- Option 79. This option adds the scaling factor to the diagonal of the Jacobian squared matrix ten times if necessary. It helped in some cases when noncondensables were present.
- Option 80. This option enables the 80-column screen output instead of the 132-column screen output.

- Option 81. This option activates the Baker-Just metal water reaction equations. It is part of the Candu package.
- Option 84. This option activates implicit gravity head and horizontal stratification force due to the void gradient in the nearly implicit scheme.
- Option 85. This option activates the PennState non-condensable updates. These updates make a prediction of the air quality in volumes in which air could possibly appear in the next time step. (This option is not recommended.)
- Option 88. This option activates the 3-D reactor kinetics coupling with the PARCS code. The heat structure heat source modeling is affected by this option. For more information, see source data card 1CCCG701 in Section A10.18.
- Option 89. This option makes the cell lengths for all the terms in the momentum equations the same. The default is to use a shortened (by the area ratio) length for the larger diameter cell when there is an area change at the junction.
- Option 90. This option turns on the e-flag (modified PV term in the energy equation) for all applicable junctions. This includes all junctions except for time-dependent junctions, accumulator junctions, and junctions associated with separator, jetmixer, pump, turbine, and eccmixer components.
- Option 91. This option creates three output files: tab1-vol.dat, tab2-jun.dat, and tab3-hs.dat. These files contain the volume data, junction data, and heat structure data, respectively, in tabular format.

A2.2 No Timing Output Option, 27

This card is optional and is equivalent to adding option 27 to card 1. It is implemented in this optional manner to simplify the automatic changing of input decks by adding a 27 card after the title card instead of having to search through the entire input deck looking for the last card 1 and adding a 27 to the list of options on that card.

W1(I) Any integer.

A2.3 Problem Type and Option, 100

This card is always required.

W1(A) Problem type. Enter one of the following: NEW, RESTART, RESET, REEDIT, STRIP, or CMPCOMS.

NEW specifies a new simulation problem.

RESTART specifies continuation from some point in a previous problem using information from the RSTPLT file.

RESET specifies the same thing as RESTART except that all information in the RSTPLT file prior to the restart point is purged.

REEDIT has not been implemented.

STRIP specifies that data are to be extracted (stripped) from the RSTPLT file, and only the data specified are written to the STRIP file.

CMPCOMS specifies that a comparison is to be made between dump records on two files written in one or two previous runs.

W2(A) Problem option. This word is needed if W1 is NEW or RESTART and is optional if W1 is STRIP.

If NEW or RESTART is entered, in W1, enter either STDY-ST or TRANSNT in W2 to specify the type of simulation. Note the cautions discussed in Section A2.6 when the problem option is changed from STDY-ST to TRANSNT or vice versa.

When STRIP is entered in W1, enter either BINARY or FMTOU or leave W2 blank. BINARY is assumed if W2 is not entered. BINARY indicates the unformatted (BUFFER OUT) file. FMTOU indicates that the same information is to be written as 80-column formatted records. One use of this option is to allow simulation results to be transmitted to a different type of computer. Formats are

STRIP Record 1. (5A8,10X,A8)

STRIP Record 2. (A10,3I10)

STRIP Record 3. (8A10)

STRIP Record 4. (A10,7I10/(8I10))

STRIP Record 5..., N. (A10, 5X,1P,4E15.6/(5E15.6)).

The STRIP record above refers to the data in one record of the unformatted file. Multiple 80-column formatted records may be written for STRIP Records 3 through N.

A2.4 Input Check or Run Option, 101

This card is optional for all types.

W1(A) Option. Enter either INP-CHK or RUN; if this card is omitted, RUN is assumed. This card has no effect on a CMPCOMS problem

If INP-CHK is entered, the problem execution stops at the end of input processing.

If RUN is entered, the problem is executed if no input errors are detected.

A2.5 Units Selection, 102

This card is optional for all problem types. If the card is omitted, SI units are assumed for both input and output. If the card is used, enter either SI or BRITISH for each word. SI units used are the basic units, kg, m, s, and the basic combined units such as $\text{Pa} = \text{kg} \cdot \text{m} / \text{s}^2 \cdot \text{m}^2$. British units are a mixture of lb (mass), ft, and s, primarily, but pressure is in $\text{lb}_f / \text{in}^2$ (lb_f is pounds force), heat energy is in Btu, and power is in MW. Thermal conductivity and heat transfer units use seconds (s), not hours (h).

W1(A) Input units. Enter either SI or BRITISH. Default is SI units.

W2(A) Output units. Enter either SI or BRITISH. Default is SI units.

A2.6 Restart Input File Control, 103

This card is required for all problem types (W1 of card 100) except NEW and is not allowed for type NEW.

When the problem option (W2 on card 100) is the same as the problem being restarted, the steady-state or transient is continued, and data on the RSTPLT file up to the point of restart are saved. If the restart continues from the point the previous problem terminated, restart and plot information is added to the end of the previous RSTPLT file. If the restart is prior to the termination point of the previous simulation, restart and plot data after the point of restart are overwritten by new results. A copy should be saved if RSTPLT files from each simulation are needed. If the problem options are different, data up to the point of restart are not saved, problem advancement time is reset to 0, and the RSTPLT file will contain information as if this problem type were NEW.

Some cautions should be observed when the problem advancement time is changed by changing the problem option from STDY-ST to TRANSNT, or vice versa, or the problem advancement time is reset through W1 on card 200. Either or both of these could be specified at restart. When the advancement time is changed, the user is responsible for ensuring that models involving problem time will operate as intended. Affected models include trips using advancement time, control systems using time as an operand (does not include differentiation or integration with respect to time), and table lookup and interpolation using time as the independent variable. If necessary, trips, control systems, general tables, time-dependent volumes, junctions, and pump speed tables can all be reentered at restart. With normal modeling practices, little use of modeling features involving advancement time is needed for runs to steady-state and accordingly little effort should be needed in switching from STDY-ST to TRANSNT. Because of the frequent use of time in logic to initiate failures, as part of safety systems, and used in establishing the delay

times allowed in most table lookup and interpolation tables, required changes to a transient run may be extensive.

The program does make a change to delay control components when the advancement time is changed. The delay control component operates by maintaining a tabular past history of the delayed functions and using table lookup and interpolating to evaluate the delayed function. The table consists of pairs of time values and the delayed function. When the problem time is changed, the time values in the history table and the time value to store the next point in the table are modified by adding the difference of the new advancement time and the old advancement time. The modified history table is as if the problem being restarted was run with the new advancement time. This may not be the desired change, and, in that case, the user can reenter the delay component.

W1(I) Restart number. This must be a number printed in one of the restart print messages and whose associated restart information is stored in the RSTPLT file. These restart numbers are conveniently printed out at the end of the output file for each run as commented out 103 cards (* in column 1). This block of cards can then be pasted into the front of the restart input file and the comment removed from the appropriate 103 card.

If the problem type (W1 on card 100) is STRIP, this number must be 0.

Alternately, a -1 may be specified instead of a specific restart number. This has the effect of restarting a problem from the last block in the restart file. Thus, the user can now set up a series of restart cases which begin from the end of a previous run without having to wait for the run to finish in order to get the actual restart number from which to restart.

W2-6(A) Restart plot file name. This optional alphanumeric entry can be used to enter the file name of the restart plot file. Up to forty characters may be entered as one alphanumeric field. (The code internally treats the field as up to five eight-character words.) The default file name for the restart plot file is rstplt. This may be overridden on Unix machines by using the -r option on the command line. Either the default name, the name from the command field, or the name from this field on a previous case may be overridden by this field.

A2.7 Restart-Plot File Control, 104

This card can be entered for NEW, RESTART, and STRIP options. For the strip option, this card controls the strip file, and the NONE option is not allowed. Only the FILENAME option can be used. If this card is omitted, the restart-plot file is rewound at the end of the problem, but no further action is taken. The user may need to provide system control cards to dispose of the file. To prevent the restart-plot file from being written, a card with NONE must be entered.

W1(A) Action. This word may not be blank. It has to be NONE or a FILENAME.

NONE, no restart-plot file is written

FILENAME, words 2-6 on this card contain the restart plot file name.

W2(A) Restart plot file name. This optional alphanumeric entry can be used to enter the file name of the restart plot file. Up to forty characters may be entered as one alphanumeric field. The default file name for the restart plot file is rstplt. This may be overridden on Unix machines by using the -r or -R option on the command line. Either the default name, the name from the command field, or the name from this field on a previous case may be overridden by this field. This information can be entered only on NEW problems; in RESTART problems, this information may be entered on the 103 card.

A2.8 CPU Time Remaining and Diagnostic Edit, 105

Card 105 controls termination of the transient advancement based on the CPU time remaining for the job. Some operating systems allow specification of the CPU time allocated for a job as part of the job control language and also provide a means to determine the CPU time remaining during job execution. As an alternative, Word 3 of this card may be entered as the CPU time allocated. An alternative CPU remaining time is computed by decrementing this quantity by the CPU used as measured by the program. If Word 3 is omitted or 0, the alternative CPU remaining time is assumed infinite. At the end of each time step, the CPU time remaining for the job is determined from the minimum of the system (if available) and alternative CPU remaining times. If the remaining CPU time is less than Word 1, the transient is immediately terminated. The advancement may not be at the end of a requested time step due to time step reduction; the hydrodynamic, heat conduction, and reactor kinetics may not be advanced to the same point; or the advancement may not be successful and the advancement is scheduled to be repeated with reduced time step. Major edits, minor edits, plot edits, and a restart record are forced. The transient can be restarted from this point as if the problem had not been interrupted. The transient is also terminated after successful advancement over a requested time step and the CPU time is less than Word 2. Word 2 should be larger than Word 1. The default values for Words 1 and 2 are 1.0 and 2.0 seconds. The default values are used if the card is not supplied or the entered numbers are less than default values. Word 2 is also forced to be 1.0 seconds larger than Word 1. The time values must include time for the final minor and major edits (very little time required), plotting, and any other processing that is to follow termination of RELAP5 execution. This card is optional, but we strongly recommend its use with Word 3 nonzero on systems that do not provide a system CPU limit.

Card 105 also controls the diagnostic edit printout through the use of Words 4 and 5. If these words are missing or 0, no debug options are in effect. If Word 4 is greater than 0, then Word 4 is the attempted advancement count number to start a diagnostic edit, and Word 5 is the attempted advancement count number to stop the diagnostic edit as well as the calculation. If Word 4 is -1, a dump file is written on the file specified by the -A option on the command line at the completion of the advancement given in Word 5. Entering 0 in Word 5 writes the dump file just before the start of transient advancement. The problem is terminated after writing the dump file. If Word 4 is -2, a dump file is written on the file given by the -A option after the advancement given in Word 5; the time advancement is then repeated and a dump file following the repeated advancement is written on the file given by the -B option. The problem is terminated after writing the second dump file. Word 5 must be greater than 0 when Word 4 is -2. The default file names are -A dumpfil1 and -B dumpfil2.

W1(R)	CPU remaining limit 1 (s).
W2(R)	CPU remaining limit 2 (s).
W3(R)	CPU time allocated (s). This quantity is optional.
W4(I)	Debug control word as described above.
W5(I)	Debug control word as described above.

If the program is compiled with compile time option CTSS defined, entering Word 1 as 0.0 will cause no testing for CPU termination and normal CTSS termination at the end of CPU time can occur. In this case, the problem can be restarted.

A2.9 Noncondensable Gas Species, 110

This card is required for all calculations that use noncondensable gas. Nitrogen must be included for any problem having accumulators. If the user wants to add additional gas types in a restart, be sure to include the gas type on this card, even though its initial mass fraction on the 115 card is 0.0. This card cannot be entered on a RESTART problem.

W1-WN(A) Noncondensable gas type. Enter up to 5 words in any order from the following list of noncondensable gas types: ARGON, HELIUM, HYDROGEN, NITROGEN, XENON, KRYPTON, or AIR.

A2.10 Noncondensable Mass Fractions

Card 115 is related to card 110. Card 115 is required if card 110 is entered unless only one gas type is entered on card 110, and then the mass fraction is set to 1.0. The number of words on card 115 must equal the number of words on card 110. The sum of the mass fractions must sum to 1.0. The mass fractions on these cards are default values and are used for initial conditions of active volumes and for values of time-dependent volumes. This card cannot be entered on a RESTART problem. However, different mass fractions can be entered in a RESTART problem via the optional CCC0301, CCC0301-399, CCC3201, and CCC3201-3299 cards as described in each component description or as additional items on the CCC0201-299 cards for time-dependent volumes. Each volume can have a different initial noncondensable mass fraction if needed in the model. The values on the 115 card are used unless a different set of values are entered on the optional CCC0301, CCC0301-399, CCC3201, or CCC3201-3299 cards or as additional items on the CCC0201-299 cards. The CCC0301 cards are for single-volume components, the additional items on the CCC0201-299 cards are for the time-dependent volume components, and the CCC3201-3299 cards are for the multiple-volume components like a pipe. The one exception is the pump, which uses the CCC3201 card because the CCC0301 card is already used for another purpose.

A2.10.1 Initial Default Mass Fraction for Each Noncondensable Gas Type, 115

W1-WN(R) Mass fraction for each noncondensable gas type. These values are used as the default values for all volumes if optional values are not entered for the component volume.

A2.11 Hydrodynamic System Control

Independent hydrodynamic systems can be described by the hydrodynamic component input. The term independent hydrodynamic systems means that there is no possibility of flow between the independent systems. A typical example would be the primary and secondary systems in a reactor where heat flows from the primary system to the secondary system in the steam generator but there is no fluid connection. If a tube rupture were modeled, the two systems would no longer be independent. Input processing lists an elevation for each volume in each independent hydrodynamic system and includes a check on elevation closure for each loop within a system. A reference volume is established for each system through input or default.

These cards are optional for each system. If not entered for a system, that system contains H₂O as the fluid unless a different fluid is specified in hydrodynamic component data, and the lowest numbered volume in each system is the reference volume. Additionally, the reference volume has a default elevation of 0. These cards should not be entered in a RESTART problem.

A2.11.1 Hydrodynamic System, 120-129

W1(I) Reference volume number of the system. This must be a volume in the hydrodynamic system.

W2(R) Reference elevation of the volume center relative to a fixed z-axis for the system (m, ft).

W3(A) Fluid type for the system. Enter H2O or H2OOLD or D2O. If H2OOLD is entered, the code uses the older steam tables and if H2O is entered, the code uses the newer thermodynamically-consistent tables. The newer tables are based on the Helmholtz functional fit in the IAPWS-1995 document.

W4(A) Optional alphanumeric name of system used in output editing. *NONE* is used if this word not entered.

W5(I) System information flag. This word is optional. If this word is not entered, g = 0 is used.

g = 0 g = 0 specifies that noncondensable gas is present for this system.

g = 1 g = 1 specifies that noncondensable gas is not present for this system. If g = 1 (no noncondensable) in a system and if the digit t = 4, 5, or 6 in the hydrodynamic volume component control word gbt, an input error will result.

g = 2 **g = 2** is like **g = 0**, except that in this case, there are only noncondensable gases in this system. No steam or water is allowed nor can any appear in the system. A reduced set of three equations is solved: gas mass, gas energy, and noncondensable mass.

W6-9(A) Thermodynamic property file name of the system. This optional alphanumeric entry can be used to enter the file name of the thermodynamic property file of the system. Up to thirty-two characters may be entered as one alphanumeric field. (The code internally treats the field as up to four eight-character words.) The default file name for the thermodynamic property file is **tpfh2onew** for H₂O, **tpfd2o** for D₂O, **tpfh2o** for H₂OOLD, etc.

A2.12 Self-Initialization Option Control

These cards are optional, are not needed, and are only used as a cross-check on the controllers specified in Section A15. Data supplied on these cards are used to invoke the self-initialization option. These data describe which and how many of each controller will be used. To retain generality and flexibility, the self-initialization option does not require that the steady-state and nearly-implicit solution scheme options be concurrently turned on. However, this is the recommended procedure. These latter options are invoked through input data cards 100 and 201 through 299. In addition to the data cards described below, the user must furnish data on the controllers to be used, as described in Section A15.

A2.12.1 Self-Initialization Control, 140

This card specifies the number and type of controllers desired.

W1(I) Number of pump controllers.

W2(I) Number of steam flow controllers.

W3(I) Number of feedwater controllers.

A2.12.2 Self-Initialization Pump Controller and Identification, 141-142

These cards establish the relationship between the pump number and the number of the pump controller. For each pump so referenced, the user *must* use the time-dependent pump velocity option. For pump component card CCC6100, Words 2 and 3 must be the alphanumeric and numeric parts for the pump controller. The time-dependent pump velocity data (pump component cards CCC6100 through CCC6199) should be input so that the search variable and pump velocity are related by a straight line through the origin with a slope of 1.

W1(I) Component number of pump number.

W2(I) Controller identification number for pump number 1.

W3(I) Component number of pump number 2.

W4(I) Controller identification number for pump number 2.

A maximum of six pump/controller pairs may be entered.

A2.12.3 Self-Initialization Steam Flow Controller Identification, 143-144

These cards establish the relationship between the steam flow control valve number and the steam flow controller number.

W1(I) Component number of steam flow control valve number 1.

W2(I) Controller number of steam flow controller for steam flow control valve number 1.

W3(I) Component number of steam flow control valve number 2.

W4(I) Controller number of steam flow controller for steam flow control valve number 2.

A maximum of six control valve/controller pairs may be entered. Note that in the above the valve component is assumed to be the control component. However, the user is not constrained to use a valve and may use a pump or a time-dependent junction. CAUTION: only a servo valve, a time-dependent junction, or a pump may be used, or a diagnostic error will result.

A2.12.4 Self-Initialization Feedwater Controller Identification, 145-146

These cards establish the relationship between the feedwater valve number and the feedwater controller number.

W1(I) Component number of feedwater valve number 1.

W2(I) Controller id number of the feedwater controller for feedwater valve number 1.

W3(I) Component number of feedwater valve number 2.

W4(I) Controller id number of the feedwater controller for feedwater valve number 2.

A maximum of six control valve/controller pairs may be entered. Note that in the above it is assumed that a valve component is the control component. However, the user is not constrained to use a valve and may use a pump or time-dependent junction. CAUTION: only a servo valve, time-dependent junction, or a pump is allowed, or a diagnostic will result, such as a time-dependent junction with the controller output used as the independent variable in place of time.

A2.12.5 Pressure and Volume Control Component Identification, 147

This card identifies the component number, connection data, and pressure level for the time-dependent volume that is to provide pressure and volume control during the self-initialization null transient.

W1(I)	Component number of time-dependent volume that replaces the pressurizer.
W2(I)	Component number to which the above time-dependent volume is connected; CAUTION: only a single-connection is allowed or an error will result.
W3(R)	Desired steady-state pressure.

A3 TIME STEP CONTROL

A3.1 Initial Time Value and User-Controlled Time Step, 200

This card is optional. See the description of each word on this card for the default values if this card is not entered.

- W1(R) Initial time. If not entered, the simulation time at the start of the advancements is 0 for a NEW problem, the advancement time at the point of restart for a RESTART problem, or 0 for a RESTART problem in which the problem option switches from STDY-ST to TRANSNT or vice versa. If this card is entered, the simulation time is set to the entered value, which must be greater than or equal to 0. Setting the simulation time with this entry can be done on any NEW or RESTART problem but with most applications should only be used in NEW or RESTART problems that switch from the STDY-ST or TRANSNT options. See the cautions discussed in Section 2-11 for this capability. When needing to enter W2 but do not wish to enter a new initial time, enter -1.0, which is a flag to ignore this word.
- W2(I) Control variable number for user-controlled time step. This word is optional. A nonzero number specifies a control variable whose value is used for user-specified time step control. The time step will be determined from the maximum of the value of the control variable and the current minimum time step entered on cards 201 through 299. The time step will be equal to or less than this value and depends on the current requested time step, the mass error and other error checks, the Courant limit, and the time-step reduction options.

A3.2 Time Step Control, 201-299

At least one card of this series is required for NEW problems. If this series is entered for RESTART problems, it replaces the series from the problem being restarted. This series is not used for other problem types. Card numbers need not be consecutive.

- W1(R) Time end for this set (s). This quantity must increase with increasing card number. The default action for the code is to stop at the first time step that is greater than or equal to this time. This means the code will probably not stop at the end time specified on this card, but will stop at some time larger than the end time. If the user wants to stop exactly at the end time, then the user needs to prefix the end time with a negative (-) sign.
- W2(R) Minimum time step (s). The code starts out with this time step and increases by a factor 1.1 for the next step if the mass error is sufficiently low.
- W3(R) Maximum time step (s). This quantity is also called the requested time step.

W4(I) Control option. This word has the packed format ssdt. It is not necessary to input leading zeros.

ss-flag The ss digits, which represent a number from 0 through 15, are used to control the printed content of the major edits. The number is treated as a four-bit binary number. To decide how to set ss, pick some appropriate numbers from the set 0, 1, 2, 3, and 4, using the description below to decide which numbers to pick, then add the numbers together and set ss to that number.

ss = 0. NORMAL OUTPUT. If no bits are set (i.e., the number is 0), all the standard major printed output is given.

ss = 1. OMIT HEAT STRUCTURE TEMPERATURE BLOCK OUTPUT. If the first bit from the right is set (i.e., ss = 1 if the other bits are not set), the heat structure temperature block is omitted.

ss = 2. OMIT SECOND PORTION OF THE JUNCTION BLOCK OUTPUT. If the second bit from the right is set (i.e., ss = 2 if the other bits are not set), the second portion of the junction block is omitted.

ss = 4. OMIT THIRD AND FOURTH PORTIONS OF VOLUME BLOCK OUTPUT. If the third bit from the right is set (i.e., ss = 4 if the other bits are not set), the third and fourth portions of the volume block is omitted.

ss = 8. OMIT STATISTICS BLOCK. If the fourth bit from the right is set (i.e., ss = 8 if the other bits are not set), the statistics block is omitted.

d-flag The d digit, which represents a number from 0 through 7, can be used to obtain extra output at every hydrodynamic time step. The number is treated as a three-bit binary number. To decide how to set d, pick some appropriate numbers from the set 0, 1, 2, and 3, using the description below to decide which numbers to pick, then add the numbers together and set d to that number.

d = 0. NORMAL OUTPUT. If no bits are set (i.e., the number is 0), the standard output at the requested frequency using the maximum time step is obtained (see words 5 and 6 of this card). If the number is nonzero, output is obtained at each successful time step; and the bits indicate which output is obtained. These options should be used carefully, since considerable output can be generated.

d = 1. MAJOR EDITS EVERY TIME STEP. If the first bit from the right is set (i.e., d = 1 if the other bits are not set), major edits are obtained every successful time step.

$\underline{d} = 2$. MINOR EDITS EVERY TIME STEP. If the second bit from the right is set (i.e., $\underline{d} = 2$ if the other bits are not set), minor edits are obtained every successful time step.

$\underline{d} = 4$. PLOT RECORDS EVERY TIME STEP. If the third bit from the right is set (i.e., $\underline{d} = 4$ if the other bits are not set), plot records are written every successful time step. As a practical example, assume you made a run with $\underline{d} = 0$ and discovered that the plot points were too far apart to tell what was happening in the transient. One solution would be to decrease the plot frequency W5, to 1 to get a plot point for each requested time step and decrease the maximum time step size to get the desired detail in the plot. An easier solution would be to set $\underline{d} = 4$ and get plot points out each successful time step, even if they are smaller than the requested time step. For example, if $\underline{tt} = 3$, then changing W4 from a 3 to a 403 would do the trick.

tt-flag

The $\underline{tt}$ digits, which represent a number from 0 through 63, are used to control the time step. The number is treated as a six-bit binary number. The effect of no bits being set, i.e., 0 being entered, and the effect of each bit are first described followed by the recommended combination of bits. To decide how to set $\underline{tt}$, pick some appropriate numbers from the set 0, 1, 2, 4, 8, 16, and 32 using the description below to decide which numbers to pick, then add the numbers together and set $\underline{tt}$ to that number.

$\underline{tt} = 0$. NORMAL OUTPUT. If no bits are set (i.e., the number is 0), no error estimate time step control is used, and the maximum time step is attempted for both hydrodynamic and heat structure advancement. The hydrodynamic time step, however, is reduced to the material Courant limit and further to the minimum time step for cases such as water property failures.

$\underline{tt} = 1$. MASS ERROR CONTROLS THE TIME STEP. If the first bit from the right is set (i.e., $\underline{tt} = 1$ if no other bits are set), the hydrodynamics advancement, in addition to the time step control when no bits are set, uses a mass error analysis to control the time step between the minimum and maximum time step.

$\underline{tt} = 2$. HYDRODYNAMIC TIME STEP SAME AS HEAT CONDUCTION TIME STEP. If the second bit from the right is set (i.e., $\underline{tt} = 2$ if the other bits are not set), the heat conduction/transfer time step is the same as the hydrodynamic time step; if the second bit from the right is not set, the heat conduction/transfer time step uses the maximum time step.

$\underline{tt} = 4$. HEAT CONDUCTION AND HYDRODYNAMICS COUPLED IMPLICITLY. If the third bit from the right is set (i.e., $\underline{tt} = 4$ if the other bits are not set), the heat conduction/transfer and hydrodynamics are coupled implicitly; if the third bit from the right is not set, the heat conduction/transfer and hydrodynamic advancements are done separately and the information between the models is coupled explicitly.

$tt = 8$. NEARLY-IMPLICIT TIME ADVANCEMENT. If the fourth bit from the right is set (i.e., $tt = 8$ if the other bits are not set), the nearly-implicit scheme is used to advance the hydrodynamics; if the fourth bit from the right is not set, the semi-implicit scheme is used to advance the hydrodynamics.

$tt = 16$. STEADY-STATE CONVERGENCE NOT CHECKED. If the fifth bit from the right is set (i.e., $tt = 16$ if the other bits are not set), the test for convergence of a steady-state calculation is not made; if the fifth bit from the right is not set, the test for convergence of a steady-state calculation is made.

$tt = 32$. AUTOMATIC SELECTION BETWEEN SEMI- AND NEARLY-IMPLICIT TIME ADVANCEMENT. If the sixth bit from the right is set (i.e., $tt = 32$ if the other bits are not set), the on-line algorithm selection of time migration is used to advance the hydrodynamics. The semi-implicit scheme will be used when the time step is below the Courant limit and the nearly-implicit will be used when a large time step can be taken.

We recommend not using tt equal to 0 except for special testing situations. The use of tt equal to 1 is possible if the maximum time step is kept sufficiently small to ensure that the explicit connection between the heat conduction/transfer and hydrodynamics calculations remains stable. If there is any doubt, use tt equal to or greater than 3 (sets first bit and second bit). Using tt equal to 3 or 11 specifies the semi-implicit or the nearly-implicit advancement scheme, respectively, with both schemes using time step control, the heat conduction and hydrodynamics use the same time step, and the heat conduction/transfer and hydrodynamics are advanced separately. Using tt equal to 7 or 15 specifies the same features as tt equal to 3 or 11 and, in addition, specifies the implicit advancement of the heat conduction/transfer with the hydrodynamics. We recommend the nearly-implicit scheme during a steady-state and/or self-initialization case problem where the time step is limited by the material Courant limit. The nearly-implicit scheme can also be used during slower phases of a transient problem, though we advise the user that the answers may change somewhat from the semi-implicit scheme answers (depending on the time step size). The nearly-implicit advancement scheme is still under development and assessment; most of the verification and assessment for the code has been done with the semi-implicit advancement scheme. We did not recommend use of the implicit coupling of the heat conduction/transfer and hydrodynamics in prior versions since the implicit coupling was only partially implemented. With the implicit coupling now complete, we encourage the use of tt equal to 7 or 15. Users should be cautioned that the implicit coupling is a recent addition to RELAP5 and is still under assessment. When using the implicit coupling, the heat conduction time step must be the same as the hydrodynamic time step. This requirement is currently not enforced by the coding. In steady-state calculations, setting the fifth bit (adding 16) for the early part of the run can ensure the calculation runs to a user-specified time; then, setting the fifth bit off can allow the steady-state convergence to test control the termination of the problem.

- W5(I) Minor edit and plot frequency. This is the number of maximum or requested time-advances per minor edit and write of plot information. Volume V, Section 3.1.3.1 contains some recommendations for this and the following two words.
- W6(I) Major edit frequency. This is the number of requested time-advances per major edit.
- W7(I) Restart frequency. This is the number of requested time-advances per write of restart information.

A4 MINOR EDITS, 301-399

These cards are optional for NEW and RESTART problems, are required for a REEDIT problem, and are not allowed for PLOT and STRIP problems. If these cards are not present, no minor edits are printed. If these cards are present, minor edits are generated, and the order of the printed quantities is given by the card number of the request card. One request is entered per card, and the card numbers need not be consecutive. For RESTART problems, if these cards are entered, all the cards from the previous problem are deleted.

W1(A) Variable code (alphanumeric).

W2(I) Parameter (numeric).

Words 1 and 2 form the variable request code pair. The quantities that can be edited and the input required are listed below. For convenience, quantities that can be used in plotting requests, in trip specifications, as search variables in tables, and as operands in control statements are listed. Units for the quantities are also given. Interactive input variables described in Section A6 can be used in batch or interactive jobs in the same manner as the variables listed below. The parameter for interactive input variables is 1000000000. Quantities compared in variable trips must have the same units, and input to tables specified by variable request codes must have the specified units. The quantities are listed in alphabetical order within each section.

The underlined quantities without an asterisk in Section A4.1 through Section A4.9 are always written to the restart-plot file (RSTPLT). Underlined quantities followed by an asterisk (*) have only some of the quantities written to the restart-plot file and the text will indicate which quantities are written. The quantities that are not underlined or some of the quantities underlined that are followed with an asterisk are written to the restart-plot file only if requested on a 2080XXXX card as described in Section A4.10.

A4.1 General Quantities

<u>Variable</u>	<u>Description</u>
<u>COUNT</u>	Current attempted advancement count number. The parameter is 0.
<u>CPUTIME</u>	Current CPU time for this problem (s). The parameter is 0.
<u>DT</u>	Current time step (s). The parameter is 0.
<u>DTCRNT</u>	Current Courant time step (s). The parameter is 0.
<u>EMASS</u>	Estimate of mass error in all the systems (kg, lb). The parameter is 0.
ERRMAX	Current estimate of the truncation mass error fraction. The parameter is 0. This is the maximum of the two types of computed mass error (ϵ_m or ϵ_{rms}) discussed in Volume I, Section 8.

<u>EXTSNN</u>	Extra system variables where NN goes from 01 to 20. These variables are output to the restart-plot file when RELAP5 is built using the “-e” flag on the “configure” command line. These variables are useful for debugging the code and for making temporary changes to the code. There is a similar set of variables for volumes called EXTVNN and for junctions called EXTJNN. Parameter is system number N. Only the extra volume and junction variables are available for minor edits and for use in a control system.
MAXVME	Maximum value of all the individual volume mass errors (kg, lb). Parameter is system number N. See VOLMER under Volume Quantities for the individual volume mass errors.
NULL	Specifies null field. Allowed only on trip cards. The parameter is 0.
<u>RKTPOW3D</u>	Total reactor power for the RELAP5/PARCS 3-D kinetics coupled code (W). The parameter is 0. It is the sum of the fission and decay powers (W) for all the core heat structures. This variable is written to the restart-plot file only when card 1 option 88 is used.
STDTRN	Steady-state/transient flag. The parameter is 0. For steady-state, the value is 0.0. For transient, the value is 1.0.
SYSMER	Estimate of mass error in system N (kg, lb) [= SYSTM C - SYSTM S]. Parameter is system number N.
SYSMERB	Estimate of boron mass error in system N (kg, lb)[= SYSTMBC - SYSTM B]. Parameter is system number N.
SYSMERF	Estimate of liquid mass error in system N (kg, lb)[= SYSTMFC - SYSTM F]. Parameter is system number N.
SYSMERG	Estimate of vapor mass error in system N (kg, lb)[= SYSTMGC - SYSTM G]. Parameter is system number N.
SYSMERN	Estimate of noncondensable mass error in system N (kg, lb)[= SYSTMNC - SYSTM N]. Parameter is system number N.
SYSTM B	Total mass of boron in system N (kg, lb) from state equation. Parameter is system number N
SYSTM C	Total mass of steam, water, and noncondensable in system N (kg, lb) from continuity equation. Parameter is system number N.
SYSTM F	Total mass of liquid in system N (kg, lb) from state equation. Parameter is system number N.

SYSTMFC	Total mass of liquid in system N (kg, lb) from continuity equation. Parameter is system number N.
SYSTMG	Total mass of vapor (steam plus noncondensable) in system N (kg, lb) from state equation. Parameter is system number N.
SYSTMGC	Total mass of vapor (steam plus noncondensable) in system N (kg, lb) from continuity equation. Parameter is system number N.
SYSTEMN	Total mass of noncondensables in system N (kg, lb) from state equation. Parameter is system number N.
SYSTEMNC	Total mass of noncondensables in system N (kg, lb) from continuity equation. Parameter is system number N.
SYSTEMS	Total mass of steam, water, and noncondensable in system N (kg, lb) from state equation. Parameter is system number N.

The following table illustrates the relationships between the system mass from the continuity equation, the system mass from the state equation, and the system mass error.

Table 4.1-1 Relationships between various system masses and mass error.

System mass from continuity equation	System mass from state equation (density times volume)	System mass error (continuity equation mass - state equation mass)	Quantity
STSTMC	STSTMS	SYSMER	Total
SYSTFC	SYSTMF	SYSMERF	Liquid
SYSTGC	SYSTMG	SYSMERG	Vapor
SYSTEMBC	SYSEMB	SYSMERB	Boron
SYSTEMNC	SYSTEMN	SYSMERN	Noncondensables

TESTDA An array, testda, of one hundred quantities, [REAL TESTDA(100)] has been defined for the convenience of program developers. This entry with a parameter ranging from 1 through 100 selects TESTDA(parameter). The testda array is initially set to 0.0, and programming must be inserted to set TESTDA values. The usual purpose of this capability is to allow a simple method for debug information to be printed in minor edits or to be plotted.

TIME Time (s). The parameter is 0. This request cannot be used for minor edit requests.

TIMEOF	Time of trip occurring (s). The parameter is the trip number. This request is allowed only on trip cards.
<u>TMASS</u>	Total mass of water, steam, and noncondensables in all the systems (kg, lb). The parameter is 0.
TOTHGEN	Total hydrogen generation from metal-water reactions in all the systems (kg, lb). The parameter is 0.
VNMAXVME	Volume number of the volume having the maximum volume mass error (see MAXVME). Parameter is system number N. It is edited in the format of CCC.NN.

A4.2 Component Quantities

The quantities listed below are unique to certain components; for example, a pump velocity can only be requested for a pump component. The parameter is the component number, i.e., the three-digit number CCC used in the input cards.

<u>Variable</u>	<u>Description</u>
ACPGTG	Accumulator vapor specific heat, C_p , at vapor temperature (J/kg•K, Btu/lb•°F).
ACPNIT	Accumulator noncondensable specific heat, C_p , at vapor temperature (J/kg•K, Btu/lb•°F).
<u>ACQTANK</u>	Total energy transport to the gas by heat and mass transfer in the accumulator (W, Btu/s).
<u>ACRHON</u>	Accumulator noncondensable density (kg/m ³ , lb/ft ³).
<u>ACTTANK</u>	Mean accumulator tank wall metal temperature (K, °F).
<u>ACVDM</u>	Gas volume in the accumulator tank, standpipe, and surge line (m ³ , ft ³).
ACVGTG	Accumulator vapor specific heat, C_v , at vapor temperature (J/kg•K, Btu/lb•°F).
<u>ACVLIQ</u>	Liquid volume in the accumulator tank, standpipe, and surge line (m ³ , ft ³).
AHFGTF	Accumulator heat of vaporization at liquid temperature (J/kg, Btu/lb).
AHFGTG	Accumulator heat of vaporization at vapor temperature (J/kg, Btu/lb).
AHFTG	Accumulator liquid enthalpy at vapor temperature (J/kg, Btu/lb).
AHGTG	Accumulator vapor enthalpy at liquid temperature (J/kg, Btu/lb).

AVGTG	Accumulator specific volume at vapor temperature (m^3/kg , ft^3/lb).
AVISCN	Accumulator noncondensable viscosity ($\text{kg}/\text{m}\cdot\text{s}$, $\text{lb}/\text{ft}\cdot\text{s}$).
BETAV	Accumulator steam saturation coefficient of expansion (K^{-1} , $^{\circ}\text{F}^{-1}$).
CDIM	GE mechanistic dryer critical inlet moisture quality.
DIM	GE mechanistic dryer inlet moisture quality.
DMGDT	Accumulator/time rate of change in dome vapor mass (kg/s , lb/s).
GDRY	GE mechanistic separator capacity factor.
OMEGA	Inertial valve disk angular velocity (rad/s , rev/min).
<u>PMPHEAD</u>	Pump head in the pump component (Pa , lb_f/in^2).
PMPMT	Pump motor torque ($\text{N}\cdot\text{m}$, $\text{lb}_f\cdot\text{ft}$).
PMPNRT	Calculated pump inertia ($\text{kg}\cdot\text{m}^2$, $\text{lb}\cdot\text{ft}^2$).
<u>PMPTRQ</u>	Pump torque in the pump component ($\text{N}\cdot\text{m}$, $\text{lb}_f\cdot\text{ft}$).
<u>PMPVEL</u>	Pump rotational velocity in the pump component (rad/s , rev/min).
PRZLVL	Pressurizer level in the PRIZER component (m , ft).
THETA	Inertial valve disk angular position (deg).
<u>TUREFF</u>	Efficiency of the turbine component.
<u>TURPOW</u>	Power developed in the turbine component (W , Btu/s).
<u>TURTRQ</u>	Torque developed in the turbine component ($\text{N}\cdot\text{m}$, $\text{lb}_f\cdot\text{ft}$).
<u>TURVEL</u>	Rotational velocity of the turbine component (rad/s , rev/min).
<u>VLVAREA</u>	Ratio of the current valve physical area to the junction area. The junction area is the fully open valve physical area for the smooth area option and the minimum of the two connecting volumes for the abrupt area change.

<u>VLVSTEM</u>	Ratio of the current valve stem position to the fully open valve stem position for the motor and servo valves when the normalized stem position option is used. For the motor and servo valves when the normalized area option is used and for all the other valves, this is the ratio of the current valve physical area to the fully open valve physical area.
XCO	GE mechanistic separator liquid carryover quality.
XCU	GE mechanistic separator vapor carryunder quality.
XI	GE mechanistic separator inlet quality.

A4.3 Volume Quantities

For most of the following variable codes, the parameter is the volume number, i.e., the nine-digit number CCCNN0000 printed in the major edit. The parameter is CCC010000 for a single-volume; CCC010000 for a time-dependent volume; CCCNN0000 for a volume in a pipe component ($01 \leq NN \leq 99$); CCC010000 for the volume in a branch, separator, jetmixer, turbine, or ECC mixer component; CCC010000 for the volume in a pump component; and CCC010000 for the volume in an accumulator component. Some of the quantities are associated with the coordinate directions in the volume, and these quantities are computed for each coordinate direction in use. The parameter for the coordinate direction-related quantities is the volume number plus F, where F is described below. The quantities requiring the volume number plus F are so identified.

Every volume has at least one coordinate direction, and some volumes may have up to three orthogonal coordinate directions. Each coordinate has an inlet face and an outlet face. Faces are numbered 1 through 6, where faces 1 and 2 are the inlet and outlet faces associated with coordinate 1 (or x), respectively, faces 3 and 4 are inlet and outlet faces associated with coordinate 2 (or y), and faces 5 and 6 are inlet and outlet faces associated with coordinate 3 (or z). All volumes use coordinate 1. The quantity F to be added to the volume number to form the parameter used with coordinate direction related quantities is 0 or the face number. When F is 0 (i.e., just the volume number) or 1 or 2, the volume velocity is for coordinate 1. When F is 3 or 4, the volume velocity is for coordinate 2, and when F is 5 or 6, the volume velocity is for coordinate 3. For the underlined quantities followed by an asterisk in the list below, the coordinate-dependent quantities for coordinate 1 are automatically written to the restart-plot records using the parameter with F equal to 0. The other coordinate-dependent quantities can be written to the plot records using the 2080XXXX card series described in Section A4.10. Input checks are made to ensure the parameter specifies a volume coordinate direction that is in use.

<u>Variable</u>	<u>Description</u>
AVOL	Area of the volume (m^2 , ft^2); the parameter is the volume number plus F.
BETAFF	Liquid isobaric coefficient of the thermal expansion, β_f , bulk conditions (K^{-1} , $^{\circ}\text{F}^{-1}$).
BETAGG	Vapor isobaric coefficient of the thermal expansion, β_g , bulk conditions (K^{-1} , $^{\circ}\text{F}^{-1}$).

<u>BORON</u>	Spatial boron density, ρ_b (kg/m ³ , lb/ft ³). Boron density is volume liquid fraction (α_f) times liquid density (ρ_f) times boron concentration (C_b). Boron concentration (kg boron/kg liquid) is used for hydrodynamic input, and boron density (kg boron/m ³ total) is used for minor edits and plots.
CSUBPF	Liquid specific heat, C_{pf} , bulk conditions (J/kg•K, Btu/lb•°F).
CSUBPG	Vapor specific heat, C_{pg} , bulk conditions (J/kg•K, Btu/lb•°F).
DRFDP	Partial derivative of ρ_f with respect to P (s ² /m ² , s ² /ft ²).
DRFDUF	Partial derivative of ρ_f with respect to U_f (kg•s ² /m ⁵ , lb•s ² /ft ⁵).
DRGDP	Partial derivative of ρ_g with respect to P (s ² /m ² , s ² /ft ²).
DRGDUG	Partial derivative of ρ_g with respect to U_g (kg•s ² /m ⁵ , lb•s ² /ft ⁵).
DRGDXA	Partial derivative of ρ_g with respect to X_n (kg/m ³ , lb/ft ³).
DTDP	Partial derivative of T^s with respect to P (K/Pa, in ² •°F/lb _f).
DTDUG	Partial derivative of T^s with respect to U_g (s ² •K/m ² , s ² •°F/ft ²).
DTDXA	Partial derivative of T^s with respect to X_n (K, °F).
DTFDP	Partial derivative of T_f with respect to pressure (K/Pa, in ² •°F/lb _f).
DTFDUF	Partial derivative of T_f with respect to U_f (s ² •K/m ² , s ² •°F/ft ²).
DTGDP	Partial derivative of T_g with respect to P (K/Pa, in ² •°F/lb _f).
DTGDUG	Partial derivative of T_g with respect to U_g (s ² •K/m ² , s ² •°F/ft ²).
DTGDXA	Partial derivative of T_g with respect to X_n (K, °F).
ENTROPYF	Liquid specific entropy, s_f , at bulk conditions (J/kg•K, Btu/lb•°F).
ENTROPYG	Vapor specific entropy, s_g , at bulk conditions (J/kg•K, Btu/lb•°F).

<u>EXTVNN</u>	Extra volume variables where NN goes from 01 to 20. These variables are output to the restart-plot file when RELAP5 is built using the “-e” flag on the “configure” command line. These variables are useful for debugging the code and for making temporary changes to the code. There is a similar set of variables for junctions called EXTJNN and for systems called EXTSNN. Only the extra volume and junction variables are available for minor edits and for use in a control system.
<u>FLOREG</u>	Flow regime number. A chart showing the meaning of each number is shown in Volume II, Section 2.
FWALF	Liquid wall frictional drag coefficient ($\text{kg/m}^3\cdot\text{s}$, $\text{lb/ft}^3\cdot\text{s}$). The parameter is the volume number plus F.
FWALG	Vapor wall frictional drag coefficient ($\text{kg/m}^3\cdot\text{s}$, $\text{lb/ft}^3\cdot\text{s}$). The parameter is the volume number plus F.
GAMMAC	For explicit coupling of heat conduction/transfer and hydrodynamics, this is 0. For implicit coupling of heat conduction/transfer and hydrodynamics, this is the mass transfer rate per unit volume at the vapor/liquid interface in the boundary layer near the wall for condensation ($\text{kg/m}^3\cdot\text{s}$, $\text{lb/ft}^3\cdot\text{s}$).
GAMMAI	Mass transfer rate per unit volume at the vapor/liquid interface in the bulk fluid for vapor generation/condensation ($\text{kg/m}^3\cdot\text{s}$, $\text{lb/ft}^3\cdot\text{s}$).
GAMMAW	For explicit coupling of heat conduction/transfer and hydrodynamics, this is the mass transfer rate per unit volume at the vapor/liquid interface in the boundary layer near the wall for vapor generation/condensation ($\text{kg/m}^3\cdot\text{s}$, $\text{lb/ft}^3\cdot\text{s}$). For implicit coupling of heat conduction/transfer and hydrodynamics, this is the mass transfer rate per unit volume at the vapor/liquid interface in the boundary layer near the wall for vapor generation ($\text{kg/m}^3\cdot\text{s}$, $\text{lb/ft}^3\cdot\text{s}$).
HGF	Direct heating heat transfer coefficient per unit volume ($\text{W/m}^3\cdot\text{K}$, $\text{Btu/s}\cdot\text{ft}^3\cdot^\circ\text{F}$).
HIF	Liquid side interfacial heat transfer coefficient per unit volume ($\text{W/m}^3\cdot\text{K}$, $\text{Btu/s}\cdot\text{ft}^3\cdot^\circ\text{F}$).
HIG	Vapor side interfacial heat transfer coefficient per unit volume ($\text{W/m}^3\cdot\text{K}$, $\text{Btu/s}\cdot\text{ft}^3\cdot^\circ\text{F}$).
HSTEAM	Steam specific enthalpy at bulk conditions using partial pressure of steam (J/kg , Btu/lb).
HVMIX	Enthalpy of the liquid and vapor (J/kg , Btu/lb).
KAPPAF	Liquid isothermal compressibility, κ_f , at bulk conditions (Pa^{-1} , in^2/lb_f).

KAPPAG	Vapor isothermal compressibility, κ_g , at bulk conditions (Pa^{-1} , in^2/lb_f).
<u>P</u>	Volume pressure (Pa , lb_f/in^2).
PECLTV	Peclet number.
<u>PPS</u>	Steam partial pressure (Pa , lb_f/in^2).
<u>Q</u>	Total volume heat source from the wall and direct moderator heating to liquid and vapor (W , Btu/s). This variable request is the same as “Q.wall.tot.” in the major edits.
<u>QUALA</u>	Volume noncondensable mass fraction. The ratio of the mass of the noncondensable gas to the total mass of the vapor phase.
QUALAIR	Volume noncondensable mass fraction for AIR. This is the ratio of the mass of AIR to the total mass of the noncondensable gas. This quantity is only available if AIR is specified on the 110 card.
QUALAR	Volume noncondensable mass fraction for ARGON. This is the ratio of the mass of ARGON to the total mass of the noncondensable gas. This quantity is only available if ARGON is specified on the 110 card.
<u>QUALE</u>	Volume equilibrium quality. This quality uses phasic enthalpies and mixture quality, with the mixture enthalpy calculated using the flow quality.
QUALHE	Volume noncondensable mass fraction for HELIUM. This is the ratio of the mass of HELIUM to the total mass of the noncondensable gas. This quantity is only available if HELIUM is specified on the 110 card.
QUALHY	Volume noncondensable mass fraction for HYDROGEN. This is the ratio of the mass of HYDROGEN to the total mass of the noncondensable gas. This quantity is only available if HYDROGEN is specified on the 110 card.
QUALKR	Volume noncondensable mass fraction for KRYPTON. This is the ratio of the mass of KRYPTON to the total mass of the noncondensable gas. This quantity is only available if KRYPTON is specified on the 110 card.
QUALNI	Volume noncondensable mass fraction for NITROGEN. This is the ratio of the mass of NITROGEN to the total mass of the noncondensable gas. This quantity is only available if NITROGEN is specified on the 110 card.
<u>QUALS</u>	Volume static quality.

Volume Quantities

QUALXE	Volume noncondensable mass fraction for XENON. This is the ratio of the mass of XENON to the total mass of the noncondensable gas. This quantity is only available if XENON is specified on the 110 card.
<u>QWG</u>	Volume heat source from the wall and direct moderator heating to vapor (W, Btu/s). This variable request is the same as “Qwg.wall.gas.” in the major edits.
<u>RHO</u>	Total density (kg/m^3 , lb/ft^3).
<u>RHOF</u>	Liquid density ρ_f (kg/m^3 , lb/ft^3).
<u>RHOG</u>	Vapor density ρ_g (kg/m^3 , lb/ft^3).
RHOM	Total density for the mass error check (kg/m^3 , lb/ft^3).
SATHF	Liquid specific enthalpy at saturation conditions using partial pressure of steam (J/kg, Btu/lb).
SATHG	Steam specific enthalpy at saturation conditions using partial pressure of steam (J/kg, Btu/lb).
<u>SATTEMP</u>	Volume saturation temperature based on the partial pressure of steam (K, °F). For saturation temperature based on total pressure, use TSATT.
SIGMA	Surface tension (N/m, lb_f/ft).
<u>SOUNDE</u>	Volume sonic velocity (m/s, ft/s).
<u>TEMPF</u>	Volume liquid temperature T_f (K, °F).
<u>TEMPG</u>	Volume vapor temperature T_g (K, °F).
THCONF	Liquid thermal conductivity ($\text{W/m}\cdot\text{K}$, $\text{Btu/s}\cdot\text{ft}\cdot^\circ\text{F}$).
THCONG	Vapor thermal conductivity ($\text{W/m}\cdot\text{K}$, $\text{Btu/s}\cdot\text{ft}\cdot^\circ\text{F}$).
TIENGV	Total internal energy (of both phases and noncondensables) in volume (J, Btu).
TMASSBV	Boron mass in volume from state equation (kg, lb).
TMASSFV	Liquid mass in volume from state equation (kg, lb).

TMASSGV	Vapor mass (includes noncondensables) in volume from state equation (kg, lb).
TMASSNV	Noncondensable mass in volume from state equation [$V \alpha_g \rho_g x_n$] (kg, lb).
TMASSV	Total fluid mass (includes both phases) in volume from state equation (kg, lb).
TSATT	Saturation temperature corresponding to total pressure (K, °F).
TSPINF	Liquid spinodal temperature (K, °F).
TSPING	Vapor spinodal temperature (K, °F).
<u>UF</u>	Liquid specific internal energy (J/kg, Btu/lb).
<u>UG</u>	Vapor specific internal energy (J/kg, Btu/lb).
<u>USATE</u>	Liquid specific internal energy at saturation (J/kg, Btu/lb).
<u>USATG</u>	Vapor specific internal energy at saturation (J/kg, Btu/lb).
<u>VAPGEN</u>	Total mass transfer rate per unit volume at the vapor/liquid interface in the bulk fluid for vapor generation/condensation and in the boundary layer near the wall for vapor generation/condensation (kg/m ³ •s, lb/ft ³ •s). [GAMMAI + GAMMAW for explicit heat transfer coupling].
<u>VELF*</u>	Volume oriented liquid velocity (m/s, ft/s); the parameter is the volume number plus F.
<u>VELG*</u>	Volume oriented vapor velocity (m/s, ft/s); the parameter is the volume number plus F.
VLEV	Velocity of level in volume (m/s, ft/s).
VISCF	Liquid viscosity (kg/m•s, lb/ft•s).
VISCG	Vapor viscosity (kg/m•s, lb/ft•s).
<u>VOIDF</u>	Volume liquid fraction.
<u>VOIDG</u>	Volume vapor fraction (void fraction).
VOIDLA	Volume vapor fraction above the level.
VOIDLB	Volume vapor fraction below the level.

Volume Quantities

VOLLEV	Elevation of the level in the volume (m, ft).
VOLMAS	Volume mass in volume from continuity equation (kg, lb).
VOLMASB	Volume boron mass from continuity equation (kg, lb).
VOLMASF	Volume liquid mass from continuity equation (kg, lb).
VOLMASG	Volume vapor mass from continuity equation (kg, lb).
VOLMASN	Volume noncondensable mass from continuity equation (kg, lb).
VOLMER	Volume mass error (kg, lb) [= VOLMAS - TMASSV]. The maximum value for each system can be requested using the variable MAXVME, and the volume number that the maximum occurs in can be requested using VNMAXVME. Both of these variables are under General Quantities above.
VOLMERB	Volume boron mass error (kg, lb) [= VOLMASB- TMASSBV].
VOLMERF	Volume liquid mass error (kg, lb) [= VOLMASF - TMASSFV].
VOLMERG	Volume vapor mass error (kg, lb) [= VOLMASG - TMASSGV].
VOLMERN	Volume noncondensable mass error (kg, lb) [= VOLMASN - TMASSNV].

The following table illustrates the relationships between the volume mass from the continuity equation, the volume mass from the state equation, and the volume mass error.

Table 4.3-1 Relationships between various volume masses and mass error.

Volume mass from continuity equation	Volume mass from state equation (density times volume)	Volume mass error (continuity equation mass - state equation mass)	Quantity
VOLMAS	TMASSV	VOLMER	Total
VOLMASF	TMASSFV	VOLMERF	Liquid
VOLMASG	TMASSGV	VOLMERG	Vapor
VOLMASB	TMASSBV	VOLMERB	Boron
VOLMASN	TMASSNV	VOLMERN	Noncondensables

VVOL Volume of the volume (m³, ft³).

A4.4 Junction Quantities

For the following variable request codes, the parameter is the junction number, i.e., the nine-digit number CCCNN0000 printed in the major edit. The parameter is CCC000000 for a single-junction; CCC000000 for a time-dependent junction; CCCMM0000 for a junction in a pipe component ($01 \leq MM \leq 99$); CCCMM0000 for a junction in a branch, separator, jetmixer, turbine, or ECC mixer component ($01 < MM \leq 9$); CCC000000 for a valve junction; CCC010000 for the inlet junction in a pump component; CCC020000 for the outlet junction in a pump component; CCCIINN00 for a junction in the multiple-junction component ($01 \leq II \leq 99$, $01 \leq NN \leq 99$); and CCC010000 for the junction in an accumulator component.

<u>Variable</u>	<u>Description</u>
BORONJ	Junction boron density (kg/m^3 , lb/ft^3).
C0J	Junction distribution coefficient. The 0 in C0J is the number zero and not the upper case letter O.
<u>CHOKEF</u>	Junction choking flag. The value is 0 if the flow is not choked, and is 1 if the flow is choked.
<u>EXTJNN</u>	Extra junction variables where NN goes from 01 to 20. These variables are output to the restart-plot file when RELAP5 is built using the “-e” flag on the “configure” command line. These variables are useful for debugging the code and for making temporary changes to the code. There is a similar set of variables for volumes called EXTVNN and for systems called EXTSNN. Only the extra volume and junction variables are available for minor edits and for use in a control system.
FIJ	Interphase friction Σ coefficient ($\text{N}\cdot\text{s}^2/\text{m}^5$, $\text{lb}_\text{f}\cdot\text{s}^2/\text{ft}^5$). This is the variable C_i in Volume I in this manual.
FJUNFT	Total forward user input form loss coefficient for irreversible losses, including Re dependence (dimensionless).
FJUNRT	Total reverse user input form loss coefficient for irreversible losses, including Re dependence (dimensionless).
FLENTH	Total enthalpy flow in junction (includes both phases and noncondensables) (J/s, Btu/s).
FLENTHF	Liquid enthalpy flow in junction (J/s, Btu/s).
FLENTHG	Vapor enthalpy flow in junction (J/s, Btu/s).

Junction Quantities

FLORGJ	Junction flow regime number. A chart showing the meaning of each number is shown in Volume II, Section 2, page 14.
FORMFJ	Liquid abrupt area change model form loss factor (dimensionless).
FORMGJ	Vapor abrupt area change model form loss factor (dimensionless).
FWALFJ	Non-dimensional liquid wall friction coefficient (dimensionless).
FWALGJ	Non-dimensional vapor wall friction coefficient (dimensionless).
IREGJ	Vertical bubbly/slug flow junction flow regime number. A chart showing the meaning of each number is shown in Volume II, Section 2, page 15.
MFLOWBJ	Boron flow rate (kg/s, lb/s).
MFLOWFJ	Liquid flow rate (kg/s, lb/s).
MFLOWGJ	Vapor flow rate (kg/s, lb/s).
<u>MFLOWJ</u>	Combined liquid and vapor flow rate (kg/s, lb/s).
MFLOWNJ	Noncondensable flow rate (kg/s, lb/s).
<u>QUALAJ</u>	Junction noncondensable mass fraction.
QUALJAIR	Junction noncondensable mass fraction for AIR. This is the ratio of the mass of AIR to the total mass of the noncondensable gas. This quantity is only available if AIR is specified on the 110 card.
QUALJAR	Junction noncondensable mass fraction for ARGON. This is the ratio of the mass of ARGON to the total mass of the noncondensable gas. This quantity is only available if ARGON is specified on the 110 card.
QUALJHE	Junction noncondensable mass fraction for HELIUM. This is the ratio of the mass of HELIUM to the total mass of the noncondensable gas. This quantity is only available if HELIUM is specified on the 110 card.
QUALJHY	Junction noncondensable mass fraction for HYDROGEN. This is the ratio of the mass of HYDROGEN to the total mass of the noncondensable gas. This quantity is only available if HYDROGEN is specified on the 110 card.

QUALJKR	Junction noncondensable mass fraction for KRYPTON. This is the ratio of the mass of KRYPTON to the total mass of the noncondensable gas. This quantity is only available if KRYPTON is specified on the 110 card.
QUALJNI	Junction noncondensable mass fraction for NITROGEN. This is the ratio of the mass of NITROGEN to the total mass of the noncondensable gas. This quantity is only available if NITROGEN is specified on the 110 card.
QUALJXE	Junction noncondensable mass fraction for XENON. This is the ratio of the mass of XENON to the total mass of the noncondensable gas. This quantity is only available if XENON is specified on the 110 card.
<u>RHOJFJ</u>	Junction liquid density (kg/m^3 , lb/ft^3).
<u>RHOJGJ</u>	Junction vapor density (kg/m^3 , lb/ft^3).
SONICJ	Junction sound speed (m/s, ft/s). This speed is based on the physical junction area. It does not include the effects of the throat ratio and the discharge coefficients. In the Henry-Fauske choking model, this is the value before the 10% new, 90% old time smoothing is applied.
<u>UFJ</u>	Junction liquid specific internal energy (J/kg, Btu/lb).
<u>UGJ</u>	Junction vapor specific internal energy (J/kg, Btu/lb).
<u>VELFJ</u>	Junction liquid velocity (m/s, ft/s). This velocity is based on the junction area A_j , which is discussed in Volume II, Section 2.4.
<u>VELGJ</u>	Junction vapor velocity (m/s, ft/s). This velocity is based on the junction area A_j , which is discussed in Volume II, Section 2.4.
VGJJ	Vapor drift velocity (m/s, ft/s).
<u>VOIDFJ</u>	Junction liquid fraction.
<u>VOIDGJ</u>	Junction vapor fraction (void fraction).
VOIDJ	Junction vapor fraction (void fraction) used in the interphase friction.
XEJ	Junction equilibrium quality. Used in the Henry-Fauske choking test for deciding between subcooled, two-phase, single phase vapor choking. Subcooled if $\text{XEJ} < 1.0\text{E-}6$, two-phase if $\text{XEJ} < 0.998$. Otherwise, single phase vapor choking.

A4.5 Heat Structure Quantities

For the request code, HTVAT, HTOXI, HTOXO, and HTHGEN, the parameter is the seven-digit heat structure number CCCG0NN. For the remaining codes, the parameter is the seven-digit heat structure number CCCG0NN with a two-digit number appended. For codes other than HTTEMP, HTVAT, HTOXI, and HTOXO, the appended number is 00 for the left boundary and 01 for the right boundary. For HTTEMP, the appended number is the mesh point number. For HTVAT, HTOXI, and HTOXO, omit the two appended digits and use only the seven digit number. Only the left and right surface temperatures are written by default in plot records on the RSTPLT file, and, thus, plot requests in plot-type problems and strip requests are limited to those temperatures unless the interior temperatures are forced to the RSTPLT file through the use of 2080XXXX cards.

<u>Variable</u>	<u>Description</u>
HFRAD	Radiation heat flux (W/m^2 , $\text{Btu/s}\cdot\text{ft}^2$). The parameter is the heat structure geometry number, CCCG0NN, with a two-digit number appended (00 for the left boundary and 01 for the right boundary).
HFTOT	Total heat flux, i.e., sum of HFRAD and HTRNR (W/m^2 , $\text{Btu/s}\cdot\text{ft}^2$). The parameter is the heat structure geometry number, CCCG0NN, with a two-digit number appended (00 for the left boundary and 01 for the right boundary).
<u>HTCHF</u>	Critical (maximum) heat flux (W/m^2 , $\text{Btu/s}\cdot\text{ft}^2$). The parameter is the heat structure geometry number, CCCG0NN, with a two-digit number appended (00 for the left boundary and 01 for the right boundary).
<u>HTCHFR</u>	Critical hat flux ratio (ratio of HTCHF to HTRNR. This variable can only be plotted and is set equal to the constant value of 12.34567 whenever HTCHF or HTRNR is equal to 0.0. The parameter is the heat structure geometry number, CCCG0NN, with a two-digit number appended (00 for the left boundary and 01 for the right boundary).
HTGAMW	Wall vapor generation rate per unit volume ($\text{kg/m}^3\cdot\text{s}$, $\text{lb/ft}^3\cdot\text{s}$). The parameter is the heat structure geometry number CCCG0NN with a two-digit number appended (00 for the left boundary and 01 for the right boundary).
HTHGEN	Total hydrogen generation at this heat structure (kg, lb). The parameter is the seven-digit heat structure number.
<u>HTHTC</u>	Heat transfer coefficient ($\text{W/m}^2\cdot\text{K}$, $\text{Btu/s}\cdot\text{ft}^2\cdot^\circ\text{F}$) The parameter is the heat structure geometry number, CCCG0NN, with a two-digit number appended (00 for the left boundary and 01 for the right boundary).

HTMODE	Boundary heat transfer mode number. The mode number indicates which heat transfer regime is currently in effect. The parameter is the seven-digit heat structure geometry number, CCCG0NN, with a two-digit number appended (00 for the left boundary and 01 for the right boundary). This same quantity is valid for the reflood heat structures. A chart showing the meaning of each number is shown in Volume II, Section 3.2.
HTOXI	Inside oxide layer thickness (m, ft). The parameter is the seven-digit heat structure number.
HTOXO	Outside oxide layer thickness (m, ft). The parameter is the seven-digit heat structure number.
HTRG	Heat flux to vapor phase (W/m^2 , $\text{Btu/s}\cdot\text{ft}^2$). The parameter is the heat structure geometry number, CCCG0NN, with a two-digit number appended (00 for the left boundary and 01 for the right boundary).
<u>HTRNR</u>	Convective heat flux (W/m^2 , $\text{Btu/s}\cdot\text{ft}^2$). The parameter is the heat structure geometry number, CCCG0NN, with a two-digit number appended (00 for the left boundary and 01 for the right boundary).
<u>HTTEMP</u>	Mesh point temperature (K, $^{\circ}\text{F}$). The parameter is the heat structure geometry number CCCG0NN with a two-digit number appended (mesh point number). The surface temperatures are written to the plot record but interior mesh point temperatures must be requested through the 2080XXXX cards.
<u>HTVAT</u>	Volume averaged temperature in the heat structure (K, $^{\circ}\text{F}$). The parameter is the seven-digit heat structure number.
PECL	Liquid Peclet number for the heat structures. The parameter is the heat structure geometry number CCCG0NN with a two-digit number appended (00 for the left boundary and 01 for the right boundary).
STANT	Stanton number. The parameter is the heat structure geometry number CCCG0NN with a two-digit number appended (00 for the left boundary and 01 for the right boundary).
HGAP	Total gap conductance, including radiation ($\text{W/m}^2\cdot\text{K}$, $\text{Btu/s}\cdot\text{ft}^2\cdot^{\circ}\text{F}$). This value is zero if the gap conductance model is turned off. The parameter is the heat structure geometry number CCCG0NN.
KGAP	Effective gap thermal conductivity ($\text{W/m}\cdot\text{K}$, $\text{Btu/s}\cdot\text{ft}\cdot^{\circ}\text{F}$). This value is zero if the gap conductance model is turned off. The parameter is the heat structure geometry number CCCG0NN.

PGAP	Pressure of the gas gap (Pa, lb_f/in^2). This value is zero if the gap conductance model is turned off. The parameter is the heat structure geometry number CCCG0NN.
TGAP	Temperature of the gas gap (K, $^{\circ}\text{F}$). This value is zero if the gap conductance model is turned off. The parameter is the heat structure geometry number CCCG0NN.
GAPWD	Width of the gas gap (m, ft). This value is zero if the gap conductance model is turned off. The parameter is the heat structure geometry number CCCG0NN.
CLSTRN	Total cladding strain. This value is zero if the gap conductance model is turned off. The parameter is the heat structure geometry number CCCG0NN.
CLHOOP	Cladding hoop stress (Pa, lb_f/in^2). This value is zero if the gap conductance model is turned off. The parameter is the heat structure geometry number CCCG0NN.

A4.6 Reflood-Related Quantities

For the following variable codes, the parameter is the heat structure geometry number, i.e., the four-digit number CCCG printed in the major edit.

<u>Variable</u>	<u>Description</u>
FINES	Current number of axial nodes on a reflood structure.
TCHFQF	Temperature at the critical (maximum) heat flux (K, $^{\circ}\text{F}$).
TREWET	Rewet, quench, Leidenfrost, or minimum film boiling temperature (K, $^{\circ}\text{F}$).
ZQBOT	Elevation of bottom quench front (m, ft).
ZQTOP	Elevation of top quench front (m, ft).

A4.7 Reactor Kinetic Quantities

The parameter is 0 for the following reactor kinetic quantities.

<u>Variable</u>	<u>Description</u>
<u>REAC</u>	Reactivity feedback total (dollars). This is the sum of REACM, REACRB, REACS, and REACTF. (Only available when the SEPARABL option is used on the 30000000 card).
<u>REACM</u>	Reactivity feedback total from moderator density changes (dollars). This is the sum of REACRM and REACTM. (Only available when the SEPARABL option is used on the 30000000 card).
<u>REACRB</u>	Reactivity feedback from boron density changes (dollars). (Only available when the SEPARABL option is used on the 30000000 card).
<u>REACRM</u>	Reactivity feedback from moderator density changes (dollars). (Only available when the SEPARABL option is used on the 30000000 card).
<u>REACS</u>	Reactivity feedback from scram curve (dollars).
<u>REACTF</u>	Reactivity feedback from fuel temperature changes (dollars). (Only available when the SEPARABL option is used on the 30000000 card).
<u>REACTM</u>	Reactivity feedback from moderator temperature (spectral) changes (dollars). (Only available when the SEPARABL option is used on the 30000000 card).
<u>RKFIPOW</u>	Reactor power from fission (W).
<u>RKGAPOW</u>	Reactor power from decay of fission products and actinides (W).
<u>RKPOWA</u>	Reactor power from decay of actinides (W).
<u>RKPOWK</u>	Reactor power from decay of fission products (W).
<u>RKREAC</u>	Reactivity (dollars).
<u>RKRECPER</u>	Reciprocal period (s^{-1}).
<u>RKTPOW</u>	Total reactor power, i.e., sum of fission and decay powers (W).

A4.8 Control System Quantities

The parameter is the control component number, i.e., the three-digit number, CCC, or the four-digit number, CCCC, used in the input cards.

<u>Variable</u>	<u>Description</u>
<u>CNTRLVAR</u>	Control component number. These quantities are assumed dimensionless except for a SHAFT component.

A4.9 Trip Quantities

Both the variable and logical trips are available for plotting purposes only. The parameter is the trip number. If the trip has not occurred or is false, i.e., trip time is equal to -1.0, then 0.0 is plotted, and if the trip has occurred or is true, i.e., trip time greater than -1.0, then 1.0 is plotted.

The previous method the user had available for plotting the trips was to use a unit trip component, TRIPUNIT, and then plot the corresponding control variable. This is a much easier way to plot the trips and does not require the user to define a control variable just for plotting the trip.

<u>Variable</u>	<u>Description</u>
<u>TRIP</u>	Trip number.

A4.10 Expanded Plot Variables, 2080XXXX

The underlined variables listed above are always available for plotting. The variables that are not underlined and some of the underlined variables followed by an asterisk are not written to the restart-plot file and are thus unavailable for plotting unless the user enters the desired variables on 2080XXXX cards. $0001 \leq XXXX \leq 9999$ The format of these cards is given below. They are only required for the additional variables that the user wants to have written on the restart-plot file. The user can specify that between 1 and 9999 of these variables be written to the restart-plot file.

For RESTART problems, if ANY 2080xxxx cards are entered, then ALL the cards from the previous problem are deleted and will have to be re-entered if these earlier 2080xxxx plot variables are desired. If NO 2080xxxx cards are requested in a RESTART problem, the cards from the previous run will continue to be used, and the variables can be plotted after the restart time. If any component is deleted in a RESTART run, and previously requested 2080xxxx plot variables included that component either explicitly or as part of the -1 expansion feature, then the 2080xxxx cards will need to be re-entered. Otherwise, an input error will occur when RELAP5 tries to access the deleted component variables.

The field XXXX need not be consecutive.

W1(A) Variable. See the previous sections for valid variable names.

W2(I) Parameter. Enter the parameter associated with the variable request code. For example, for a single volume 567, the user would enter, 567010000.

-1 may be entered to plot a volume variable for all VOLUMES or to plot a junction variable for all JUNCTIONS. This feature was added to reduce the number of 2080XXXX cards the user has to enter. For example, to get the volume mass error, VOLMER, at all the volumes, instead of adding 2080XXXX cards for each volume, the user only has to enter one card:

```
20800001 VOLMER -1
```

A4.10.1 Plotting Additional 2080XXXX Variables from a Restart Run

If new 2080XXXX cards are added in a restart run, these new variables are added to the restart-plot file at the time of the restart, not at the beginning of the restart-plot file. This is shown in **Figure 4.10-1**.

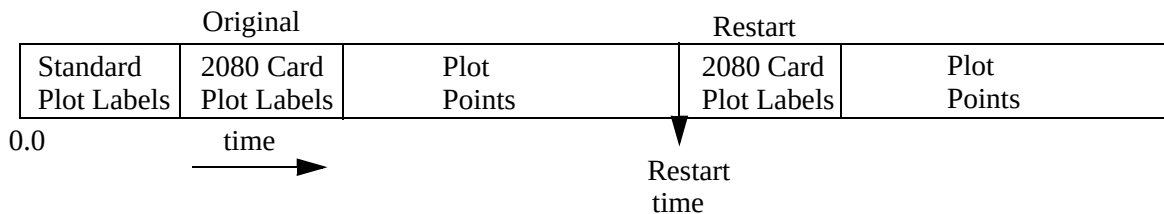


Figure 4.10-1 Layout of the restart-plot file in a restart run.

When XMGR5/ACGRACE opens this restart-plot file for plotting, it initially only reads the first set of plot labels. Thus, the new variables requested on the 2080xxxx cards in the restart run are unknown to XMGR5/ACGRACE. It is only after XMGR5/ACGRACE plots a variable known at the beginning of the run, e.g., DT-0, which requires reading the entire file to get all the values, that it finds the new 2080xxxx variables that are in the restart input deck. Once XMGR5/ACGRACE finds these new variables, it adds them to its list of plot variables and they will show up in the XMGR5/ACGRACE Edit RELAP5 data window.

The same problem occurs if the user adds a new component in the restart. The standard plot labels for this new component will be written to the restart-plot file at the time of the restart and can only be plotted after the user has plotted a variable that exists at the beginning of the run.

A different kind of problem occurs if the user changes from a transient run to a steady-state run or vice versa in order to reset the time to zero. In this case, RELAP5 ignores the information on the restart-plot file prior to the restart time, which means that the 2080xxxx card plot labels in the original run are discarded along with everything else ahead of the restart point. If the user wants to plot these variables in the restart run, then the original 2080xxxx cards have to be included in the restart input deck.

A5 TRIPS, 400-799 or 20600010-20620000 or 20600001-20620000

These cards are optional for NEW and RESTART type problems and are not used for other problem types. Three different card series are available for entering trip data, but only one series type may be used in a problem.

- Cards 401-799 199 variable trips and 199 logical trips (increment card numbers by 1)
- Cards 20600010 -20620000 1000 variable trips and 1000 logical trips (increment card numbers by 10)
- Cards 20600001 -20620000 10000 variable trips and 10000 logical trips (increment card numbers by 1)

A5.1 Trips Cancellation, 400

This card is allowed only for RESTART problems. The card causes all trips in the problem being restarted to be deleted. Any desired trips must be reentered.

W1(A) DISCARD. Any other entry is an error.

A5.2 Trip Series Type, 20600000

This card is optional. If this card is not supplied, then card numbers 401-599 are used for the variable trips, card numbers 601-799 are used for the logical trips, and the trip numbers are equal to the card numbers.

If this card is entered, with EXPANDED as word 1, then card numbers 20600010-20610000 are used for the variable trips, card numbers 20610010-20620000 are used for the logical trips, and the trip numbers are equal to the last 5 digits of the card number divided by 10, i.e., the last digit of the card number must be a 0. This allows 1000 variable trips and 1000 logical trips.

If this card is entered, with EXPBIG as word 1, then card numbers 20600001-20610000 are used for the variable trips, card numbers 20610001-20620000 are used for the logical trips, and the trip numbers are equal to the last 5 digits of the card number. This allows 10,000 variable trips and 10,000 logical trips.

Trip numbers do not have to be consecutive.

W1(A) EXPANDED or EXPBIG. EXPANDED allows 1000 variable and 1000 logical trips, while EXPBIG allows 10,000 variable and 10,000 logical trips.

A5.3 Variable Trips, 401-599 or 20600010-20610000 or 20600001-20610000

Each card defines a logical statement or trip condition concerned with the quantities being advanced in time. A trip is false or not set if the trip condition is not met, and true if it is met. On restart, new trips can be introduced, old trips can be deleted, and a new trip with the same number as an old trip replaces the old trip.

The variables and parameters are the same as described for minor edits, Section A4. NULL is allowed for the right side when only a comparison to the constant is desired. The variable code TIMEOF, with the parameter set to the number of another trip, indicates the time when that other trip was last set to true. Whenever that other trip is set to false, its TIMEOF value is set to -1.0, the comparison with the TIMEOF value within the variable trip is NOT made, and the variable trip is set to false.

- W1(A) Variable. On RESTART problems, this word can also contain DISCARD or RESET. DISCARD deletes the trip; RESET sets the trip to false. If DISCARD or RESET are entered, no further words are entered on the card.

- W2(I) Parameter.

- W3(A) Relationship. This may be either EQ, NE, GT, GE, LT, or LE, where the symbols have the standard FORTRAN meaning. Do not enter periods as part of the designator. For example, use GE rather than “.GE.” to specify *greater than* or *equal to*.

- W4(A) Variable.

- W5(I) Parameter.

- W6(R) Additive constant.

- W7(A) Latch indicator. If L, the trip once set true remains true, even if the condition later is not met. If N, the trip is tested each time advancement.

- W8(R) Timeof quantity (s). This word is optional. If it is not entered, the trip is initialized as false and the associated TIMEOF quantity is set to -1.0. If -1.0 is entered, the trip is initialized as false. If 0 or a positive number is entered for TIMEOF, the trip is initialized as true. TIMEOF must not be greater than 0 for NEW problems and must not be greater than the time of restart for RESTART problems.

The logical statement is “Does the quantity given by Words 1 and 2 have the relationship given by Word 3 with the quantity given by Words 4 and 5 plus Word 6?” If the relationship is false, the trip is false or not set. If the relationship is true, the trip is true or set. The TIMEOF variable is -1.0 if the trip is false. If the trip is true, this variable is the time the trip was last set true. A latched trip is never reset, so the trip time never changes once it changes from -1.0. For the nonlatched trips, the trip time when set remains constant until the trip condition becomes false and then the trip time is -1.0 again. If the trip condition becomes true again, the process is repeated. For trips such as a time test, L should be used to eliminate repeated testing, although no error or difference in results will occur if N is used.

A5.4 Logical Trips, 601-799 or 20610010-20620000 or 20610001-20620000

If these cards are entered, at least one of the variable trips must have been entered. Each card defines a logical relationship with the trips defined on these cards or on the variable trips.

- W1(I) Trip number. The absolute value of this number must be one of the trip numbers defined by the variable or logical trips. A negative trip number indicates that the complement of the trip is to be used in the test.
- W2(A) Operator. The operator may be AND, OR, or XOR. For RESTART problems, this quantity may also contain DISCARD or RESET. DISCARD deletes the trip and RESET sets the trip to false. If DISCARD or RESET are entered, no further words are entered on the card and Word 1 (W1) may be 0.
- W3(I) Trip number. This is similar to Word 1 (W1).
- W4(A) Latch indicator. If L, the trip when set remains set. If N, the trip is tested each time advancement.
- W5(R) Timeof quantity (s). This word is optional. If not entered, the trip is initialized as false, and the associated TIMEOF quantity is set to -1.0. If -1.0 is entered, the trip is initialized as false. If 0 or a positive number is entered for TIMEOF, the trip is initialized as true. TIMEOF must not be greater than 0 for NEW problems and must not be greater than the time of restart for RESTART problems.

The trip condition is given by the result of the following logical expression:

CONDITION OF TRIP IN W1 OPERATOR IN W2 CONDITION OF TRIP IN W3.

A5.5 Trip Stop Advancement, 600

This card can be entered in NEW and RESTART problems. One or two trip numbers may be entered. If either of the indicated trips are true, the problem advancement is terminated. These trips are tested only at the end of a requested advancement. If the trips can cycle true and false, they should be latched-type trips to ensure being true at the test time.

- W1(I) Trip number.
- W2(I) Trip number. This second trip number is optional.

A6 INTERACTIVE INPUT, 801-999

An interactive and color display capability exists when the code is interfaced with Nuclear Plant Analyzer (NPA) software. This capability allows a user to view selected results on a color graphics terminal and to modify user-defined input quantities. A user can view RELAP5/MOD3.3Patch04 output in a format that enhances understanding of the transient phenomena and enter commands during the simulation. This input, coupled with trip and control system capability, allows a user to initiate operator-like actions, such as opening/closing valves, starting/stopping/changing speed on pumps, and changing operating power settings.

These data may be entered for either batch or interactive jobs. These cards may be used in a NEW or RESTART job; in a restart job, they add to or replace data in the restarted problem.

These cards define variables that may be changed during execution by data input from a computer terminal if the job is being run interactively. The card input defines input variable names and initial values. These variables are completely independent from the Fortran variable names used in the RELAP5 coding, even if they are spelled the same. These user-defined variables can appear wherever variables listed in Section A4 can be used. Thus, the user-defined variables can be used in trips, control variable statements, search arguments for some tables, edited in minor edits, and plotted. With appropriate input, an interactive user can effect changes similar to those made by a reactor operator, such as opening/closing/repositioning valves or setting new operating points in controllers. When entering these user-defined variables, the variable name is the alphanumeric part of the variable request code, and 1000000000 is the numeric part.

W1(A) Variable name. Enter the variable name or DELETE in a RESTART job to delete the variable.

W2(R) Initial value. This is not needed if DELETE is entered in Word 1.

In interactive execution, the initial value is used until changed by a terminal entry. The value can be changed at any time and as often as needed. One or more variables can be changed by entering the variable name and value pairs on the computer terminal. An example is VLV1 = 0 VLV2,1 VLV3,0, POWER = 3050.+6, where VLV1, VLV2, VLV3, and POWER are user-defined variable names. The format is identical to data input on cards. An equal sign is treated as a terminating comma. The values should be floating-point quantities, but integers are converted to floating point values. The NPA interface also allows other more convenient methods for entering new values during the simulation.

W3(R) Conversion factor. Word 2 or any terminal-entered replacement value is entered in user-defined units. These quantities should be converted to SI units if they are to be involved in comparisons or computations with quantities advanced in time. User units can be used only if these input interactive variables are used with control variables defined in compatible units. This word, if nonzero, is the conversion factor. If this word is positive, the conversion is $V(\text{converted}) = V(\text{input}) \cdot W3$. If negative, $V(\text{converted}) = V(\text{input}) / 1.8 - W3$. For temperature conversion from °F to K, Word 3 should be -255.3722222. If this

word is missing, the conversion factor defaults to 1.0. If this word is 0, the next two words must contain a variable request code, and the conversion factor appropriate for this quantity is supplied by the code. If SI units are in use, the supplied conversion factor is 1.0. If British units are in use, the appropriate conversion factor is supplied.

A7 STRIP REQUESTS, 1001-1999

These cards are required only in STRIP-type problems. One or more cards are entered, each card containing one variable request. Card numbers need not be consecutive. Variables are ordered on the STRIPF file in the order of increasing card numbers. If an incorrect variable request code is entered, the value will be 0.0. It is not flagged as an input error, since at some later time in the transient, a renodalization may result in the variable request code becoming correct.

W1(A) Alphanumeric part of the variable request code.

W2(I) Integer part of the variable request code.

A8 COMPARE DUMP FILES, 1001-1999

These cards are required only in CMPCOMS problems. One or more cards are entered, each card containing one request to compare dump blocks on the files specified with the -A and -B options on the command line. Card numbers need not be consecutive.

W1(I) Dump file number from file specified on -A command line option.

W2(I) Dump file number from file specified on -B command line option.

The values in Words 1 and 2 on a succeeding card must be greater than the values on the preceding card. The values in Words 1 and 2 are the advancement number when the dump block was written. This information is written as a line in the printed output of the run writing the dump file. The form of the line is, “---Dmpcom no. nnn written, block no. mmm on unit u---,” where nnn is the advancement count number, mmm is the count of the number of blocks written, and u is A or B indicating the file specified by the -A or -B option.

A9 HYDRODYNAMIC COMPONENTS

These cards are required for NEW type problems and may be entered for RESTART problems. Hydrodynamic systems are described in a NEW problem. In a RESTART problem, the hydrodynamic systems may be modified by deleting, adding, or replacing components. The resultant problem must describe at least two volumes and one junction. The hydrodynamic card numbers are divided into fields, where CCC is the component number (the component numbers need not be consecutive), XX is the card type, and NN is the card number within type. When a range is indicated, the numbers need not be consecutive.

For junctions, an old format has been used in the past to connect volumes. In the old format, CCC000000 was used if the connection was to the inlet side of the component and CCC010000 was used if the connection was to the outlet side of the component. The use of this format is discouraged and the expanded connection format is preferred because it is more general and easier to understand. In the expanded format, the connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. A nonzero N specifies the expanded format. The number N equal to 1 and 2 specifies the inlet and outlet faces, respectively, for the volume's coordinate direction (see Section 2.1 of Volume II). The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For example, in a 5 volume pipe component, the old format inlet connection code of CCC000000 is equivalent to CCC010001 in the expanded format, and the old outlet connection code of CCC010000 is equivalent to CCC050002 in the expanded format. In the following component descriptions, only the expanded format connection code will be used.

A9.1 Components

A9.2 General Information

A9.2.1 Component Name and Type, CCC0000

This card is required for each component.

- | | |
|-------|--|
| W1(A) | Component name. Use a name descriptive of the component's use in system. A limit of 8 characters is allowed for most computers, e.g., workstations, CRAY, and IBM computers. |
| W2(A) | Component type. Enter one of the following component types, SNGLVOL, TMDPVOL, SNGLJUN, TMDPJUN, PIPE, ANNULUS, PRIZER, CANCHAN, BRANCH, SEPARATR, JETMIXER, TURBINE, ECCMIX, VALVE, PUMP, MTPLJUN, ACCUM, or the command DELETE. |

The command DELETE is allowed only in RESTART problems, and the component number must be an existing component at the time of restart. The DELETE command deletes the component.

If the input deck that created the restart file contained any 2080xxxx expanded plot variable cards with a volume or junction minor edit variable and a -1 parameter, which plots this variable for all the volumes or junctions, then these 2080xxxx cards must be reentered in the restart input deck in order for the code to recompute the component numbers to plot. Any 2080xxxx cards that do not have the -1 parameter do not have to be reentered and will be available in the restart file to be plotted.

The remaining cards for each component depend on the type of component.

A9.3 Single-Volume Component, SNGLVOL

A single-volume component is indicated by SNGLVOL for Word 2 on card CCC0000. The junction connection code determines the placement of the volume within the system. More than one junction may be connected to an inlet or outlet. If an end has no junctions, that end is considered a closed end. Normally, only a branch has more than one junction connected to a volume end. For major edits, minor edits, and plot variables, the volume in the single-volume component is numbered as CCC010000.

A9.3.1 Single-Volume X-Coordinate Volume Data, CCC0101-0109

This card (or cards) is required for a single-volume component. The nine words can be entered on one or more cards, and the card numbers need not be consecutive.

W1(R)	Area (m ² , ft ²).
W2(R)	Length (m, ft).
W3(R)	Volume (m ³ , ft ³). The program requires that the volume equals the area times the length (W3 = W1•W2). At least two of the three quantities, W1, W2, and W3, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the words are 0, the volume must equal the x-direction area times the x-direction length within a relative error of 0.000001. The same relative error check is done for the y- and z-directions.
W4(R)	Azimuthal angle (degrees). The absolute value of this angle must be ≤ 360 degrees and is defined as a positional quantity. This quantity is not used in the calculation but is specified for possible automated drawing of nodalization diagrams.
W5(R)	Inclination angle (degrees). The absolute value of this angle must be ≤ 90 degrees. The angle 0 degrees is horizontal; and positive angles have an upward inclination, i.e., the inlet is at the lowest elevation. This angle is used in the interphase drag calculation.
W6(R)	Elevation change (m, ft). A positive value is an increase in elevation. The absolute value of this quantity must be less than or equal to the length. If the vertical angle orientation is 0, this quantity must be 0. If the vertical angle is nonzero, this quantity must also be

nonzero and have the same sign. See Section 2.4.1 of Volume II of the manual for further discussion. When the absolute value of the elevation angle determined by the ratio of the elevation change (this Word 6) and the length (Word 2) is less than or equal to 45 degrees, the horizontal flow regime map is used. When the ratio is greater than 45 degrees, the vertical flow regime map is used.

W7(R)	Wall roughness (m, ft).
W8(R)	Hydraulic diameter (m, ft). This should be computed from $4.0 \bullet \frac{\text{area}}{\text{wetted perimeter}}$. If 0, the hydraulic diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi}\right)^{0.5}$. A check is made to ensure the pipe roughness is less than half the hydraulic diameter. See Word 1 for area.
W9(I)	Volume control flags. This word has the format <u>tlpvbf</u> . It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags. Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f-flag is the only coordinate direction oriented flag. This word enters the scalar oriented flags and the x-coordinate flag.
t-flag	The digit <u>t</u> specifies whether the thermal front tracking model is to be used; <u>t</u> = 0 specifies that the front tracking model is not to be used for the volume, and <u>t</u> = 1 specifies that the front tracking model is to be used for the volume.
l-flag	The digit <u>l</u> specifies whether the mixture level tracking model is to be used; <u>l</u> = 0 specifies that the level model not be used for the volume, and <u>l</u> = 1 specifies that the level model be used for the volume.
p-flag	The digit <u>p</u> specifies whether the water packing scheme is to be used. <u>p</u> = 0 specifies that the water packing scheme is to be used for the volume, and <u>p</u> = 1 specifies that the water packing scheme is not to be used for the volume.
v-flag	The digit <u>v</u> specifies whether the vertical stratification model is to be used. <u>v</u> = 0 specifies that the vertical stratification model is to be used for the volume, and <u>v</u> = 1 specifies that the vertical stratification model is not to be used for the volume.
b-flag	The digit <u>b</u> specifies the interphase friction that is used. <u>b</u> = 0 means that the pipe interphase friction model will be applied, <u>b</u> = 1 means that the rod bundle interphase friction model will be applied, <u>b</u> = 2 means that the ORNL ANS interphase friction model will be applied (see card CCC0111 in Section A9.3.2).

f-flag	The digit $\underline{f}$ specifies whether wall friction is to be computed. $\underline{f} = 0$ specifies that wall friction effects are to be computed along the x-coordinate of the volume, and $\underline{f} = 1$ specifies that wall friction effects are not to be computed along the x-coordinate.
e-flag	The digit $\underline{e}$ specifies if nonequilibrium or equilibrium is to be used. $\underline{e} = 0$ specifies that a nonequilibrium (unequal temperature) calculation is to be used, and $\underline{e} = 1$ specifies that an equilibrium (equal temperature) calculation is to be used. Equilibrium volumes should not be connected to nonequilibrium volumes. The equilibrium option is provided only for comparison with other codes.

A9.3.2 Single-Volume ORNL ANS Interphase Model Values, CCC0111

This card is required if the interphase friction flag $\underline{b}$ in Word 9 on cards CCC0101-0109 is set to 2.

W1(R)	Gap (distance between side walls, short length, pitch, channel width) (m, ft).
W2(R)	Span (distance from one end to the other, long length) (m, ft).
W3(I)	Volume number.

A9.3.3 Single-Volume Additional Wall Friction, CCC0131

This card is optional. If this card is not entered, the default values are 1.0 for the laminar shape factor and 0.0 for the viscosity ratio exponent. Two, four, or six quantities may be entered on the card, and the data not entered are set to default values. A description of this input is presented in Section 3 of Volume I.

W1(R)	Shape factor for x-coordinate.
W2(R)	Viscosity ratio exponent for x-coordinate.
W3(R)	Shape factor for y-coordinate.
W4(R)	Viscosity ratio exponent for y-coordinate.
W5(R)	Shape factor for z-coordinate.
W6(R)	Viscosity ratio exponent for z-coordinate.

A9.3.4 Single-Volume Y-Coordinate Volume Data, CCC0181-0189

These cards are optional. These cards are used when the user specifies the y-direction connection with the crossflow model. The volume is the same for the x-, y-, and z-directions. If these cards are entered, either W1 or W2 must be nonzero.

W1(R)	Area (m ² , ft ²). If these cards are missing or if this word is 0, this y-direction area is computed from $\frac{\text{volume}}{\text{y-direction length}}$.
W2(R)	Length (m, ft). If these cards are missing, this y-direction length is computed from $2.0 \bullet \left(\frac{\text{x-direction area}}{\pi} \right)^{0.5}$. If this word is 0, this y-direction length is computed from $\frac{\text{volume}}{\text{y-direction area}}$.
W3(R)	Roughness (m, ft).
W4(R)	Hydraulic diameter (m, ft). If these cards are missing or if this word is 0, this y-direction hydraulic diameter is computed from $\frac{4.0 \bullet \text{y-direction volume flow area}}{(\pi \bullet \text{x-direction volume flow area})^{0.5}}$. A check is made to ensure the y-direction roughness is less than half the y-direction hydraulic diameter.
W5(I)	Volume control flags. This word has the general format <u>tlpybfe</u> , but this word is limited to <u>00000f0</u> since it only enters the coordinate oriented flags for the y-direction.
f-flag	The digit <u>f</u> specifies whether wall friction is to be computed. <u>f</u> = 0 specifies that wall friction effects are to be computed along the y-coordinate direction in the volume, and <u>f</u> = 1 specifies that wall friction effects are not to be computed along the y-coordinate direction.
W6(R)	This word is not used. Enter 0.
W7(R)	This word is not used. Enter 0.
W8(R)	This word is the position change in the z-fixed (vertical) direction as the flow passes from the y-inlet face to the y-outlet face (m, ft). This quantity affects problems if connections are made to the y-faces.

A9.3.5 Single-Volume Z-Coordinate Volume Data, CCC0191-0199

These cards are optional. These cards are used when the user specifies the z-direction connection with the crossflow model. The volume is the same for the x-, y-, and z-directions. If these cards are entered, either W1 or W2 must be nonzero.

W1(R)	Area (m ² , ft ²). If these cards are missing or if this word is 0, this z-direction area is computed from $\frac{\text{volume}}{\text{z-direction length}}$.
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- W2(R) Length (m, ft). If these cards are missing, this z-direction length is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi}\right)^{0.5}$. If this word is 0, this z-direction length is computed from $\frac{\text{volume}}{\text{z-direction area}}$.
- W3(R) Roughness (m, ft).
- W4(R) Hydraulic diameter (m, ft). If these cards are missing or if this word is 0, this z-direction hydraulic diameter is computed from $\frac{4.0 \bullet \text{z-direction volume flow area}}{(\pi \bullet \text{x-direction volume flow area})^{0.5}}$. A check is made to ensure the z-direction roughness is less than half the z-direction hydraulic diameter.
- W5(I) Volume control flags. This word has the general format tlpybf, but this word is limited to 00000f0 since it only enters the coordinate oriented flags for the z-direction.
- f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed along the z-coordinate direction in the volume, and f = 1 specifies that wall friction effects are not to be computed along the z-coordinate direction.
- W6(R) This word is not used. Enter 0.
- W7(R) This word is not used. Enter 0.
- W8(R) This word is the position change in the z-fixed (vertical) direction as the flow passes from the z-inlet face to the z-outlet face (m, ft). This quantity affects problems if connections are made to the z-faces.

A9.3.6 Single-Volume Initial Conditions, CCC0200

This card is required for a single-volume.

- W1(I) Control word. This word has the format gbt. It is not necessary to input leading zeros.
- ε -flag The digit ε specifies the fluid, where ε = 0 is the default fluid. The value for ε > 0 corresponds to the position number of the fluid type indicated on the 120-9 cards (i.e., ε = 1 specifies H₂O, ε = 2 specifies D₂O, etc.). The default fluid is that set for the hydrodynamic system by cards 120-9 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-9 or these control words must be consistent (i.e., not specify different fluids). If cards 120-9 are not entered and all control

words use the default $\varepsilon = 0$, then H_2O is assumed as the fluid. It is recommended that the user use the 120-9 cards to set the fluid type in the systems.

b-flag The digit $\underline{b}$ specifies whether boron is present or not. The digit $\underline{b} = 0$ specifies that the volume liquid does not contain boron; $\underline{b} = 1$ specifies that a boron concentration in mass of boron per mass of liquid (which may be 0) is being entered after the other required thermodynamic information.

t-flag The digit $\underline{t}$ specifies how the following words are to be used to determine the initial thermodynamic state. Entering $\underline{t} = 0-3$ or 7 specifies only one component (steam/water). Entering $\underline{t} = 4-6$ allows the specification of two components (steam/water and noncondensable gas).

With options $\underline{t}$ equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.

One Component (Steam/Water), Equilibrium or Non-equilibrium

$[P, U_f, U_g, \alpha_g]$ If $\underline{t} = 0$, the next four words are interpreted as pressure (Pa, lb_f/in^2), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. Enter boron in W6 if needed.

One Component (Steam/Water), Equilibrium

$[T, x_s]$ If $\underline{t} = 1$, the next two words are interpreted as temperature (K, $^{\circ}F$) and static quality in equilibrium condition. Enter boron in W4 if needed.

$[P, x_s]$ If $\underline{t} = 2$, the next two words are interpreted as pressure (Pa, lb_f/in^2) and static quality in equilibrium condition. Enter boron in W4 if needed.

$[P, T]$ If $\underline{t} = 3$, the next two words are interpreted as pressure (Pa, lb_f/in^2) and temperature (K, $^{\circ}F$) in equilibrium condition. Enter boron in W4 if needed.

$[P, T_f, T_g, \alpha_g]$ If $\underline{t} = 7$, the next four words are interpreted as pressure (Pa, lb_f/in^2), liquid temperature (K, $^{\circ}F$), vapor temperature (K, $^{\circ}F$), and vapor void fraction. These quantities will be interpreted as equilibrium conditions. The liquid temperature has to be equal to or less than the saturation temperature, and the vapor temperature has to be equal to or greater than the

saturation temperature. Enter boron in W6 if needed. This input option was added to allow easier comparisons with TRACE calculations.

Two Components (Steam/Water/Air), Equilibrium or Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0\text{E-}9 \leq x_s \leq 0.99999999$,

Single phase if $x_s < 1.0 \text{ E-}9$ or $x_s > 0.99999999$

The static quality is given by $x_s = M_g/(M_g + M_f)$, where $M_g = M_s + M_n$. Section 3.2 of Volume I of the manual discusses this further.

[P,T, x_s]

If $t = 4$, the next three words are interpreted as pressure (Pa, lb_f/in²), temperature (K, °F), and static quality in equilibrium condition. Using this input option with static quality > 0.0 and ≤ 1.0 , saturated noncondensables will result. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. If the pressure is greater than the critical pressure, $P > P_{\text{crit}}$, then the static quality, x_s (mass steam + air)/(mass steam + air + water) has to be large enough to give a partial steam pressure that is less than the critical pressure, $P_s < P_{\text{crit}}$. Otherwise an input error will occur. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensable (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option.

[T, x_s , x_n]

If $t = 5$, the next three words are interpreted as steam saturation temperature (K, °F), static quality, and noncondensable quality in equilibrium condition. Both the static and noncondensable qualities are restricted to be between $1.0 \text{ E-}9$ and 0.99999999 . The liquid will be subcooled. The input temperature determines the steam partial pressure. However, the total pressure may be greater than the critical pressure, which will cause an input error.

[P, U_f , U_g , α_g , x_n]

If $t = 6$, the next five words are interpreted as pressure (Pa, lb_f/in²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing ($t = 0$). If noncondensables are present (noncondensable quality greater than

0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then the vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor specific internal energy.

W2-W7(R) Quantities as described under Word 1 (W1). Depending on the control word, 2-6 quantities may be required. Enter only the minimum number required. If entered, boron concentration (mass of boron per mass liquid) follows the last required word for thermodynamic conditions.

A9.4 Single Volume Initial Noncondensable Mass Fractions, CCC0301

This card is optional for a single volume and is related to card 110. If card CCC0301 is not supplied, then the values on the card 115 are used for the noncondensable mass fractions. The number of words on card CCC0301 must equal the number of words on card 110. The mass fractions must sum to 1.0 to within a tolerance of 1.0E-10.

W1-WN(R) Initial mass fraction for each noncondensable gas type as specified on card 110 where N is the number of gas types on card 110.

A9.5 Time-Dependent Volume Component, TMDPVOL

This component is indicated by TMDPVOL for Word 2 on card CCC0000. For major edits, minor edits, and plot variables, the volume in the time-dependent volume component is numbered as CCC010000.

A9.5.1 Time-Dependent Volume Geometry, CCC0101-0109

This card (or cards) is required for a time-dependent volume component. The nine words can be entered on one or more cards, and the card numbers need not be consecutive.

W1(R) Area (m^2 , ft^2). When a time-dependent volume is used to model a pressure boundary condition (i.e., the time-dependent volume is connected to the system through a normal junction), it is generally recommended that the cross-sectional area of the time-dependent volume be large compared to the area of the normal junction.

W2(R) Length (m, ft). After initialization, the length is set to 0.

W3(R) Volume (m^3 , ft^3). The program requires that the volume equals the area times the length ($W3 = W1 \bullet W2$). At least two of the three quantities, W1, W2, and W3, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the words are 0, the volume must equal the area times the length within a relative error of 0.000001. After initialization, the volume is set to 0.

- W4(R) Azimuthal angle (degrees). The absolute value of this angle must be ≤ 360 degrees. This quantity is not used in the calculation but is specified for possible automated drawing of nodalization diagrams.
- W5(R) Inclination angle (degrees). The absolute value of this angle must be ≤ 90 degrees. The angle 0 degrees is horizontal, and positive angles have an upward inclination, i.e., the inlet is at the lowest elevation. This angle is used in the interphase drag calculation.
- W6(R) Elevation change (m, ft). A positive value is an increase in elevation. The absolute value of this quantity must be less than or equal to the length. If the vertical angle orientation is 0, this quantity must be 0. If the vertical angle is nonzero, this quantity must also be nonzero and have the same sign. See Section 2.4.1 of Volume II of the manual for further discussion. As with the other components, this Word 6 is compared to the length (Word 2) to determine if the horizontal or vertical flow regime map is used. This is not important for this component, since the correlations that depend on the flow regime maps are not needed for this component. The volume conditions are prescribed through input cards CCC0201-0299. After initialization, the elevation change is set to 0.
- W7(R) Wall roughness (m, ft).
- W8(R) Hydraulic diameter (m, ft). This should be computed from $4.0 \cdot \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If 0, the hydraulic diameter is computed from $2.0 \cdot \left(\frac{\text{area}}{\pi} \right)^{0.5}$. A check is made to ensure the pipe roughness is less than half the hydraulic diameter. See Word 1 for the area.
- W9(I) Volume control flags. This word has the format tlpvbfe. For the time-dependent volume component, the volume control flag is restricted to the format 0000000. So enter 0 for this word. Internally, the p (no water packing) and v (no vertical stratification) flags are set to 1 and will show up that way in the output file.

A9.5.2 Time-Dependent Volume Data Control Word, CCC0200

This card is required for a time-dependent volume.

- W1(I) Control word for time-dependent data on CCC02NN cards. This word has the format n**e**b**t**. It is not necessary to input leading zeros.
- n-flag The digit n specifies whether noncondensable mass fractions are input on the CCC0201-299 cards. The digit n = 0 specifies that no noncondensable mass fractions are on the CCC0201-299 cards; n = 1 specifies that noncondensable mass fractions are on the CCC0201-299 cards. The ordering of the mass fractions follows the same order as the gas

types on the 110 card and are entered after the other required thermodynamic information, and if $\underline{b} = 1$, the boron concentration.

- ε -flag The digit $\underline{\varepsilon}$ specifies the fluid, where $\underline{\varepsilon} = 0$ is the default fluid. The value for $\underline{\varepsilon} > 0$ corresponds to the position number of the fluid type indicated on the 120-9 cards (i.e., $\underline{\varepsilon} = 1$ specifies H_2O , $\underline{\varepsilon} = 2$ specifies D_2O , etc.). The default fluid is that set for the hydrodynamic system by cards 120-9 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-9 or these control words within the hydrodynamic system must be consistent (i.e., not specify different fluids). If cards 120-129 are not entered and all control words use the default $\underline{\varepsilon} = 0$, then H_2O is assumed as the fluid. It is recommended that the user use the 120-129 cards to set the fluid type in the systems.
- $\underline{b}$ -flag The digit $\underline{b}$ specifies whether boron is present or not. The digit $\underline{b} = 0$ specifies that the volume liquid does not contain boron; $\underline{b} = 1$ specifies that a boron concentration in mass of boron per mass of liquid water (which may be 0) is being entered after the other required thermodynamic information.
- $\underline{t}$ -flag The digit $\underline{t}$ specifies how the words of the time-dependent volume data in cards CCC0201-0299 are to be used to determine the initial thermodynamic state. Entering $\underline{t}$ equal to 0-3 and 7 specifies one component (steam/water). Entering $\underline{t}$ equal to 4-6 allows the specification of two components (steam/water and noncondensable gas).
- With options $\underline{t}$ equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.
- W2(I) Table trip number. This word is optional. If missing or 0 and Word 3 is missing, no trip is used, and the time argument is the advancement time. If nonzero and Word 3 is missing, this number is the trip number, and the time argument is -1.0 if the trip is false, and the advancement time minus the trip time if the trip is true. (internally *ncttrp*)
- W3(A) Alphanumeric part of variable request code. This quantity is optional. If not present, time is the search argument. If present, this word and the next are a variable request code that specifies the search argument for the table lookup and interpolation. If the trip number is 0, the specified argument is used. If the trip number is nonzero, -1.0E+75 is used if the trip is false, and the specified argument is used if the trip is true. TIME can be selected, but note that the trip logic is different than if this word were omitted. The variable MFLOWJ can not be used as a search variable; a user-initialized control variable that uses MFLOWJ should be used instead (see Volume V). (internally *nctalf*)

W4(I) Numeric part of variable request code. This is assumed 0 if missing. For example, to use the pressure in volume 567010000 for the search argument, the user would enter P for W3 and 567010000 in W4. (internally *nctdpv*)

A9.5.3 Time-Dependent Volume Data, CCC0201-0299

These cards are required for time-dependent volume components. A set of data is made up of the search variable (e.g., time) followed by the required data indicated by control Word 1 in card CCC0200. The card numbers need not be consecutive, but the value of the search variable in a succeeding set must be equal to or greater than the value in the previous set. One or more sets of data, up to 5,000 sets, are allowed. Enter only the minimum number of words required. If entered, boron concentration follows the last required word for thermodynamic conditions. If entered, the noncondensable mass fractions follow the boron concentration (if present). Linear interpolation is used if the search argument lies between the search variable entries. End-point values are used if the argument lies outside the table values. Only one set is needed if constant values are desired, and computer time is reduced when only one set is entered. Step changes can be accommodated by entering the two adjacent sets with the same search variable values or an extremely small difference between them. Given two identical argument values, the set selected will be the closest to the previous argument value. Sets may be entered one or more per card and may be split across cards. The total number of words must be a multiple of the set size.

Inputting time-dependent volume tables where the search variable is a thermodynamic variable from some other component can run into difficulties if the component numbering is such that the time-dependent volume is initialized before the component providing the needed search variable. A reliable fix for this is to make the search variable a control system output in the desired units, while the thermodynamic variable is the control system input in code internal (SI) units. The control system initial value can be set to the desired initial value of the search variable, and this will be used by the time-dependent table.

W1(R) Search variable (e.g., time).

W2-W12(R) Quantities as described below. Depending on the control word on card CCC0200, 2-5 quantities may be required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions. If entered, noncondensable mass fractions follow the last required word for thermodynamic conditions, and if entered, the boron concentration. Ordering of the noncondensable mass fractions follows the same order as the gas types on the 110 card. Enter as many mass fractions as there are gas types on the 110 card (5 maximum).

Table 9.5-1 shows the parameters entered on the CCC0201-299 cards for the various values of the t-flag. Remember, word 1 is the search parameter, e.g., time. Only t-flag values equal to 4, 5, or 6 can have

Table 9.5-1 Parameters entered on the CCC0201-299 cards for various values of the t-flag

t-flag	Word 2	Word 3	Word 4	Word 5	Word 6	Word 7	Word 8
0	pressure	uf	ug	voidg	(boron)		

Table 9.5-1 Parameters entered on the CCC0201-299 cards for various values of the t-flag

t-flag	Word 2	Word 3	Word 4	Word 5	Word 6	Word 7	Word 8
1	temperature	quals	(boron)				
2	pressure	quals	(boron)				
3	pressure	temperature	(boron)				
4	pressure	temperature	quals	(boron)	(quala1-5)		
5	temperature	quals	quala	(boron)	(quala1-5)		
6	pressure	uf	ug	voidg	quala	(boron)	(quala1-5)
7	pressure	tempf	tempg	voidg	(boron)		

noncondensables. This is why the (quala1-5) entry is only shown for these three cases. The parentheses indicate optional entries. For example, if the b-flag on the CCC0200 card is 1, then the boron concentration is entered, otherwise, it is not entered. If the n-flag on the CCC0200 card is 1, then up to five noncondensable mass fractions are entered after thermodynamic (and boron) conditions, otherwise they are not entered. In the latter case, the noncondensable mass fractions on the 115 card will be used for the time-dependent volume. The t-flag = 7 option was added for compatibility with TRACE thermodynamic input. It has not been fully checked out and should be used with caution.

One Component (Steam/Water), Equilibrium or Non-equilibrium

[P,U_f,U_g,α_g] If $t = 0$, the second, third, fourth, and fifth words of the time-dependent volume data on cards CCC0201-0299 are interpreted as pressure (Pa, lb_f/in²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. Enter boron in W6 if needed.

One Component (Steam/Water), Equilibrium

[T,x_s] If $t = 1$, the second and third words of the time-dependent volume data on cards CCC0201-0299 are interpreted as temperature (K, °F) and static quality in equilibrium condition. Enter only the minimum number of words required. If entered, boron concentration follows the last required word for thermodynamic conditions. Enter boron in W4 if needed.

[P,x_s] If $t = 2$, the second and third words of the time-dependent volume data on cards CCC0201-0299 are interpreted as pressure (Pa, lb_f/in²) and static quality in equilibrium condition. Enter only the minimum number of words required. If entered, boron concentration

follows the last required word for thermodynamic conditions. Enter boron in W4 if needed.

[P,T] If $t = 3$, the second and third words of the time-dependent volume data on cards CCC0201-0299 are interpreted as pressure (Pa, lb_f/in²) and temperature (K, °F) in equilibrium conditions. Enter only the minimum number of words required. If entered, boron concentration follows the last required word for thermodynamic conditions. Enter boron in W4 if needed.

[P,T_f,T_g,α_g] If $t = 7$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid temperature (K, F), vapor temperature (K, F), and vapor void fraction. These quantities will be interpreted as equilibrium conditions. The liquid temperature has to be equal to or less than the saturation temperature, and the vapor temperature has to be equal to or greater than the saturation temperature. Enter boron in W6 if needed. This input option was added to allow easier comparisons with TRACE calculations.

Two Components (Steam/Water/Air), Equilibrium or Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0\text{E-}9 \leq x_s \leq 0.99999999$,

Single phase if $x_s < 1.0 \text{ E-}9$ or $x_s > 0.99999999$

The static quality is given by $x_s = M_g/(M_g + M_f)$, where $M_g = M_s + M_n$. Section 3.2 of Volume I of the manual discusses this further.

[P,T,x_s] If $t = 4$, the second, third, and fourth words of the time-dependent data on cards CCC0201-0299 are interpreted as pressure (Pa, lb_f/in.²), temperature (K, °F), and static quality in equilibrium condition. Using this input option with static quality > 0.0 and ≤ 1.0 , saturated noncondensables will result. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. If the pressure is greater than the critical pressure, $P > P_{\text{crit}}$, then the static quality, x_s (mass steam + air)/(mass steam + air + water) has to be large enough to give a partial steam pressure that is less than the critical pressure, $P_s < P_{\text{crit}}$. Otherwise an input error will occur. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensable (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option. Enter boron in W5 if needed. If entered, the noncondensable mass fractions follow the boron concentration (if present).

- [T,x_s,x_n] If $\underline{t} = 5$, the second, third, and fourth words are interpreted as steam saturation temperature (K, °F), static quality, and noncondensable quality in equilibrium condition. Both the static and noncondensable qualities are restricted to be between 1.0 E-9 and 0.99999999. The liquid will be subcooled. The input temperature determines the steam partial pressure. However, the total pressure may be greater than the critical pressure, which will cause an input error. Enter boron in W5 if needed. If entered, the noncondensable mass fractions follow the boron concentration (if present).
- [P,U_f,U_g,α_g,x_n] If $\underline{t} = 6$, the second, third, fourth, fifth, and sixth words of the time-dependent data on cards CCC0201-0299 are interpreted as pressure (Pa, lb_f/in.²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present, and the input processing branches to that type of processing ($\underline{t} = 0$). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then the vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor specific internal energy. Enter only the minimum number of words required. If entered, boron concentration follows the last required word for thermodynamic conditions. If entered, the noncondensable mass fractions follow the boron concentration (if present).
- W2-W7(R) Quantities as described under Word 1 in card 200. Depending on the control word, 2-5 quantities may be required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions.

As described above, sets may be entered one or more per card.

A9.6 Single-Junction Component, SNGLJUN

A single-junction component is indicated by SNGLJUN for Word 2 on card CCC0000. For major edits, minor edits, and plot variables, the junction in the single-junction component is numbered CCC000000.

A9.6.1 Single-Junction Geometry, CCC0101-0109

This card (or cards) is required for single-junction components.

W1(I) From connection code to a component. This refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces, respectively, for the volume's coordinate direction (see Section 2.1 of Volume II). The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, N is restricted to 1 or 2.

W2(I) To connection code to a component. This refers to the component at which the junction coordinate direction ends. See the description for W1 above.

W3(R) Area (m^2 , ft^2). If 0, the area is set to the minimum area of the adjoining volumes. For abrupt area changes, the junction area must be equal to or smaller than the minimum of the adjoining volume areas. For smooth area changes, there are no restrictions.

W4(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. Note: a variable loss coefficient may be specified (see Section A9.6.3). The interpretation and use of this loss coefficient vary based upon the "a" junction control flag selected in W6.

If the smooth area change option is selected (a = 0), the entry here is the loss coefficient applied for the junction flow area resulting from the entry made in W3.

If the abrupt area change option is selected (a = 1 or 2), the loss coefficient entered here is adjusted by the square of the ratio of the junction area and the minimum flow area of the adjacent connected volumes. This adjusted loss coefficient is applied to the junction flow area.

If the a = 1 option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

Example: Two volumes with volume average flow areas of 0.3 m^2 and 0.4 m^2 are connected with a single junction ($A_{\text{jun}} = 0.075 \text{ m}^2$). The user supplied Kloss is 10.0. For the smooth area change option (a = 0), the Kloss of 10 is applied directly to the junction. For the abrupt area change option (a = 1 or 2), the Kloss applied at the junction is $\text{Kloss}_{\text{jun}} = \text{Kloss}_{\text{user}} * (A_{\text{jun}}/A_{\text{min}})^2$. For this example: $\text{Kloss}_{\text{jun}} = 10.0 * (0.075/0.3)^2 = 0.625$.

- W5(R) Reynolds number independent reverse flow energy loss coefficient A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. Note: a variable loss coefficient may be specified (see Section A9.6.3). See information for the previous word regarding interpretation and use of this loss coefficient.
- W6(I) Junction control flags. This word has the format jefvcahs. It is not necessary to input leading zeros.
- j-flag (*ijt*) The digit j specifies that this junction is a jet junction. Pool surface condensation is enhanced in the volume above the junction when this model is activated. This junction must be underneath the to volume. (internally *isjetjunflg25*)
- e-flag (*ief*) The digit e specifies the modified PV term in the energy equations. e = 0 means that the modified PV term will not be applied, and e = 1 means that it will be applied. (internally *isdonpresspvwrk15*)
- f-flag (*icc*) The digit f specifies CCFL options. f = 0 means that the CCFL model will not be applied, and f = 1 means that it will be applied. (internally *isnpccflflg2*)
- v-flag (*ist*) The digit v specifies horizontal stratification entrainment/pullthrough options. This model is for junctions connecting a horizontal stratification volume to an offtake volume. If v = 0, the model is not applied. When v = 1, 2, 3, the offtake junction is located on the top, bottom, or side, respectively, of the horizontal stratification volume. When v = 4, the offtake junction is located at an angle to the horizontal (The angle is specified as Word 9 on this card). The horizontally stratified volume can be either the “from” or “to” volume. For liquid entrainment, the vapor velocity direction determines the upstream or horizontal stratification volume. For vapor pull-through, the liquid velocity direction determines the upstream or horizontal stratification volume.
- [v = 1] For an upward-oriented offtake junction, v = 1, and the offtake volume must be vertical (the offtake junction connects face 6 of the horizontal stratification volume to face 1 or 2 of the offtake volume). **Figure 9.6-1** shows an example for the v = 1 case.
- [v = 2] For a downward-oriented offtake junction, v = 2, and the offtake volume must be vertical (the offtake junction connects face 5 of the horizontal stratification volume to face 1 or 2 of the offtake volume).
- [v = 3 or 4] For a side, v = 3, or angled, v = 4, offtake junction the offtake volume can be either horizontal or vertical. If the offtake volume is horizontal, the offtake junction connects face 1, 2, 3, or 4 of the horizontal stratification volume with face 1, 2, 3, or 4 of the offtake volume. If the offtake volume is vertical, the offtake junction connects face 1, 2, 3, or 4 of the horizontal stratification volume with any face of the offtake volume. The offtake

volume can be a time-dependent volume. (This checking of the offtake junction can be turned off by Card 1, Option 29. This is useful when running old decks). (internally *isstratinpd7*, *isstratinpd8*, *isstratinpd17*, *isstratinpd18*)

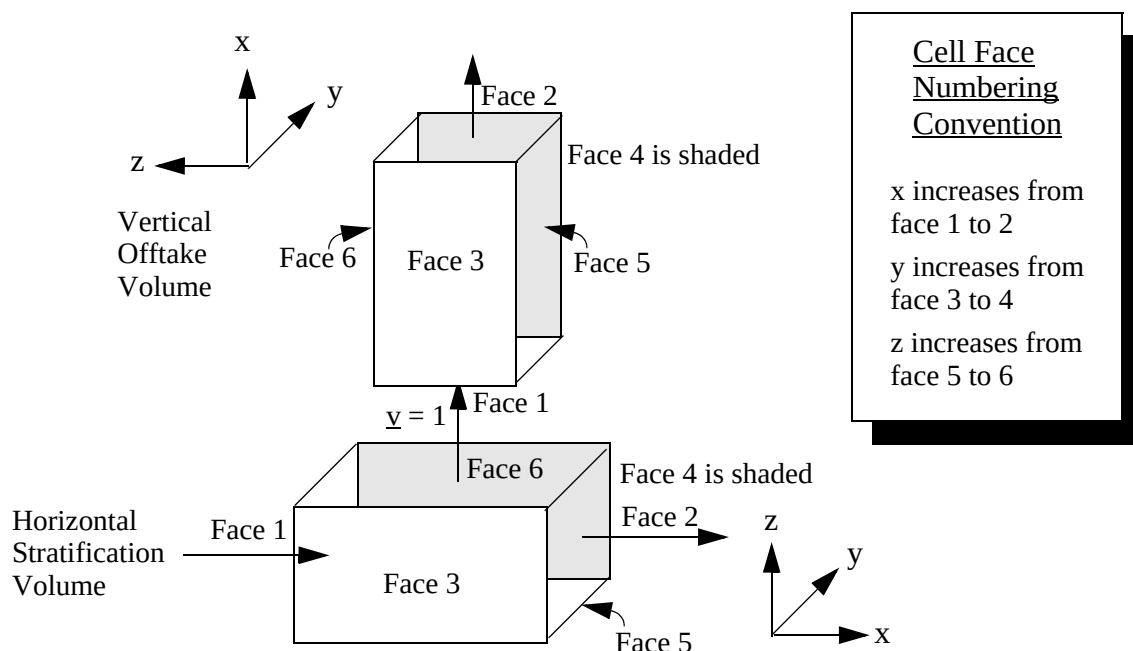


Figure 9.6-1 Example of a top offtake junction, $v = 1$.

- c-flag (*ick*) The digit $\underline{c}$ specifies choking options. $\underline{c} = 0$ means that the choking model will be applied, and $\underline{c} = 1$ means that the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske or Moody (option 54) critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active. (internally *isnochokflg4*)
- a-flag (*irf*) The digit $\underline{a}$ specifies area change options. $\underline{a} = 0$ means a smooth area change and only the user supplied K_{loss} is applied.
- [$\underline{a} = 1$] $\underline{a} = 1$ means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta.
- [$\underline{a} = 2$] $\underline{a} = 2$ means a partial abrupt area change model and it is the same as $\underline{a} = 1$ except no code calculated forward and reverse contraction/expansion K_{loss} is applied. See discussion for W4(R) regarding the K_{loss} flow area reference, which varies based on the a-flag option selected here. (internally *isabrareachgflg8*, *isnolosscoefabrjun29*)

- h-flag (*ihf*)** The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two velocity momentum equations) option, and h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit printout will show a 1. (internally *is2vel1velfag9*)
- s-flag (*isf*)** The digit s specifies momentum flux options. s = 0 uses momentum flux in both the to volume and the from volume. s = 1 uses momentum flux in the from volume, but not in the to volume. s = 2 uses momentum flux in the to volume, but not in the from volume. s = 3 does not use momentum flux in either the to or the from volume. (internally *istomomflxoffopt12, isfrmmomflxopt13*)
- W7(R)** The meaning of this word is different, depending on which the critical flow model has been selected (see the c-flag under W6 above).
- If the Henry-Fauske or Moody (option 54) critical flow model is active, the discharge coefficient is entered. If W7 and W8 are missing, W7 is set to 1.0, and W8 is set to 0.14.
- If the original RELAP5 critical flow model is active, the subcooled discharge coefficient is entered. This entry applies only to subcooled liquid choked flow calculations. The entry must be > 0.0 and < 2.0. If W7, W8, and W9 are missing, they are each set to 1.0.
- W8(R)** The meaning of this word is different, depending on which the critical flow model has been selected (see the c-flag under W6 above).
- If the Henry-Fauske or Moody critical flow model is active, the thermal nonequilibrium constant is entered. If W7 is entered, and W8 is missing, W8 is set to 0.14. If W8 is < 0.01, the equilibrium option is used, and W8 is reset to 0.0. If W8 is >1000.0, the frozen option is used, and W8 is reset to 100.0.
- If the original RELAP5 critical flow model is active, the two-phase discharge coefficient is entered. This entry applies only to two-phase choked flow calculations. The entry must be > 0.0 and < 2.0. If W7 is entered, and W8 and W9 are missing, W8 and W9 are each set to 1.0.
- W9(R)** The meaning of this word is different, depending on which critical flow model has been selected (see the c-flag under W6 above) and on the type of offtake model selected (see the y-flag under W6 above).
- If the Henry-Fauske or Moody critical flow model is active and the horizontal offtake mode with y = 4 is not used, this word is not used and must not be entered. If it is entered, an input error results. This drastic action is to prevent the user from using an older deck with the new code without changing the three discharge coefficients from their values for

the original RELAP5 critical flow model to an appropriate set for the Henry-Fauske or Moody critical flow model. W8 has a completely different meaning in the two models.

If the Henry-Fauske or Moody critical flow model is active and the horizontal offtake mode with $\underline{v} = 4$ is specified, this word is the angle between the horizontal and the offtake junction (-90 to +90 degrees).

If the original RELAP5 critical flow model is active, the superheated discharge coefficient is entered. This entry applies only to superheated vapor choked flow calculations. The entry must be > 0.0 and < 2.0 . If W7 and W8 are entered, and W9 is missing, then W9 is set to 1.0.

A9.6.2 Single-Junction Diameter and CCFL Data, CCC0110

This card is optional. The defaults indicated for each word are used if the card is not entered. This card can be used for three purposes: (1) interphase drag, (2) CCFL model, and (3) lumped offtake model.

1. If this card is being used to specify only the junction hydraulic diameter for the interphase drag calculation (i.e., $\underline{f} = 0$ in Word 6 of cards CCC0101-0109), then, the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (they will not be used).
2. If this card is being used for the CCFL model (i.e., $\underline{f} = 1$ in Word 6 of cards CCC0101-0109), then enter all four words for the appropriate CCFL model if values different from the default values are desired.
3. If this card is being used to specify the diameter of a single offtake junction when lumping two or more offtake pipes into a single pipe, (i.e., $\underline{v} = 1, 2, 3,$ or 4 in Word 6 of cards CCC0101-0109), then the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (they will not be used).

W1(R) Junction hydraulic diameter, D_j (m, ft). This quantity is the junction hydraulic diameter used in the CCFL correlation equation and interphase drag and must be ≥ 0 . This number should be computed from $4.0 \bullet \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If 0 is entered or if the default is used, the junction diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi} \right)^{0.5}$. See Word 3 of cards CCC0101-0109 for the area.

W2(R) Flooding correlation form, β . If 0, the Wallis CCFL form is used. If 1, the Kutateladze CCFL form is used. If between 0.0 and 1.0, Bankoff weighting between the Wallis and Kutateladze CCFL forms is used. This number must be ≥ 0 and ≤ 1 . The default value is 0 (Wallis form). See Section 3 of Volume I for details of the model.

W3(R) Gas intercept, c . This quantity is the gas intercept used in the CCFL correlation (when $H_f^{1/2} = 0$) and must be > 0 . The default value is 1.

W4(R) Slope, m. This quantity is the slope used in the CCFL correlation and must be > 0 . The default value is 1.

A9.6.3 Single-Junction Form Loss Data, CCC0111

This card is optional. The user-specified form loss is given in Words 4 and 5 of card CCC0101 if this card is not entered. If this card is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F \text{Re}^{-C_F} + \text{KFJUNCV}$$

$$K_R = A_R + B_R \text{Re}^{-C_R} + \text{KRJUNCV}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 4 and 5 of card CCC0101. Re is the Reynolds number based on mixture fluid properties. KFJUNCV and KRJUNCV are the contributions to the forward and reverse form loss coefficients obtained from control variables. If this card is being used for the form loss calculation, then enter at least the first four words for the appropriate expression. The fifth and sixth words are optional. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R .

W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.

W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.

W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.

W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

W5(I) Control variable number for specifying a portion of the forward form loss (KFJUNCV). This quantity must be greater than or equal to 0. (*Optional*)

W6(I) Control variable number for specifying a portion of the reverse form loss (KRJUNCV). This quantity must be greater than or equal to 0. (*Optional*)

A9.6.4 Single-Junction Initial Conditions, CCC0201

This card is required for single-junction components.

W1(I) Control word. If 0, the next two words are velocities; if 1, the next two words are mass flows.

W2(R) Initial liquid velocity or mass flow. This quantity is either velocity (m/s, ft/s) [$W1 = 0$] or mass flow (kg/s, lb/s) [$W1 = 1$].

W3(R) Initial vapor velocity or mass flow. This quantity is either velocity (m/s, ft/s) [W1 = 0] or mass flow (kg/s, lb/s) [W1 = 1].

W4(R) Interface velocity (m/s, ft/s). Enter 0.

A9.7 Time-Dependent Junction Component, TMDPJUN

This component is indicated by TMDPJUN for Word 2 on card CCC0000. For major edits, minor edits, and plot variables, the junction in the time-dependent junction component is numbered as CCC000000.

A9.7.1 Time-Dependent Junction Geometry, CCC0101

This card is required for time-dependent junction components.

W1(I) From connection code to a component. This refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. A nonzero N specifies the expanded format. The number N equal to 1 and 2 specifies the inlet and outlet faces, respectively, for the volume's coordinate direction (see Section 2.1). The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, N is restricted to 1 or 2.

W2(I) To connection code to a component. This refers to the component at which the junction coordinate direction ends. See the description for W1 above.

W3(R) Area (m², ft²). If 0, the area is set to the minimum flow area of the adjoining volumes. There are no junction area restrictions for time-dependent junctions.

W4(I) Junction control flag. This word is optional and has the format jefvcahs. It is not necessary to input leading zeros. Only the e-flag can be set. The other flags have to be 0, so the only two allowable entries for this word are 0 or 1000000.

e-flag The digit e specifies the modified PV term in the energy equations. e = 0 means that the modified PV term will not be applied, and e = 1 means that it will be applied.

A9.7.2 Time-Dependent Junction Data Control Word, CCC0200

This card is optional. If this card is missing, the second and third words of the time-dependent data are assumed to be velocities.

- W1(I) Control word. If 0, the second and third words of the time-dependent junction data in cards CCC0201-0299 are velocities. If 1, the second and third words of the time-dependent junction data in cards CCC0201-0299 are mass flows. In both cases, the fourth word is interface velocity and should be entered as 0.
- W2(I) Table trip number. This word is optional. If missing or 0 and Word 3 is missing, no trip is used, and the time argument is the advancement time. If nonzero and Word 3 is missing, this number is the trip number and the time argument is -1.0 if the trip is false, and the advancement time minus the trip time if the trip is true.
- W3(A) Alphanumeric part of variable request code. This quantity is optional. If present, this word and the next are a variable request code that specifies the search argument for the table lookup and interpolation. If the trip number is 0, the specified argument is always used. If the trip number is nonzero, -1.0E75 is used if the trip is false, and the specified argument is used if the trip is true. TIME can be selected, but note that the trip logic is different than when this word is omitted. The variable MFLOWJ should not be used as a search variable; a user-initialized control variable that uses MFLOWJ should be used instead.
- W4(I) Numeric part of variable request code. This is assumed 0 if missing.

A9.7.3 Time-Dependent Junction Data, CCC0201-0299

These cards are required for time-dependent junction components. A set of data consists of the search variable (e.g., time) followed by the required data indicated by control Word 1 on card CCC0200. The card numbers need not be consecutive, but the value of the search variable in a succeeding set must be equal to or greater than the value in the previous set. One or more sets of data up to 5000 sets may be entered. If the phase or material is not present, 0 may be entered for a velocity or flow. The interpolation and card formats for the time-dependent data are identical to that in Section A9.5.3 (cards CCC0201-0209, Time-Dependent Volume Data cards).

When doing a single-phase problem and entering velocities here, the same value should be entered for both liquid and vapor velocities. If entering mass flows, the correct value should be entered for either liquid or vapor (whichever single-phase is being modeled) and the other entry should be 0.

If the user wants to specify the vapor void fraction as a function of time in the time-dependent volume, and the total mass flow as a function of time in the time-dependent junction, then both the phasic (gas and liquid) mass flow rates must be calculated and entered in these cards.

- W1(R) Search variable (e.g., time).
- W2(R) Liquid velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s), depending on control Word 1 on card CCC0200.

W3(R) Vapor velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s), depending on control Word 1 on card CCC0200.

W4(R) Interface velocity (m/s, ft/s). This feature has not been implemented yet, so enter 0.

As described above, sets may be entered one or more per card.

A9.8 Pipe Component, PIPE

A pipe component is indicated by PIPE. An annulus component is similar to a pipe component and is covered in the next section. The annulus component must be vertical and all the water is in the film on the wall (i.e., no drops) when in the annular mist flow regime.

More than one junction may be connected to the inlet or outlet of a pipe. If an end has no junctions, that end is considered a closed end. For major edits, minor edits, and plot variables, the volumes in the pipe component are numbered as CCCNN0000, where NN is the volume number ($01 \leq NN \leq 99$). The junctions in the pipe component are numbered as CCCJJ0000, where JJ is the junction number ($01 \leq JJ \leq 98$).

The general input for a pipe component assumes that the pipe has at least two volumes with one junction separating the two volumes. It is possible to input a one-volume pipe. In order to implement this special case, the user must set the number of volumes and the volume number on the volume cards to 1. In addition, the user should not input any of the junction cards. A single volume is much better suited for this type of component.

The volumes in a pipe are usually considered one-dimensional components and flow in the volumes is along the x-coordinate. Crossflow junctions can connect to any of the pipe volumes in the y- and z-coordinate directions (faces 3, 4, 5, or 6) using a form of the momentum equation that does or does not include momentum flux terms. It is also possible to connect external junctions to the x-coordinate direction faces (1 or 2) of any of the pipe volumes using a form of the momentum equation that does or does not include the momentum flux terms. It is also possible to include or not include the momentum flux terms in internal pipe junctions.

A9.8.1 Pipe Information, CCC0001

This card is required.

W1(I) Number of volumes, nv. The number nv must be greater than 0 and less than 100. The number of associated junctions internal to these components is nv-1. The outer junctions are described by other components.

A9.8.2 Pipe X-Coordinate Areas, CCC0101-0199

These cards are required, and the card numbers need not be consecutive. The format is two words per set in sequential expansion format for nv sets. (The corresponding y- and z-coordinate cards are CCC1601-1699 and CCC1701-1799). The words for one set are

W1(R) Area (m^2 , ft^2).

W2(I) Volume number.

A9.8.3 Pipe Junction Areas, CCC0201-0299

These cards are optional, and, if entered, the card numbers need not be consecutive. The format is two words per set in sequential expansion format for nv-1 sets.

W1(R) Internal junction flow area (m^2 , ft^2). If cards are missing or a word is 0, the junction flow area is set to the minimum area of the adjoining volumes. For abrupt area changes, the junction area must be equal to or less than the minimum of the adjacent volume areas. There is no restriction for smooth area changes.

W2(I) Junction number.

A9.8.4 Pipe X-Coordinate Lengths, CCC0301-0399

These cards are required. The format is two words per set in sequential expansion format for nv sets. Card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC1801-1899 and CCC1901-1999).

W1(R) Length (m, ft).

W2(I) Volume number.

A9.8.5 Pipe Volumes, CCC0401-0499

These cards are optional, and if not entered, the volumes are set to 0. The format is two words per set in sequential expansion format for nv sets. Card numbers need not be consecutive.

W1(R) Volume (m^3 , ft^3). The code requires that each volume equal the x-direction flow area times the x-direction length. If activated, the code also requires each volume equal the y-direction flow area times the y-direction length, and each volume equal the z-direction flow area times the z-direction length. For any volume, at least two of the three quantities, x-direction area, the x-direction length, or volume, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the quantities are 0, the volume must equal the x-direction area times the x-direction length within a relative error of 0.000001. The same relative error check is done for the y- and z-directions. If both the y-direction area and y-direction length are not entered or are 0, the y-direction length is

computed from $2.0 \bullet \left(\frac{x - \text{direction area}}{\pi} \right)^{0.5}$ and the y-direction area is computed from $\frac{\text{volume}}{\text{y-direction length}}$. The same is true for the z-direction.

W2(I) Volume number.

A9.8.6 Pipe Volume Azimuthal Angles, CCC0501-0599

These cards are optional, and if not entered, the angles are set to 0. The angles on these cards are used for the 3-D drawing programs, so it is recommended that the user enter these values if they are known. The format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive.

W1(R) Azimuthal angle (degrees). The absolute value of the angle must be ≤ 360 degrees.

W2(I) Volume number.

A9.8.7 Pipe Volume Vertical Angles, CCC0601-0699

These cards are required. The format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive.

W1(R) Vertical angle (degrees). The absolute value of the angle must be less than or equal to 90 degrees. This angle is used in the interphase drag calculation.

W2(I) Volume number.

A9.8.8 Pipe X-Coordinate (Elevation) Changes, CCC0701-0799

These cards are optional. If these cards are missing, the coordinate changes or elevation changes are computed from the x-coordinate length and a rotation matrix computed from the angle information. If these cards are entered, the entered data becomes the x-coordinate change or elevation change data. Two formats entering 1 or 3 coordinate changes per volume are provided. The card format is two or four words per set in sequential expansion format up to nv sets, and card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC2101-2199 and CCC2201-2299).

1 coordinate change per volume format:

W1(R) Elevation change. This is the coordinate change along the fixed x-axis due to the traverse from inlet to outlet along the local x-coordinate, Δ_{zx} (m, ft). A positive value is an increase in elevation. The magnitude must be equal to or less than the length. See Section 2.4.1 of Volume II of the manual for further discussion. When the absolute value of the elevation angle determined by the ratio of the elevation change (this Word 6) and the length (Word

2) is less than or equal to 45 degrees, the horizontal flow regime map is used; when the ratio is greater than 45 degrees, the vertical flow regime map is used.

W2(I) Volume number.

Three coordinate changes per volume format:

W1(R) Coordinate change along the fixed x-axis due to traverse from inlet to outlet along the local x-coordinate, Δ_{xx} (m, ft).

W2(R) Coordinate change along the fixed y-axis due to traverse from inlet to outlet along the local x-coordinate, Δ_{yx} (m, ft).

W3(R) Coordinate change along the fixed z-axis due to traverse from inlet to outlet along the local x-coordinate, Δ_{zx} (m, ft).

W4(I) Volume number.

A9.8.9 Pipe Volume X-Coordinate Friction Data, CCC0801-0899

These cards are required. The card format is three words per set for nv sets, and card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC2301-2399 and CCC2401-2499).

W1(R) Wall roughness (m, ft).

W2(R) Hydraulic diameter (m, ft). This should be computed from $4.0 \bullet \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If 0, the hydraulic diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi} \right)^{0.5}$. A check is made to ensure that the roughness is less than half the hydraulic diameter. See Word 1 on cards CCC0101-0109 for the area.

W3(I) Volume number.

A9.8.10 Pipe Junction Loss Coefficients, CCC0901-0999

These cards are optional and if missing, the energy loss coefficients are set to 0. The card format is three words per set in sequential expansion format for nv-1 sets, and card numbers need not be consecutive.

W1(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that

phase is positive or 0. Note: a variable loss coefficient may be specified (see Section A9.8.29). The interpretation and use of this loss coefficient vary based upon the “a” junction control flag selected for this junction on cards CCC1101-1199. If the smooth area change option is selected ($a = 0$), the entry here is the loss coefficient applied to the junction flow area resulting (from the entries on cards CCC0201-0299). If one of the abrupt area change options is selected ($a = 1$ or 2), the loss coefficient entered here is adjusted by the square of the ratio of the junction area and the minimum flow area between the two hydrodynamic volumes connected by this junction (from cards CCC0101-0199). This adjusted loss coefficient is applied to the junction flow area. If the $a = 1$ option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

Example: Two volumes with volume average flow areas of 0.3 m^2 and 0.4 m^2 are connected with a junction ($A_{\text{jun}} = 0.075 \text{ m}^2$). The user supplied $K_{\text{loss_user}}$ is 10.0. For the smooth area change option ($a = 0$), the $K_{\text{loss_user}}$ of 10.0 is applied directly to the junction. For the abrupt area change option ($a = 1$ or 2), the K_{loss} applied at the junction is $K_{\text{loss_jun}} = K_{\text{loss_user}} * (A_{\text{jun}}/A_{\text{min}})^2$. For this example, $K_{\text{loss_jun}} = 10.0 * (0.075/0.3)^2 = 0.625$.

W2(R) Reynolds number independent reverse flow energy loss coefficient, A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. Note: a variable loss coefficient may be specified (see Section A9.8.29). See information for the previous word regarding interpretation and use of this loss coefficient.

W3(I) Junction number.

A9.8.11 Pipe Volume X-Coordinate Control Flags, CCC1001-1099

These cards are required. The card format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC2701-2799 and CCC2801-2899).

W1(I) Volume control flags. This word has the format tl

v

bfe. It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags. Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f-flag is the only coordinate direction oriented flag. These words enter the scalar oriented flags and the x-coordinate flags for each volume in the component.

t-flag. The digit t specifies whether the thermal front tracking model is to be used; t = 0 specifies that the front tracking model is not to be used for the volume, and t = 1 specifies that the

front tracking model is to be used for the volume. The thermal front tracking model can only be applied to vertically-oriented components.

l-flag	The digit <u>l</u> specifies whether the mixture level tracking model is to be used; <u>l</u> = 0 specifies that the level model not be used for the volume, and <u>l</u> = 1 specifies that the level model be used for the volume. The mixture level tracking model can only be applied to vertically-oriented components.
p-flag	The digit <u>p</u> specifies whether the water packing scheme is to be used. <u>p</u> = 0 specifies that the water packing scheme is to be used for the volume, and <u>p</u> = 1 specifies that the water packing scheme is not to be used for the volume.
v-flag	The digit <u>v</u> specifies whether the vertical stratification model is to be used. <u>v</u> = 0 specifies that the vertical stratification model is to be used for the volume, and <u>v</u> = 1 specifies that the vertical stratification model is not to be used for the volume.
b-flag	The digit <u>b</u> specifies the interphase friction that is used. <u>b</u> = 0 means that the pipe interphase friction model will be applied, <u>b</u> = 1 means that the rod bundle interphase friction model will be applied, <u>b</u> = 2 means that the ORNL ANS interphase friction model will be applied (see card ccc3101 in Section A9.8.30).
f-flag	The digit <u>f</u> specifies if wall friction is to be computed. <u>f</u> = 0 specifies that wall friction effects are to be computed along the x-coordinate of the volume, and <u>f</u> = 1 specifies that wall friction effects are NOT to be computed along the x-coordinate.
e-flag	The digit <u>e</u> specifies if nonequilibrium or equilibrium is to be used. <u>e</u> = 0 specifies that a nonequilibrium (unequal temperature) calculation is to be used, and <u>e</u> = 1 specifies that an equilibrium (equal temperature) calculation is to be used. Equilibrium volumes should not be connected to nonequilibrium volumes. The equilibrium option is provided only for comparison to other codes.
W2(I)	Volume number.

A9.8.12 Pipe Junction Control Flags, CCC1101-1199

These cards are required. The card format is two words per set in sequential expansion format for nv-1 sets, and card numbers need not be consecutive.

W1(I)	Junction control flags. This word has the format <u>jefvcahs</u> . It is not necessary to input leading zeros. For the pipe component, the junction control flag is limited to the format <u>0ef0cahs</u> .
e-flag	The digit <u>e</u> specifies the modified PV term in the energy equations. <u>e</u> = 0 means that the modified PV term will not be applied, and <u>e</u> = 1 means that it will be applied.

f-flag	The digit <u>f</u> specifies CCFL options. <u>f</u> = 0 means that the CCFL model will not be applied, and <u>f</u> = 1 means that the CCFL model will be applied.
c-flag	The digit <u>c</u> specifies choking options. <u>c</u> = 0 means that the choking model will be applied, and <u>c</u> = 1 means that the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske or Moody (option 54) critical flow model is active. If card 1 is present and contains Option 50, the original RELAP5 critical flow model is active. Unlike junctions for certain other hydrodynamic components, only the default values of discharge coefficients and factors used in the critical flow models are allowed for junctions in PIPE components. Problem renodalization should be used to circumvent this limitation should it prove to be significant. For the Henry-Fauske or Moody critical flow model, the default value for the discharge coefficient is 1.0 and the default value for the thermal nonequilibrium constant is 0.14. For the original RELAP5 critical flow model, the default values for the subcooled, two-phase and superheated discharge coefficients each are 1.0.
a-flag	The digit <u>a</u> specifies area change options. <u>a</u> = 0 means a smooth area change and only the user supplied K_{loss} is applied. <u>a</u> = 1 means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. <u>a</u> = 2 means a partial abrupt area change model and it is the same as <u>a</u> = 1 except no code calculated forward and reverse contraction/expansion K_{loss} is applied. See discussion for W1(R) regarding the K_{loss} flow area reference, which varies based on the a-flag option selected here.
h-flag	The digit <u>h</u> specifies nonhomogeneous or homogeneous. <u>h</u> = 0 specifies the nonhomogeneous (two-velocity momentum equations) option, and <u>h</u> = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (<u>h</u> = 1 or 2), the major edit printout will show a 1.
s-flag	The digit <u>s</u> specifies momentum flux options. <u>s</u> = 0 uses momentum flux in both the <u>to</u> volume and the <u>from</u> volume. <u>s</u> = 1 uses momentum flux in the <u>from</u> volume, but not in the <u>to</u> volume. <u>s</u> = 2 uses momentum flux in the <u>to</u> volume but not in the <u>from</u> volume. <u>s</u> = 3 does not use momentum flux in either the <u>to</u> or the <u>from</u> volume. For this component, the option <u>s</u> = 0 is the usual recommendation (momentum flux in both volumes). The other options <u>s</u> = 1, 2, and 3 are included to allow consistency for this flag for other components (single-junction, branch junction, etc.).
W2(I)	Junction number.

A9.8.13 Pipe Volume Initial Conditions, CCC1201-1299

These cards are required. The card format is seven words per set in sequential expansion format for nv sets, and card numbers need not be consecutive.

W1(I) Control word. This word has the format $\underline{\epsilon}\underline{b}\underline{t}$. It is not necessary to input leading zeros.

ϵ -flag The digit $\underline{\epsilon}$ specifies the fluid, where $\underline{\epsilon} = 0$ is the default fluid. The value for $\underline{\epsilon} > 0$ corresponds to the position number of the fluid type indicated on the 120-9 cards (i.e., $\underline{\epsilon} = 1$ specifies H_2O , $\underline{\epsilon} = 2$ specifies D_2O , etc.). The default fluid is that set for the hydrodynamic system by cards 120-129 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-9 or these control words must be consistent (i.e., not specify different fluids). If cards 120-9 are not entered and all control words use the default $\underline{\epsilon} = 0$, then H_2O is assumed as the fluid. It is recommended that the user use the 120-9 cards to set the fluid type in the systems.

b-flag The digit $\underline{b}$ specifies whether boron is present or not. $\underline{b} = 0$ specifies that the volume liquid does not contain boron; $\underline{b} = 1$ specifies that a boron concentration in mass of boron per mass of liquid (which may be 0) is being entered on the CCC2001-2099 cards.

t-flag The digit $\underline{t}$ specifies how the following words are to be used to determine the initial thermodynamic state. Entering $\underline{t}$ equal to 0-3 and 7 specifies one component (steam/water). Entering $\underline{t}$ equal to 4-6 allows the specification of two components (steam/water and noncondensable gas).

With options $\underline{t}$ equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.

One Component (Steam/Water), Equilibrium or Non-equilibrium

$[P, U_f, U_g, \alpha_g]$ If $\underline{t} = 0$, the next four words are interpreted as pressure (Pa, lb_f/in^2), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions, depending on the internal energies used to define the thermodynamic state. W6 should be 0.0.

One Component (Steam/Water), Equilibrium

$[T, x_s]$ If $\underline{t} = 1$, the next two words are interpreted as temperature (K, $^{\circ}F$) and static quality in equilibrium condition. W4, W5, and W6 should be 0.0.

- [P,x_s] If $\underline{t} = 2$, the next two words are interpreted as pressure (Pa, lb_f/in²) and static quality in equilibrium condition. W4, W5, and W6 should be 0.0.
- [P,T] If $\underline{t} = 3$, the next two words are interpreted as pressure (Pa, lb_f/in²) and temperature (K, °F) in equilibrium condition. W4, W5, and W6 should be 0.0.
- [P,T_f,T_g,α_g] If $\underline{t} = 7$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid temperature (K, F), vapor temperature (K, F), and vapor void fraction. These quantities will be interpreted as equilibrium conditions. The liquid temperature has to be equal to or less than the saturation temperature, and the vapor temperature has to be equal to or greater than the saturation temperature. Enter boron in W6 if needed. This input option was added to allow easier comparisons with TRACE calculations.

Two Components (Steam/Water/Air), Equilibrium or Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0\text{E-}9 \leq x_s \leq 0.99999999$,

Single phase if $x_s < 1.0 \text{ E-}9$ or $x_s > 0.99999999$

The static quality is given by $x_s = M_g/(M_g + M_f)$, where $M_g = M_s + M_n$. Section 3.2 of Volume I of the manual discusses this further.

- [P,T,x_s] If $\underline{t} = 4$, the next three words are interpreted as pressure (Pa, lb_f/in²), temperature (K, °F), and static quality in equilibrium condition. Using this input option with static quality > 0.0 and ≤ 1.0, saturated noncondensables will result. W5 and W6 should be 0.0. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. If the pressure is greater than the critical pressure, $P > P_{\text{crit}}$, then the static quality, x_s (mass steam + air)/(mass steam + air + water) has to be large enough to give a partial steam pressure that is less than the critical pressure, $P_s < P_{\text{crit}}$. Otherwise an input error will occur. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensable (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option.
- [T,x_s,x_n] If $\underline{t} = 5$, the next three words are interpreted as steam saturation temperature (K, °F), static quality, and noncondensable quality in equilibrium condition. W5 and W6 should be 0.0. Both the static and noncondensable qualities are restricted to be between 1.0 E-9 and 0.99999999. The liquid will be subcooled. The input temperature determines the steam

partial pressure. However, the total pressure may be greater than the critical pressure, which will cause an input error.

[P,U_f,U_g,α_g,x_n] If $t = 6$, the next five words are interpreted as pressure (Pa, lb_f/in²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing ($t = 0$). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor-specific internal energy.

W2-W6(R) Quantities as described under Word 1. Five quantities must be entered, and zeros should be entered for unused quantities. If any control word (Word 1) indicates that boron is present, cards CCC2001-2099 must be entered to define the initial boron concentrations. Boron concentrations are not entered in Words 2-6 for pipe components.

W7(I) Volume number.

A9.8.14 Pipe Junction Conditions Control Words, CCC1300

This card is optional, and, if missing, velocities are assumed on cards CCC1301-1399.

W1(I) Control word. If 0, the first and second words of each set on cards CCC1301-1399 are velocities. If 1, the first and second words of each set on cards CCC1301-1399 are mass flows.

A9.8.15 Pipe Junction Initial Conditions, CCC1301-1399

These cards are required.

W1(R) Initial liquid velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W2(R) Initial vapor velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W3(R) Interface velocity (m/s, ft/s). This capability has not yet been implemented, so enter 0.

W4(I) Junction number.

A9.8.16 Pipe Junction Diameter and CCFL Data, CCC1401-1499

These cards are optional. The defaults indicated for each word are used if the card is not entered. If this card is being used to specify only the junction hydraulic diameter for the interphase drag calculation, (i.e., $f = 0$ in Word 1 of cards CCC1101-1199) then the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (will not be used). If this card is being used for the CCFL model (i.e., $f = 1$ in Word 1 of cards CCC1101-1199), then enter all four words for the appropriate CCFL model if values different from the default value are desired.

- W1(R) Junction hydraulic diameter, D_j (m, ft). This quantity is the junction hydraulic diameter used in the CCFL correlation equation and interphase drag and must be ≥ 0 . The number should be computed from $4.0 \bullet \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If a 0 is entered or if the default is used, the junction diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi} \right)^{0.5}$. See Word 1 of cards CCC0201-0299 for the area.
- W2(R) Flooding correlation form, β . If 0, the Wallis CCFL form is used. If 1, the Kutateladze CCFL form is used. If between 0.0 and 1.0, Bankoff weighting between the Wallis and Kutateladze CCFL forms is used. This number must be ≥ 0 and ≤ 1 . The default value is 0 (Wallis form).
- W3(R) Gas intercept, c . This quantity is the gas intercept used in the CCFL correlation (when $H_f^{1/2} = 0$) and must be > 0 . The default value is 1.
- W4(R) Slope, m . This quantity is the slope used in the CCFL correlation and must be > 0 . The default value is 1.
- W5(I) Junction number.

A9.8.17 Pipe Y-Coordinate Areas, CCC1601-1699

These cards are optional and if entered activate the y-coordinate for each volume and allow the full one-dimensional momentum equations to be used in connections to the y-faces. (The corresponding x-coordinate cards are CCC0101-0199). The format is two words per set in sequential expansion format for n_v sets. The card numbers need not be consecutive. The words for one set are:

- W1(R) Area (m^2 , ft^2). If these cards are missing, this y-direction area is computed from (volume)/(y-direction length).
- W2(I) Volume number.

A9.8.18 Pipe Z-Coordinate Areas, CCC1701-1799

These cards are optional and if entered activate the z-coordinate for each volume and allow the full one-dimensional momentum equations to be used in connections to the z-faces. (The corresponding x-coordinate cards are CCC0101-0199). The format is two words per set in sequential expansion format for nv sets. The card numbers need not be consecutive. The words for one set are:

W1(R) Area (m^2 , ft^2). If these cards are missing, this z-direction area is computed from (volume)/(z-direction length).

W2(I) Volume number.

A9.8.19 Pipe Y-Coordinate Lengths, CCC1801-1899

These cards are optional and, if entered, activate the y-coordinate for each volume and allow the full one-dimensional momentum equation for connections to the y-faces. The format is two words per set in sequential expansion format for nv sets. Card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC0301-0399).

W1(R) Length (m, ft). If these cards are missing and cards CCC1601-1699 are missing, this y-direction length is computed from $2.0 \cdot \left(\frac{\text{x-direction area}}{\pi} \right)^{0.5}$. If these cards are missing and cards CCC1601-1699 are present, this y-direction length is computed from (volume)/(y-direction area).

W2(I) Volume number.

A9.8.20 Pipe Z-Coordinate Lengths, CCC1901-1999

These cards are optional and, if entered, activate the z-coordinate for each volume and allow the full one-dimensional momentum equation for connections to the z-faces. The format is two words per set in sequential expansion format for nv sets. Card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC0301-0399).

W1(R) Length (m, ft). If these cards are missing and cards CCC1701-1799 are missing, this z-direction length is computed from $2.0 \cdot \left(\frac{\text{x-direction area}}{\pi} \right)^{0.5}$. If these cards are missing and cards CCC1701-1799 are present, this z-direction length is computed from (volume)/(z-direction area).

W2(I) Volume number.

A9.8.21 Pipe Initial Boron Concentrations, CCC2001-2099

These cards are required only if boron is specified in one of the control words (Word 1) in cards CCC1201-1299. The card format is two words per set in sequential expansion format for nv sets. Boron concentrations must be entered for each volume, and 0 should be entered for those volumes whose associated control word did not specify boron.

W1(R) Boron concentration (mass of boron per mass of liquid).

W2(I) Volume number.

A9.8.22 Pipe Y-Coordinate (Elevation) Changes, CCC2101-2199

These cards are optional. If these cards are missing, the coordinate changes are computed from the y-coordinate length and a rotation matrix computed from the angle information. If these cards are entered, the entered data overwrites the previously computed coordinate change data. The card format is four words per set in sequential expansion format up to nv sets, and card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC0701-0799).

W1(R) Coordinate change along fixed x-axis due to traverse from inlet to outlet along the local y-coordinate, Δ_{xy} (m, ft).

W2(R) Coordinate change along fixed y-axis due to traverse from inlet to outlet along the local y-coordinate, Δ_{yy} (m, ft).

W3(R) Coordinate change along fixed z-axis due to traverse from inlet to outlet along the local y-coordinate, Δ_{zy} (m, ft).

W4(I) Volume number.

A9.8.23 Pipe Z-Coordinate (Elevation) Changes, CCC2201-2299

These cards are optional. If these cards are missing, the coordinate changes are computed from the z-coordinate length and a rotation matrix computed from the angle information. If these cards are entered, the entered data overwrites the previously computed coordinate change data. The card format is four words per set in sequential expansion format up to nv sets, and card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC0701-0799).

W1(R) Coordinate change along fixed x-axis due to traverse from inlet to outlet along the local z-coordinate, Δ_{xz} (m, ft).

W2(R) Coordinate change along fixed y-axis due to traverse from inlet to outlet along the local z-coordinate, Δ_{yz} (m, ft).

W3(R) Coordinate change along fixed z-axis due to traverse from inlet to outlet along the local z-coordinate, Δ_{zz} (m, ft).

W4(I) Volume number.

A9.8.24 Pipe Volume Y-Coordinate Friction Data, CCC2301-2399

These cards are optional and may be entered if area or length data was entered for the y-coordinate. If the cards are not entered, the wall roughness defaults to 0 and the default hydraulic diameter is computed from $\frac{4.0 \bullet \text{y-direction volume flow area}}{(\pi \bullet \text{x-direction volume flow area})^{0.5}} = \frac{4A_y}{\sqrt{\pi A_x}}$. The corresponding x-coordinate cards are CCC0801-0899.

W1(R) Wall roughness (m, ft).

W2(R) Hydraulic diameter (m, ft). A check is made to ensure that the y-direction roughness is less than half the y-direction hydraulic diameter. See Word 1 on cards CCC0101-0109 for the area.

W3(I) Volume number.

A9.8.25 Pipe Volume Z-Coordinate Friction Data, CCC2401-2499

These cards are optional and have to be entered if area or length data was entered for the z-coordinate. If the cards are not entered, the wall roughness defaults to 0 and the default hydraulic diameter is computed from $\frac{4.0 \bullet \text{z-direction volume flow area}}{(\pi \bullet \text{x-direction volume flow area})^{0.5}} = \frac{4A_z}{\sqrt{\pi A_x}}$. The corresponding x-coordinate cards are CCC0801-0899.

W1(R) Wall roughness (m, ft).

W2(R) Hydraulic diameter (m, ft). A check is made to ensure that the z-direction roughness is less than half the z-direction hydraulic diameter. See Word 1 on cards CCC0101-0109 for the area.

W3(I) Volume number.

A9.8.26 Pipe Volume Additional Wall Friction Data, CCC2501-2599

These cards are optional. If these cards are not entered, the default values are 1.0 for the laminar shape factor and 0.0 for the viscosity ratio exponent. The card format is seven words per set in sequential expansion format for nv sets and card numbers need not be consecutive. A description of this input is presented in Section 3 of Volume I.

W1(R)	Shape factor for x-coordinate.
W2(R)	Viscosity ratio exponent for x-coordinate.
W3(R)	Shape factor for y-coordinate.
W4(R)	Viscosity ratio exponent for y-coordinate.
W5(R)	Shape factor for z-coordinate.
W6(R)	Viscosity ratio exponent for z-coordinate.
W7(I)	Volume number.

A9.8.27 Pipe Y-Coordinate Control Flags, CCC2701-2799

These cards are optional. The card format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC1001-1099).

W1(I)	Volume control flags. This word has the general format <u>tlpvbfe</u> , but this word is limited to <u>00000f0</u> since it only enters the coordinate oriented flags for the y-direction.
f-flag	The digit <u>f</u> specifies whether wall friction is to be computed. <u>f</u> = 0 specifies that wall friction effects are to be computed along the y-coordinate direction in the volume, and <u>f</u> = 1 specifies that wall friction effects are not to be computed along the y-coordinate direction.
W2(I)	Volume number.

A9.8.28 Pipe Z-Coordinate Control Flags, CCC2801-2899

These cards are optional. The card format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC1001-1099).

W1(I)	Volume control flags. This word has the general format <u>tlpvbfe</u> , but this word is limited to <u>00000f0</u> since it only enters the coordinate oriented flags for the z-direction.
f-flag	The digit <u>f</u> specifies whether wall friction is to be computed. <u>f</u> = 0 specifies that wall friction effects are to be computed along the z-coordinate direction in the volume, and <u>f</u> = 1 specifies that wall friction effects are not to be computed along the z-coordinate direction.

W2(I) Volume number.

A9.8.29 Pipe Junction Form Loss Data, CCC3001-3099

These cards are optional. The user-specified form loss is given in Words 1 and 2 of cards CCC0901-0999 if this card is not entered. If this card is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F \text{Re}^{-C_F}$$

$$K_R = A_R + B_R \text{Re}^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 1 and 2 of cards CCC0901-0999. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all five words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R .

W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.

W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.

W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.

W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

W5(I) Junction number.

A9.8.30 Pipe ORNL ANS Interphase Model Values, CCC3101-3199

These cards are required of any of the interphase friction flags b in the volume control flags entered on cards CCC1001-1099 are set to 2.

W1(R) Gap (distance between side walls, short length, pitch, channel width) (m, ft).

W2(R) Span (distance from one end to the other, long length) (m, ft).

W3(I) Volume number.

A9.8.31 Pipe Initial Noncondensable Mass Fractions, CCC3201-3299

These cards are optional. They are required only if the user desires to use initial noncondensable mass fractions that are different than the default values specified on card 115. The card format is M = N+1 words per set in sequential expansion format for **nv** sets. The plus one is for the volume number. Initial noncondensable mass fractions must be entered for each volume, and the default set on card 115 should be entered for those volumes whose values are the same as the default set on card 115.

W1-WN(R) Initial mass fraction for each noncondensable gas type as specified on card 110 where N is the number of gas types specified on card 110.

WM(I) Volume number.

A9.9 Annulus Component, ANNULUS

An annulus component is indicated by ANNULUS for Word 2 on card CCC0000. The annulus and pipe components are similar, except that the annulus component must be vertical and all the water is in the film (i.e., no drops) when in the annular mist flow regime. The remaining input for both components is identical.

More than one junction may be connected to the inlet or outlet. If an end has no junctions, that end is considered a closed end. For major edits, minor edits, and plot variables, the volumes in the annulus component are numbered as CCCNN0000, where NN is the volume number ($01 \leq NN \leq 99$). The junctions in the annulus component are numbered as CCCJJ0000, where JJ is the junction number ($01 \leq JJ \leq 98$).

The general input for a annulus component assumes that the annulus has at least two volumes with one junction separating the two volumes. It is possible to input a one-volume annulus. In order to implement this special case, the user must set the number of volumes and the volume number on the volume cards to 1. In addition, the user should not input any of the junction cards.

The volumes in an annulus are usually considered one-dimensional components and flow in the volumes is along the x-coordinate (face 1 or 2). Crossflow junctions can connect to any of the annulus volumes in the y- and z-coordinate directions (faces 3, 4, 5, or 6) using a form of the momentum equation that does or does not include momentum flux terms. It is also possible to connect external junctions to the x-coordinate direction faces of any of the annulus volumes using a form of the momentum equation that does or does not include the momentum flux terms. It is also possible to include or not include the momentum flux terms in internal annulus junctions.

A9.9.1 Annulus Information, CCC0001

This card is required.

W1(I) Number of volumes, nv. The number nv must be greater than 0 and less than 100. The number of associated junctions internal to these components is nv-1. The outer junctions are described by other components.

A9.9.2 Annulus X-Coordinate Areas, CCC0101-0199

These cards are required, and the card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC1601-1699 and CCC1701-1799). The format is two words per set in sequential expansion format for nv sets. The words for one set are

W1(R) Area (m^2 , ft^2).

W2(I) Volume number.

A9.9.3 Annulus Junction Areas, CCC0201-0299

These cards are optional, and, if entered, the card numbers need not be consecutive. The format is two words per set in sequential expansion format for nv-1 sets.

W1(R) Internal junction flow area (m^2 , ft^2). If cards are missing or a word is 0, the junction flow area is set to the minimum area of the adjoining volumes. For abrupt area changes, the junction area must be equal to or less than the minimum of the adjacent volume areas. There is no restriction for smooth area changes.

W2(I) Junction number.

A9.9.4 Annulus X-Coordinate Lengths, CCC0301-0399

These cards are required. The format is two words per set in sequential expansion format for nv sets. Card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC1801-1899 and CCC1901-99).

W1(R) Length (m, ft).

W2(I) Volume number.

A9.9.5 Annulus Volumes, CCC0401-0499

These cards are optional, and, if not entered, the volumes are set to 0. The format is two words per set in sequential expansion format for nv sets. Card numbers need not be consecutive.

W1(R) Volume (m^3 , ft^3). The code requires that each volume equal the x-direction flow area times the x-direction length. If activated, the code also requires each volume equal the y-direction flow area times the y-direction length, and each volume equal the z-direction flow area times the z-direction length. For any volume, at least two of the three quantities, x-direction area, the x-direction length, or volume, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the quantities are 0, the volume must equal the x-direction area times the x-direction length within a relative error

of 0.000001. The same relative error check is done for the y- and z-directions. If both the y-direction area and y-direction length are not entered or are 0, the y-direction length is computed from $2.0 \bullet \left(\frac{x - \text{direction area}}{\pi} \right)^{0.5}$ and the y-direction area is computed from $\frac{\text{volume}}{\text{y-direction length}}$. The same is true for the z-direction.

W2(I) Volume number.

A9.9.6 Annulus Volume Azimuthal Angles, CCC0501-0599

These cards are optional, and, if not entered, the angles are set to 0. The angles on these cards are used for the 3-D drawing programs, so it is recommended that the user enter these values if they are known. The format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive.

W1(R) Azimuthal angle (degrees). The absolute value of the angle must be ≤ 360 degrees.

W2(I) Volume number.

A9.9.7 Annulus Volume Vertical Angles, CCC0601-0699

These cards are required. The format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive.

W1(R) Vertical angle (degrees). Since an annulus component is vertical, the angle must be ± 90.0 . This angle is used in the interphase drag calculation.

W2(I) Volume number.

A9.9.8 Annulus X-Coordinate (Elevation) Changes, CCC0701-0799

These cards are optional. If these cards are missing, the coordinate changes or elevation changes are computed from the x-coordinate length and a rotation matrix computed from the angle information. If these cards are entered, the entered data becomes the x-coordinate change or elevation change data. Since an annulus component is vertical, any elevation changes entered on these cards have to be consistent with the length of the component and the vertical angle entered on the previous card. Two formats entering 1 or 3 coordinate changes per volume are provided. The card format is two or four words per set in sequential expansion format up to nv sets, and card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC2101-2199 and CCC2201-2299).

1 coordinate change per volume format:

W1(R) Elevation change. This is the coordinate change along the fixed x-axis due to the traverse from inlet to outlet along the local x-coordinate, Δ_{zx} (m, ft). A positive value is an increase in elevation. The magnitude must be equal to the length.

W2(I) Volume number.

Three coordinate changes per volume format:

W1(R) Coordinate change along the fixed x-axis due to traverse from inlet to outlet along the local x-coordinate, Δ_{xx} (m, ft).

W2(R) Coordinate change along the fixed y-axis due to traverse from inlet to outlet along the local x-coordinate, Δ_{yx} (m, ft).

W3(R) Coordinate change along the fixed z-axis due to traverse from inlet to outlet along the local x-coordinate, Δ_{zx} (m, ft).

W4(I) Volume number.

A9.9.9 Annulus Volume X-Coordinate Friction Data, CCC0801-0899

These cards are required. The card format is three words per set for nv sets, and card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC2301-2399 and CCC2401-2499).

W1(R) Wall roughness (m, ft).

W2(R) Hydraulic diameter (m, ft). This should be computed from $4.0 \cdot \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If 0, the hydraulic diameter is computed from $2.0 \cdot \left(\frac{\text{area}}{\pi} \right)^{0.5}$. A check is made to ensure that the roughness is less than half the hydraulic diameter. See Word 1 on cards CCC0101-0109 for the area.

W3(I) Volume number.

A9.9.10 Annulus Junction Loss Coefficients, CCC0901-0999

These cards are optional and if missing, the energy loss coefficients are set to 0. The card format is three words per set in sequential expansion format for nv-1 sets, and card numbers need not be consecutive.

W1(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. Note: a variable loss coefficient may be specified (see Section A9.8.29). The interpretation and use of this loss coefficient vary based upon the “ $\underline{a}$ ” junction control flag selected for this junction on cards CCC1101-1199. If the smooth area change option is selected ($\underline{a} = 0$), the entry here is the loss coefficient applied to the junction flow area resulting from the entries on cards CCC0201-0299). If the abrupt area change option is selected ($\underline{a} = 1$ or 2), the loss coefficient entered here is adjusted by the square of the ratio of the junction area and the minimum flow area of the adjacent connected volumes (from cards CCC0101-0199). This adjusted loss coefficient is applied to the junction flow area). If the $\underline{a} = 1$ option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

Example: Two volumes with volume average flow areas of 0.3 m^2 and 0.4 m^2 are connected with a junction ($A_{\text{jun}} = 0.075 \text{ m}^2$). The user supplied $K_{\text{loss}_{\text{user}}}$ is 10.0. For the smooth area change option ($\underline{a} = 0$), the $K_{\text{loss}_{\text{user}}}$ of 10.0 is applied directly to the junction. For the abrupt area change option ($\underline{a} = 1$ or 2), the K_{loss} applied at the junction is $K_{\text{loss}_{\text{jun}}} = K_{\text{loss}_{\text{user}}} * (A_{\text{jun}}/A_{\text{min}})^2$. For this example, $K_{\text{loss}_{\text{jun}}} = 10.0 * (0.075/0.3)^2 = 0.625$.

W2(R) Reynolds number independent reverse flow energy loss coefficient, A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. Note: a variable loss coefficient may be specified (see Section A9.8.29). See information for the previous word regarding interpretation and use of this loss coefficient.

W3(I) Junction number.

A9.9.11 Annulus Volume X-Coordinate Control Flags, CCC1001-1099

These cards are required. The card format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC2701-2799 and CCC2801-2899).

W1(I) Volume control flags. This word has the format tlpvbfe. It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags. Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f-flag is the only coordinate direction oriented flag. These words enter the scalar oriented flags and the x-coordinate flags for each volume in the component.

t-flag.	The digit <u>t</u> specifies whether the thermal front tracking model is to be used; <u>t</u> = 0 specifies that the front tracking model is not to be used for the volume, and <u>t</u> = 1 specifies that the front tracking model is to be used for the volume. The thermal front tracking model can only be applied to vertically-oriented components.
l-flag	The digit <u>l</u> specifies whether the mixture level tracking model is to be used; <u>l</u> = 0 specifies that the level model not be used for the volume, and <u>l</u> = 1 specifies that the level model be used for the volume. The mixture level tracking model can only be applied to vertically-oriented components.
p-flag	The digit <u>p</u> specifies whether the water packing scheme is to be used. <u>p</u> = 0 specifies that the water packing scheme is to be used for the volume, and <u>p</u> = 1 specifies that the water packing scheme is not to be used for the volume.
v-flag	The digit <u>v</u> specifies whether the vertical stratification model is to be used. <u>v</u> = 0 specifies that the vertical stratification model is to be used for the volume, and <u>v</u> = 1 specifies that the vertical stratification model is not to be used for the volume.
b-flag	The digit <u>b</u> specifies the interphase friction that is used. <u>b</u> = 0 means that the pipe interphase friction model will be applied, <u>b</u> = 1 means that the rod bundle interphase friction model will be applied, <u>b</u> = 2 means that the ORNL ANS interphase friction model will be applied (see card ccc3101 in Section A9.8.30).
f-flag	The digit <u>f</u> specifies whether wall friction is to be computed. <u>f</u> = 0 specifies that wall friction effects are to be computed along the x-coordinate of the volume, and <u>f</u> = 1 specifies that wall friction effects are not to be computed along the x-coordinate.
e-flag	The digit <u>e</u> specifies if nonequilibrium or equilibrium is to be used. <u>e</u> = 0 specifies that a nonequilibrium (unequal temperature) calculation is to be used, and <u>e</u> = 1 specifies that an equilibrium (equal temperature) calculation is to be used. Equilibrium volumes should not be connected to nonequilibrium volumes. The equilibrium option is provided only for comparison to other codes.
W2(I)	Volume number.

A9.9.12 Annulus Junction Control Flags, CCC1101-1199

These cards are required. The card format is two words per set in sequential expansion format for nv-1 sets, and card numbers need not be consecutive.

W1(I)	Junction control flags. This word has the format <u>jefvcahs</u> . It is not necessary to input leading zeros. For the annulus component, the junction control flag is limited to the format <u>0ef0cahs</u> .
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e-flag	The digit <u>e</u> specifies the modified PV term in the energy equations. <u>e</u> = 0 means that the modified PV term will not be applied, and <u>e</u> = 1 means that it will be applied.
f-flag	The digit <u>f</u> specifies CCFL options. <u>f</u> = 0 means that the CCFL model will not be applied, and <u>f</u> = 1 means that the CCFL model will be applied.
c-flag	The digit <u>c</u> specifies choking options. <u>c</u> = 0 means that the choking model will be applied, and <u>c</u> = 1 means that the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske or Moody (option 54) critical flow model is active. If card 1 is present and contains Option 50, the original RELAP5 critical flow model is active. Unlike junctions for certain other hydrodynamic components, only the default values of discharge coefficients and factors used in the critical flow models are allowed for junctions in ANNULUS components. Problem renodalization should be used to circumvent this limitation should it prove to be significant. For the Henry-Fauske or Moody critical flow model, the default value for the discharge coefficient is 1.0 and the default value for the thermal nonequilibrium constant is 0.14. For the original RELAP5 critical flow model, the default values for the subcooled, two-phase and superheated discharge coefficients each are 1.0.
a-flag	The digit <u>a</u> specifies area change options. <u>a</u> = 0 means a smooth area change and only the user supplied K_{loss} is applied. <u>a</u> = 1 means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. <u>a</u> = 2 means a partial abrupt area change model and it is the same as <u>a</u> = 1 except no code calculated forward and reverse contraction/expansion K_{loss} is applied. See discussion for W1(R) regarding the K_{loss} flow area reference, which varies based on the a-flag option selected here.
h-flag	The digit <u>h</u> specifies nonhomogeneous or homogeneous. <u>h</u> = 0 specifies the nonhomogeneous (two-velocity momentum equations) option, and <u>h</u> = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (<u>h</u> = 1 or 2), the major edit printout will show a 1.
s-flag	The digit <u>s</u> specifies momentum flux options. <u>s</u> = 0 uses momentum flux in both the <u>to</u> volume and the <u>from</u> volume. <u>s</u> = 1 uses momentum flux in the <u>from</u> volume, but not in the <u>to</u> volume. <u>s</u> = 2 uses momentum flux in the <u>to</u> volume but not in the <u>from</u> volume. <u>s</u> = 3 does not use momentum flux in either the <u>to</u> or the <u>from</u> volume. For this component, the option <u>s</u> = 0 is the usual recommendation (momentum flux in both volumes). The other options <u>s</u> = 1, 2, and 3 are included to allow consistency for this flag for other components (single-junction, branch junction, etc.).
W2(I)	Junction number.

A9.9.13 Annulus Volume Initial Conditions, CCC1201-99

These cards are required. The card format is seven words per set in sequential expansion format for nv sets, and card numbers need not be consecutive.

- W1(I) Control word. This word has the format $\underline{\epsilon}\underline{b}\underline{t}$. It is not necessary to input leading zeros.
- ϵ -flag The digit $\underline{\epsilon}$ specifies the fluid, where $\underline{\epsilon} = 0$ is the default fluid. The value for $\underline{\epsilon} > 0$ corresponds to the position number of the fluid type indicated on the 120-9 cards (i.e., $\underline{\epsilon} = 1$ specifies H_2O , $\underline{\epsilon} = 2$ specifies D_2O , etc.). The default fluid is that set for the hydrodynamic system by cards 120-129 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-9 or these control words must be consistent (i.e., not specify different fluids). If cards 120-9 are not entered and all control words use the default $\underline{\epsilon} = 0$, then H_2O is assumed as the fluid. It is recommended that the user use the 120-9 cards to set the fluid type in the systems.
- b-flag The digit $\underline{b}$ specifies whether boron is present or not. $\underline{b} = 0$ specifies that the volume liquid does not contain boron; $\underline{b} = 1$ specifies that a boron concentration in mass of boron per mass of liquid (which may be 0) is being entered on the CCC2001-2099 cards.
- t-flag The digit $\underline{t}$ specifies how the following words are to be used to determine the initial thermodynamic state. Entering $\underline{t}$ equal to 0-3 and 7 specifies one component (steam/water). Entering $\underline{t}$ equal to 4-6 allows the specification of two components (steam/water and noncondensable gas).

With options $\underline{t}$ equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.

One Component (Steam/Water), Equilibrium or Non-equilibrium

- $[P, U_f, U_g, \alpha_g]$ If $\underline{t} = 0$, the next four words are interpreted as pressure (Pa, lb_f/in^2), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions, depending on the internal energies used to define the thermodynamic state. W6 should be 0.0.

One Component (Steam/Water), Equilibrium

- $[T, x_s]$ If $\underline{t} = 1$, the next two words are interpreted as temperature (K, $^{\circ}F$) and static quality in equilibrium condition. W4, W5, and W6 should be 0.0.

- [P,x_s] If $\underline{t} = 2$, the next two words are interpreted as pressure (Pa, lb_f/in²) and static quality in equilibrium condition. W4, W5, and W6 should be 0.0.
- [P,T] If $\underline{t} = 3$, the next two words are interpreted as pressure (Pa, lb_f/in²) and temperature (K, °F) in equilibrium condition. W4, W5, and W6 should be 0.0.
- [P,T_f,T_g,α_g] If $\underline{t} = 7$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid temperature (K, F), vapor temperature (K, F), and vapor void fraction. These quantities will be interpreted as equilibrium conditions. The liquid temperature has to be equal to or less than the saturation temperature, and the vapor temperature has to be equal to or greater than the saturation temperature. Enter boron in W6 if needed. This input option was added to allow easier comparisons with TRACE calculations.

Two Components (Steam/Water/Air), Equilibrium or Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0\text{E-}9 \leq x_s \leq 0.99999999$,

Single phase if $x_s < 1.0 \text{ E-}9$ or $x_s > 0.99999999$

The static quality is given by $x_s = M_g/(M_g + M_f)$, where $M_g = M_s + M_n$. Section 3.2 of Volume I of the manual discusses this further.

- [P,T,x_s] If $\underline{t} = 4$, the next three words are interpreted as pressure (Pa, lb_f/in²), temperature (K, °F), and static quality in equilibrium condition. Using this input option with static quality > 0.0 and ≤ 1.0 , saturated noncondensables will result. W5 and W6 should be 0.0. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. If the pressure is greater than the critical pressure, $P > P_{\text{crit}}$, then the static quality, x_s (mass steam + air)/(mass steam + air + water) has to be large enough to give a partial steam pressure that is less than the critical pressure, $P_s < P_{\text{crit}}$. Otherwise an input error will occur. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensable (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option.
- [T,x_s,x_n] If $\underline{t} = 5$, the next three words are interpreted as steam saturation temperature (K, °F), static quality, and noncondensable quality in equilibrium condition. W5 and W6 should be 0.0. Both the static and noncondensable qualities are restricted to be between $1.0 \text{ E-}9$ and

0.99999999. The liquid will be subcooled. However, the total pressure may be greater than the critical pressure, which will cause an input error.

[P,U_f,U_g,α_g,x_n] If $t = 6$, the next five words are interpreted as pressure (Pa, lb_f/in²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing ($t = 0$). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor-specific internal energy.

W2-W6(R) Quantities as described under Word 1. Five quantities must be entered, and zeros should be entered for unused quantities. If any control word (Word 1) indicates that boron is present, cards CCC2001-2099 must be entered to define the initial boron concentrations. Boron concentrations are not entered in Words 2-6 for annulus components.

W7(I) Volume number.

A9.9.14 Annulus Junction Conditions Control Words, CCC1300

This card is optional, and if missing, velocities are assumed on cards CCC1301-99.

W1(I) Control word. If 0, the first and second words of each set on cards CCC1301-1399 are velocities. If 1, the first and second words of each set on cards CCC1301-1399 are mass flows.

A9.9.15 Annulus Junction Initial Conditions, CCC1301-1399

These cards are required.

W1(R) Initial liquid velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W2(R) Initial vapor velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W3(R) Interface velocity (m/s, ft/s). The feature is not implemented yet, so enter 0.

W4(I) Junction number.

A9.9.16 Annulus Junction Diameter and CCFL Data, CCC1401-1499

These cards are optional. The defaults indicated for each word are used if the card is not entered. If this card is being used to specify only the junction hydraulic diameter for the interphase drag calculation, (i.e., $f = 0$ in Word 1 of cards CCC1101-1199) then the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (will not be used). If this card is being used for the CCFL model (i.e., $f = 1$ in Word 1 of cards CCC1101-1199), then enter all four words for the appropriate CCFL model if values different from the default value are desired.

- W1(R) Junction hydraulic diameter, D_j (m, ft). This quantity is the junction hydraulic diameter used in the CCFL correlation equation and interphase drag and must be ≥ 0 . The number should be computed from $4.0 \bullet \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If a 0 is entered or if the default is used, the junction diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi} \right)^{0.5}$. See Word 1 of cards CCC0201-0299 for the area.
- W2(R) Flooding correlation form, β . If 0, the Wallis CCFL form is used. If 1, the Kutateladze CCFL form is used. If between 0.0 and 1.0, Bankoff weighting between the Wallis and Kutateladze CCFL forms is used. This number must be ≥ 0 and ≤ 1 . The default value is 0 (Wallis form).
- W3(R) Gas intercept, c . This quantity is the gas intercept used in the CCFL correlation (when $H_f^{1/2} = 0$) and must be > 0 . The default value is 1.
- W4(R) Slope, m . This quantity is the slope used in the CCFL correlation and must be > 0 . The default value is 1.
- W5(I) Junction number.

A9.9.17 Annulus Y-Coordinate Areas, CCC1601-1699

These cards are optional, and if entered, activate the y-coordinate for each volume and allow the full one-dimensional momentum equations to be used in connections to the y-faces. The format is two words per set in sequential expansion format for n_v sets. (The corresponding x-coordinate cards are CCC0101-0199). The card numbers need not be consecutive. The words for one set are:

- W1(R) Area (m^2 , ft^2). If these cards are missing, this y-direction area is computed from (volume)/(y-direction length).
- W2(I) Volume number.

A9.9.18 Annulus Z-Coordinate Areas, CCC1701-1799

These cards are optional, and if entered, activate the z-coordinate for each volume and allow the full one-dimensional momentum equations to be used in connections to the z-faces. The format is two words per set in sequential expansion format for nv sets. (The corresponding x-coordinate cards are CCC0101-0199). The card numbers need not be consecutive. The words for one set are:

W1(R) Area (m^2 , ft^2). If these cards are missing, this z-direction area is computed from (volume)/(z-direction length).

W2(I) Volume number.

A9.9.19 Annulus Y-Coordinate Lengths, CCC1801-1899

These cards are optional, and if entered, activate the y-coordinate for each volume and allow the full one-dimensional momentum equation for connections to the y-faces. The format is two words per set in sequential expansion format for nv sets. Card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC0301-0399).

W1(R) Length (m, ft). If these cards are missing and cards CCC1601-1699 are missing, this y-direction length is computed from $2.0 \cdot \left(\frac{\text{x-direction area}}{\pi} \right)^{0.5}$. If these cards are missing and cards CCC1601-1699 are present, this y-direction length is computed from (volume)/(y-direction area).

W2(I) Volume number.

A9.9.20 Annulus Z-Coordinate Lengths, CCC1901-1999

These cards are optional, and if entered, activate the z-coordinate for each volume and allow the full one-dimensional momentum equation for connections to the z-faces. The format is two words per set in sequential expansion format for nv sets. Card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC0301-0399).

W1(R) Length (m, ft). If these cards are missing and cards CCC1701-1799 are missing, this z-direction length is computed from $2.0 \cdot \left(\frac{\text{x-direction area}}{\pi} \right)^{0.5}$. If these cards are missing and cards CCC1701-1799 are present, this z-direction length is computed from (volume)/(z-direction area).

W2(I) Volume number.

A9.9.21 Annulus Initial Boron Concentrations, CCC2001-2099

These cards are required only if boron is specified in one of the control words (Word 1) in cards CCC1201-1299. The card format is two words per set in sequential expansion format for nv sets. Boron concentrations must be entered for each volume, and 0 should be entered for those volumes whose associated control word did not specify boron.

W1(R) Boron concentration (mass of boron per mass of liquid).

W2(I) Volume number.

A9.9.22 Annulus Y-Coordinate (Elevation) Changes, CCC2101-2199

These cards are optional. If these cards are missing, the coordinate changes are computed from the y-coordinate length and a rotation matrix computed from the angle information. If these cards are entered, the entered data overwrites the previously computed coordinate change data. The card format is four words per set in sequential expansion format up to nv sets, and card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC0701-0799).

W1(R) Coordinate change along fixed x-axis due to traverse from inlet to outlet along the local y-coordinate, Δ_{xy} (m, ft).

W2(R) Coordinate change along fixed y-axis due to traverse from inlet to outlet along the local y-coordinate, Δ_{yy} (m, ft).

W3(R) Coordinate change along fixed z-axis due to traverse from inlet to outlet along the local y-coordinate, Δ_{zy} (m, ft).

W4(I) Volume number.

A9.9.23 Annulus Z-Coordinate (Elevation) Changes, CCC2201-2299

These cards are optional. If these cards are missing, the coordinate changes are computed from the z-coordinate length and a rotation matrix computed from the angle information. If these cards are entered, the entered data overwrites the previously computed coordinate change data. The card format is four words per set in sequential expansion format up to nv sets, and card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC0701-0799).

W1(R) Coordinate change along fixed x-axis due to traverse from inlet to outlet along the local z-coordinate, Δ_{xz} (m, ft).

W2(R) Coordinate change along fixed y-axis due to traverse from inlet to outlet along the local z-coordinate, Δ_{yz} (m, ft).

W3(R) Coordinate change along fixed z-axis due to traverse from inlet to outlet along the local z-coordinate, Δ_{zz} (m, ft).

W4(I) Volume number.

A9.9.24 Annulus Volume Y-Coordinate Friction Data, CCC2301-2399

These cards are optional and may be entered if area or length data was entered for the y-coordinate. If the cards are not entered, the wall roughness defaults to 0 and the default hydraulic diameter is computed from $\frac{4.0 \bullet \text{y-direction volume flow area}}{(\pi \bullet \text{x-direction volume flow area})^{0.5}}$. The corresponding x-coordinate cards are CCC0801-0899.

W1(R) Wall roughness (m, ft).

W2(R) Hydraulic diameter (m, ft). A check is made to ensure that the y-direction roughness is less than half the y-direction hydraulic diameter. See Word 1 on cards CCC0101-0109 for the area.

W3(I) Volume number.

A9.9.25 Annulus Volume Z-Coordinate Friction Data, CCC2401-2499

These cards are optional and have to be entered if area or length data was entered for the z-coordinate. If the cards are not entered, the wall roughness defaults to 0 and the default hydraulic diameter is computed from $\frac{4.0 \bullet \text{z-direction volume flow area}}{(\pi \bullet \text{x-direction volume flow area})^{0.5}}$. The corresponding x-coordinate cards are CCC0801-0899.

W1(R) Wall roughness (m, ft).

W2(R) Hydraulic diameter (m, ft). A check is made to ensure that the z-direction roughness is less than half the z-direction hydraulic diameter. See Word 1 on cards CCC0101-0109 for the area.

W3(I) Volume number.

A9.9.26 Annulus Volume Additional Wall Friction Data, CCC2501-2599

These cards are optional. If these cards are not entered, the default values are 1.0 for the laminar shape factor and 0.0 for the viscosity ratio exponent. The card format is seven words per set in sequential expansion format for nv sets and card numbers need not be consecutive. A description of this input is presented in Section 3 of Volume I.

W1(R) Shape factor for x-coordinate.

W2(R)	Viscosity ratio exponent for x-coordinate.
W3(R)	Shape factor for y-coordinate.
W4(R)	Viscosity ratio exponent for y-coordinate.
W5(R)	Shape factor for z-coordinate.
W6(R)	Viscosity ratio exponent for z-coordinate.
W7(I)	Volume number.

A9.9.27 Annulus Y-Coordinate Control Flags, CCC2701-2799

These cards are optional. The card format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC1001-1099).

W1(I)	Volume control flags. This word has the general format <u>tlpvbfe</u> , but this word is limited to <u>00000f0</u> since it only enters the coordinate oriented flags for the y-direction.
f-flag	The digit <u>f</u> specifies whether wall friction is to be computed. <u>f</u> = 0 specifies that wall friction effects are to be computed along the y-coordinate direction in the volume, and <u>f</u> = 1 specifies that wall friction effects are not to be computed along the y-coordinate direction.
W2(I)	Volume number.

A9.9.28 Annulus Z-Coordinate Control Flags, CCC2801-2899

These cards are optional. The card format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC1001-1099).

W1(I)	Volume control flags. This word has the general format <u>tlpvbfe</u> , but this word is limited to <u>00000f0</u> since it only enters the coordinate oriented flags for the z-direction.
f-flag	The digit <u>f</u> specifies whether wall friction is to be computed. <u>f</u> = 0 specifies that wall friction effects are to be computed along the z-coordinate direction in the volume, and <u>f</u> = 1 specifies that wall friction effects are not to be computed along the z-coordinate direction.
W2(I)	Volume number.

A9.9.29 Annulus Junction Form Loss Data, CCC3001-3099

This card is optional. The user-specified form loss is given in Words 1 and 2 of cards CCC0901-0999 if this card is not entered. If this card is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F \text{Re}^{-C_F}$$

$$K_R = A_R + B_R \text{Re}^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 1 and 2 of cards CCC0901-0999. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all five words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R .

W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.

W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.

W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.

W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

W5(I) Junction number.

A9.9.30 Annulus ORNL ANS Interphase Model Values, CCC3101-3199

These cards are required if any of the interphase friction flags b in the volume control flags entered on cards CCC1001-1099 are set to 2.

W1(R) Gap (distance between side walls, short length, pitch, channel width) (m, ft).

W2(R) Span (distance from one end to the other, long length) (m, ft).

W3(I) Volume number.

A9.9.31 Annulus Initial Noncondensable Mass Fractions, CCC3201-3299

These cards are optional. They are required only if the user desires to use initial noncondensable mass fractions that are different than the default values specified on card 115. The card format is $M = N+1$ words per set in sequential expansion format for nv sets. The plus one is for the volume number. Initial noncondensable mass fractions must be entered for each volume, and the default set on card 115 should be entered for those volumes whose values are the same as the default set on card 115.

- W1-WN(R) Initial mass fraction for each noncondensable gas type as specified on card 110 where N is the number of gas types specified on card 110.
- WM(I) Volume number.

A9.10 Pressurizer Component, PRIZER

A pressurizer component is indicated by PRIZER for Word 2 on CCC0000.

More than one junction may be connected to the inlet or outlet of a pressurizer. If an end has no junctions, that end is considered a closed end. For major edits, minor edits, and plot variables, the volumes in the pressurizer component are numbered as CCCNN0000, where NN is the volume number ($01 \leq NN \leq 99$). The junctions in the pressurizer component are numbered as CCCJJ0000, where JJ is the junction number ($01 \leq JJ \leq 98$).

The volumes in a pressurizer are usually considered one-dimensional components and flow in the volumes is along the x-coordinate. Crossflow junctions can connect to any of the pressurizer volumes in the y- and z-coordinate directions (faces 3, 4, 5, or 6) using a form of the momentum equation that does or does not include momentum flux terms. It is also possible to connect external junctions to the x-coordinate direction faces (1 or 2) of any of the pressurizer volumes using a form of the momentum equation that does or does not include the momentum flux terms. It is also possible to include or not include the momentum flux terms in internal pressurizer junctions.

A9.10.1 Pressurizer Information, CCC0001

This card is required.

- W1(I) Number of volumes, nv. The number nv must be greater than 0 and less than 100. The number of associated junctions internal to these components is nv-1. The outer junctions are described by other components.
- W2(I) Surgeline junction connection number. This word must the same 9-digit format as printed in the output.
- W3(R) User specified interfacial heat transfer coefficient from liquid to saturation state (W/m²-K, Btu/hr-ft²-F). This word is optional. If it is not supplied, a value equal to $\frac{k_f[\max(0.15\text{Ray}^{0.333}, 2.0\text{Nu})]}{D_{\text{Hyd}}}$ is used.
- W4(R) User specified interfacial heat transfer coefficient from vapor to saturation state (W/m²-K, Btu/hr-ft²-F). This word is optional. If it is not supplied, a value equal to $\frac{k_g[\max(0.15\text{Ray}^{0.333}, 2.0\text{Nu})]}{D_{\text{Hyd}}}$ is used.

W5(R) Spray droplet diameter (m, ft). This is used to compute the interphase drag term for bubbles and droplets. This word is optional and the code computed value is used if this word is not input.

A9.10.2 Pressurizer X-Coordinate Areas, CCC0101-0199

These cards are required, and the card numbers need not be consecutive. The format is two words per set in sequential expansion format for nv sets. (The corresponding y- and z-coordinate cards are CCC1601-1699 and CCC1701-1799). The words for one set are

W1(R) Area (m^2 , ft^2).

W2(I) Volume number.

A9.10.3 Pressurizer Junction Areas, CCC0201-0299

These cards are optional, and, if entered, the card numbers need not be consecutive. The format is two words per set in sequential expansion format for nv-1 sets.

W1(R) Internal junction flow area (m^2 , ft^2). If cards are missing or a word is 0, the junction flow area is set to the minimum area of the adjoining volumes. For abrupt area changes, the junction area must be equal to or less than the minimum of the adjacent volume areas. There is no restriction for smooth area changes.

W2(I) Junction number.

A9.10.4 Pressurizer X-Coordinate Lengths, CCC0301-0399

These cards are required. The format is two words per set in sequential expansion format for nv sets. Card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC1801-1899 and CCC1901-1999).

W1(R) Length (m, ft).

W2(I) Volume number.

A9.10.5 Pressurizer Volumes, CCC0401-0499

These cards are optional, and if not entered, the volumes are set to 0. The format is two words per set in sequential expansion format for nv sets. Card numbers need not be consecutive.

W1(R) Volume (m^3 , ft^3). The code requires that each volume equal the x-direction flow area times the x-direction length. If activated, the code also requires each volume equal the y-direction flow area times the y-direction length, and each volume equal the z-direction

flow area times the z-direction length. For any volume, at least two of the three quantities, x-direction area, the x-direction length, or volume, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the quantities are 0, the volume must equal the x-direction area times the x-direction length within a relative error of 0.000001. The same relative error check is done for the y- and z-directions. If both the y-direction area and y-direction length are not entered or are 0, the y-direction length is computed from $2.0 \bullet \left(\frac{x - \text{direction area}}{\pi} \right)^{0.5}$ and the y-direction area is computed from $\frac{\text{volume}}{\text{y-direction length}}$. The same is true for the z-direction.

W2(I) Volume number.

A9.10.6 Pressurizer Volume Azimuthal Angles, CCC0501-0599

These cards are optional, and if not entered, the angles are set to 0. The angles on these cards are used for the 3-D drawing programs, so it is recommended that the user enter these values if they are known. The format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive.

W1(R) Azimuthal angle (degrees). The absolute value of the angle must be ≤ 360 degrees.

W2(I) Volume number.

A9.10.7 Pressurizer Volume Vertical Angles, CCC0601-0699

These cards are required. The format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive.

W1(R) Vertical angle (degrees). The absolute value of the angle must be less than or equal to 90 degrees. This angle is used in the interphase drag calculation.

W2(I) Volume number.

A9.10.8 Pressurizer X-Coordinate (Elevation) Changes, CCC0701-0799

These cards are optional. If these cards are missing, the coordinate changes or elevation changes are computed from the x-coordinate length and a rotation matrix computed from the angle information. If these cards are entered, the entered data becomes the x-coordinate change or elevation change data. Two formats entering 1 or 3 coordinate changes per volume are provided. The card format is two or four words per set in sequential expansion format up to nv sets, and card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC2101-2199 and CCC2201-2299).

1 coordinate change per volume format:

W1(R) Elevation change. This is the coordinate change along the fixed x-axis due to the traverse from inlet to outlet along the local x-coordinate, Δ_{zx} (m, ft). A positive value is an increase in elevation. The magnitude must be equal to or less than the length. See Section 2.4.1 of Volume II of the manual for further discussion. When the absolute value of the elevation angle determined by the ratio of the elevation change (this Word 6) and the length (Word 2) is less than or equal to 45 degrees, the horizontal flow regime map is used; when the ratio is greater than 45 degrees, the vertical flow regime map is used.

W2(I) Volume number.

Three coordinate changes per volume format:

W1(R) Coordinate change along the fixed x-axis due to traverse from inlet to outlet along the local x-coordinate, Δ_{xx} (m, ft).

W2(R) Coordinate change along the fixed y-axis due to traverse from inlet to outlet along the local x-coordinate, Δ_{yx} (m, ft).

W3(R) Coordinate change along the fixed z-axis due to traverse from inlet to outlet along the local x-coordinate, Δ_{zx} (m, ft).

W4(I) Volume number.

A9.10.9 Pressurizer Volume X-Coordinate Friction Data, CCC0801-0899

These cards are required. The card format is three words per set for nv sets, and card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC2301-2399 and CCC2401-2499).

W1(R) Wall roughness (m, ft).

W2(R) Hydraulic diameter (m, ft). This should be computed from $4.0 \cdot \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If 0, the hydraulic diameter is computed from $2.0 \cdot \left(\frac{\text{area}}{\pi} \right)^{0.5}$. A check is made to ensure that the roughness is less than half the hydraulic diameter. See Word 1 on cards CCC0101-0109 for the area.

W3(I) Volume number.

A9.10.10 Pressurizer Junction Loss Coefficients, CCC0901-0999

These cards are optional and if missing, the energy loss coefficients are set to 0. The card format is three words per set in sequential expansion format for nv-1 sets, and card numbers need not be consecutive.

W1(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. Note: a variable loss coefficient may be specified (see Section A9.8.29). The interpretation and use of this loss coefficient vary based upon the “a” junction control flag selected for this junction on cards CCC1101-1199. If the smooth area change option is selected ($\underline{a} = 0$), the entry here is the loss coefficient applied to the junction flow area entries on cards CCC0201-0299. If the abrupt area change option is selected ($\underline{a} = 1$ or 2), the loss coefficient entered here is adjusted by the square of the ratio of the junction area and the minimum flow area of the adjacent connected volumes (from cards CCC0101-0199). This adjusted loss coefficient is applied to the junction flow area. If the $\underline{a} = 1$ option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

Example: Two volumes with volume average flow areas of 0.3 m^2 and 0.4 m^2 are connected with a junction ($A_{\text{jun}} = 0.075 \text{ m}^2$). The user supplied $K_{\text{loss_user}}$ is 10.0. For the smooth area change option ($\underline{a} = 0$), the $K_{\text{loss_user}}$ of 10.0 is applied directly to the junction. For the abrupt area change option ($\underline{a} = 1$ or 2), the K_{loss} applied at the junction is $K_{\text{loss_jun}} = K_{\text{loss_user}} * (A_{\text{jun}}/A_{\text{min}})^2$. For this example, $K_{\text{loss_jun}} = 10.0 * (0.075/0.3)^2 = 0.625$.

W2(R) Reynolds number independent reverse flow energy loss coefficient, A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. Note: a variable loss coefficient may be specified (see Section A9.8.29). See information for the previous word regarding interpretation and use of this loss coefficient.

W3(I) Junction number.

A9.10.11 Pressurizer Volume X-Coordinate Control Flags, CCC1001-1099

These cards are required. The card format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC2701-2799 and CCC2801-2899).

W1(I) Volume control flags. This word has the format tlpvbfe. It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags.

Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f-flag is the only coordinate direction oriented flag. These words enter the scalar oriented flags and the x-coordinate flags for each volume in the component.

t-flag	The digit <u>t</u> specifies whether the thermal front tracking model is to be used; <u>t</u> = 0 specifies that the front tracking model is not to be used for the volume, and <u>t</u> = 1 specifies that the front tracking model is to be used for the volume. The thermal front tracking model can only be applied to vertically-oriented components.
l-flag	The digit <u>l</u> specifies whether the mixture level tracking model is to be used; <u>l</u> = 0 specifies that the level model not be used for the volume, and <u>l</u> = 1 specifies that the level model be used for the volume. The mixture level tracking model can only be applied to vertically-oriented components.
p-flag	The digit <u>p</u> specifies whether the water packing scheme is to be used. <u>p</u> = 0 specifies that the water packing scheme is to be used for the volume, and <u>p</u> = 1 specifies that the water packing scheme is not to be used for the volume.
v-flag	The digit <u>v</u> specifies whether the vertical stratification model is to be used. <u>v</u> = 0 specifies that the vertical stratification model is to be used for the volume, and <u>v</u> = 1 specifies that the vertical stratification model is not to be used for the volume.
b-flag	The digit <u>b</u> specifies the interphase friction that is used. <u>b</u> = 0 means that the pressurizer interphase friction model will be applied, <u>b</u> = 1 means that the rod bundle interphase friction model will be applied, <u>b</u> = 2 means that the ORNL ANS interphase friction model will be applied (see card ccc3101 in Section A9.8.30).
f-flag	The digit <u>f</u> specifies whether wall friction is to be computed. <u>f</u> = 0 specifies that wall friction effects are to be computed along the x-coordinate of the volume, and <u>f</u> = 1 specifies that wall friction effects are not to be computed along the x-coordinate.
e-flag	The digit <u>e</u> specifies if nonequilibrium or equilibrium is to be used. <u>e</u> = 0 specifies that a nonequilibrium (unequal temperature) calculation is to be used, and <u>e</u> = 1 specifies that an equilibrium (equal temperature) calculation is to be used. Equilibrium volumes should not be connected to nonequilibrium volumes. The equilibrium option is provided only for comparison to other codes.
W2(I)	Volume number.

A9.10.12 Pressurizer Junction Control Flags, CCC1101-1199

These cards are required. The card format is two words per set in sequential expansion format for nv-1 sets, and card numbers need not be consecutive.

W1(I)	Junction control flags. This word has the format <u>jefvcahs</u> . It is not necessary to input leading zeros. For the pressurizer component, the junction control flag is limited to the format <u>0ef0cahs</u> .
e-flag	The digit <u>e</u> specifies the modified PV term in the energy equations. <u>e</u> = 0 means that the modified PV term will not be applied, and <u>e</u> = 1 means that it will be applied.
f-flag	The digit <u>f</u> specifies CCFL options. <u>f</u> = 0 means that the CCFL model will not be applied, and <u>f</u> = 1 means that the CCFL model will be applied.
c-flag	The digit <u>c</u> specifies choking options. <u>c</u> = 0 means that the choking model will be applied, and <u>c</u> = 1 means that the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske or Moody (option 54) critical flow model is active. If card 1 is present and contains Option 50, the original RELAP5 critical flow model is active. Unlike junctions for certain other hydrodynamic components, only the default values of discharge coefficients and factors used in the critical flow models are allowed for junctions in PRESSURIZER components. Problem renodalization should be used to circumvent this limitation should it prove to be significant. For the Henry-Fauske or Moody critical flow model, the default value for the discharge coefficient is 1.0 and the default value for the thermal nonequilibrium constant is 0.14. For the original RELAP5 critical flow model, the default values for the subcooled, two-phase and superheated discharge coefficients each are 1.0.
a-flag	The digit <u>a</u> specifies area change options. <u>a</u> = 0 means a smooth area change and only the user supplied K_{loss} is applied. <u>a</u> = 1 means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. <u>a</u> = 2 means a partial abrupt area change model and it is the same as <u>a</u> = 1 except no code calculated forward and reverse contraction/expansion K_{loss} is applied. See discussion for W1(R) regarding the K_{loss} flow area reference, which varies based on the a-flag option selected here.
h-flag	The digit <u>h</u> specifies nonhomogeneous or homogeneous. <u>h</u> = 0 specifies the nonhomogeneous (two-velocity momentum equations) option, and <u>h</u> = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (<u>h</u> = 1 or 2), the major edit printout will show a 1.

- s-flag The digit $\underline{s}$ specifies momentum flux options. $\underline{s} = 0$ uses momentum flux in both the to volume and the from volume. $\underline{s} = 1$ uses momentum flux in the from volume, but not in the to volume. $\underline{s} = 2$ uses momentum flux in the to volume but not in the from volume. $\underline{s} = 3$ does not use momentum flux in either the to or the from volume. For this component, the option $\underline{s} = 0$ is the usual recommendation (momentum flux in both volumes). The other options $\underline{s} = 1, 2$, and 3 are included to allow consistency for this flag for other components (single-junction, branch junction, etc.).
- W2(I) Junction number.

A9.10.13 Pressurizer Volume Initial Conditions, CCC1201-1299

These cards are required. The card format is seven words per set in sequential expansion format for nv sets, and card numbers need not be consecutive.

- W1(I) Control word. This word has the format gbt. It is not necessary to input leading zeros.
- ε -flag The digit $\underline{\varepsilon}$ specifies the fluid, where $\underline{\varepsilon} = 0$ is the default fluid. The value for $\underline{\varepsilon} > 0$ corresponds to the position number of the fluid type indicated on the 120-9 cards (i.e., $\underline{\varepsilon} = 1$ specifies H_2O , $\underline{\varepsilon} = 2$ specifies D_2O , etc.). The default fluid is that set for the hydrodynamic system by cards 120-129 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-9 or these control words must be consistent (i.e., not specify different fluids). If cards 120-9 are not entered and all control words use the default $\underline{\varepsilon} = 0$, then H_2O is assumed as the fluid. It is recommended that the user use the 120-9 cards to set the fluid type in the systems.
- b-flag The digit $\underline{b}$ specifies whether boron is present or not. $\underline{b} = 0$ specifies that the volume liquid does not contain boron; $\underline{b} = 1$ specifies that a boron concentration in mass of boron per mass of liquid (which may be 0) is being entered on the CCC2001-2099 cards.
- t-flag The digit $\underline{t}$ specifies how the following words are to be used to determine the initial thermodynamic state. Entering $\underline{t}$ equal to 0-3 and 7 specifies one component (steam/water). Entering $\underline{t}$ equal to 4-6 allows the specification of two components (steam/water and noncondensable gas).
- With options $\underline{t}$ equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.

One Component (Steam/Water), Equilibrium or Non-equilibrium

[P,U_f,U_g,α_g] If $t = 0$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions, depending on the internal energies used to define the thermodynamic state. W6 should be 0.0.

One Component (Steam/Water), Equilibrium

[T,x_s] If $t = 1$, the next two words are interpreted as temperature (K, °F) and static quality in equilibrium condition. W4, W5, and W6 should be 0.0.

[P,x_s] If $t = 2$, the next two words are interpreted as pressure (Pa, lb_f/in²) and static quality in equilibrium condition. W4, W5, and W6 should be 0.0.

[P,T] If $t = 3$, the next two words are interpreted as pressure (Pa, lb_f/in²) and temperature (K, °F) in equilibrium condition. W4, W5, and W6 should be 0.0.

[P,T_f,T_g,α_g] If $t = 7$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid temperature (K, F), vapor temperature (K, F), and vapor void fraction. These quantities will be interpreted as equilibrium conditions. The liquid temperature has to be equal to or less than the saturation temperature, and the vapor temperature has to be equal to or greater than the saturation temperature. Enter boron in W6 if needed. This input option was added to allow easier comparisons with TRACE calculations.

Two Components (Steam/Water/Air), Equilibrium or Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0\text{E-}9 \leq x_s \leq 0.99999999$,

Single phase if $x_s < 1.0 \text{ E-}9$ or $x_s > 0.99999999$

The static quality is given by $x_s = M_g / (M_g + M_f)$, where $M_g = M_s + M_n$. Section 3.2 of Volume I of the manual discusses this further.

[P,T,x_s] If $t = 4$, the next three words are interpreted as pressure (Pa, lb_f/in²), temperature (K, °F), and static quality in equilibrium condition. Using this input option with static quality > 0.0 and ≤ 1.0 , saturated noncondensables will result. W5 and W6 should be 0.0. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. If

the pressure is greater than the critical pressure, $P > P_{crit}$, then the static quality, x_s (mass steam + air)/(mass steam + air + water) has to be large enough to give a partial steam pressure that is less than the critical pressure, $P_s < P_{crit}$. Otherwise an input error will occur. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensable (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option.

[T,x_s,x_n] If $\underline{t} = 5$, the next three words are interpreted as steam saturation temperature (K, °F), static quality, and noncondensable quality in equilibrium condition. W5 and W6 should be 0.0. Both the static and noncondensable qualities are restricted to be between 1.0 E-9 and 0.99999999. The liquid will be subcooled. The input temperature determines the steam partial pressure. However, the total pressure may be greater than the critical pressure, which will cause an input error.

[P,U_f,U_g,α_g,x_n] If $\underline{t} = 6$, the next five words are interpreted as pressure (Pa, lb_f/in²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing ($\underline{t} = 0$). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor-specific internal energy.

W2-W6(R) Quantities as described under Word 1. Five quantities must be entered, and zeros should be entered for unused quantities. If any control word (Word 1) indicates that boron is present, cards CCC2001-2099 must be entered to define the initial boron concentrations. Boron concentrations are not entered in Words 2-6 for pressurizer components.

W7(I) Volume number.

A9.10.14 Pressurizer Junction Conditions Control Words, CCC1300

This card is optional, and, if missing, velocities are assumed on cards CCC1301-99.

W1(I) Control word. If 0, the first and second words of each set on cards CCC1301-1399 are velocities. If 1, the first and second words of each set on cards CCC1301-1399 are mass flows.

A9.10.15 Pressurizer Junction Initial Conditions, CCC1301-1399

These cards are required.

- W1(R) Initial liquid velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).
- W2(R) Initial vapor velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).
- W3(R) Interface velocity (m/s, ft/s). This capability has not yet been implemented, so enter 0.
- W4(I) Junction number.

A9.10.16 Pressurizer Junction Diameter and CCFL Data, CCC1401-1499

These cards are optional. The defaults indicated for each word are used if the card is not entered. If this card is being used to specify only the junction hydraulic diameter for the interphase drag calculation, (i.e., $f = 0$ in Word 1 of cards CCC1101-1199) then the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (will not be used). If this card is being used for the CCFL model (i.e., $f = 1$ in Word 1 of cards CCC1101-1199), then enter all four words for the appropriate CCFL model if values different from the default value are desired.

- W1(R) Junction hydraulic diameter, D_j (m, ft). This quantity is the junction hydraulic diameter used in the CCFL correlation equation and interphase drag and must be ≥ 0 . The number should be computed from $4.0 \bullet \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If a 0 is entered or if the default is used, the junction diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi} \right)^{0.5}$. See Word 1 of cards CCC0201-0299 for the area.
- W2(R) Flooding correlation form, β . If 0, the Wallis CCFL form is used. If 1, the Kutateladze CCFL form is used. If between 0.0 and 1.0, Bankoff weighting between the Wallis and Kutateladze CCFL forms is used. This number must be ≥ 0 and ≤ 1 . The default value is 0 (Wallis form).
- W3(R) Gas intercept, c . This quantity is the gas intercept used in the CCFL correlation (when $H_f^{1/2} = 0$) and must be > 0 . The default value is 1.
- W4(R) Slope, m . This quantity is the slope used in the CCFL correlation and must be > 0 . The default value is 1.
- W5(I) Junction number.

A9.10.17 Pressurizer Y-Coordinate Areas, CCC1601-1699

These cards are optional and if entered activate the y-coordinate for each volume and allow the full one-dimensional momentum equations to be used in connections to the y-faces. (The corresponding x-coordinate cards are CCC0101-0199). The format is two words per set in sequential expansion format for nv sets. The card numbers need not be consecutive. The words for one set are:

W1(R) Area (m^2 , ft^2). If these cards are missing, this y-direction area is computed from (volume)/(y-direction length).

W2(I) Volume number.

A9.10.18 Pressurizer Z-Coordinate Areas, CCC1701-1799

These cards are optional and if entered activate the z-coordinate for each volume and allow the full one-dimensional momentum equations to be used in connections to the z-faces. (The corresponding x-coordinate cards are CCC0101-0199). The format is two words per set in sequential expansion format for nv sets. The card numbers need not be consecutive. The words for one set are:

W1(R) Area (m^2 , ft^2). If these cards are missing, this z-direction area is computed from (volume)/(z-direction length).

W2(I) Volume number.

A9.10.19 Pressurizer Y-Coordinate Lengths, CCC1801-1899

These cards are optional and, if entered, activate the y-coordinate for each volume and allow the full one-dimensional momentum equation for connections to the y-faces. The format is two words per set in sequential expansion format for nv sets. Card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC0301-0399).

W1(R) Length (m, ft). If these cards are missing and cards CCC1601-1699 are missing, this y-direction length is computed from $2.0 \cdot \left(\frac{x - \text{direction area}}{\pi} \right)^{0.5}$. If these cards are missing and cards CCC1601-1699 are present, this y-direction length is computed from (volume)/(y-direction area).

W2(I) Volume number.

A9.10.20 Pressurizer Z-Coordinate Lengths, CCC1901-1999

These cards are optional and, if entered, activate the z-coordinate for each volume and allow the full one-dimensional momentum equation for connections to the z-faces. The format is two words per set in

sequential expansion format for nv sets. Card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC0301-0399).

W1(R) Length (m, ft). If these cards are missing and cards CCC1701-1799 are missing, this z-direction length is computed from $2.0 \cdot \left(\frac{x - \text{direction area}}{\pi} \right)^{0.5}$. If these cards are missing and cards CCC1701-1799 are present, this z-direction length is computed from (volume)/(z-direction area).

W2(I) Volume number.

A9.10.21 Pressurizer Initial Boron Concentrations, CCC2001-2099

These cards are required only if boron is specified in one of the control words (Word 1) in cards CCC1201-1299. The card format is two words per set in sequential expansion format for nv sets. Boron concentrations must be entered for each volume, and 0 should be entered for those volumes whose associated control word did not specify boron.

W1(R) Boron concentration (mass of boron per mass of liquid).

W2(I) Volume number.

A9.10.22 Pressurizer Y-Coordinate (Elevation) Changes, CCC2101-2199

These cards are optional. If these cards are missing, the coordinate changes are computed from the y-coordinate length and a rotation matrix computed from the angle information. If these cards are entered, the entered data overwrites the previously computed coordinate change data. The card format is four words per set in sequential expansion format up to nv sets, and card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC0701-0799).

W1(R) Coordinate change along fixed x-axis due to traverse from inlet to outlet along the local y-coordinate, Δ_{xy} (m, ft).

W2(R) Coordinate change along fixed y-axis due to traverse from inlet to outlet along the local y-coordinate, Δ_{yy} (m, ft).

W3(R) Coordinate change along fixed z-axis due to traverse from inlet to outlet along the local y-coordinate, Δ_{zy} (m, ft).

W4(I) Volume number.

A9.10.23 Pressurizer Z-Coordinate (Elevation) Changes, CCC2201-2299

These cards are optional. If these cards are missing, the coordinate changes are computed from the z-coordinate length and a rotation matrix computed from the angle information. If these cards are entered, the entered data overwrites the previously computed coordinate change data. The card format is four words per set in sequential expansion format up to nv sets, and card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC0701-0799).

- W1(R) Coordinate change along fixed x-axis due to traverse from inlet to outlet along the local z-coordinate, Δ_{xz} (m, ft).
- W2(R) Coordinate change along fixed y-axis due to traverse from inlet to outlet along the local z-coordinate, Δ_{yz} (m, ft).
- W3(R) Coordinate change along fixed z-axis due to traverse from inlet to outlet along the local z-coordinate, Δ_{zz} (m, ft).
- W4(I) Volume number.

A9.10.24 Pressurizer Volume Y-Coordinate Friction Data, CCC2301-2399

These cards are optional and may be entered if area or length data was entered for the y-coordinate. If the cards are not entered, the wall roughness defaults to 0 and the default hydraulic diameter is computed from $\frac{4.0 \bullet \text{y-direction volume flow area}}{(\pi \bullet \text{x-direction volume flow area})^{0.5}}$. The corresponding x-coordinate cards are CCC0801-0899.

- W1(R) Wall roughness (m, ft).
- W2(R) Hydraulic diameter (m, ft). A check is made to ensure that the y-direction roughness is less than half the y-direction hydraulic diameter. See Word 1 on cards CCC0101-0109 for the area.
- W3(I) Volume number.

A9.10.25 Pressurizer Volume Z-Coordinate Friction Data, CCC2401-2499

These cards are optional and have to be entered if area or length data was entered for the z-coordinate. If the cards are not entered, the wall roughness defaults to 0 and the default hydraulic diameter is computed from $\frac{4.0 \bullet \text{z-direction volume flow area}}{(\pi \bullet \text{x-direction volume flow area})^{0.5}}$. The corresponding x-coordinate cards are CCC0801-0899.

- W1(R) Wall roughness (m, ft).

W2(R) Hydraulic diameter (m, ft). A check is made to ensure that the z-direction roughness is less than half the z-direction hydraulic diameter. See Word 1 on cards CCC0101-0109 for the area.

W3(I) Volume number.

A9.10.26 Pressurizer Volume Additional Wall Friction Data, CCC2501-2599

These cards are optional. If these cards are not entered, the default values are 1.0 for the laminar shape factor and 0.0 for the viscosity ratio exponent. The card format is seven words per set in sequential expansion format for nv sets and card numbers need not be consecutive. A description of this input is presented in Section 3 of Volume I.

W1(R) Shape factor for x-coordinate.

W2(R) Viscosity ratio exponent for x-coordinate.

W3(R) Shape factor for y-coordinate.

W4(R) Viscosity ratio exponent for y-coordinate.

W5(R) Shape factor for z-coordinate.

W6(R) Viscosity ratio exponent for z-coordinate.

W7(I) Volume number.

A9.10.27 Pressurizer Y-Coordinate Control Flags, CCC2701-2799

These cards are optional. The card format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC1001-1099).

W1(I) Volume control flags. This word has the general format tlpvbfe, but this word is limited to 00000f0 since it only enters the coordinate oriented flags for the y-direction.

f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed along the y-coordinate direction in the volume, and f = 1 specifies that wall friction effects are not to be computed along the y-coordinate direction.

W2(I) Volume number.

A9.10.28 Pressurizer Z-Coordinate Control Flags, CCC2801-2899

These cards are optional. The card format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC1001-1099).

W1(I) Volume control flags. This word has the general format tlpybfe, but this word is limited to 00000f0 since it only enters the coordinate oriented flags for the z-direction.

f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed along the z-coordinate direction in the volume, and f = 1 specifies that wall friction effects are not to be computed along the z-coordinate direction.

W2(I) Volume number.

A9.10.29 Pressurizer Junction Form Loss Data, CCC3001-3099

These cards are optional. The user-specified form loss is given in Words 1 and 2 of cards CCC0901-0999 if this card is not entered. If this card is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F \text{Re}^{-C_F}$$

$$K_R = A_R + B_R \text{Re}^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 1 and 2 of cards CCC0901-0999. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all five words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R .

W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.

W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.

W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.

W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

W5(I) Junction number.

A9.10.30 Pressurizer ORNL ANS Interphase Model Values, CCC3101-3199

These cards are required if any of the interphase friction flags *b* in the volume control flags entered on cards CCC1001-1099 are set to 2.

W1(R) Gap (distance between side walls, short length, pitch, channel width) (m, ft).

W2(R) Span (distance from one end to the other, long length) (m, ft).

W3(I) Volume number.

A9.10.31 Pressurizer Initial Noncondensable Mass Fractions, CCC3201-3299

These cards are optional. They are required only if the user desires to use initial noncondensable mass fractions that are different than the default values specified on card 115. The card format is $M = N+1$ words per set in sequential expansion format for *nv* sets. The plus one is for the volume number. Initial noncondensable mass fractions must be entered for each volume, and the default set on card 115 should be entered for those volumes whose values are the same as the default set on card 115.

W1-WN(R) Initial mass fraction for each noncondensable gas type as specified on card 110 where *N* is the number of gas types specified on card 110.

WM(I) Volume number.

A9.11 CANDU Channel Component, CANCHAN

A CANDU channel component is indicated by CANCHAN for Word 2 on CCC0000.

More than one junction may be connected to the inlet or outlet of a CANDU channel. If an end has no junctions, that end is considered a closed end. For major edits, minor edits, and plot variables, the volumes in the CANDU channel component are numbered as CCCNN0000, where *NN* is the volume number ($01 \leq NN \leq 99$). The junctions in the CANDU channel component are numbered as CCCJJ0000, where *JJ* is the junction number ($01 \leq JJ \leq 98$).

The volumes in a CANDU channel are usually considered one-dimensional components and flow in the volumes is along the x-coordinate. Crossflow junctions can connect to any of the CANDU channel volumes in the y- and z-coordinate directions (faces 3, 4, 5, or 6) using a form of the momentum equation that does or does not include momentum flux terms. It is also possible to connect external junctions to the x-coordinate direction faces (1 or 2) of any of the CANDU channel volumes using a form of the momentum equation that does or does not include the momentum flux terms. It is also possible to include or not include the momentum flux terms in internal CANDU channel junctions.

A9.11.1 CANDU Channel Information, CCC0001

This card is required.

- W1(I) Number of volumes, nv. The number nv must be greater than 0 and less than 100. The number of associated junctions internal to these components is nv-1. The outer junctions are described by other components.
- W2(I) Surgeline connection number. This word must the same format as printed in the output.
- W3(R) User specified interfacial heat transfer coefficient from liquid to saturation state (W/m²-K, Btu/hr-ft²-F). This word is optional.
- W4(R) User specified interfacial heat transfer coefficient from vapor to saturation state (W/m²-K, Btu/hr-ft²-F). This word is optional.

A9.11.2 CANDU Channel X-Coordinate Areas, CCC0101-0199

These cards are required, and the card numbers need not be consecutive. The format is two words per set in sequential expansion format for nv sets. (The corresponding y- and z-coordinate cards are CCC1601-1699 and CCC1701-1799). The words for one set are

- W1(R) Area (m², ft²).
- W2(I) Volume number.

A9.11.3 CANDU Channel Junction Areas, CCC0201-0299

These cards are optional, and, if entered, the card numbers need not be consecutive. The format is two words per set in sequential expansion format for nv-1 sets.

- W1(R) Internal junction flow area (m², ft²). If cards are missing or a word is 0, the junction flow area is set to the minimum area of the adjoining volumes. For abrupt area changes, the junction area must be equal to or less than the minimum of the adjacent volume areas. There is no restriction for smooth area changes.
- W2(I) Junction number.

A9.11.4 CANDU Channel X-Coordinate Lengths, CCC0301-0399

These cards are required. The format is two words per set in sequential expansion format for nv sets. Card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC1801-1899 and CCC1901-1999).

- W1(R) Length (m, ft).
- W2(I) Volume number.

A9.11.5 CANDU Channel Volumes, CCC0401-0499

These cards are optional, and if not entered, the volumes are set to 0. The format is two words per set in sequential expansion format for nv sets. Card numbers need not be consecutive.

W1(R) Volume (m³, ft³). The code requires that each volume equal the x-direction flow area times the x-direction length. If activated, the code also requires each volume equal the y-direction flow area times the y-direction length, and each volume equal the z-direction flow area times the z-direction length. For any volume, at least two of the three quantities, x-direction area, the x-direction length, or volume, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the quantities are 0, the volume must equal the x-direction area times the x-direction length within a relative error of 0.000001. The same relative error check is done for the y- and z-directions. If both the y-direction area and y-direction length are not entered or are 0, the y-direction length is computed from $2.0 \bullet \left(\frac{\text{x-direction area}}{\pi} \right)^{0.5}$ and the y-direction area is computed from $\frac{\text{volume}}{\text{y-direction length}}$. The same is true for the z-direction.

W2(I) Volume number.

A9.11.6 CANDU Channel Volume Azimuthal Angles, CCC0501-0599

These cards are optional, and if not entered, the angles are set to 0. The angles on these cards are used for the 3-D drawing programs, so it is recommended that the user enter these values if they are known. The format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive.

W1(R) Azimuthal angle (degrees). The absolute value of the angle must be ≤ 360 degrees.

W2(I) Volume number.

A9.11.7 CANDU Channel Volume Vertical Angles, CCC0601-0699

These cards are required. The format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive.

W1(R) Vertical angle (degrees). The absolute value of the angle must be less than or equal to 90 degrees. This angle is used in the interphase drag calculation.

W2(I) Volume number.

A9.11.8 CANDU Channel X-Coordinate (Elevation) Changes, CCC0701-0799

These cards are optional. If these cards are missing, the coordinate changes or elevation changes are computed from the x-coordinate length and a rotation matrix computed from the angle information. If these cards are entered, the entered data becomes the x-coordinate change or elevation change data. Two formats entering 1 or 3 coordinate changes per volume are provided. The card format is two or four words per set in sequential expansion format up to nv sets, and card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC2101-2199 and CCC2201-2299).

1 coordinate change per volume format:

W1(R) Elevation change. This is the coordinate change along the fixed x-axis due to the traverse from inlet to outlet along the local x-coordinate, Δ_{zx} (m, ft). A positive value is an increase in elevation. The magnitude must be equal to or less than the length. See Section 2.4.1 of Volume II of the manual for further discussion. When the absolute value of the elevation angle determined by the ratio of the elevation change over the length is less than or equal to 45 degrees, the horizontal flow regime map is used; when the ratio is greater than 45 degrees, the vertical flow regime map is used.

W2(I) Volume number.

Three coordinate changes per volume format:

W1(R) Coordinate change along the fixed x-axis due to traverse from inlet to outlet along the local x-coordinate, Δ_{xx} (m, ft).

W2(R) Coordinate change along the fixed y-axis due to traverse from inlet to outlet along the local x-coordinate, Δ_{yx} (m, ft).

W3(R) Coordinate change along the fixed z-axis due to traverse from inlet to outlet along the local x-coordinate, Δ_{zx} (m, ft).

W4(I) Volume number.

A9.11.9 CANDU Channel Volume X-Coordinate Friction Data, CCC0801-0899

These cards are required. The card format is three words per set for nv sets, and card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC2301-2399 and CCC2401-2499).

W1(R) Wall roughness (m, ft).

W2(R) Hydraulic diameter (m, ft). This should be computed from $4.0 \bullet \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If 0, the hydraulic diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi} \right)^{0.5}$. A check is made to ensure that the roughness is less than half the hydraulic diameter. See Word 1 on cards CCC0101-0109 for the area.

W3(I) Volume number.

A9.11.10 CANDU Channel Junction Loss Coefficients, CCC0901-0999

These cards are optional and if missing, the energy loss coefficients are set to 0. The card format is three words per set in sequential expansion format for nv-1 sets, and card numbers need not be consecutive.

W1(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. Note: a variable loss coefficient may be specified (see Section A9.8.29). The interpretation and use of this loss coefficient vary based upon the "a" junction control flag selected for this junction on cards CCC1101-1199. If the smooth area change option is selected (a = 0), the entry here is the loss coefficient applied to the junction flow areas entered on cards CCC0201-0299. If the abrupt area change option is selected (a = 1 or 2), the loss coefficient entered here is adjusted by the loss square of the ratio of the junction area and the minimum flow area of the adjacent connected volumes (from cards CCC0101-0199). This adjusted loss coefficient is applied to the junction flow area. If the a = 1 option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

Example: Two volumes with volume average flow areas of 0.3 m² and 0.4 m² are connected with a junction ($A_{\text{jun}} = 0.075 \text{ m}^2$). The user supplied $K_{\text{loss_user}}$ is 10.0. For the smooth area change option (a = 0), the $K_{\text{loss_user}}$ of 10.0 is applied directly to the junction. For the abrupt area change option (a = 1 or 2), the K_{loss} applied at the junction is $K_{\text{loss_jun}} = K_{\text{loss_user}} \bullet (A_{\text{jun}}/A_{\text{min}})^2$. For this example, $K_{\text{loss_jun}} = 10.0 \bullet (0.075/0.3)^2 = 0.625$.

W2(R) Reynolds number independent reverse flow energy loss coefficient, A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. Note: a variable loss coefficient may be specified (see Section A9.8.29). See information for the previous word regarding interpretation and use of this loss coefficient.

W3(I) Junction number.

A9.11.11 CANDU Channel Volume X-Coordinate Control Flags, CCC1001-1099

These cards are required. The card format is two words per set in sequential expansion format for *nv* sets, and card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC2701-2799 and CCC2801-2899).

W1(I) Volume control flags. This word has the format tlpvbfe. It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags. Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f-flag is the only coordinate direction oriented flag. These words enter the scalar oriented flags and the x-coordinate flags for each volume in the component.

t-flag. The digit t specifies whether the thermal front tracking model is to be used; t = 0 specifies that the front tracking model is not to be used for the volume, and t = 1 specifies that the front tracking model is to be used for the volume. The thermal front tracking model can only be applied to vertically-oriented components.

l-flag The digit l specifies whether the mixture level tracking model is to be used; l = 0 specifies that the level model not be used for the volume, and l = 1 specifies that the level model be used for the volume. The mixture level tracking model can only be applied to vertically-oriented components.

p-flag The digit p specifies whether the water packing scheme is to be used. p = 0 specifies that the water packing scheme is to be used for the volume, and p = 1 specifies that the water packing scheme is not to be used for the volume.

v-flag The digit v specifies whether the vertical stratification model is to be used. v = 0 specifies that the vertical stratification model is to be used for the volume, and v = 1 specifies that the vertical stratification model is not to be used for the volume.

b-flag The digit b specifies the interphase friction that is used. b = 0 means that the CANDU channel interphase friction model will be applied, b = 1 means that the rod bundle interphase friction model will be applied, b = 2 means that the ORNL ANS interphase friction model will be applied (see card ccc3101 in Section A9.8.30).

f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed along the x-coordinate of the volume, and f = 1 specifies that wall friction effects are not to be computed along the x-coordinate.

e-flag The digit e specifies if nonequilibrium or equilibrium is to be used. e = 0 specifies that a nonequilibrium (unequal temperature) calculation is to be used, and e = 1 specifies that an

equilibrium (equal temperature) calculation is to be used. Equilibrium volumes should not be connected to nonequilibrium volumes. The equilibrium option is provided only for comparison to other codes.

W2(I) Volume number.

A9.11.12 CANDU Channel Junction Control Flags, CCC1101-1199

These cards are required. The card format is two words per set in sequential expansion format for nv-1 sets, and card numbers need not be consecutive.

W1(I) Junction control flags. This word has the format jefvcahs. It is not necessary to input leading zeros. For the CANDU channel component, the junction control flag is limited to the format 0ef0cahs.

e-flag The digit e specifies the modified PV term in the energy equations. e = 0 means that the modified PV term will not be applied, and e = 1 means that it will be applied.

f-flag The digit f specifies CCFL options. f = 0 means that the CCFL model will not be applied, and f = 1 means that the CCFL model will be applied.

c-flag The digit c specifies choking options. c = 0 means that the choking model will be applied, and c = 1 means that the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske or Moody (option 54) critical flow model is active. If card 1 is present and contains Option 50, the original RELAP5 critical flow model is active. Unlike junctions for certain other hydrodynamic components, only the default values of discharge coefficients and factors used in the critical flow models are allowed for junctions in CANDU channel components. Problem renodalization should be used to circumvent this limitation should it prove to be significant. For the Henry-Fauske or Moody critical flow model, the default value for the discharge coefficient is 1.0 and the default value for the thermal nonequilibrium constant is 0.14. For the original RELAP5 critical flow model, the default values for the subcooled, two-phase and superheated discharge coefficients each are 1.0.

a-flag The digit a specifies area change options. a = 0 means a smooth area change and only the user supplied K_{loss} is applied. a = 1 means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. a = 2 means a partial abrupt area change model and it is the same as a = 1 except no code calculated forward and reverse contraction/expansion K_{loss} is applied. See discussion for W1(R) regarding the K_{loss} flow area reference, which varies based on the a-flag option selected here.

- h-flag** The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two-velocity momentum equations) option, and h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit printout will show a 1.
- s-flag** The digit s specifies momentum flux options. s = 0 uses momentum flux in both the to volume and the from volume. s = 1 uses momentum flux in the from volume, but not in the to volume. s = 2 uses momentum flux in the to volume but not in the from volume. s = 3 does not use momentum flux in either the to or the from volume. For this component, the option s = 0 is the usual recommendation (momentum flux in both volumes). The other options s = 1, 2, and 3 are included to allow consistency for this flag for other components (single-junction, branch junction, etc.).

W2(I) Junction number.

A9.11.13 CANDU Channel Volume Initial Conditions, CCC1201-1299

These cards are required. The card format is seven words per set in sequential expansion format for nv sets, and card numbers need not be consecutive.

- W1(I)** Control word. This word has the format εbt. It is not necessary to input leading zeros.
- ε-flag** The digit ε specifies the fluid, where ε = 0 is the default fluid. The value for ε > 0 corresponds to the position number of the fluid type indicated on the 120-9 cards (i.e., ε = 1 specifies H₂O, ε = 2 specifies D₂O, etc.). The default fluid is that set for the hydrodynamic system by cards 120-129 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-9 or these control words must be consistent (i.e., not specify different fluids). If cards 120-9 are not entered and all control words use the default ε = 0, then H₂O is assumed as the fluid. It is recommended that the user use the 120-9 cards to set the fluid type in the systems.
- b-flag** The digit b specifies whether boron is present or not. b = 0 specifies that the volume liquid does not contain boron; b = 1 specifies that a boron concentration in mass of boron per mass of liquid (which may be 0) is being entered on the CCC2001-2099 cards.
- t-flag** The digit t specifies how the following words are to be used to determine the initial thermodynamic state. Entering t equal to 0-3 and 7 specifies one component (steam/water). Entering t equal to 4-6 allows the specification of two components (steam/water and noncondensable gas).
- With options t equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.

One Component (Steam/Water), Equilibrium or Non-equilibrium

[P,U_f,U_g,α_g] If $t = 0$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions, depending on the internal energies used to define the thermodynamic state. W6 should be 0.0.

One Component (Steam/Water), Equilibrium

[T,x_s] If $t = 1$, the next two words are interpreted as temperature (K, °F) and static quality in equilibrium condition. W4, W5, and W6 should be 0.0.

[P,x_s] If $t = 2$, the next two words are interpreted as pressure (Pa, lb_f/in²) and static quality in equilibrium condition. W4, W5, and W6 should be 0.0.

[P,T] If $t = 3$, the next two words are interpreted as pressure (Pa, lb_f/in²) and temperature (K, °F) in equilibrium condition. W4, W5, and W6 should be 0.0.

[P,T_f,T_g,α_g] If $t = 7$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid temperature (K, F), vapor temperature (K, F), and vapor void fraction. These quantities will be interpreted as equilibrium conditions. The liquid temperature has to be equal to or less than the saturation temperature, and the vapor temperature has to be equal to or greater than the saturation temperature. Enter boron in W6 if needed. This input option was added to allow easier comparisons with TRACE calculations.

Two Components (Steam/Water/Air), Equilibrium or Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0\text{E-}9 \leq x_s \leq 0.99999999$,

Single phase if $x_s < 1.0 \text{ E-}9$ or $x_s > 0.99999999$

The static quality is given by $x_s = M_g/(M_g + M_f)$, where $M_g = M_s + M_n$. Section 3.2 of Volume I of the manual discusses this further.

[P,T,x_s] If $t = 4$, the next three words are interpreted as pressure (Pa, lb_f/in²), temperature (K, °F), and static quality in equilibrium condition. Using this input option with static quality > 0.0

and ≤ 1.0 , saturated noncondensables will result. W5 and W6 should be 0.0. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. If the pressure is greater than the critical pressure, $P > P_{\text{crit}}$, then the static quality, x_s (mass steam + air)/(mass steam + air + water) has to be large enough to give a partial steam pressure that is less than the critical pressure, $P_s < P_{\text{crit}}$. Otherwise an input error will occur. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensable (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option.

[T, x_s , x_n] If $t = 5$, the next three words are interpreted as steam saturation temperature (K, °F), static quality, and noncondensable quality in equilibrium condition. W5 and W6 should be 0.0. Both the static and noncondensable qualities are restricted to be between 1.0 E-9 and 0.99999999. The liquid will be subcooled. The input temperature determines the steam partial pressure. However, the total pressure may be greater than the critical pressure, which will cause an input error.

[P, U_f , U_g , α_g , x_n] If $t = 6$, the next five words are interpreted as pressure (Pa, lb_f/in²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing ($t = 0$). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor-specific internal energy.

W2-W6(R) Quantities as described under Word 1. Five quantities must be entered, and zeros should be entered for unused quantities. If any control word (Word 1) indicates that boron is present, cards CCC2001-2099 must be entered to define the initial boron concentrations. Boron concentrations are not entered in Words 2-6 for CANDU channel components.

W7(I) Volume number.

A9.11.14 CANDU Channel Junction Conditions Control Words, CCC1300

This card is optional, and, if missing, velocities are assumed on cards CCC1301-1399.

W1(I) Control word. If 0, the first and second words of each set on cards CCC1301-1399 are velocities. If 1, the first and second words of each set on cards CCC1301-1399 are mass flows.

A9.11.15 CANDU Channel Junction Initial Conditions, CCC1301-1399

These cards are required.

W1(R) Initial liquid velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W2(R) Initial vapor velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W3(R) Interface velocity (m/s, ft/s). This capability has not yet been implemented, so enter 0.

W4(I) Junction number.

A9.11.16 CANDU Channel Junction Diameter and CCFL Data, CCC1401-1499

These cards are optional. The defaults indicated for each word are used if the card is not entered. If this card is being used to specify only the junction hydraulic diameter for the interphase drag calculation, (i.e., $f = 0$ in Word 1 of cards CCC1101-1199) then the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (will not be used). If this card is being used for the CCFL model (i.e., $f = 1$ in Word 1 of cards CCC1101-1199), then enter all four words for the appropriate CCFL model if values different from the default value are desired.

W1(R) Junction hydraulic diameter, D_j (m, ft). This quantity is the junction hydraulic diameter used in the CCFL correlation equation and interphase drag and must be ≥ 0 . The number should be computed from $4.0 \bullet \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If a 0 is entered or if the default is used, the junction diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi} \right)^{0.5}$. See Word 1 of cards CCC0201-0299 for the area.

W2(R) Flooding correlation form, β . If 0, the Wallis CCFL form is used. If 1, the Kutateladze CCFL form is used. If between 0.0 and 1.0, Bankoff weighting between the Wallis and Kutateladze CCFL forms is used. This number must be ≥ 0 and ≤ 1 . The default value is 0 (Wallis form).

W3(R) Gas intercept, c . This quantity is the gas intercept used in the CCFL correlation (when $H_f^{1/2} = 0$) and must be > 0 . The default value is 1.

W4(R) Slope, m . This quantity is the slope used in the CCFL correlation and must be > 0 . The default value is 1.

W5(I) Junction number.

A9.11.17 CANDU Channel Y-Coordinate Areas, CCC1601-1699

These cards are optional and if entered activate the y-coordinate for each volume and allow the full one-dimensional momentum equations to be used in connections to the y-faces. (The corresponding x-coordinate cards are CCC0101-0199). The format is two words per set in sequential expansion format for nv sets. The card numbers need not be consecutive. The words for one set are:

W1(R) Area (m^2 , ft^2). If these cards are missing, this y-direction area is computed from (volume)/(y-direction length).

W2(I) Volume number.

A9.11.18 CANDU Channel Z-Coordinate Areas, CCC1701-1799

These cards are optional and if entered activate the z-coordinate for each volume and allow the full one-dimensional momentum equations to be used in connections to the z-faces. (The corresponding x-coordinate cards are CCC0101-0199). The format is two words per set in sequential expansion format for nv sets. The card numbers need not be consecutive. The words for one set are:

W1(R) Area (m^2 , ft^2). If these cards are missing, this z-direction area is computed from (volume)/(z-direction length).

W2(I) Volume number.

A9.11.19 CANDU Channel Y-Coordinate Lengths, CCC1801-1899

These cards are optional and, if entered, activate the y-coordinate for each volume and allow the full one-dimensional momentum equation for connections to the y-faces. The format is two words per set in sequential expansion format for nv sets. Card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC0301-0399).

W1(R) Length (m, ft). If these cards are missing and cards CCC1601-1699 are missing, this y-direction length is computed from $2.0 \bullet \left(\frac{\text{x-direction area}}{\pi} \right)^{0.5}$. If these cards are missing and cards CCC1601-1699 are present, this y-direction length is computed from (volume)/(y-direction area).

W2(I) Volume number.

A9.11.20 CANDU Channel Z-Coordinate Lengths, CCC1901-1999

These cards are optional and, if entered, activate the z-coordinate for each volume and allow the full one-dimensional momentum equation for connections to the z-faces. The format is two words per set in sequential expansion format for nv sets. Card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC0301-0399).

W1(R) Length (m, ft). If these cards are missing and cards CCC1701-1799 are missing, this z-direction length is computed from $2.0 \cdot \left(\frac{x - \text{direction area}}{\pi} \right)^{0.5}$. If these cards are missing and cards CCC1701-1799 are present, this z-direction length is computed from (volume)/(z-direction area).

W2(I) Volume number.

A9.11.21 CANDU Channel Initial Boron Concentrations, CCC2001-2099

These cards are required only if boron is specified in one of the control words (Word 1) in cards CCC1201-1299. The card format is two words per set in sequential expansion format for nv sets. Boron concentrations must be entered for each volume, and 0 should be entered for those volumes whose associated control word did not specify boron.

W1(R) Boron concentration (mass of boron per mass of liquid).

W2(I) Volume number.

A9.11.22 CANDU Channel Y-Coordinate (Elevation) Changes, CCC2101-2199

These cards are optional. If these cards are missing, the coordinate changes are computed from the y-coordinate length and a rotation matrix computed from the angle information. If these cards are entered, the entered data overwrites the previously computed coordinate change data. The card format is four words per set in sequential expansion format up to nv sets, and card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC0701-0799).

W1(R) Coordinate change along fixed x-axis due to traverse from inlet to outlet along the local y-coordinate, Δ_{xy} (m, ft).

W2(R) Coordinate change along fixed y-axis due to traverse from inlet to outlet along the local y-coordinate, Δ_{yy} (m, ft).

W3(R) Coordinate change along fixed z-axis due to traverse from inlet to outlet along the local y-coordinate, Δ_{zy} (m, ft).

W4(I) Volume number.

A9.11.23 CANDU Channel Z-Coordinate (Elevation) Changes, CCC2201-2299

These cards are optional. If these cards are missing, the coordinate changes are computed from the z-coordinate length and a rotation matrix computed from the angle information. If these cards are entered, the entered data overwrites the previously computed coordinate change data. The card format is four words per set in sequential expansion format up to nv sets, and card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC0701-1799).

- W1(R) Coordinate change along fixed x-axis due to traverse from inlet to outlet along the local z-coordinate, Δ_{xz} (m, ft).
- W2(R) Coordinate change along fixed y-axis due to traverse from inlet to outlet along the local z-coordinate, Δ_{yz} (m, ft).
- W3(R) Coordinate change along fixed z-axis due to traverse from inlet to outlet along the local z-coordinate, Δ_{zz} (m, ft).
- W4(I) Volume number.

A9.11.24 CANDU Channel Volume Y-Coordinate Friction Data, CCC2301-2399

These cards are optional and may be entered if area or length data was entered for the y-coordinate. If the cards are not entered, the wall roughness defaults to 0 and the default hydraulic diameter is computed from $\frac{4.0 \bullet \text{y-direction volume flow area}}{(\pi \bullet \text{x-direction volume flow area})^{0.5}}$. The corresponding x-coordinate cards are CCC0801-0899.

- W1(R) Wall roughness (m, ft).
- W2(R) Hydraulic diameter (m, ft). A check is made to ensure that the y-direction roughness is less than half the y-direction hydraulic diameter. See Word 1 on cards CCC0101-0109 for the area.
- W3(I) Volume number.

A9.11.25 CANDU Channel Volume Z-Coordinate Friction Data, CCC2401-2499

These cards are optional and have to be entered if area or length data was entered for the z-coordinate. If the cards are not entered, the wall roughness defaults to 0 and the default hydraulic diameter is computed from $\frac{4.0 \bullet \text{z-direction volume flow area}}{(\pi \bullet \text{x-direction volume flow area})^{0.5}}$. The corresponding x-coordinate cards are CCC0801-0899.

- W1(R) Wall roughness (m, ft).

W2(R) Hydraulic diameter (m, ft). A check is made to ensure that the z-direction roughness is less than half the z-direction hydraulic diameter. See Word 1 on cards CCC0101-0109 for the area.

W3(I) Volume number.

A9.11.26 CANDU Channel Volume Additional Wall Friction Data, CCC2501-2599

These cards are optional. If these cards are not entered, the default values are 1.0 for the laminar shape factor and 0.0 for the viscosity ratio exponent. The card format is seven words per set in sequential expansion format for nv sets and card numbers need not be consecutive. A description of this input is presented in Section 3 of Volume I.

W1(R) Shape factor for x-coordinate.

W2(R) Viscosity ratio exponent for x-coordinate.

W3(R) Shape factor for y-coordinate.

W4(R) Viscosity ratio exponent for y-coordinate.

W5(R) Shape factor for z-coordinate.

W6(R) Viscosity ratio exponent for z-coordinate.

W7(I) Volume number.

A9.11.27 CANDU Channel Y-Coordinate Control Flags, CCC2701-2799

These cards are optional. The card format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC1001-1099).

W1(I) Volume control flags. This word has the general format tlpvbfe, but this word is limited to 00000f0 since it only enters the coordinate oriented flags for the y-direction.

f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed along the y-coordinate direction in the volume, and f = 1 specifies that wall friction effects are not to be computed along the y-coordinate direction.

W2(I) Volume number.

A9.11.28 CANDU Channel Z-Coordinate Control Flags, CCC2801-2899

These cards are optional. The card format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive. (The corresponding x-coordinate cards are CCC1001-1099).

W1(I) Volume control flags. This word has the general format tlpybfe, but this word is limited to 00000f0 since it only enters the coordinate oriented flags for the z-direction.

f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed along the z-coordinate direction in the volume, and f = 1 specifies that wall friction effects are not to be computed along the z-coordinate direction.

W2(I) Volume number.

A9.11.29 CANDU Channel Junction Form Loss Data, CCC3001-3099

These cards are optional. The user-specified form loss is given in Words 1 and 2 of cards CCC0901-0999 if this card is not entered. If this card is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F \text{Re}^{-C_F}$$

$$K_R = A_R + B_R \text{Re}^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 1 and 2 of cards CCC0901-0999. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all five words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R .

W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.

W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.

W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.

W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

W5(I) Junction number.

A9.11.30 CANDU Channel ORNL ANS Interphase Model Values, CCC3101-3199

These cards are required of any of the interphase friction flags b in the volume control flags entered on cards CCC1001-1099 are set to 2.

W1(R) Gap (distance between side walls, short length, pitch, channel width) (m, ft).

W2(R) Span (distance from one end to the other, long length) (m, ft).

W3(I) Volume number.

A9.11.31 CANDU Channel Initial Noncondensable Mass Fractions, CCC3201-3299

These cards are optional. They are required only if the user desires to use initial noncondensable mass fractions that are different than the default values specified on card 115. The card format is M = N+1 words per set in sequential expansion format for *nv* sets. The plus one is for the volume number. Initial noncondensable mass fractions must be entered for each volume, and the default set on card 115 should be entered for those volumes whose values are the same as the default set on card 115.

W1-WN(R) Initial mass fraction for each noncondensable gas type as specified on card 110 where N is the number of gas types specified on card 110.

WM(I) Volume number.

A9.12 Branch Component, BRANCH

A branch component is indicated by BRANCH for Word 2 on card CCC0000. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number (which must be 01 for the branch component) and N is the face number. More than one junction may be connected to the inlet or outlet. If an end has no junctions, that end is considered a closed end. If more than one junction is connected on one end of a branch, each junction should be modeled as an abrupt area change. For major edits, minor edits, and plot variables, the volume in the branch component is numbered as CCC010000. For edits and plot variables, the junctions are numbered as CCC0J0000, where J is the junction number ($1 \leq J \leq 9$).

A9.12.1 Branch Component Information, CCC0001

This card is required.

W1(I) Number of junctions, nj. The variable nj is the number of junctions described in the input data for this component ($0 \leq nj \leq 9$). Not all junctions connecting to the branch need be described with this component input, and nj is not necessarily the total number of junctions connecting to the branch. Junctions described in single-junctions, time-dependent junctions, pumps, separators, jetmixers, ECC mixers, and other branches can be connected to this branch.

W2(I) Initial condition control. This word is optional and, if missing, the junction initial velocities in the first and second words on card CCCN201 are assumed to be velocities. If 0, velocities are assumed; if nonzero, mass flows are assumed.

A9.12.2 Branch X-Coordinate Volume Data, CCC0101-0109

This card (or cards) is required. The nine words can be entered on one or more cards, and the card numbers need not be consecutive.

W1(R) Area (m², ft²).

W2(R) Length (m, ft).

W3(R) Volume (m³, ft³). The code requires that the volume equals the area times the length ($W3 = W1 \bullet W2$). At least two of the three quantities, W1, W2, and W3, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the words are 0, the volume must equal the x-direction area times the x-direction length within a relative error of 0.000001. The same relative error check is done for the y- and z-directions.

W4(R) Azimuthal angle (degrees). The absolute value of this angle must be ≤ 360 degrees. This quantity is not used in the calculation but is specified for possible automated drawing of nodalization diagrams.

W5(R) Inclination angle (degrees). The absolute value of this angle must be ≤ 90 degrees. The angle 0 degrees is horizontal, and positive angles have an upward inclination, i.e., the inlet is at the lowest elevation. This angle is used in the interphase drag calculation.

W6(R) Elevation change (m, ft). A positive value is an increase in elevation. The absolute value of this quantity must be less than or equal to the length. If the vertical angle orientation is 0, this quantity must be 0. If the vertical angle is nonzero, this quantity must also be nonzero and have the same sign. See Section 2.4.1 of Volume II of the manual for further discussion. When the absolute value of the elevation angle determined by the ratio of the elevation change (this Word 6) and the length (Word 2) is less than or equal to 45 degrees, the horizontal flow regime map is used. When the ratio is greater than 45 degrees, the vertical flow regime map is used.

W7(R) Wall roughness (m, ft).

W8(R) Hydraulic diameter (m, ft). This should be computed from $4.0 \bullet \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If 0, the hydraulic diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi} \right)^{0.5}$. A check is made that the pressurizer roughness is less than half the hydraulic diameter. See Word 1 for the area.

W9(I)	Volume control flags. This word has the format <u>t</u> <u>p</u> <u>v</u> <u>b</u> <u>f</u> <u>e</u> . It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags. Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f-flag is the only coordinate direction oriented flag. This word enters the scalar oriented flags and the x-coordinate flag.
t-flag	The digit <u>t</u> specifies whether the thermal front tracking model is to be used; <u>t</u> = 0 specifies that the front tracking model is not to be used for the volume, and <u>t</u> = 1 specifies that the front tracking model is to be used for the volume.
l-flag	The digit <u>l</u> specifies whether the mixture level tracking model is to be used; <u>l</u> = 0 specifies that the level model not be used for the volume, and <u>l</u> = 1 specifies that the level model be used for the volume.
p-flag	The digit <u>p</u> specifies whether the water packing scheme is to be used. <u>p</u> = 0 specifies that the water packing scheme is to be used for the volume, and <u>p</u> = 1 specifies that the water packing scheme is not to be used for the volume.
v-flag	The digit <u>v</u> specifies whether the vertical stratification model is to be used. <u>v</u> = 0 specifies that the vertical stratification model is to be used for the volume, and <u>v</u> = 1 specifies that the vertical stratification model is not to be used for the volume.
b-flag	The digit <u>b</u> specifies the interphase friction that is used. <u>b</u> = 0 means that the pressurizer interphase friction model will be applied, <u>b</u> = 1 means that the rod bundle interphase friction model will be applied, <u>b</u> = 2 means that the ORNL ANS interphase friction model will be applied (see card ccc0111 in Section A9.12.3).
f-flag	The digit <u>f</u> specifies whether wall friction is to be computed. <u>f</u> = 0 specifies that wall friction effects are to be computed along the x-coordinate direction in the volume, and <u>f</u> = 1 specifies that wall friction effects are not to be computed along the x-coordinate.
e-flag	The digit <u>e</u> specifies if nonequilibrium or equilibrium is to be used. <u>e</u> = 0 specifies that a nonequilibrium (unequal temperature) calculation is to be used, and <u>e</u> = 1 specifies that an equilibrium (equal temperature) calculation is to be used. Equilibrium volumes should not be connected to nonequilibrium volumes. The equilibrium option is provided only for comparison to other codes.

A9.12.3 Branch ORNL ANS Interphase Model Values, CCC0111

This card is required if the interphase friction flag b in Word 9 of card CCC0101-0109 is set to 2.

W1(R)	Gap (distance between side walls, short length, pitch, channel width) (m, ft).
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W2(R) Span (distance from one end to the other, long length) (m, ft).

W3(I) Volume number.

A9.12.4 Branch Additional Wall Friction, CCC0131

This card is optional. If this card is not entered, the default values are 1.0 for the laminar shape factor and 0.0 for the viscosity ratio exponent. Two, four, or six quantities may be entered on the card, and the data not entered are set to default values. A description of this input is presented in Section 3 of Volume I.

W1(R) Shape factor for x-coordinate.

W2(R) Viscosity ratio exponent for x-coordinate.

W3(R) Shape factor for y-coordinate.

W4(R) Viscosity ratio exponent for y-coordinate.

W5(R) Shape factor for z-coordinate.

W6(R) Viscosity ratio exponent for z-coordinate.

A9.12.5 Branch Y-Coordinate Volume Data, CCC0181-0189

These cards are optional. These cards are used when the user specifies the y-direction connection with the crossflow model. The volume of the volume is the same for the x-, y-, and z-directions. If these cards are entered, either W1 or W2 must be nonzero.

W1(R) Area (m², ft²). If these cards are missing or if this word is 0, this y-direction area is computed from $\frac{\text{volume}}{\text{y-direction length}}$.

W2(R) Length (m, ft). If these cards are missing, this y-direction length is computed from $2.0 \cdot \left(\frac{\text{x-direction area}}{\pi} \right)^{0.5}$. If this word is 0, this y-direction length is computed from $\frac{\text{volume}}{\text{y-direction area}}$.

W3(R) Roughness (m, ft).

W4(R) Hydraulic diameter (m, ft). If these cards are missing or if this word is 0, this y-direction hydraulic diameter is computed from $\frac{4.0 \cdot \text{y-direction volume flow area}}{(\pi \cdot \text{x-direction volume flow area})^{0.5}}$. A check is

made to ensure the y-direction roughness is less than half the y-direction hydraulic diameter.

W5(I) Volume control flags. This word has the general format tlpvbfe, but this word is limited to 00000f0 since it only enters the coordinate oriented flags for the y-direction.

f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed along the y-coordinate direction in the volume, and f = 1 specifies that wall friction effects are not to be computed along the y-coordinate direction.

W6(R) This word is not used. Enter 0.

W7(R) This word is not used. Enter 0.

W8(R) This word is the position change in the z-fixed (vertical) direction as the flow passes from the y-inlet face to the y-outlet face (m, ft). This quantity affects problems if connections are made to the y-faces.

A9.12.6 Branch Z-Coordinate Volume Data, CCC0191-0199

These cards are optional. These cards are used when the user specifies the z-direction connection with the crossflow model. The volume of the volume is the same for the x-, y-, and z-directions. If these cards are entered, either W1 or W2 must be nonzero.

W1(R) Area (m², ft²). If these cards are missing or if this word is 0, this z-direction area is computed from $\frac{\text{volume}}{\text{z-direction length}}$.

W2(R) Length (m, ft). If these cards are missing, this z-direction length is computed from $2.0 \bullet \left(\frac{\text{x-direction area}}{\pi} \right)^{0.5}$. If this word is 0, this z-direction length is computed from $\frac{\text{volume}}{\text{z-direction area}}$.

W3(R) Roughness (m, ft).

W4(R) Hydraulic diameter (m, ft). If these cards are missing or if this word is 0, this z-direction hydraulic diameter is computed from $\frac{4.0 \bullet \text{z-direction volume flow area}}{(\pi \bullet \text{x-direction volume flow area})^{0.5}}$. A check is made to ensure the z-direction roughness is less than half the z-direction hydraulic diameter.

- W5(I) Volume control flags. This word has the general format tlpybfe, but this word is limited to 00000f0 since it only enters the coordinate oriented flags for the z-direction.
- f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed along the z-coordinate direction in the volume, and f = 1 specifies that wall friction effects are not to be computed along the z-coordinate direction.
- W6(R) This word is not used. Enter 0.
- W7(R) This word is not used. Enter 0.
- W8(R) This word is the position change in the z-fixed (vertical) direction as the flow passes from the z-inlet face to the z-outlet face (m, ft). This quantity affects problems if connections are made to the z-faces.

A9.12.7 Branch Volume Initial Conditions, CCC0200

This card is required.

- W1(I) Control word. This word has the format gbt. It is not necessary to input leading zeros.
- ε -flag The digit ε specifies the fluid, where ε = 0 is the default fluid. The value for ε > 0 corresponds to the position number of the fluid type indicated on the 120-129 cards (i.e., ε = 1 specifies H₂O, ε = 2 specifies D₂O etc.). The default fluid is that set for the hydrodynamic system by cards 120-129 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-129 or these control words must be consistent (i.e., not specify different fluids). If cards 120-129 are not entered and all control words use the default ε = 0, then water is assumed to be the fluid. It is recommended that the user use the 120-129 cards to set the fluid type in the systems.
- b-flag The digit b specifies whether boron is present. b = 0 specifies that the volume liquid does not contain boron, and b = 1 specifies that a boron concentration in mass of boron per mass of liquid (which may be 0) is being entered after the other required thermodynamic information.
- t-flag The digit t specifies how the following words are to be used to determine the initial thermodynamic state. t = 0-3 and 7 specifies one component (steam/water); t = 4-6 allows the specification of two components (steam/water and noncondensable gas).
- With options t equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.

One Component (Steam/Water), Equilibrium or Non-equilibrium

[P,U_f,U_g,α_g] If $t = 0$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. Enter boron in W6 if needed.

One Component (Steam/Water), Equilibrium

[T,x_s] If $t = 1$, the next two words are interpreted as temperature (K, °F) and static quality in equilibrium condition. Enter boron in W4 if needed.

[P,x_s] If $t = 2$, the next two words are interpreted as pressure (Pa, lb_f/in²) and static quality in equilibrium condition. Enter boron in W4 if needed.

[P,T] If $t = 3$, the next two words are interpreted as pressure (Pa, lb_f/in²) and temperature (K, °F) in equilibrium condition. Enter boron in W4 if needed.

[P,T_f,T_g,α_g] If $t = 7$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid temperature (K, F), vapor temperature (K, F), and vapor void fraction. These quantities will be interpreted as equilibrium conditions. The liquid temperature has to be equal to or less than the saturation temperature, and the vapor temperature has to be equal to or greater than the saturation temperature. Enter boron in W6 if needed. This input option was added to allow easier comparisons with TRACE calculations.

Two Components (Steam/Water/Air), Equilibrium or Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0\text{E-}9 \leq x_s \leq 0.99999999$,

Single phase if $x_s < 1.0 \text{ E-}9$ or $x_s > 0.99999999$

The static quality is given by $x_s = M_g/(M_g + M_f)$, where $M_g = M_s + M_n$. Section 3.2 of Volume I of the manual discusses this further.

[P,T,x_s] If $t = 4$, the next three words are interpreted as pressure (Pa, lb_f/in²), temperature (K, °F), and static quality in equilibrium condition. Using this input option with static quality

greater than 0.0 and less than or equal to 1.0, saturated noncondensables will result. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. If the pressure is greater than the critical pressure, $P > P_{crit}$, then the static quality, x_s (mass steam + air)/(mass steam + air + water) has to be large enough to give a partial steam pressure that is less than the critical pressure, $P_s < P_{crit}$. Otherwise an input error will occur. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensable (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option. Enter boron in W5 if needed.

[T, x_s , x_n] If $t = 5$, the next three words are interpreted as steam saturation temperature (K, °F), static quality, and noncondensable quality in equilibrium condition. Enter boron in W5 if needed. Both the static and noncondensable qualities are restricted to be between 1.0×10^{-9} and 0.99999999. The liquid will be subcooled. The input temperature determines the steam partial pressure. However, the total pressure may be greater than the critical pressure, which will cause an input error.

[P, U_f , U_g , α_g , x_n] If $t = 6$, the next five words are interpreted as pressure (Pa, lb_f/in.²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing ($t = 0$). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor-specific internal energy. Enter boron in W7 if needed.

W2-W7(R) Quantities as described under Word 1. Depending on the control word, 2-6 quantities may be required. Enter only the minimum number required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions.

A9.12.8 Branch Initial Noncondensable Mass Fractions, CCC0301

This card is optional for a branch and is related to card 110. If card CCC0301 is not supplied, then the values on the card 115 are used for the noncondensable mass fractions. The number of words on card CCC0301 must equal the number of words on card 110. The mass fractions must sum to 1.0 to within a tolerance of 1.0×10^{-10} .

W1-WN(R) Initial mass fraction for each noncondensable gas type as specified on card 110 where N is the number of gas types on card 110.

A9.12.9 Branch Junction Geometry, CCCJ101-J109

These cards are required if n_j is greater than 0. Cards with J equal to 1-9 are entered, one for each junction. For a BRANCH component, J need not be consecutive, but n_j cards must be entered. The card format for Words 1-6 is listed below and is identical to Words 1-6 on card CCC0101 of the Single-Junction Geometry Card, except that J instead of 0 is used in the fourth digit.

W1(I) From connection code to a component. This refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction (see Section 2.1). The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, N is restricted to 1 or 2.

W2(I) To connection code to a component. This refers to the component at which the junction coordinate direction ends. See the description for W1 above.

W3(R) Area (m^2 , ft^2). If 0, the area is set to the minimum area of the adjoining volumes. For abrupt area changes, the junction area must be equal to or smaller than the minimum of the adjoining areas. For smooth area changes, there are no restrictions.

W4(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. A variable loss coefficient may be specified (see Section A9.12.11). The interpretation and use of this loss coefficient vary based upon the "a" junction control flag selected in W6. If the smooth area change option is selected ($a = 0$), the entry here is the loss coefficient applied to the junction flow area resulting from the entry made in W3. If the abrupt area change option is selected ($a = 1$ or 2), the loss coefficient entered here is adjusted by the square of the ratio of the junction area and the minimum flow area of the adjacent connected volumes. This adjusted loss coefficient is applied to the junction flow area. If the $a = 1$ option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

Example: Two volumes with volume average flow areas of 0.3 m^2 and 0.4 m^2 are connected with a junction ($A_{\text{jun}} = 0.075 \text{ m}^2$). The user supplied $K_{\text{loss, user}}$ is 10.0. For the smooth area change option ($a = 0$), the $K_{\text{loss, user}}$ of 10.0 is applied directly to the junction.

For the abrupt area change option ($\underline{a} = 1$ or 2), the K_{loss} applied at the junction is $K_{loss_{jun}} = K_{loss_{user}} * (A_{jun}/A_{min})^2$. For this example, $K_{loss_{jun}} = 10.0 * (0.075/0.3)^2 = 0.625$.

- W5(R) Reynolds number independent reverse flow energy loss coefficient, A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. A variable loss coefficient may be specified (see Section A9.12.11). See information for the previous word regarding interpretation and use of this loss coefficient.
- W6(I) Junction control flags. This word has the format jefvcahs. For the branch component, the junction control flag is limited to the format 0efvcahs.
- e-flag (*ief*) The digit $\underline{e}$ specifies the modified PV term in the energy equations. $\underline{e} = 0$ means that the modified PV term will not be applied, and $\underline{e} = 1$ means that it will be applied.
- f-flag (*icc*) The digit $\underline{f}$ specifies CCFL options. $\underline{f} = 0$ means that the CCFL model will not be applied, and $\underline{f} = 1$ means that the CCFL model will be applied.
- v-flag (*ist*) The digit $\underline{v}$ specifies horizontal stratification entrainment/pullthrough options. This model is for junctions connected to a horizontal volume. $\underline{v} = 0$ means the model is not applied; $\underline{v} = 1$ means an upward oriented junction (offtake volume must be vertical); $\underline{v} = 2$ means a downward oriented junction (offtake volume must be vertical); and $\underline{v} = 3$ means a centrally (side) located junction. When $\underline{v} = 4$, the offtake junction is located at an angle to the horizontal (The angle is specified as Word 9). (See the v-flag description for the single junction component for a more detailed list of conditions that have to be satisfied in order to use this model.)
- c-flag (*ick*) The digit $\underline{c}$ specifies choking options. $\underline{c} = 0$ means that the choking model will be applied, and $\underline{c} = 1$ means that the choking model will not be applied.
- The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske or Moody (option 54) critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active.
- a-flag (*irf*) The digit $\underline{a}$ specifies area change options. $\underline{a} = 0$ means a smooth area change and only the user supplied K_{loss} is applied. $\underline{a} = 1$ means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. $\underline{a} = 2$ means a partial abrupt area change model and it is the same as $\underline{a} = 1$ except no code calculated forward and reverse contraction/expansion K_{loss} is

applied. See discussion for W4(R) regarding the Kloss flow area reference, which varies based on the a-flag option selected here.

h-flag (*ihf*) The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two-velocity momentum equations) option and h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit printout will show h = 1.

s-flag (*isf*) The digit s specifies momentum flux options. s = 0 uses momentum flux in both the to and the from volume. s = 1 uses momentum flux in the from volume, but not in the to volume. s = 2 uses momentum flux in the to volume, but not in the from volume. s = 3 does not use momentum flux in either the to volume or the from volume.

W7(R) The meaning of this word is different, depending on the critical flow model which has been selected (see W6).

If the Henry-Fauske or Moody (option 54) critical flow model is active, the discharge coefficient is entered. If W7 and W8 are missing, W7 is set to 1.0 and W8 is set to 0.14.

If the original RELAP5 critical flow model is active, the subcooled discharge coefficient is entered. This entry applies only to subcooled liquid choked flow calculations. The entry must be > 0.0 and < 2.0. If W7, W8 and W9 are missing, they are each set to 1.0.

W8(R) The meaning of this word is different, depending on the critical flow model which has been selected (see W6).

If the Henry-Fauske or Moody critical flow model is active, the thermal nonequilibrium constant is entered. If W7 is entered and W8 is missing, W8 is set to 0.14. If the entry is < 0.01, the equilibrium option is used and W8 is reset to 0.0. If the entry is >1000.0, the frozen option is used and W8 is reset to 100.0.

If the original RELAP5 critical flow model is active, the two-phase discharge coefficient is entered. This entry applies only to two-phase choked flow calculations. The entry must be > 0.0 and < 2.0. If W7 is entered and W8 and W9 are missing, W8 and W9 are each set to 1.0.

W9(R) If the Henry-Fauske or Moody critical flow model is active and the horizontal offtake mode with y = 4 is not used, this word is not used and must not be entered (see the v-flag under W6 above).

If the Henry-Fauske or Moody critical flow model is active and the horizontal offtake mode with $\underline{y} = 4$ is specified, this word is the angle between the horizontal and the offtake junction (-90 to +90 degrees).

If the original RELAP5 critical flow model is active, the superheated discharge coefficient is entered. This entry applies only to superheated vapor choked flow calculations. The entry must be > 0.0 and < 2.0 . If W7 and W8 are entered and W9 is missing, then W9 is set to 1.0.

A9.12.10 Branch Junction Diameter and CCFL Data, CCCJ110

This card is optional. The value J should follow the same approach as used in cards CCCJ101-J109. The defaults indicated for each word are used if the card is not entered. This card can be used for three purposes: (1) interphase drag, (2) CCFL model, and (3) lumped offtake model.

1. If this card is being used to specify only the junction hydraulic diameter for the interphase drag calculation (i.e., $\underline{f} = 0$ in Word 6 of cards CCCJ101-J109), then, the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (will not be used).
2. If this card is being used for the CCFL model (i.e., $\underline{f} = 1$ in Word 6 of cards CCCJ101-J109), then enter all four words for the appropriate CCFL model if values different from the default values are desired.
3. If this card is being used to specify the diameter of a single offtake junction when lumping two or more offtake pipes into a single pipe, (i.e., $\underline{y} = 1, 2, 3,$ or 4 in Word 6 of cards CCCJ101-J109), then the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (will not be used).

W1(R) Junction hydraulic diameter, D_j (m, ft). This quantity is the junction hydraulic diameter used in the CCFL correlation equation and interphase drag. This number must be ≥ 0 . This number should be computed from $4.0 \cdot \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If a 0 is entered or if the default is used, the junction diameter is computed from $2.0 \cdot \left(\frac{\text{area}}{\pi} \right)^{0.5}$ of the respective junction. See Word 3 of cards CCCN101-N109 for the junction area.

W2(R) Flooding correlation form, β . If 0, the Wallis CCFL form is used. If 1, the Kutateladze CCFL form is used. β between 0.0 and 1.0, Bankoff weighting between the Wallis and Kutateladze CCFL forms is used. This number must be ≥ 0 and ≤ 1 . The default value is 0 (Wallis form).

W3(R) Gas intercept, c . This quantity is the gas intercept used in the CCFL correlation (when $H_f^{1/2} = 0$) and must be > 0 . The default value is 1.

W4(R) Slope, m . This quantity is the slope used in the CCFL correlation and must be > 0 . The default value is 1.

A9.12.11 Branch Junction Form Loss Data, CCCJ112

These cards are optional. The values of J should follow the same approach as used in cards CCCJ101-J109. The user-specified form loss is given in Words 4 and 5 of cards CCCJ101-J109 if these cards are not entered. If these cards are entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F \text{Re}^{-C_F} + \text{KFJUNCV}$$

$$K_R = A_R + B_R \text{Re}^{-C_R} + \text{KRJUNCV}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 4 and 5 of cards CCCN101-N109. Re is the Reynolds number based on mixture fluid properties. KFJUNCV and KRJUNCV are the contributions to the forward and reverse form loss coefficients obtained from control variables. If these cards are being used for the form loss calculation, then enter at least four words for the appropriate expression. The fifth and sixth words are optional. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R .

W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.

W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.

W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.

W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

W5(I) Control variable number for specifying a portion of the forward form loss (KFJUNCV). This quantity must be greater than or equal to 0. (*Optional*)

W6(I) Control variable number for specifying a portion of the reverse form loss (KRJUNCV). This quantity must be greater than or equal to 0. (*Optional*)

A9.12.12 Branch Junction Initial Conditions, CCCJ201

These cards are required depending on the value of n_j as described for cards CCCJ101-J109. The values of J should follow the same approach as used in cards CCCJ101-J109.

W1(R) Initial liquid or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

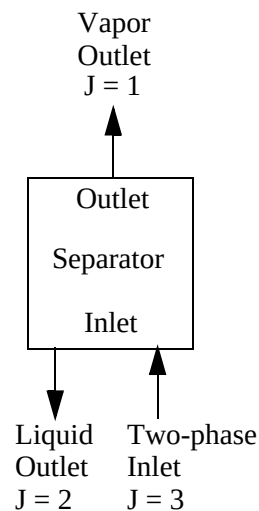
W2(R) Initial vapor velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W3(R) Interface velocity (m/s, ft/s). Enter 0.

A9.13 Separator Component, SEPARATR

A steam separator component is indicated by SEPARATR for Word 2 on card CCC0000. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number (which must be 01 for the separator), and N is the face number. For major edits, minor edits, and plot variables, the volume is numbered as CCC010000. The edits and plot variables, the junctions are numbered as CCC0J0000, where J is the junction number ($1 \leq J \leq 3$).

A separator component is a specialized branch component having three junctions. The number of junctions n_j must be 3, and no junctions may connect to this component. The variable N defined below must have values of 1, 2, and 3. For the junctions, $J = 1$ is the vapor outlet junction, $J = 2$ is the liquid fall back junction, and $J = 3$ is the separator inlet junction. The from part of the vapor outlet junction must refer to the outlet of the separator (CCC010002). The from part of the liquid fall back and the to part of the two-phase inlet must refer to the inlet of the separator (CCC010001). To include the direct path from a steam generator downcomer to the steam dome, a bypass volume is recommended. The smooth or abrupt junction option can be used for the three junctions. Appropriate user-input energy loss coefficients may be needed to match a known pressure drop across the separator. We recommend that choking be turned off for all three junctions. The vapor outlet and liquid fall back junctions should use the nonhomogeneous option.



A9.13.1 Separator Component Information, CCC0001

This card is required.

- W1(I) Number of junctions, n_j . The variable n_j is the number of junctions described in the input data for this component and must be 3.
- W2(I) Initial condition control. This word is optional and, if missing, the junction initial velocities in the first and second words on card CCCN201 are assumed to be velocities. If 0, velocities are assumed; if nonzero, mass flows are assumed.

A9.13.2 Separator Options, CCC0002

This card is optional. The first word specifies the separator option while the second word specifies the number of actual separator components represented by this RELAP5 SEPARATOR component. The second word is needed if the user uses the General Electric separator options

- W1(I) Separator option, ISEPST. A value of 0 specifies the simple separator contained in previous versions of RELAP5 (default), a value of 1 specifies the General Electric dryer model, a value of 2 specifies a General Electric two-stage separator, and a value of 3 specifies a three-stage General Electric separator.

W2(I) Number of separator components represented by this RELAP5 component. The number is needed only if Word 1 has a value of 2 or 3.

A9.13.3 Separator X-Coordinate Volume Data, CCC0101-0109

This card (or cards) is required. The nine words can be entered on one or more cards, and the card numbers need not be consecutive.

W1(R) Area (m², ft²).

W2(R) Length (m, ft).

W3(R) Volume (m³, ft³). The code requires that the volume equals the area times the length (W3 = W1•W2). At least two of the three quantities, W1, W2, and W3, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the words are 0, the volume must equal the x-direction area times the x-direction length within a relative error of 0.000001. The same relative error check is done for the y- and z-directions.

W4(R) Azimuthal angle (degrees). The absolute value of this angle must be ≤ 360 degrees. This quantity is not used in the calculation but is specified for possible automated drawing of nodalization diagrams.

W5(R) Inclination angle (degrees). The absolute value of this angle must be ≤ 90 degrees. The angle 0 degrees is horizontal, and positive angles have an upward inclination, i.e., the inlet is at the lowest elevation. This angle is used in the interphase drag calculation.

W6(R) Elevation change (m, ft). A positive value is an increase in elevation. The absolute value of this quantity must be less than or equal to the length. If the vertical angle orientation is 0, this quantity must be 0. If the vertical angle is nonzero, this quantity must also be nonzero and have the same sign. See Section 2.4.1 of Volume II of the manual for further discussion. When the absolute value of the elevation angle determined by the ratio of the elevation change (this Word 6) and the length (Word 2) is less than or equal to 45 degrees, the horizontal flow regime map is used. When the ratio is greater than 45 degrees, the vertical flow regime map is used.

W7(R) Wall roughness (m, ft).

W8(R) Hydraulic diameter (m, ft). This should be computed from $4.0 \cdot \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If 0, the hydraulic diameter is computed from $2.0 \cdot \left(\frac{\text{area}}{\pi} \right)^{0.5}$. A check is made that the pipe roughness is less than half the hydraulic diameter. See Word 1 for the area.

W9(I) Volume control flags. This word has the format tlpvbfe. It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags. Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f-flag is the only coordinate direction oriented flag. This word enters the scalar oriented flags and the x-coordinate flag. For the SEPARATR component, the volume control flag is limited to the format 000001e.

e-flag The digit e specifies if nonequilibrium or equilibrium is to be used. e = 0 specifies that a nonequilibrium (unequal temperature) calculation is to be used, and e = 1 specifies that an equilibrium (equal temperature) calculation is to be used. Equilibrium volumes should not be connected to nonequilibrium volumes. The equilibrium option is provided only for comparison to other codes.

A9.13.4 Separator Additional Wall Friction, CCC0131

This card is optional. If this card is not entered, the default values are 1.0 for the laminar shape factor and 0.0 for the viscosity ratio exponent. Two, four, or six quantities may be entered on the card, and the data not entered are set to default values. A description of this input is presented in Section 3 of Volume I.

W1(R) Shape factor for x-coordinate.

W2(R) Viscosity ratio exponent for x-coordinate.

W3(R) Shape factor for y-coordinate.

W4(R) Viscosity ratio exponent for y-coordinate.

W5(R) Shape factor for z-coordinate.

W6(R) Viscosity ratio exponent for z-coordinate.

A9.13.5 Separator Y-Coordinate Volume Data, CCC0181-0189

These cards are optional. These cards are used when the user specifies the y-direction connection with the crossflow model. The volume of the volume is the same for the x-, y-, and z-directions. If these cards are entered, either W1 or W2 must be nonzero.

W1(R) Area (m^2 , ft^2). If these cards are missing or if this word is 0, this y-direction area is computed from $\frac{\text{volume}}{\text{y-direction length}}$.

- W2(R) Length (m, ft). If these cards are missing, this y-direction length is computed from $2.0 \bullet \left(\frac{x - \text{direction area}}{\pi} \right)^{0.5}$. If this word is 0, this y-direction length is computed from $\frac{\text{volume}}{\text{y-direction area}}$.
- W3(R) Roughness (m, ft).
- W4(R) Hydraulic diameter (m, ft). If these cards are missing or if this word is 0, this y-direction hydraulic diameter is computed from $\frac{4.0 \bullet \text{y-direction volume flow area}}{(\pi \bullet \text{x-direction volume flow area})^{0.5}}$. A check is made to ensure the y-direction roughness is less than half the y-direction hydraulic diameter.
- W5(I) Volume control flags. This word has the general format tlpybfe, but this word is limited to 00000f0 since it only enters the coordinate oriented flags for the y-direction.
- f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed along the y-coordinate direction in the volume, and f = 1 specifies that wall friction effects are not to be computed along the y-coordinate direction.
- W6(R) This word is not used. Enter 0.
- W7(R) This word is not used. Enter 0.
- W8(R) This word is the position change in the z-fixed (vertical) direction as the flow passes from the y-inlet face to the y-outlet face (m, ft). This quantity affects problems if connections are made to the y-faces.

A9.13.6 Separator Z-Coordinate Volume Data, CCC0191-0199

These cards are optional. These cards are used when the user specifies the z-direction connection with the crossflow model. The volume of the volume is the same for the x-, y-, and z-directions. If these cards are entered, either W1 or W2 must be nonzero.

- W1(R) Area (m², ft²). If these cards are missing or if this word is 0, this z-direction area is computed from $\frac{\text{volume}}{\text{z-direction length}}$.

- W2(R) Length (m, ft). If these cards are missing, this z-direction length is computed from $2.0 \bullet \left(\frac{x - \text{direction area}}{\pi} \right)^{0.5}$. If this word is 0, this z-direction length is computed from $\frac{\text{volume}}{\text{z-direction area}}$.
- W3(R) Roughness (m, ft).
- W4(R) Hydraulic diameter (m, ft). If these cards are missing or if this word is 0, this z-direction hydraulic diameter is computed from $\frac{4.0 \bullet \text{z-direction volume flow area}}{(\pi \bullet \text{x-direction volume flow area})^{0.5}}$. A check is made to ensure the z-direction roughness is less than half the z-direction hydraulic diameter.
- W5(I) Volume control flags. This word has the general format tlpybfe, but this word is limited to 00000f0 since it only enters the coordinate oriented flags for the z-direction.
- f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed along the z-coordinate direction in the volume, and f = 1 specifies that wall friction effects are not to be computed along the z-coordinate direction.
- W6(R) This word is not used. Enter 0.
- W7(R) This word is not used. Enter 0.
- W8(R) This word is the position change in the z-fixed (vertical) direction as the flow passes from the z-inlet face to the z-outlet face (m, ft). This quantity affects problems if connections are made to the z-faces.

A9.13.7 Separator Volume Initial Conditions, CCC0200

This card is required.

- W1(I) Control word. This word has the format gbt. It is not necessary to input leading zeros.
- ε-flag The digit ε specifies the fluid, where ε = 0 is the default fluid. The value for ε > 0 corresponds to the position number of the fluid type indicated on the 120-129 cards (i.e., ε = 1 specifies H₂O, ε = 2 specifies D₂O etc.). The default fluid is that set for the hydrodynamic system by cards 120-129 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-129 or these control words must be consistent (i.e., not specify different fluids). If cards 120-129 are not entered and all

control words use the default $\underline{\varepsilon} = 0$, then water is assumed to be the fluid. It is recommended that the user use the 120-129 cards to set the fluid type in the systems.

b-flag The digit $\underline{b}$ specifies whether boron is present. $\underline{b} = 0$ specifies that the volume liquid does not contain boron, and $\underline{b} = 1$ specifies that a boron concentration in mass of boron per mass of liquid (which may be 0) is being entered after the other required thermodynamic information.

t-flag The digit $\underline{t}$ specifies how the following words are to be used to determine the initial thermodynamic state. $\underline{t} = 0-3$ and 7 specifies one component (steam/water); $\underline{t} = 4-6$ allows the specification of two components (steam/water and noncondensable gas).

With options $\underline{t}$ equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.

One Component (Steam/Water), Equilibrium or Non-equilibrium

$[P, U_f, U_g, \alpha_g]$ If $\underline{t} = 0$, the next four words are interpreted as pressure (Pa, lb_f/in^2), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. Enter boron in W6 if needed.

One Component (Steam/Water), Equilibrium

$[T, x_s]$ If $\underline{t} = 1$, the next two words are interpreted as temperature (K, $^{\circ}\text{F}$) and static quality in equilibrium condition. Enter boron in W4 if needed.

$[P, x_s]$ If $\underline{t} = 2$, the next two words are interpreted as pressure (Pa, lb_f/in^2) and static quality in equilibrium condition. Enter boron in W4 if needed.

$[P, T]$ If $\underline{t} = 3$, the next two words are interpreted as pressure (Pa, lb_f/in^2) and temperature (K, $^{\circ}\text{F}$) in equilibrium condition. Enter boron in W4 if needed.

$[P, T_f, T_g, \alpha_g]$ If $\underline{t} = 7$, the next four words are interpreted as pressure (Pa, lb_f/in^2), liquid temperature (K, $^{\circ}\text{F}$), vapor temperature (K, $^{\circ}\text{F}$), and vapor void fraction. These quantities will be interpreted as equilibrium conditions. The liquid temperature has to be equal to or less than the saturation temperature, and the vapor temperature has to be equal to or greater than the saturation temperature. Enter boron in W6 if needed. This input option was added to allow easier comparisons with TRACE calculations.

Two Components (Steam/Water/Air), Equilibrium or Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0\text{E-}9 \leq x_s \leq 0.99999999$,

Single phase if $x_s < 1.0\text{E-}9$ or $x_s > 0.99999999$

The static quality is given by $x_s = M_g / (M_g + M_f)$, where $M_g = M_s + M_n$. Section 3.2 of Volume I of the manual discusses this further.

[P,T, x_s] If $t = 4$, the next three words are interpreted as pressure (Pa, lb_f/in.²), temperature (K, °F), and static quality in equilibrium condition. Using this input option with static quality greater than 0.0 and less than or equal to 1.0, saturated noncondensables will result. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. If the pressure is greater than the critical pressure, $P > P_{\text{crit}}$, then the static quality, x_s (mass steam + air)/(mass steam + air + water) has to be large enough to give a partial steam pressure that is less than the critical pressure, $P_s < P_{\text{crit}}$. Otherwise an input error will occur. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensable (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option. Enter boron in W5 if needed.

[T, x_s , x_n] If $t = 5$, the next three words are interpreted as steam saturation temperature (K, °F), static quality, and noncondensable quality in equilibrium condition. Enter boron in W5 if needed. Both the static and noncondensable qualities are restricted to be between $1.0\text{E-}9$ and 0.99999999 . The liquid will be subcooled. The input temperature determines the steam partial pressure. However, the total pressure may be greater than the critical pressure, which will cause an input error.

[P, U_f , U_g , α_g , x_n] If $t = 6$, the next five words are interpreted as pressure (Pa, lb_f/in.²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing ($t = 0$). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then vapor void fraction must also be 1.0.

When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor-specific internal energy. Enter boron in W7 if needed.

W2-W7(R) Quantities as described under Word 1. Depending on the control word, 2-6 quantities may be required. Enter only the minimum number required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions.

A9.13.8 Separator Initial Noncondensable Mass Fractions, CCC0301

This card is optional for a separator and is related to card 110. If card CCC0301 is not supplied, then the values on the card 115 are used for the noncondensable mass fractions. The number of words on card CCC0301 must equal the number of words on card 110. The mass fractions must sum to 1.0 to within a tolerance of 1.0E-10.

W1-WN(R) Initial mass fraction for each noncondensable gas type as specified on card 110 where N is the number of gas types on card 110.

A9.13.9 Separator Junction Geometry, CCCJ101-J109

These cards are required. Cards with J equal to 1, 2 and 3 are entered, one for each junction.

W1(I) From connection code to a component. This refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction (see Section 2.1).

W2(I) To connection code to a component. This refers to the component at which the junction coordinate direction ends. See the description for W1 above.

W3(R) Area (m^2 , ft^2). If 0, the area is set to the minimum area of the adjoining volumes. For abrupt area changes, the junction area must be equal to or smaller than the minimum of the adjoining areas. For smooth area changes, there are no restrictions.

W4(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. A variable loss coefficient may be specified (see Section A9.12.11). The interpretation and use of this loss coefficient vary based upon the "a" junction control flag selected in W6. If the smooth area change option is selected ($a = 0$), the entry here is the loss coefficient applied to the junction flow area resulting from the entry made in W3. If the abrupt area change option is selected ($a = 1$ or 2), the loss coefficient entered here is

adjusted by the square of the ratio of the junction area and the minimum flow area of the adjacent connected volumes. This adjusted loss coefficient is applied to the junction flow area. If the $a = 1$ option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

Example: Two volumes with volume average flow areas of 0.3 m^2 and 0.4 m^2 are connected with a junction ($A_{\text{jun}} = 0.075 \text{ m}^2$). The user supplied $K_{\text{loss_user}}$ is 10.0. For the smooth area change option ($a = 0$), the $K_{\text{loss_user}}$ of 10.0 is applied directly to the junction. For the abrupt area change option ($a = 1$ or 2), the K_{loss} applied at the junction is $K_{\text{loss_jun}} = K_{\text{loss_user}} * (A_{\text{jun}}/A_{\text{min}})^2$. For this example, $K_{\text{loss_jun}} = 10.0 * (0.075/0.3)^2 = 0.625$.

- W5(R) Reynolds number independent reverse flow energy loss coefficient, A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. A variable loss coefficient may be specified (see Section A9.12.11). See information for the previous word regarding interpretation and use of this loss coefficient.
- W6(I) Junction control flags. This word has the format jefvcahs. For this component, the junction control flag format is limited to 0000cahs.
- c-flag The digit c specifies choking options. $\underline{c} = 0$ means that the choking model will be applied, and $\underline{c} = 1$ means that the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske or Moody (option 54) critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active.
- a-flag The digit a specifies area change options. $\underline{a} = 0$ means a smooth area change and only the user supplied K_{loss} is applied. $\underline{a} = 1$ means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. $\underline{a} = 2$ means a partial abrupt area change model and it is the same as $\underline{a} = 1$ except no code calculated forward and reverse contraction/expansion K_{loss} is applied. See discussion for W4(R) regarding the K_{loss} flow area reference, which varies based on the a-flag option selected here.
- h-flag The digit h specifies nonhomogeneous or homogeneous. $\underline{h} = 0$ specifies the nonhomogeneous (two-velocity momentum equations) option and $\underline{h} = 1$ or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option ($\underline{h} = 1$ or 2), the major edit printout will show $\underline{h} = 1$.

s-flag The digit $\underline{s}$ specifies momentum flux options. $\underline{s} = 0$ uses momentum flux in both the to and the from volume. $\underline{s} = 1$ uses momentum flux in the from volume, but not in the to volume. $\underline{s} = 2$ uses momentum flux in the to volume, but not in the from volume. $\underline{s} = 3$ does not use momentum flux in either the to volume or the from volume.

W7(R) Void fraction limit. For the vapor exit junction ($J = 1$), this quantity (VOVER) is the vapor void fraction above which flow out of the vapor outlet is pure vapor. If the word is missing, a default value of 0.5 is used. For the liquid fall back junction ($J = 2$), this quantity (VUNDER) is the liquid volume fraction, α_f , above which flow out of the liquid fall back is pure liquid. If the word is missing, a default value of 0.15 is used. For the separator inlet, this word is not used.

A9.13.10 Separator Junction Diameter and CCFL Data, CCCJ110

These cards are optional. The value J should follow the same approach as used in cards CCCJ101-J109. The defaults indicated for each word are used if the card is not entered. If these cards are being used to specify only the junction hydraulic diameter for the interphase drag calculations (i.e., $\underline{f} = 0$ in Word 6 of cards CCCJ101-J109), then the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (will not be used). If these cards are being used for the CCFL model (i.e., $\underline{f} = 1$ in Word 6 of cards CCCJ101-J109), then enter all four words for the appropriate CCFL model if values different from the default values are desired.

W1(R) Junction hydraulic diameter, D_j (m, ft). This quantity is the junction hydraulic diameter used in the CCFL correlation equation and interphase drag. This number must be ≥ 0 . This number should be computed from $4.0 \cdot \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If a 0 is entered or if the default is used, the junction diameter is computed from $2.0 \cdot \left(\frac{\text{area}}{\pi} \right)^{0.5}$ of the respective junction. See Word 3 of cards CCCN101-N109 for the junction area.

W2(R) Flooding correlation form, β . If 0, the Wallis CCFL form is used. If 1, the Kutateladze CCFL form is used. β between 0.0 and 1.0, Bankoff weighting between the Wallis and Kutateladze CCFL forms is used. This number must be ≥ 0 and ≤ 1 . The default value is 0 (Wallis form).

W3(R) Gas intercept, c . This quantity is the gas intercept used in the CCFL correlation (when $H_f^{1/2} = 0$) and must be > 0 . The default value is 1.

W4(R) Slope, m . This quantity is the slope used in the CCFL correlation and must be > 0 . The default value is 1.

A9.13.11 Separator Junction Form Loss Data, CCCJ112

These cards are optional. The values of J should follow the same approach as used in cards CCCJ101-J109. The user-specified form loss is given in Words 4 and 5 of cards CCCJ101-J109 if these cards are not entered. If these cards are entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F \text{Re}^{-C_F}$$

$$K_R = A_R + B_R \text{Re}^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 4 and 5 of cards CCCN101-N109. Re is the Reynolds number based on mixture fluid properties. If these cards are being used for the form loss calculation, then enter all four words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R .

W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.

W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.

W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.

W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

A9.13.12 Separator Junction Initial Conditions, CCCJ201

These cards are required depending on the value of n_j as described for cards CCCJ101-J109. The values of J should follow the same approach as used in cards CCCJ101-J109. A 90% extraction limit during input processing is tested for the vapor at the vapor outlet junction and for the liquid fall back junction. If greater than 90%, an input error occurs.

W1(R) Initial liquid or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W2(R) Initial vapor velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W3(R) Interface velocity (m/s, ft/s). Enter 0.

A9.13.13 GE Separator Data, CCC0500

This card is optional for the GE separator. If this card is missing and the GE separator has been specified on card CCC0002, the default values will be used. If the card is present, all eight values must be specified.

W1(R)	Radius of larger pickoff ring at first stage of a two-stage separator (m, ft). (Default = 0.0857208 m).
W2(R)	Standpipe flow area (m ² , ft ²). Default = 0.018637 m ² .
W3(R)	Separator nozzle exit area (m ² , ft ²). Default = 0.01441 m ² .
W4(R)	Radius of separator hub at inlet (m, ft). Default = 0.0809585 m.
W5(R)	Swirl vane angle relative to the horizontal (deg). Default = 48 deg.
W6(R)	Liquid carryover coefficient for upper separating stages. Default = 0.009 for two-stage separator and 0.110 for three-stage separator.
W7(R)	Vapor carryunder coefficient for upper separating stages. Default = 0.0004.
W8(R)	Axial distance between exit of first stage discharge passage and swirl vanes (m, ft). Default = 0.2127 m for two-stage separator and 0.45083 m for three-stage separator.

A9.13.14 GE Separator First Stage Data, CCC0501

This card is optional for the GE separator. If this card is missing and the GE separator has been specified on card CCC0002, the default values will be used. If the card is present, all nine values must be specified.

W1(R)	Liquid film void profile coefficient. Default = 110.0.
W2(R)	Vapor core void profile coefficient. Default = 0.5.
W3(R)	Separator wall inner radius (m, ft). Default = 0.10794 m.
W4(R)	Pickoff ring inner radius (m, ft). Default = 0.069875 m for two-stage separator and 0.0857208 m for three-stage separator.
W5(R)	Discharge passage flow area (m ² , ft ²). Default = 0.0415776 m ² for two-stage separator and 0.0096265 m ² for three-stage separator.
W6(R)	Discharge passage hydraulic diameter (m, ft). Default = 0.045558 m for two-stage separator and 0.025399 m for three-stage separator.
W7(R)	Separating barrel length (m, ft). Default = 0.877845 m for two-stage separator and 1.0699 m for three-stage separator.

W8(R) Discharge passage loss coefficient. Default = 10.0 for two-stage separator and 2.5 for three-stage separator.

W9(R) Discharge passage effective L/D coefficient. Default = 450.0 for two-stage separator and 53.44 for three-stage separator.

A9.13.15 GE Separator Second Stage Data, CCC0502

This card is optional for the GE separator. If this card is missing and the GE separator has been specified on card CCC0002, the default values will be used. If the card is present, all nine values must be specified.

W1(R) Liquid film void profile coefficient. Default = 20.0.

W2(R) Vapor core void profile coefficient. Default = 0.25.

W3(R) Separator wall inner radius (m, ft). Default = 0.06985 m for two-stage separator and 0.10794 m for three-stage separator.

W4(R) Pickoff ring inner radius (m, ft). Default = 0.06032 m for two-stage separator and 0.0952453 m for three-stage separator.

W5(R) Discharge passage flow area (m², ft²). Default = 0.0029133 m² for two-stage separator and 0.0096265 m² for three-stage separator.

W6(R) Discharge passage hydraulic diameter (m, ft). Default = 0.0121699 m for two-stage separator and 0.025399 m for three-stage separator.

W7(R) Separating barrel length (m, ft). Default = 0.16255 m for two-stage separator and 0.384156 m for three-stage separator.

W8(R) Discharge passage loss coefficient. Default = 0.5 for two-stage separator and 1.429 for three-stage separator.

W9(R) Discharge passage effective $\frac{L}{D}$ coefficient. Default = 95.85 for two-stage separator and 194.64 for three-stage separator.

A9.13.16 GE Separator Third Stage Data, CCC0503

This card is optional for the GE separator. If this card is missing and the GE three-stage separator has been specified on card CCC0002, the default values will be used. If the card is present, all nine values must be specified.

W1(R)	Liquid film void profile coefficient. Default = 20.0.
W2(R)	Vapor core void profile coefficient. Default = 0.55.
W3(R)	Separator wall inner radius (m, ft). Default = 0.10794 m.
W4(R)	Pickoff ring inner radius (m, ft). Default = 0.0984201 m.
W5(R)	Discharge passage flow area (m ² , ft ²). Default = 0.0096265 m ² .
W6(R)	Discharge passage hydraulic diameter (m, ft). Default = 0.025399 m.
W7(R)	Separating barrel length (m, ft). Default = 0.384156 m.
W8(R)	Discharge passage loss coefficient. Default = 2.563.
W9(R)	Discharge passage effective L/D coefficient. Default = 424.96.

A9.13.17 GE Dryer Data, CCC0600

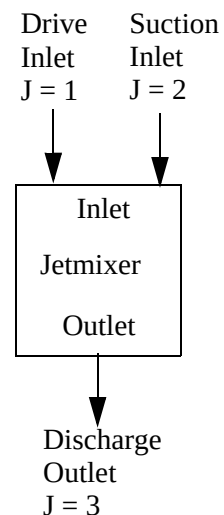
This card is optional for the GE separator. If this card is missing and the GE dryer has been specified on card CCC0002, the default values will be used. If the card is present, all three values must be specified.

W1(R)	Vapor velocity at dryer inlet below which there is 0% liquid carryover (m/s, ft/s). Default = 1.5 m/s.
W2(R)	Vapor velocity at dryer inlet above which there is 100% liquid carryover (m/s, ft/s). Default = 6.0 m/s.
W3(R)	Range of dryer inlet quality where dryer carryover changes from 0 to 100% when dryer inlet vapor velocity is between lower and upper values. Default = 0.05.

A9.14 Jetmixer Component, JETMIXER

A jetmixer component is indicated by JETMIXER for Word 2 on card CCC0000. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number (which must be 01 for the jetmixer), and N is the face number. Each junction should be modeled as an abrupt area change. For major edits, minor edits, and plot variables, the volume is numbered as CCC010000. For edit and plot variables, the junctions are numbered as CCC0J0000, where J is the junction number ($1 \leq J \leq 3$).

A JETMIXER component is a specialized branch using three junctions. For the junctions, J = 1 represents the drive junction, J = 2 represents the suction junction, and J = 3 represents the discharge junction. The to part of the drive and suction junctions (J = 1 and 2) must refer to the inlet end of the jetmixer (CCC010001), and the from part of the discharge junction (J = 3) must refer to the outlet end of the jetmixer (CCC010002). To model a jet pump properly, the junction flow areas of the drive and suction should equal the volume flow area.



A9.14.1 Jetmixer Component Information, CCC0001

This card is required.

- | | |
|-------|---|
| W1(I) | Number of junctions, nj. Enter 3. |
| W2(I) | Initial condition control. This word is optional and, if missing, the junction initial velocities in the first and second words on card CCCJ201 are assumed to be velocities. If 0, velocities are assumed; if nonzero, mass flows are assumed. |

A9.14.2 Jetmixer X-Coordinate Volume Data, CCC0101-0109

This card (or cards) is required. The nine words can be entered on one or more cards, and the card numbers need not be consecutive.

- | | |
|-------|--|
| W1(R) | Area (m ² , ft ²). |
| W2(R) | Length (m, ft). |
| W3(R) | Volume (m ³ , ft ³). The code requires that the volume equals the area times the length (W3 = W1•W2). At least two of the three quantities, W1, W2, and W3, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the words are 0, the volume must equal the x-direction area times the x-direction length within a relative error of 0.000001. The same relative error check is done for the y- and z-directions. |

- W4(R) Azimuthal angle (degrees). The absolute value of this angle must be ≤ 360 degrees. This quantity is not used in the calculation but is specified for possible automated drawing of nodalization diagrams.
- W5(R) Inclination angle (degrees). The absolute value of this angle must be ≤ 90 degrees. The angle 0 degrees is horizontal, and positive angles have an upward inclination, i.e., the inlet is at the lowest elevation. This angle is used in the interphase drag calculation.
- W6(R) Elevation change (m, ft). A positive value is an increase in elevation. The absolute value of this quantity must be less than or equal to the length. If the vertical angle orientation is 0, this quantity must be 0. If the vertical angle is nonzero, this quantity must also be nonzero and have the same sign. See Section 2.4.1 of Volume II of the manual for further discussion. When the absolute value of the elevation angle determined by the ratio of the elevation change (this Word 6) and the length (Word 2) is less than or equal to 45 degrees, the horizontal flow regime map is used. When the ratio is greater than 45 degrees, the vertical flow regime map is used.
- W7(R) Wall roughness (m, ft).
- W8(R) Hydraulic diameter (m, ft). This should be computed from $4.0 \cdot \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If 0, the hydraulic diameter is computed from $2.0 \cdot \left(\frac{\text{area}}{\pi} \right)^{0.5}$. A check is made that the pipe roughness is less than half the hydraulic diameter. See Word 1 for the area.
- W9(I) Volume control flags. This word has the format tlpybfe. It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags. Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f-flag is the only coordinate direction oriented flag. This word enters the scalar oriented flags and the x-coordinate flag. The jetmixer volume control flag is restricted to the format 00000fe.
- f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed along the x-coordinate direction in the volume, and f = 1 specifies that wall friction effects are not to be computed along the x-coordinate.
- e-flag The digit e specifies if nonequilibrium or equilibrium is to be used. e = 0 specifies that a nonequilibrium (unequal temperature) calculation is to be used, and e = 1 specifies that an equilibrium (equal temperature) calculation is to be used. Equilibrium volumes should not be connected to nonequilibrium volumes. The equilibrium option is provided only for comparison to other codes.

A9.14.3 Jetmixer Additional Wall Friction, CCC0131

This card is optional. If this card is not entered, the default values are 1.0 for the laminar shape factor and 0.0 for the viscosity ratio exponent. Two, four, or six quantities may be entered on the card, and the data not entered are set to default values. A description of this input is presented in Section 3 of Volume I.

W1(R)	Shape factor for x-coordinate.
W2(R)	Viscosity ratio exponent for x-coordinate.
W3(R)	Shape factor for y-coordinate.
W4(R)	Viscosity ratio exponent for y-coordinate.
W5(R)	Shape factor for z-coordinate.
W6(R)	Viscosity ratio exponent for z-coordinate.

A9.14.4 Jetmixer Y-Coordinate Volume Data, CCC0181-0189

These cards are optional. These cards are used when the user specifies the y-direction connection with the crossflow model. The volume of the volume is the same for the x-, y-, and z-directions. If these cards are entered, either W1 or W2 must be nonzero.

W1(R)	Area (m ² , ft ²). If these cards are missing or if this word is 0, this y-direction area is computed from $\frac{\text{volume}}{\text{y-direction length}}$.
W2(R)	Length (m, ft). If these cards are missing, this y-direction length is computed from $2.0 \cdot \left(\frac{\text{x-direction area}}{\pi} \right)^{0.5}$. If this word is 0, this y-direction length is computed from $\frac{\text{volume}}{\text{y-direction area}}$.
W3(R)	Roughness (m, ft).
W4(R)	Hydraulic diameter (m, ft). If these cards are missing or if this word is 0, this y-direction hydraulic diameter is computed from $\frac{4.0 \cdot \text{y-direction volume flow area}}{(\pi \cdot \text{x-direction volume flow area})^{0.5}}$. A check is made to ensure the y-direction roughness is less than half the y-direction hydraulic diameter.

- W5(I) Volume control flags. This word has the general format tlpybfe, but this word is limited to 00000f0 since it only enters the coordinate oriented flags for the y-direction.
- f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed along the y-coordinate direction in the volume, and f = 1 specifies that wall friction effects are not to be computed along the y-coordinate direction.
- W6(R) This word is not used. Enter 0.
- W7(R) This word is not used. Enter 0.
- W8(R) This word is the position change in the z-fixed (vertical) direction as the flow passes from the y-inlet face to the y-outlet face (m, ft). This quantity affects problems if connections are made to the y-faces.

A9.14.5 Jetmixer Z-Coordinate Volume Data, CCC0191-0199

These cards are optional. These cards are used when the user specifies the z-direction connection with the crossflow model. The volume of the volume is the same for the x-, y-, and z-directions. If these cards are entered, either W1 or W2 must be nonzero.

- W1(R) Area (m², ft²). If these cards are missing or if this word is 0, this z-direction area is computed from $\frac{\text{volume}}{\text{z-direction length}}$.
- W2(R) Length (m, ft). If these cards are missing, this z-direction length is computed from $2.0 \cdot \left(\frac{\text{x-direction area}}{\pi} \right)^{0.5}$. If this word is 0, this z-direction length is computed from $\frac{\text{volume}}{\text{z-direction area}}$.
- W3(R) Roughness (m, ft).
- W4(R) Hydraulic diameter (m, ft). If these cards are missing or if this word is 0, this z-direction hydraulic diameter is computed from $\frac{4.0 \cdot \text{z-direction volume flow area}}{(\pi \cdot \text{x-direction volume flow area})^{0.5}}$. A check is made to ensure the z-direction roughness is less than half the z-direction hydraulic diameter.
- W5(I) Volume control flags. This word has the general format tlpybfe, but this word is limited to 00000f0 since it only enters the coordinate oriented flags for the z-direction.

f-flag	The digit $\underline{f}$ specifies whether wall friction is to be computed. $\underline{f} = 0$ specifies that wall friction effects are to be computed along the z-coordinate direction in the volume, and $\underline{f} = 1$ specifies that wall friction effects are not to be computed along the z-coordinate direction.
W6(R)	This word is not used. Enter 0.
W7(R)	This word is not used. Enter 0.
W8(R)	This word is the position change in the z-fixed (vertical) direction as the flow passes from the z-inlet face to the z-outlet face (m, ft). This quantity affects problems if connections are made to the z-faces.

A9.14.6 Jetmixer Volume Initial Conditions, CCC0200

This card is required.

W1(I)	Control word. This word has the format $\underline{gbt}$. It is not necessary to input leading zeros.
$\underline{\epsilon}$ -flag	The digit $\underline{\epsilon}$ specifies the fluid, where $\underline{\epsilon} = 0$ is the default fluid. The value for $\underline{\epsilon} > 0$ corresponds to the position number of the fluid type indicated on the 120-129 cards (i.e., $\underline{\epsilon} = 1$ specifies H_2O , $\underline{\epsilon} = 2$ specifies D_2O etc.). The default fluid is that set for the hydrodynamic system by cards 120-129 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-129 or these control words must be consistent (i.e., not specify different fluids). If cards 120-129 are not entered and all control words use the default $\underline{\epsilon} = 0$, then water is assumed to be the fluid. It is recommended that the user use the 120-129 cards to set the fluid type in the systems.
b-flag	The digit $\underline{b}$ specifies whether boron is present. $\underline{b} = 0$ specifies that the volume liquid does not contain boron, and $\underline{b} = 1$ specifies that a boron concentration in mass of boron per mass of liquid (which may be 0) is being entered after the other required thermodynamic information.
t-flag	The digit $\underline{t}$ specifies how the following words are to be used to determine the initial thermodynamic state. $\underline{t} = 0-3$ and 7 specifies one component (steam/water); $\underline{t} = 4-6$ allows the specification of two components (steam/water and noncondensable gas). With options $\underline{t}$ equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.

One Component (Steam/Water), Equilibrium or Non-equilibrium

[P,U_f,U_g,α_g] If $t = 0$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. Enter boron in W6 if needed.

One Component (Steam/Water), Equilibrium

[T,x_s] If $t = 1$, the next two words are interpreted as temperature (K, °F) and static quality in equilibrium condition. Enter boron in W4 if needed.

[P,x_s] If $t = 2$, the next two words are interpreted as pressure (Pa, lb_f/in²) and static quality in equilibrium condition. Enter boron in W4 if needed.

[P,T] If $t = 3$, the next two words are interpreted as pressure (Pa, lb_f/in²) and temperature (K, °F) in equilibrium condition. Enter boron in W4 if needed.

[P,T_f,T_g,α_g] If $t = 7$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid temperature (K, F), vapor temperature (K, F), and vapor void fraction. These quantities will be interpreted as equilibrium conditions. The liquid temperature has to be equal to or less than the saturation temperature, and the vapor temperature has to be equal to or greater than the saturation temperature. Enter boron in W6 if needed. This input option was added to allow easier comparisons with TRACE calculations.

Two Components (Steam/Water/Air), Equilibrium or Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0\text{E-}9 \leq x_s \leq 0.99999999$,

Single phase if $x_s < 1.0 \text{ E-}9$ or $x_s > 0.99999999$

The static quality is given by $x_s = M_g/(M_g + M_f)$, where $M_g = M_s + M_n$. Section 3.2 of Volume I of the manual discusses this further.

[P,T,x_s] If $t = 4$, the next three words are interpreted as pressure (Pa, lb_f/in²), temperature (K, °F), and static quality in equilibrium condition. Using this input option with static quality

greater than 0.0 and less than or equal to 1.0, saturated noncondensables will result. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. If the pressure is greater than the critical pressure, $P > P_{crit}$, then the static quality, x_s (mass steam + air)/(mass steam + air + water) has to be large enough to give a partial steam pressure that is less than the critical pressure, $P_s < P_{crit}$. Otherwise an input error will occur. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensable (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option. Enter boron in W5 if needed.

[T, x_s , x_n] If $t = 5$, the next three words are interpreted as steam saturation temperature (K, °F), static quality, and noncondensable quality in equilibrium condition. Enter boron in W5 if needed. Both the static and noncondensable qualities are restricted to be between $1.0 \text{ E-}9$ and 0.99999999 . The liquid will be subcooled. The input temperature determines the steam partial pressure. However, the total pressure may be greater than the critical pressure, which will cause an input error.

[P, U_f , U_g , α_g , x_n] If $t = 6$, the next five words are interpreted as pressure (Pa, $\text{lb}_f/\text{in.}^2$), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing ($t = 0$). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor-specific internal energy. Enter boron in W7 if needed.

W2-W7(R) Quantities as described under Word 1. Depending on the control word, 2-6 quantities may be required. Enter only the minimum number required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions.

A9.14.7 Jetmixer Initial Noncondensable Mass Fractions, CCC0301

This card is optional for a jetmixer and is related to card 110. If card CCC0301 is not supplied, then the values on the card 115 are used for the noncondensable mass fractions. The number of words on card CCC0301 must equal the number of words on card 110. The mass fractions must sum to 1.0 to within a tolerance of $1.0\text{E-}10$.

W1-WN(R) Initial mass fraction for each noncondensable gas type as specified on card 110 where N is the number of gas types on card 110.

A9.14.8 Jetmixer Junction Geometry, CCCJ101-J109

These cards are required. Cards with J equal to 1, 2 and 3 are required, one for each junction.

W1(I) From connection code to a component. This refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction (see Section 2.1).

W2(I) To connection code to a component. This refers to the component at which the junction coordinate direction ends. See the description for W1 above.

W3(R) Area (m^2 , ft^2). If 0, the area is set to the minimum area of the adjoining volumes. For abrupt area changes, the junction area must be equal to or smaller than the minimum of the adjoining areas. For smooth area changes, there are no restrictions.

W4(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. A variable loss coefficient may be specified (see Section A9.12.11). The interpretation and use of this loss coefficient vary based upon the "a" junction control flag selected in W6. If the smooth area change option is selected ($\underline{a} = 0$), the entry here is the loss coefficient applied to the junction flow area resulting from the entry made in W3. If the abrupt area change option is selected ($\underline{a} = 1$ or 2), the loss coefficient entered here is adjusted by the square of the ratio of the junction area and the minimum flow area of the adjacent connected volumes. This adjusted loss coefficient is applied to the junction flow area. If the $\underline{a} = 1$ option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

Example: Two volumes with volume average flow areas of 0.3 m^2 and 0.4 m^2 are connected with a junction ($A_{\text{jun}} = 0.075 \text{ m}^2$). The user supplied $K_{\text{loss_user}}$ is 10.0. For the smooth area change option ($\underline{a} = 0$), the $K_{\text{loss_user}}$ of 10.0 is applied directly to the junction. For the abrupt area change option ($\underline{a} = 1$ or 2), the K_{loss} applied at the junction is $K_{\text{loss_jun}} = K_{\text{loss_user}} * (A_{\text{jun}}/A_{\text{min}})^2$. For this example, $K_{\text{loss_jun}} = 10.0 * (0.075/0.3)^2 = 0.625$.

W5(R) Reynolds number independent reverse flow energy loss coefficient, A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that

phase is negative. A variable loss coefficient may be specified (see Section A9.12.11). See information for the previous word regarding interpretation and use of this loss coefficient.

- W6(I) Junction control flags. This word has the format jefvcahs. For the JETMIXER, the junction control flag format is limited to 0000cah0.
- c-flag The digit c specifies choking options. c = 0 means that the choking model will be applied, and c = 1 means that the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske or Moody (option 54) critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active.
- a-flag The digit a specifies area change options. a = 0 means a smooth area change and only the user supplied K_{loss} is applied. a = 1 means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. a = 2 means a partial abrupt area change model and it is the same as a = 1 except no code calculated forward and reverse contraction/expansion K_{loss} is applied. See discussion for W4(R) regarding the K_{loss} flow area reference, which varies based on the a-flag option selected here.
- h-flag The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two-velocity momentum equations) option and h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit printout will show h = 1.

A9.14.9 Jetmixer Junction Form Loss Data, CCCJ112

These cards are optional. The values of J should follow the same approach as used in cards CCCJ101-J109. The user-specified form loss is given in Words 4 and 5 of cards CCCJ101-J109 if these cards are not entered. If these cards are entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F Re^{-C_F}$$

$$K_R = A_R + B_R Re^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 4 and 5 of cards CCCJ101-J109. Re is the Reynolds number based on mixture fluid properties. If these cards are being used for the form loss calculation, then enter all four words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R .

W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.

W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.

W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.

W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

A9.14.10 Jetmixer Junction Initial Conditions, CCCJ201

These cards are required depending on the value of n_j as described for cards CCCJ101-J109. The values of J should follow the same approach as used in cards CCCJ101-J109.

W1(R) Initial liquid or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W2(R) Initial vapor velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W3(R) Interface velocity (m/s, ft/s). Enter 0.

A9.15 Turbine Component, TURBINE

A turbine component is indicated by TURBINE for Word 2 on card CCC0000. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number (which must be 01 for the turbine), and N is the face number. More than one junction may be connected to the inlet or outlet. If an end has no junctions, that end is considered a closed end.

A TURBINE component is a specialized branch with additional input to describe the turbine characteristics. A simple turbine might use only one turbine component. A multistage turbine with steam extraction points might require several turbine components (see **Figure 9.15-1**). The number of junctions in each turbine component, n_j , must be equal to 1 or 2. Junction $J = 1$ is the turbine inlet junction, and junction $J = 2$ is the steam extraction (bleed) junction that should be a crossflow junction. Junction $J = 1$ must be the only entrance junction, and other than junction 2, there must be only one exit junction, which is part of another component. The to part of the steam inlet junction ($J = 1$) must refer to the inlet end of the turbine volume (CCC010001). A current code restriction is that the volume and junction component numbers (CCC) of each succeeding turbine component must be the numerically increasing, e.g. CCC = 400 for turbine 1, CCC = 500 for turbine 2, and CCC = 600 for turbine 3, etc. There is no numbering restriction on the component upstream of turbine 1. The first turbine component in the set must be an artificial turbine having a constant efficiency = 0 and an $\underline{h} = 0$. The inertia and the friction of this artificial turbine should be somewhat less than that of the succeeding turbine components. If several turbine components are in series, the choking flag should be left on ($\underline{c} = 0$) for the first turbine component but

turned off for the other components ($\underline{c} = 1$). The smooth junction option ($\underline{a} = 0$) should be used at both the inlet and outlet junctions. The inlet and outlet junctions must be input as homogeneous junctions ($\underline{h} = 1$).

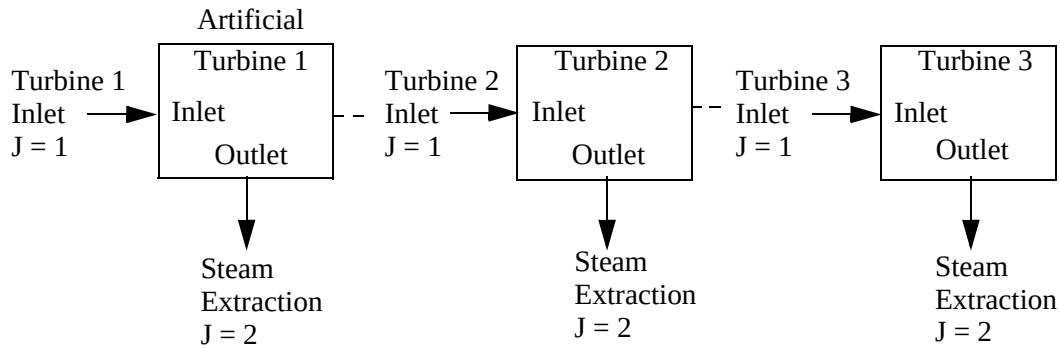


Figure 9.15-1 Multistage turbine with intermediate steam extraction points.

A9.15.1 Turbine Component Information, CCC0001

This card is required.

- W1(I) Number of junctions, nj. The variable nj must be 1 or 2.
- W2(I) Initial condition control. This word is optional and, if missing, the junction initial velocities in the first and second words on card CCCJ201 are assumed to be velocities. If 0, velocities are assumed; if nonzero, mass flows are assumed.

A9.15.2 Turbine X-Coordinate Volume Data, CCC0101-0109

This card (or cards) is required. The nine words can be entered on one or more cards, and the card numbers need not be consecutive.

- W1(R) Area (m^2 , ft^2).
- W2(R) Length (m, ft).
- W3(R) Volume (m^3 , ft^3). The code requires that the volume equals the area times the length ($W3 = W1 \bullet W2$). At least two of the three quantities, W1, W2, and W3, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the words are 0, the volume must equal the x-direction area times the x-direction length within a relative error of 0.000001. The same relative error check is done for the y- and z-directions.
- W4(R) Azimuthal angle (degrees). The absolute value of this angle must be ≤ 360 degrees. This quantity is not used in the calculation but is specified for possible automated drawing of nodalization diagrams.

- W5(R) Inclination angle (degrees). The absolute value of this angle must be ≤ 90 degrees. The angle 0 degrees is horizontal, and positive angles have an upward inclination, i.e., the inlet is at the lowest elevation. This angle is used in the interphase drag calculation.
- W6(R) Elevation change (m, ft). A positive value is an increase in elevation. The absolute value of this quantity must be less than or equal to the length. If the vertical angle orientation is 0, this quantity must be 0. If the vertical angle is nonzero, this quantity must also be nonzero and have the same sign. See Section 2.4.1 of Volume II of the manual for further discussion. When the absolute value of the elevation angle determined by the ratio of the elevation change (this Word 6) and the length (Word 2) is less than or equal to 45 degrees, the horizontal flow regime map is used. When the ratio is greater than 45 degrees, the vertical flow regime map is used.
- W7(R) Wall roughness (m, ft).
- W8(R) Hydraulic diameter (m, ft). This should be computed from $4.0 \cdot \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If 0, the hydraulic diameter is computed from $2.0 \cdot \left(\frac{\text{area}}{\pi} \right)^{0.5}$. A check is made that the pipe roughness is less than half the hydraulic diameter. See Word 1 for the area.
- W9(I) Volume control flags. This word has the format tlpvbfe. It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags. Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f-flag is the only coordinate direction oriented flag. This word enters the scalar oriented flags and the x-coordinate flag. For the TURBINE component the volume control flag is limited to the format 000001e.
- e-flag The digit e specifies if nonequilibrium or equilibrium is to be used. e = 0 specifies that a nonequilibrium (unequal temperature) calculation is to be used, and e = 1 specifies that an equilibrium (equal temperature) calculation is to be used. Equilibrium volumes should not be connected to nonequilibrium volumes. The equilibrium option is provided only for comparison to other codes.

A9.15.3 Turbine Y-Coordinate Volume Data, CCC0181-0189

These cards are optional. These cards are used when the user specifies the y-direction connection with the crossflow model. The volume of the volume is the same for the x-, y-, and z-directions. If these cards are entered, either W1 or W2 must be nonzero.

- W1(R) Area (m^2 , ft^2). If these cards are missing or if this word is 0, this y-direction area is computed from $\frac{\text{volume}}{\text{y-direction length}}$.
- W2(R) Length (m, ft). If these cards are missing, this y-direction length is computed from $2.0 \bullet \left(\frac{\text{x-direction area}}{\pi} \right)^{0.5}$. If this word is 0, this y-direction length is computed from $\frac{\text{volume}}{\text{y-direction area}}$.
- W3(R) Roughness (m, ft).
- W4(R) Hydraulic diameter (m, ft). If these cards are missing or if this word is 0, this y-direction hydraulic diameter is computed from $\frac{4.0 \bullet \text{y-direction volume flow area}}{(\pi \bullet \text{x-direction volume flow area})^{0.5}}$. A check is made to ensure the y-direction roughness is less than half the y-direction hydraulic diameter.
- W5(I) Volume control flags. This word has the general format tlpvbfe, but this word is limited to 00000f0 since it only enters the coordinate oriented flags for the y-direction.
- f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed along the y-coordinate direction in the volume, and f = 1 specifies that wall friction effects are not to be computed along the y-coordinate direction.
- W6(R) This word is not used. Enter 0.
- W7(R) This word is not used. Enter 0.
- W8(R) This word is the position change in the z-fixed (vertical) direction as the flow passes from the y-inlet face to the y-outlet face (m, ft). This quantity affects problems if connections are made to the y-faces.

A9.15.4 Turbine Z-Coordinate Volume Data, CCC0191-0199

These cards are optional. These cards are used when the user specifies the z-direction connection with the crossflow model. The volume of the volume is the same for the x-, y-, and z-directions. If these cards are entered, either W1 or W2 must be nonzero.

- W1(R) Area (m^2 , ft^2). If these cards are missing or if this word is 0, this z-direction area is computed from $\frac{\text{volume}}{\text{z-direction length}}$.

- W2(R) Length (m, ft). If these cards are missing, this z-direction length is computed from $2.0 \bullet \left(\frac{x - \text{direction area}}{\pi} \right)^{0.5}$. If this word is 0, this z-direction length is computed from $\frac{\text{volume}}{\text{z-direction area}}$.
- W3(R) Roughness (m, ft).
- W4(R) Hydraulic diameter (m, ft). If these cards are missing or if this word is 0, this z-direction hydraulic diameter is computed from $\frac{4.0 \bullet \text{z-direction volume flow area}}{(\pi \bullet \text{x-direction volume flow area})^{0.5}}$. A check is made to ensure the z-direction roughness is less than half the z-direction hydraulic diameter.
- W5(I) Volume control flags. This word has the general format tlpybfe, but this word is limited to 00000f0 since it only enters the coordinate oriented flags for the z-direction.
- f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed along the z-coordinate direction in the volume, and f = 1 specifies that wall friction effects are not to be computed along the z-coordinate direction.
- W6(R) This word is not used. Enter 0.
- W7(R) This word is not used. Enter 0.
- W8(R) This word is the position change in the z-fixed (vertical) direction as the flow passes from the z-inlet face to the z-outlet face (m, ft). This quantity affects problems if connections are made to the z-faces.

A9.15.5 Turbine Volume Initial Conditions, CCC0200

This card is required.

- W1(I) Control word. This word has the format gbt. It is not necessary to input leading zeros.
- ε-flag The digit ε specifies the fluid, where ε = 0 is the default fluid. The value for ε > 0 corresponds to the position number of the fluid type indicated on the 120-129 cards (i.e., ε = 1 specifies H₂O, ε = 2 specifies D₂O etc.). The default fluid is that set for the hydrodynamic system by cards 120-129 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-129 or these control words must be consistent (i.e., not specify different fluids). If cards 120-129 are not entered and all

control words use the default $\underline{\varepsilon} = 0$, then water is assumed to be the fluid. It is recommended that the user use the 120-129 cards to set the fluid type in the systems.

b-flag The digit $\underline{b}$ specifies whether boron is present. $\underline{b} = 0$ specifies that the volume liquid does not contain boron, and $\underline{b} = 1$ specifies that a boron concentration in mass of boron per mass of liquid (which may be 0) is being entered after the other required thermodynamic information.

t-flag The digit $\underline{t}$ specifies how the following words are to be used to determine the initial thermodynamic state. $\underline{t} = 0-3$ and 7 specifies one component (steam/water); $\underline{t} = 4-6$ allows the specification of two components (steam/water and noncondensable gas).

With options $\underline{t}$ equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.

One Component (Steam/Water), Equilibrium or Non-equilibrium

$[P, U_f, U_g, \alpha_g]$ If $\underline{t} = 0$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. Enter boron in W6 if needed.

One Component (Steam/Water), Equilibrium

$[T, x_s]$ If $\underline{t} = 1$, the next two words are interpreted as temperature (K, °F) and static quality in equilibrium condition. Enter boron in W4 if needed.

$[P, x_s]$ If $\underline{t} = 2$, the next two words are interpreted as pressure (Pa, lb_f/in²) and static quality in equilibrium condition. Enter boron in W4 if needed.

$[P, T]$ If $\underline{t} = 3$, the next two words are interpreted as pressure (Pa, lb_f/in²) and temperature (K, °F) in equilibrium condition. Enter boron in W4 if needed.

$[P, T_f, T_g, \alpha_g]$ If $\underline{t} = 7$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid temperature (K, F), vapor temperature (K, F), and vapor void fraction. These quantities will be interpreted as equilibrium conditions. The liquid temperature has to be equal to or less than the saturation temperature, and the vapor temperature has to be equal to or greater than the saturation temperature. Enter boron in W6 if needed. This input option was added to allow easier comparisons with TRACE calculations.

Two Components (Steam/Water/Air), Equilibrium or Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0\text{E-}9 \leq x_s \leq 0.99999999$,

Single phase if $x_s < 1.0\text{ E-}9$ or $x_s > 0.99999999$

The static quality is given by $x_s = M_g/(M_g + M_f)$, where $M_g = M_s + M_n$. Section 3.2 of Volume I of the manual discusses this further.

[P,T, x_s] If $t = 4$, the next three words are interpreted as pressure (Pa, $\text{lb}_f/\text{in.}^2$), temperature (K, $^{\circ}\text{F}$), and static quality in equilibrium condition. Using this input option with static quality greater than 0.0 and less than or equal to 1.0, saturated noncondensables will result. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. If the pressure is greater than the critical pressure, $P > P_{\text{crit}}$, then the static quality, x_s (mass steam + air)/(mass steam + air + water) has to be large enough to give a partial steam pressure that is less than the critical pressure, $P_s < P_{\text{crit}}$. Otherwise an input error will occur. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensable (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option. Enter boron in W5 if needed.

[T, x_s , x_n] If $t = 5$, the next three words are interpreted as steam saturation temperature (K, $^{\circ}\text{F}$), static quality, and noncondensable quality in equilibrium condition. Enter boron in W5 if needed. Both the static and noncondensable qualities are restricted to be between $1.0\text{ E-}9$ and 0.99999999 . The liquid will be subcooled. The input temperature determines the steam partial pressure. However, the total pressure may be greater than the critical pressure, which will cause an input error.

[P, U_f , U_g , α_g , x_n] If $t = 6$, the next five words are interpreted as pressure (Pa, $\text{lb}_f/\text{in.}^2$), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing ($t = 0$). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then vapor void fraction must also be 1.0.

When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor-specific internal energy. Enter boron in W7 if needed.

W2-W7(R) Quantities as described under Word 1. Depending on the control word, 2-6 quantities may be required. Enter only the minimum number required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions.

A9.15.6 Turbine Initial Noncondensable Mass Fractions, CCC0301

This card is optional for a turbine and is related to card 110. If card CCC0301 is not supplied, then the values on the card 115 are used for the noncondensable mass fractions. The number of words on card CCC0301 must equal the number of words on card 110. The mass fractions must sum to 1.0 to within a tolerance of 1.0E-10.

W1-WN(R) Initial mass fraction for each noncondensable gas type as specified on card 110 where N is the number of gas types on card 110.

A9.15.7 Turbine Junction Geometry, CCCJ101-J109

Two or three cards are required. Cards with J equal to 1 and/or 2 are entered, one for each junction.

W1(I) From connection code to a component. This refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction (see Section 2.1). The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, N is restricted to 1 or 2.

W2(I) To connection code to a component. This refers to the component at which the junction coordinate direction ends. See the description for W1 above.

W3(R) Area (m^2 , ft^2). If 0, the area is set to the minimum area of the adjoining volumes. For abrupt area changes, the junction area must be equal to or smaller than the minimum of the adjoining areas. For smooth area changes, there are no restrictions.

W4(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. A variable loss coefficient may be specified (see Section A9.12.11). The interpretation and use of this loss coefficient vary based upon the "a" junction control

flag selected in W6. If the smooth area change option is selected ($\underline{a} = 0$), the entry here is the loss coefficient applied to the junction flow area resulting from the entry made in W3. If the abrupt area change option is selected ($\underline{a} = 1$ or 2), the loss coefficient entered here is adjusted by the square of the ratio of the junction area and the minimum flow area of the adjacent connected volumes. This adjusted loss coefficient is applied to the junction flow area. If the $\underline{a} = 1$ option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

Example: Two volumes with volume average flow areas of 0.3 m^2 and 0.4 m^2 are connected with a junction ($A_{\text{jun}} = 0.075 \text{ m}^2$). The user supplied $K_{\text{loss_user}}$ is 10.0. For the smooth area change option ($\underline{a} = 0$), the $K_{\text{loss_user}}$ of 10.0 is applied directly to the junction. For the abrupt area change option ($\underline{a} = 1$ or 2), the K_{loss} applied at the junction is $K_{\text{loss_jun}} = K_{\text{loss_user}} * (A_{\text{jun}}/A_{\text{min}})^2$. For this example, $K_{\text{loss_jun}} = 10.0 * (0.075/0.3)^2 = 0.625$.

- W5(R) Reynolds number independent reverse flow energy loss coefficient, A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. A variable loss coefficient may be specified (see Section A9.12.11). See information for the previous word regarding interpretation and use of this loss coefficient.
- W6(I) Junction control flags. This word has the format jefvcahs. For the TURBINE component, the junction control flag format is limited to 0000cahs.
- c-flag The digit $\underline{c}$ specifies choking options. $\underline{c} = 0$ means that the choking model will be applied, and $\underline{c} = 1$ means that the choking model will not be applied. Turn off the choking ($\underline{c} = 1$) for the second and succeeding turbines in a series of turbines. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske or Moody (option 54) critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active.
- a-flag The digit $\underline{a}$ specifies area change options. $\underline{a} = 0$ means a smooth area change and only the user supplied K_{loss} is applied. $\underline{a} = 1$ means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. $\underline{a} = 2$ means a partial abrupt area change model and it is the same as $\underline{a} = 1$ except no code calculated forward and reverse contraction/expansion K_{loss} is applied. See discussion for W4(R) regarding the K_{loss} flow area reference, which varies based on the a-flag option selected here.

- h-flag The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two-velocity momentum equations) option and h = 1 specifies the homogeneous (single-velocity momentum equation) option.
- s-flag The digit s specifies momentum flux options. s = 0 uses momentum flux in both the to and the from volume. s = 1 uses momentum flux in the from volume, but not in the to volume. s = 2 uses momentum flux in the to volume, but not in the from volume. s = 3 does not use momentum flux in either the to volume or the from volume.

A9.15.8 Turbine Junction Form Loss Data, CCCJ112

These cards are optional. The values of J should follow the same approach as used in cards CCCJ101-J109. The user-specified form loss is given in Words 4 and 5 of cards CCCJ101-J109 if these cards are not entered. If these cards are entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F \text{Re}^{-C_F}$$

$$K_R = A_R + B_R \text{Re}^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 4 and 5 of cards CCCJ101-J109. Re is the Reynolds number based on mixture fluid properties. If these cards are being used for the form loss calculation, then enter all four words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R .

- W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.
- W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.
- W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.
- W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

A9.15.9 Turbine Junction Initial Conditions, CCCJ201

These cards are required depending on the value of n_j as described for cards CCCJ101-J109. The values of J should follow the same approach as used in cards CCCJ101-J109.

- W1(R) Initial liquid or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).
- W2(R) Initial vapor velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W3(R) Interface velocity (m/s, ft/s). Enter 0.

A9.15.10 Turbine Shaft Geometry, CCC0300

This card is required.

W1(R) Turbine stage shaft speed, w (rad/s, rev/min). This speed should equal the shaft speed used in the SHAFT component.

W2(R) Inertia of rotating stages in stage group, I_i , ($\text{kg}\cdot\text{m}^2$, $\text{lb}_f\cdot\text{ft}\cdot\text{s}$).

W3(R) Shaft friction coefficient, f_i ($\text{N}\cdot\text{m}\cdot\text{s}$, $\text{lb}\cdot\text{ft}\cdot\text{s}$). The frictional torque equals $f_i\omega$. This fractional torque is used by the SHAFT component.

W4(I) Shaft component number to which the turbine stage is connected.

W5(I) Disconnect trip number. If 0, the turbine is always connected to the shaft. If nonzero, the turbine is connected to the shaft when the trip is false and disconnected when the trip is true.

W6(I) Drain flag. At the present time, this is not used and can be neglected or set to 0.

A9.15.11 Turbine Performance Data, CCC0400

This card is required.

W1(I) Turbine type

0 = Two-row impulse stage group.

1 = General impulse-reaction stage group.

2 = Constant efficiency stage group.

W2(R) Actual efficiency h_o at the maximum efficiency design point.

W3(R) Design reaction fraction, r . This is the fraction of the enthalpy decrease that takes place in the rotating blade system.

W4(R) Mean stage radius, r (m, ft).

A9.16 ECC Mixer Component, ECCMIX

An ECC mixer is indicated by ECCMIX for Word 2 on card CCC0000. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number (which must be 01 for the ECC mixer), and N is the face number.

An ECC mixer (ECCMIX) component is a specialized branch that requires three junctions with a certain numbering order. The physical extent of the ECC mixer is a length of the cold leg, or any other horizontal pipe, centered around the position of the ECC injection location. The length of this pipe segment should be equal to three times the inside diameter of the pipe (if the physical arrangement of the system permits). Junction number 1 is the ECC connection.

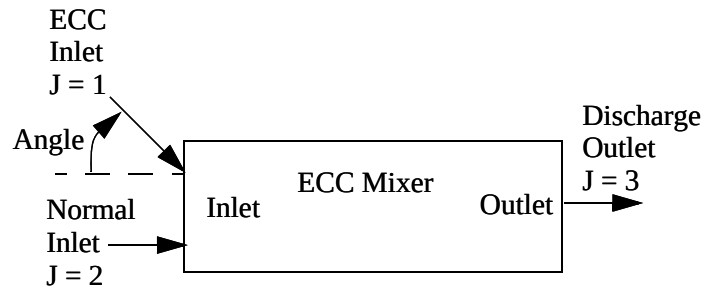


Figure 9.16-1 ECC Mixer model.

Junction number 2 is the flow inlet to the ECC mixer component in normal operation. The geometrical angle between the axis of junctions 1 and 2 is input on the ECC Mixer Junction Geometry Card CCCJ101-J109 for junction number 1. Junction number 3 is the discharge, junction in normal operation. The to part of junctions 1 and 2 must refer to the inlet end of the ECC mixer (CCC010001), and the from part of the discharge junction must refer to the outlet end of the ECC mixer (CCC010002). Note that junction 1 is not a crossflow junction. Two or more ECCMIX components may be considered in modeling some piping. These may be connected in tandem and require at least one normal volume between them.

ECC Mixer Component Information, CCC0001

This card is required.

- | | |
|-------|---|
| W1(I) | Number of junctions, nj. This number must be 3 for ECCMIX components. |
| W2(I) | Initial condition control. This word is optional and, if missing, the junction initial velocities in the first and second words on card CCCN201 are assumed to be velocities. If 0, velocities are assumed; if nonzero, mass flows are assumed. |

A9.16.1 ECC Mixer X-Coordinate Volume Data, CCC0101-0109

This card (or cards) is required. The nine words can be entered on one or more cards, and the card numbers need not be consecutive.

- | | |
|-------|---|
| W1(R) | Area (m ² , ft ²). |
| W2(R) | Length (m, ft). |

- W3(R) Volume (m^3 , ft^3). The code requires that the volume equals the area times the length ($W3 = W1 \bullet W2$). At least two of the three quantities, W1, W2, and W3, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the words are 0, the volume must equal the x-direction area times the x-direction length within a relative error of 0.000001. The same relative error check is done for the y- and z-directions.
- W4(R) Azimuthal angle (degrees). The absolute value of this angle must be ≤ 360 degrees. This quantity is not used in the calculation but is specified for possible automated drawing of nodalization diagrams.
- W5(R) Inclination angle (degrees). The value of this angle must be less than ± 15 degrees. Any other value will be considered an input error. The angle 0 degrees is horizontal, and positive angles have an upward inclination, i.e., the inlet is at the lowest elevation. This angle is used in the interphase drag calculation.
- W6(R) Elevation change (m, ft). A positive value is an increase in elevation. The absolute value of this quantity must be less than or equal to the length. If the vertical angle orientation is 0, this quantity must be 0. If the vertical angle is nonzero, this quantity must also be nonzero and have the same sign. See Section 2.4.1 of Volume II of the manual for further discussion. For ECCMIX, the ECC mixer flow regimes are used.
- W7(R) Wall roughness (m, ft).
- W8(R) Hydraulic diameter (m, ft). This should be computed from $4.0 \bullet \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If 0, the hydraulic diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi} \right)^{0.5}$. A check is made that the pipe roughness is less than half the hydraulic diameter. See Word 1 for the area.
- W9(I) Volume control flags. This word has the format tlpybfe. It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags. Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f-flag is the only coordinate direction oriented flag. This word enters the scalar oriented flags and the x-coordinate flag. For the ECCMIX component, the volume control flags are limited to a format of 00p00fe.
- p-flag The digit p specifies whether the water packing scheme is to be used. p = 0 specifies that the water packing scheme is to be used for the volume, and p = 1 specifies that the water packing scheme is not to be used for the volume.

f-flag	The digit $\underline{f}$ specifies whether wall friction is to be computed. $\underline{f} = 0$ specifies that wall friction effects are to be computed along the x-coordinate direction in the volume, and $\underline{f} = 1$ specifies that wall friction effects are not to be computed along the x-coordinate.
e-flag	The digit $\underline{e}$ specifies if nonequilibrium or equilibrium is to be used. $\underline{e} = 0$ specifies that a nonequilibrium (unequal temperature) calculation is to be used, and $\underline{e} = 1$ specifies that an equilibrium (equal temperature) calculation is to be used. Equilibrium volumes should not be connected to nonequilibrium volumes. The equilibrium option is provided only for comparison to other codes.

A9.16.2 ECC Mixer Additional Wall Friction, CCC0131

This card is optional. If this card is not entered, the default values are 1.0 for the laminar shape factor and 0.0 for the viscosity ratio exponent. Two, four, or six quantities may be entered on the card, and the data not entered are set to default values. A description of this input is presented in Section 3 of Volume I.

W1(R)	Shape factor for x-coordinate.
W2(R)	Viscosity ratio exponent for x-coordinate.
W3(R)	Shape factor for y-coordinate.
W4(R)	Viscosity ratio exponent for y-coordinate.
W5(R)	Shape factor for z-coordinate.
W6(R)	Viscosity ratio exponent for z-coordinate.

A9.16.3 ECC Mixer Y-Coordinate Volume Data, CCC0181-0189

These cards are optional. These cards are used when the user specifies the y-direction connection with the crossflow model. The volume of the volume is the same for the x-, y-, and z-directions. If these cards are entered, either W1 or W2 must be nonzero.

W1(R)	Area (m^2 , ft^2). If these cards are missing or if this word is 0, this y-direction area is computed from $\frac{\text{volume}}{\text{y-direction length}}$.
W2(R)	Length (m, ft). If these cards are missing, this y-direction length is computed from $2.0 \cdot \left(\frac{\text{x-direction area}}{\pi} \right)^{0.5}$. If this word is 0, this y-direction length is computed from $\frac{\text{volume}}{\text{y-direction area}}$.

W3(R)	Roughness (m, ft).
W4(R)	Hydraulic diameter (m, ft). If these cards are missing or if this word is 0, this y-direction hydraulic diameter is computed from $\frac{4.0 \bullet \text{y-direction volume flow area}}{(\pi \bullet \text{x-direction volume flow area})^{0.5}}$. A check is made to ensure the y-direction roughness is less than half the y-direction hydraulic diameter.
W5(I)	Volume control flags. This word has the general format <u>tlpvbfe</u> , but this word is limited to <u>00000f0</u> since it only enters the coordinate oriented flags for the y-direction.
f-flag	The digit <u>f</u> specifies whether wall friction is to be computed. <u>f</u> = 0 specifies that wall friction effects are to be computed along the y-coordinate direction in the volume, and <u>f</u> = 1 specifies that wall friction effects are not to be computed along the y-coordinate direction.
W6(R)	This word is not used. Enter 0.
W7(R)	This word is not used. Enter 0.
W8(R)	This word is the position change in the z-fixed (vertical) direction as the flow passes from the y-inlet face to the y-outlet face (m, ft). This quantity affects problems if connections are made to the y-faces.

A9.16.4 ECC Mixer Z-Coordinate Volume Data, CCC0191-0199

These cards are optional. These cards are used when the user specifies the z-direction connection with the crossflow model. The volume of the volume is the same for the x-, y-, and z-directions. If these cards are entered, either W1 or W2 must be nonzero.

W1(R)	Area (m ² , ft ²). If these cards are missing or if this word is 0, this z-direction area is computed from $\frac{\text{volume}}{\text{z-direction length}}$.
W2(R)	Length (m, ft). If these cards are missing, this z-direction length is computed from $2.0 \bullet \left(\frac{\text{x-direction area}}{\pi} \right)^{0.5}$. If this word is 0, this z-direction length is computed from $\frac{\text{volume}}{\text{z-direction area}}$.
W3(R)	Roughness (m, ft).

- W4(R) Hydraulic diameter (m, ft). If these cards are missing or if this word is 0, this z-direction hydraulic diameter is computed from $\frac{4.0 \bullet \text{z-direction volume flow area}}{(\pi \bullet \text{x-direction volume flow area})^{0.5}}$. A check is made to ensure the z-direction roughness is less than half the z-direction hydraulic diameter.
- W5(I) Volume control flags. This word has the general format tlpvbfe, but this word is limited to 00000f0 since it only enters the coordinate oriented flags for the z-direction.
- f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed along the z-coordinate direction in the volume, and f = 1 specifies that wall friction effects are not to be computed along the z-coordinate direction.
- W6(R) This word is not used. Enter 0.
- W7(R) This word is not used. Enter 0.
- W8(R) This word is the position change in the z-fixed (vertical) direction as the flow passes from the z-inlet face to the z-outlet face (m, ft). This quantity affects problems if connections are made to the z-faces.

A9.16.5 ECC Mixer Volume Initial Conditions, CCC0200

This card is required.

- W1(I) Control word. This word has the format gbt. It is not necessary to input leading zeros.
- ε -flag The digit ε specifies the fluid, where ε = 0 is the default fluid. The value for ε > 0 corresponds to the position number of the fluid type indicated on the 120-129 cards (i.e., ε = 1 specifies H₂O, ε = 2 specifies D₂O etc.). The default fluid is that set for the hydrodynamic system by cards 120-129 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-129 or these control words must be consistent (i.e., not specify different fluids). If cards 120-129 are not entered and all control words use the default ε = 0, then water is assumed to be the fluid. It is recommended that the user use the 120-129 cards to set the fluid type in the systems.
- b-flag The digit b specifies whether boron is present. b = 0 specifies that the volume liquid does not contain boron, and b = 1 specifies that a boron concentration in mass of boron per mass of liquid (which may be 0) is being entered after the other required thermodynamic information.

t-flag The digit t specifies how the following words are to be used to determine the initial thermodynamic state. $t = 0-3$ and 7 specifies one component (steam/water); $t = 4-6$ allows the specification of two components (steam/water and noncondensable gas).

With options t equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.

One Component (Steam/Water), Equilibrium or Non-equilibrium

[P,U_f,U_g, α_g] If $t = 0$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. Enter boron in W6 if needed.

One Component (Steam/Water), Equilibrium

[T,x_s] If $t = 1$, the next two words are interpreted as temperature (K, °F) and static quality in equilibrium condition. Enter boron in W4 if needed.

[P,x_s] If $t = 2$, the next two words are interpreted as pressure (Pa, lb_f/in²) and static quality in equilibrium condition. Enter boron in W4 if needed.

[P,T] If $t = 3$, the next two words are interpreted as pressure (Pa, lb_f/in²) and temperature (K, °F) in equilibrium condition. Enter boron in W4 if needed.

[P,T_f,T_g, α_g] If $t = 7$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid temperature (K, F), vapor temperature (K, F), and vapor void fraction. These quantities will be interpreted as equilibrium conditions. The liquid temperature has to be equal to or less than the saturation temperature, and the vapor temperature has to be equal to or greater than the saturation temperature. Enter boron in W6 if needed. This input option was added to allow easier comparisons with TRACE calculations.

Two Components (Steam/Water/Air), Equilibrium or Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0E-9 \leq x_s \leq 0.99999999$,

Single phase if $x_s < 1.0 \text{ E-9}$ or $x_s > 0.99999999$

The static quality is given by $x_s = M_g / (M_g + M_f)$, where $M_g = M_s + M_n$. Section 3.2 of Volume I of the manual discusses this further.

- [P,T,x_s] If $t = 4$, the next three words are interpreted as pressure (Pa, lb_f/in.²), temperature (K, °F), and static quality in equilibrium condition. Using this input option with static quality greater than 0.0 and less than or equal to 1.0, saturated noncondensables will result. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. If the pressure is greater than the critical pressure, $P > P_{\text{crit}}$, then the static quality, x_s (mass steam + air)/(mass steam + air + water) has to be large enough to give a partial steam pressure that is less than the critical pressure, $P_s < P_{\text{crit}}$. Otherwise an input error will occur. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensable (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option. Enter boron in W5 if needed.
- [T,x_s,x_n] If $t = 5$, the next three words are interpreted as steam saturation temperature (K, °F), static quality, and noncondensable quality in equilibrium condition. Enter boron in W5 if needed. Both the static and noncondensable qualities are restricted to be between 1.0 E-9 and 0.99999999 . The liquid will be subcooled. The input temperature determines the steam partial pressure. However, the total pressure may be greater than the critical pressure, which will cause an input error.
- [P,U_f,U_g,α_g,x_n] If $t = 6$, the next five words are interpreted as pressure (Pa, lb_f/in.²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing ($t = 0$). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor-specific internal energy. Enter boron in W7 if needed.
- W2-W7(R) Quantities as described under Word 1. Depending on the control word, 2-6 quantities may be required. Enter only the minimum number required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions.

A9.16.6 ECC Mixer Initial Noncondensable Mass Fractions, CCC0301

This card is optional for a ECC mixer and is related to card 110. If card CCC0301 is not supplied, then the values on the card 115 are used for the noncondensable mass fractions. The number of words on card CCC0301 must equal the number of words on card 110. The mass fractions must sum to 1.0 to within a tolerance of $1.0\text{E-}10$.

W1-WN(R) Initial mass fraction for each noncondensable gas type as specified on card 110 where N is the number of gas types on card 110.

A9.16.7 ECC Mixer Junction Geometry, CCCJ101-J109

These cards are required. Cards with J equal to 1, 2, and 3 are entered, one for each junction. For the first junction, the ECC injection connection, enter 7 words, and for junctions 2 and 3, enter 6 words.

W1(I) From connection code to a component. This refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction (see Section 2.1). The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, N is restricted to 1 or 2.

W2(I) To connection code to a component. This refers to the component at which the junction coordinate direction ends. See the description for W1 above.

W3(R) Area (m^2 , ft^2). If 0, the area is set to the minimum area of the adjoining volumes. For abrupt area changes, the junction area must be equal to or smaller than the minimum of the adjoining areas. For smooth area changes, there are no restrictions.

W4(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. A variable loss coefficient may be specified (see Section A9.12.11). The interpretation and use of this loss coefficient vary based upon the "a" junction control flag selected in W6. If the smooth area change option is selected (a = 0), the entry here is the loss coefficient applied to the junction flow area resulting from the entry made in W3. If the abrupt area change option is selected (a = 1 or 2), the loss coefficient entered here is adjusted by the square of the ratio of the junction area and the minimum flow area of the adjacent connected volumes. This adjusted loss coefficient is applied to the junction flow area. If the a = 1 option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

Example: Two volumes with volume average flow areas of 0.3 m^2 and 0.4 m^2 are connected with a junction ($A_{\text{jun}} = 0.075 \text{ m}^2$). The user supplied $K_{\text{loss_user}}$ is 10.0. For the smooth area change option ($\underline{a} = 0$), the $K_{\text{loss_user}}$ of 10.0 is applied directly to the junction. For the abrupt area change option ($\underline{a} = 1$ or 2), the K_{loss} applied at the junction is $K_{\text{loss_jun}} = K_{\text{loss_user}} * (A_{\text{jun}}/A_{\text{min}})^2$. For this example, $K_{\text{loss_jun}} = 10.0 * (0.075/0.3)^2 = 0.625$.

- W5(R) Reynolds number independent reverse flow energy loss coefficient, A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. A variable loss coefficient may be specified (see Section A9.12.11). See information for the previous word regarding interpretation and use of this loss coefficient.
- W6(I) Junction control flags. This word has the format jefvcahs. For the ECCMIX component, the format is restricted to 0000cah0.
- c-flag The digit c specifies choking options. c = 0 means that the choking model will be applied, and c = 1 means that the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske or Moody (option 54) critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active.
- a-flag The digit a specifies area change options. a = 0 means a smooth area change and only the user supplied K_{loss} is applied. a = 1 means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. a = 2 means a partial abrupt area change model and it is the same as a = 1 except no code calculated forward and reverse contraction/expansion K_{loss} is applied. See discussion for W4(R) regarding the K_{loss} flow area reference, which varies based on the a-flag option selected here.
- h-flag The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two-velocity momentum equations) option and h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit printout will show h = 1.
- W7(R) Angle (degrees) between the axis of the ECC injection line and the main pipe (or the angle between Junctions 1 and 2). This angle must be between 0 and 180 degrees. If missing, a 90-degree connection for the ECC pipe is assumed. Enter W7 only for the first junction.

A9.16.8 ECC Mixer Junction Form Loss Data, CCCJ112

These cards are optional. The values of J should follow the same approach as used in cards CCCJ101-J109. The user-specified form loss is given in Words 4 and 5 of cards CCCJ101-J109 if these cards are not entered. If these cards are entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F \text{Re}^{-C_F}$$

$$K_R = A_R + B_R \text{Re}^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 4 and 5 of cards CCCJ101-J109. Re is the Reynolds number based on mixture fluid properties. If these cards are being used for the form loss calculation, then enter all four words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R .

W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.

W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.

W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.

W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

A9.16.9 ECC Mixer Junction Initial Conditions, CCCJ201

Three cards are required. The values of J should be 1, 2, and 3.

W1(R) Initial liquid or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W2(R) Initial vapor velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W3(R) Interface velocity (m/s, ft/s). Enter 0.

A9.17 Valve Component, VALVE

A valve junction component is indicated by VALVE for Word 2 on card CCC0000. For major edits, minor edits, and plot variables, the junction in the valve junction component is numbered CCC000000.

A9.17.1 Valve Junction Geometry, CCC0101-0109

This card (or cards) is required for valve junction components.

- W1(I) From connection code to a component. This refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction (see Section 2.1). The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, N is restricted to 1 or 2.
- W2(I) To connection code to a component. This refers to the component at which the junction coordinate direction ends. See the description for W1 above.
- W3(R) Area (m^2 , ft^2). This quantity is the full open area of the valve except in the case of a relief valve. For valves other than relief valves, if this area is input as 0, the area is set to the minimum area of adjoining volumes. If nonzero, this area is used. For relief valves, this term is the valve inlet throat area. If this term is input as 0, it will default to the area calculated from the inlet diameter term input on cards CCC0301-0309, in which case the inlet diameter term cannot be input as 0. If both this area and the inlet diameter are input as nonzero, this area will be used but must agree with the area calculated from the inlet diameter within 10^{-5} m^2 . However, if this area is input as nonzero and the inlet diameter is input as 0, the inlet diameter will default to the diameter calculated from this area. When an abrupt area change model is specified, the area must be less than or equal to the minimum of the adjoining areas.
- W4(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. A variable loss coefficient may be specified (see Section A9.17.3). The interpretation and use of this loss coefficient vary based upon the "a" junction control flag selected in W6. If the smooth area change option is selected (a = 0), the entry here is the loss coefficient applied to the junction flow area resulting from the entry made in W3. If the abrupt area change option is selected (a = 1 or 2), the loss coefficient entered here is adjusted by the square of the ratio of the junction area and the minimum flow area of the adjacent connected volumes. This adjusted loss coefficient is applied to the junction flow area. If the a = 1 option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

Example: Two volumes with volume average flow areas of 0.3 m^2 and 0.4 m^2 are connected with a junction ($A_{\text{jun}} = 0.075 \text{ m}^2$). The user supplied $K_{\text{loss_user}}$ is 10.0. For the smooth area change option (a = 0), the $K_{\text{loss_user}}$ of 10.0 is applied directly to the junction.

For the abrupt area change option ($\underline{a} = 1$ or 2), the K_{loss} applied at the junction is $K_{loss_{jun}} = K_{loss_{user}} * (A_{jun}/A_{min})^2$. For this example, $K_{loss_{jun}} = 10.0 * (0.075/0.3)^2 = 0.625$.

- W5(R) Reynolds number independent reverse flow energy loss coefficient, A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. A variable loss coefficient may be specified (see Section A9.17.3). See information for the previous word regarding interpretation and use of this loss coefficient.
- W6(I) Junction control flags. This word has the format je fvcahs. It is not necessary to input leading zeros. For the VALVE component, the junction control flag is limited to the format 0efvcahs
- e-flag The digit $\underline{e}$ specifies the modified PV term in the energy equations. $\underline{e} = 0$ means that the modified PV term will not be applied, and $\underline{e} = 1$ means that it will be applied.
- f-flag The digit $\underline{f}$ specifies CCFL options. $\underline{f} = 0$ means that the CCFL model will not be applied, and $\underline{f} = 1$ means that the CCFL model will be applied.
- v-flag The digit $\underline{v}$ specifies horizontal stratification entrainment/pullthrough options. This model is for junctions connected to a horizontal volume. $\underline{v} = 0$ means the model is not applied; $\underline{v} = 1$ means an upward-oriented junction (offtake volume must be vertical); $\underline{v} = 2$ means a downward-oriented junction (offtake volume must be vertical); and $\underline{v} = 3$ means a centrally (side) located junction. When $\underline{v} = 4$, the offtake junction is located at an angle to the horizontal (The angle is specified as Word 9).(See the v-flag description for the single junction component for a more detailed list of conditions that have to be satisfied in order to use this model.)
- c-flag The digit $\underline{c}$ specifies choking options. $\underline{c} = 0$ means that the choking model will be applied, and $\underline{c} = 1$ means that the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske or Moody (option 54) critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active.
- a-flag The digit $\underline{a}$ specifies area change options. $\underline{a} = 0$ means a smooth area change and only the user supplied K_{loss} is applied. $\underline{a} = 1$ means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. $\underline{a} = 2$ means a partial abrupt area change model and it is the same as $\underline{a} = 1$ except no code calculated forward and reverse contraction/expansion K_{loss} is applied. All options may be input for a motor or servo valve. If the smooth area change option is input, then a C_v table must be input; or, if no C_v table is input, then one of the abrupt area change options ($\underline{a} = 1$ or 2) must be input. All options may be input for check

valves, but if a nonzero leak ratio is entered, then one of the abrupt area change options ($\underline{a} = 1$ or 2) must be input. All options may be input for a trip valve. For inertial and relief valves, one of the abrupt area change options ($\underline{a} = 1$ or 2) must be input. It is recommended that the $\underline{a} = 1$ option be used for these valves since this option computes a variable-area K-loss and these valves have variable areas during a transient. If the $\underline{a} = 2$ option is used, the variable-area K-loss will be zero, which is not correct. See discussion for W4(R) regarding the K-loss flow area reference, which varies based on the a-flag option selected here.

h-flag The digit $\underline{h}$ specifies nonhomogeneous or homogeneous. $\underline{h} = 0$ specifies the nonhomogeneous (two-velocity momentum equations) option; $\underline{h} = 1$ or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option ($\underline{h} = 1$ or 2), the major edit printout will show $\underline{h} = 1$.

s-flag The digit $\underline{s}$ specifies momentum flux options. $\underline{s} = 0$ uses momentum flux in both the to volume and the from volume. $\underline{s} = 1$ uses momentum flux in the from volume, but not in the to volume. $\underline{s} = 2$ uses momentum flux in the to volume, but not in the from volume. $\underline{s} = 3$ does not use momentum flux in either the to or the from volume.

W7(R) The meaning of this word is different, depending on the critical flow model which has been selected (see W6).

If the Henry-Fauske or Moody (option 54) critical flow model is active, the discharge coefficient is entered. If W7 and W8 are missing, W7 is set to 1.0 and W8 is set to 0.14.

If the original RELAP5 critical flow model is active, the subcooled discharge coefficient is entered. This entry applies only to subcooled liquid choked flow calculations. The entry must be > 0.0 and < 2.0 . If W7, W8 and W9 are missing, they are each set to 1.0.

W8(R) The meaning of this word is different, depending on the critical flow model which has been selected (see W6).

If the Henry-Fauske or Moody critical flow model is active, the thermal nonequilibrium constant is entered. If W7 is entered and W8 is missing, W8 is set to 0.14. If the entry is < 0.01 , the equilibrium option is used and the entry is reset to 0.0. If the entry is > 1000.0 , the frozen option is used and the entry is reset to 100.0.

If the original RELAP5 critical flow model is active, the two-phase discharge coefficient is entered. This entry applies only to two-phase choked flow calculations. The entry must be > 0.0 and < 2.0 . If W7 is entered and W8 and W9 are missing, W8 and W9 are each set to 1.0.

W9(R) If the Henry-Fauske or Moody critical flow model is active and the horizontal offtake mode with $\underline{v} = 4$ is not used, this word is not used and must not be entered (see the v-flag under W6 above).

If the Henry-Fauske or Moody critical flow model is active and the horizontal offtake mode with $\underline{v} = 4$ is specified, this word is the angle between the horizontal and the offtake junction (-90 to +90 degrees).

If the original RELAP5 critical flow model is active, the superheated discharge coefficient is entered. This entry applies only to superheated vapor choked flow calculations. The entry must be > 0.0 and < 2.0 . If W7 and W8 are entered and W9 is missing, then W9 is set to 1.0.

A9.17.2 Valve Junction Diameter and CCFL Data, CCC0110

This card is optional. The defaults indicated for each word are used if the card is not entered. This card can be used for three purposes: (1) interphase drag, (2) CCFL model, and (3) lumped offtake model.

1. If this card is being used to specify only the junction hydraulic diameter for the interphase drag calculation (i.e., $\underline{f} = 0$ in Word 6 of cards CCC0101-0109), then, the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (will not be used).
2. If this card is being used for the CCFL model (i.e., $\underline{f} = 1$ in Word 6 of cards CCC0101-0109), then enter all four words for the appropriate CCFL model if values different from the default values are desired.
3. If this card is being used to specify the diameter of a single offtake junction when lumping two or more offtake pipes into a single pipe, (i.e., $\underline{v} = 1, 2, 3,$ or 4 in Word 6 of cards CCC0101-0109), then the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (will not be used).

W1(R) Junction hydraulic diameter, D_j (m, ft). This is the junction hydraulic diameter used in the CCFL correlation equation and interphase drag and must be ≥ 0 . This number should be computed from $4.0 \cdot \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If a 0 is entered or if the default is used, the junction diameter is computed from $2.0 \cdot \left(\frac{\text{area}}{\pi} \right)^{0.5}$. See Word 3 of cards CCC0101-0109 for the area.

W2(R) Flooding correlation form, β . If 0, the Wallis CCFL form is used. If 1, the Kutateladze CCFL form is used. If between 0.0 and 1.0, Bankoff weighting between the Wallis and Kutateladze CCFL forms is used. This number must be ≥ 0 and ≤ 1 . The default value is 0 (Wallis form).

W3(R) Gas intercept, c . This is the gas intercept used in the CCFL correlation (when $H_f^{1/2} = 0$) and must be > 0 . The default value is 1.

W4(R) Slope, m. This is the slope used in the CCFL correlation and must be > 0 . The default value is 1.

A9.17.3 Valve Junction Form Loss Data, CCC0111

This card is optional. The user-specified form loss is given in Words 4 and 5 of cards CCC0101-0109 if card CCC0111 is not entered. If card CCC0111 is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F \text{Re}^{-C_F}$$

$$K_R = A_R + B_R \text{Re}^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 4 and 5 of cards CCC0101-0109. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all four words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R .

W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.

W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.

W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.

W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

A9.17.4 Valve Junction Initial Conditions, CCC0201

This card is required for valve junction components.

W1(I) Control word. If 0, the next two words are velocities; if 1, the next two words are mass flows.

W2(R) Initial liquid velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s), depending on the control word.

W3(R) Initial vapor velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s), depending on the control word.

W4(R) Interface velocity (m/s, ft/s). Enter 0.

A9.17.5 Valve Type, CCC0300

This card is required to specify the valve type.

W1(A) Valve type. This word must contain one of the variables from the following table.

Table 9.17-1 Valve Types

Valve Name	Type of Valve
CHKVLV	Check valve
TRPVLV	Trip valve
INRVLV	Inertial swing check valve
RLFVLV	Relief valve
MTRVLV	Motor valve
ENDVLV	End valve
SRVVLV	Servo valve

A9.17.6 Check Valve Data and Initial Conditions, CCC0301-0399

This card is required for a check valve. The following words may be placed on one or more cards, and the card numbers need not be consecutive. The check valve behaves as an on, off switch. If the valve is on, then it is fully open; and if the valve is off, it is fully closed.

W1(I) Check valve type. Enter +1 for a static pressure-controlled check valve (no hysteresis), 0 for a static pressure/flow-controlled check valve (has hysteresis effect), or -1 for a static/dynamic pressure-controlled check valve (has hysteresis effect). It is recommended that 0 be used for most calculations, as it is more stable (i.e., less noisy and less oscillations) than +1 or -1.

W2(I) Check valve initial position. The valve is initially open if 0, closed if 1.

W3(R) Additive closing ΔP to model spring-loaded valves (Pa, lb_f/in^2).

W4(R) Leak ratio. This defines the flow area for leakage when the valve is nominally closed. If omitted or input as 0, then any of the area change models ($a = 0, 1$ or 2) may be specified for the check valve. If input as nonzero, then one of the abrupt area change models ($a = 1$ or 2) must be specified for the check valve. The leak ratio is specified as the fraction of the “junction area” through which leakage occurs. Because nonzero leak ratios require use of an abrupt area change model, the “junction area” is taken to mean the minimum of the flow areas in the two hydrodynamic volumes connected by the valve.

A9.17.7 Trip Valve Data and Initial Conditions, CCC0301-0399

This card is required for a trip valve. The trip valve behaves as an on, off switch. If the valve is on, then it is fully open; and if the valve is off, it is fully closed.

W1(I) Trip number. This must be a valid trip number. If the trip is false, the valve is closed; if the trip is true, the valve is open.

A9.17.8 Inertial Valve Data and Initial Conditions, CCC0301-0399

These cards are required for an inertial valve. The following words may be placed on one or more cards, and the card numbers need not be consecutive. The inertial valve behaves realistically in that the valve area varies considering the hydrodynamic forces and the flapper inertia, momentum, and angular acceleration. The abrupt area change model must be specified.

W1(I) Latch option. The valve can open and close repeatedly if the latch option is 0. When W1 = 1, the valve either opens or closes only once if the initial angle is between the maximum and minimum. If the flapper starts at either the maximum or minimum angle it will not move. When W1 = 2, the flapper will latch only at the maximum position. If it starts at the maximum, it will not move.

W2(I) Valve initial condition. The valve is initially open if 0, initially closed if 1.

W3(R) Cracking pressure (Pa, lb_f/in²).

W4(R) Leakage fraction. Fraction of the “junction area” for leakage when the valve is nominally closed. Because an abrupt area change option must be used, the “junction area” is the minimum of the flow areas in the two hydrodynamic volumes connected by the valve.

W5(R) Initial flapper angle (degrees). The flapper angle must be within the minimum and maximum angles specified in Words 6 and 7.

W6(R) Minimum flapper angle (degrees). This must be greater than or equal to 0.

W7(R) Maximum flapper angle (degrees).

W8(R) Moment of inertia of valve flapper (kg•m², lb•ft²).

W9(R) Initial angular velocity (rad/s).

W10(R) Moment length of flapper (m, ft).

W11(R) Radius of flapper (m, ft).

W12(R) Mass of flapper (kg, lb).

A9.17.9 Relief Valve Data and Initial Conditions, CCC0301-0399

These cards are required for a relief valve. The following words may be placed on one or more cards, and the card numbers need not be consecutive. The relief valve area varies, considering the hydrodynamic forces and the valve mass, momentum, and acceleration. One of the abrupt area change models ($\alpha = 1$ or 2) must be specified. The junction area input by cards CCC0101-0199 is the valve inlet area.

W1(I) Valve initial condition. The valve is initially closed if 0, open if 1.

W2(R) Inlet diameter (m, ft). This is the inside diameter of the valve inlet. If this term is input as 0, it will default to the diameter calculated from the junction area input on cards CCC0101-0109. If both this diameter and the junction area are input as nonzero, care must be taken that these terms are input with enough significant digits so that the areas agree within 10^{-5} m^2 . If the junction area is input as 0, then this diameter must be input as nonzero.

W3(R) Valve seat diameter (m, ft). Nonzero input is required. This term is the outside diameter of the valve seat, including the minimum diameter of the inner adjustment ring. This term must also be greater than or equal to the inlet diameter.

W4(R) Valve piston diameter (m, ft). If input as 0, the default is to the valve seat diameter.

W5(R) Valve lift (m, ft). Nonzero input is required. This is the distance the valve piston rises above the valve seat at the fully open position.

W6(R) Maximum outside diameter of the inner adjustment ring (m, ft). If this input is 0, it will default to the valve seat diameter; in which case W7, following, must be input as 0. If this input is nonzero, the value must be greater than or equal to the valve seat diameter. If input is greater than the valve seat diameter, a nonzero input of W7, is allowed. Also refer to the warning stated for W9.

W7(R) Height of outside shoulder relative to the valve seat for inner adjustment ring (m, ft). Input of a positive, nonzero value is not allowed. Input of a 0 value is required if W6 preceding is defaulted or input equal to the valve seat diameter. If the shoulder is below the seat, this distance is negative. Also refer to the warning stated for W9.

W8(R) Minimum inside diameter of the outer adjustment ring (m, ft). If this input is 0, it will default to the valve piston diameter, in which case W9 must be input as positive and nonzero. If this input is nonzero, the value must be greater than or equal to the valve piston diameter. Input of a negative W9 is allowed only if this diameter is greater than the valve piston diameter. Also refer to the warning stated for W9.

- W9(R) Height of inside bottom edge relative to the valve seat for outer adjustment ring (m, ft). This may be input as positive, 0, or negative. If this input is negative, then W8 preceding must be greater than the valve piston diameter. If the bottom edge is below the valve seat, this distance is negative. WARNING: Input of this term and terms W6, W7, and W8 preceding must be done with care to ensure that the resultant gap between the adjustment rings is positive and nonzero; otherwise, an input error will result.
- W10(R) Bellows average diameter (m, ft). If this term is input as 0, it will default to the valve piston diameter, resulting in a model not containing a bellows for which the valve bonnet region is vented to the atmosphere.
- W11(R) Valve spring constant (N/m, lb_f/ft). Positive, nonzero input is required.
- W12(R) Valve setpoint pressure (Pa, lb_f/in²). Positive input is required.
- W13(R) Valve piston, rod, spring, bellows mass (kg, lb). Nonzero input is required.
- W14(R) Valve damping coefficient (N•s/m, lb_f•s/ft).
- W15(R) Bellows inside pressure (Pa, lb_f/in²). Defaults to standard atmospheric pressure if omitted or input as 0.
- W16(R) Initial stem position. This is the fraction of total lift and is required if W1 is input as 1. Total lift is input as W5.
- W17(R) Initial valve piston velocity (m/s, ft/s). This must be 0 or omitted if W1 is input as 0

A9.17.10 Motor Valve Data and Initial Conditions, CCC0301-0399

These cards are required for a motor valve. The following words may be placed on one or more cards, and the card numbers need not be consecutive. The motor valve behaves realistically in that the valve area varies as a function of time by either of two models specified by the user. The user must also select the model for valve hydrodynamic losses by specifying either the smooth or the abrupt area change model. If the smooth area change model is selected ($\underline{a} = 0$), a table of flow coefficients must also be input as described in cards CCC0400-0499, CSUBV Table Section A9.17.13. If an abrupt area change model is selected ($\underline{a} = 1$ or 2), a flow coefficient table cannot be input.

- W1(I) Open trip number.
- W2(I) Close trip number. Both the open and close trip numbers must be valid trips. When both trips are false, the valve remains at its current position. When one of the trips is true, the valve opens or closes depending on which trip is true. The transient will be terminated if both trips are true at the same time.

- W3(R) Valve opening (or closing) change rate (s^{-1}). If Word 5 is not entered, this quantity is the rate of change of the normalized valve area as the valve opens (or closes). If Word 5 is entered, this quantity is the rate of change of the normalized valve stem position. If Word 6 is entered, this quantity is the rate of change of the normalized valve area as the valve opens only. This word must be greater than 0.
- W4(R) Initial position. This number is the initial normalized valve area or the initial normalized stem position depending on Word W5. This quantity must be between 0.0 and 1.0.
- W5(I) Valve table number. If this word is omitted or input as 0, the valve area is determined by the valve change rate and the trips. If this word is input as nonzero, the valve stem position is determined by the valve change rate and the trips; and the valve area is determined from a general table containing normalized valve area versus normalized stem position.
- W6(R) Valve closing change rate (s^{-1}). This word is optional and if it is not entered, the valve closing change rate is the same as the valve opening change rate.

Input for general tables is discussed in cards 202TTTNN, General Table Data, Section A13. For this case, the normalized stem position is input as the *argument value* and the normalized valve area is input as the *function value*.

A9.17.11 End Valve Data and Initial Conditions, CCC0301-0399

These cards are required for a end valve, which is just a motor valve that once it starts opening or closing, it goes all the way to the end. The following words may be placed on one or more cards, and the card numbers need not be consecutive. The end valve behaves realistically in that the valve area varies as a function of time by either of two models specified by the user. The user must also select the model for valve hydrodynamic losses by specifying either the smooth or the abrupt area change model. If the smooth area change model is selected ($\underline{a} = 0$), a table of flow coefficients must also be input as described in cards CCC0400-0499, CSUBV Table Section A9.17.13. If an abrupt area change model is selected ($\underline{a} = 1$ or 2), a flow coefficient table cannot be input.

- W1(I) Open trip number.
- W2(I) Close trip number. Both the open and close trip numbers must be valid trips. When both trips are false, the valve remains at its current position. When one of the trips is true, the valve opens or closes depending on which trip is true. The transient will be terminated if both trips are true at the same time.
- W3(R) Valve opening (or closing) change rate (s^{-1}). If Word 5 is not entered, this quantity is the rate of change of the normalized valve area as the valve opens (or closes). If Word 5 is entered, this quantity is the rate of change of the normalized valve stem position. If Word 6 is entered, this quantity is the rate of change of the normalized valve area as the valve opens only. This word must be greater than 0.

- W4(R) Initial position. This number is the initial normalized valve area or the initial normalized stem position depending on Word W5. This quantity must be between 0.0 and 1.0.
- W5(I) Valve table number. If this word is omitted or input as 0, the valve area is determined by the valve change rate and the trips. If this word is input as nonzero, the valve stem position is determined by the valve change rate and the trips; and the valve area is determined from a general table containing normalized valve area versus normalized stem position.
- W6(R) Valve closing change rate (s^{-1}). This word is optional and if it is not entered, the valve closing change rate is the same as the valve opening change rate.

Input for general tables is discussed in cards 202TTTNN, General Table Data, Section A13. For this case, the normalized stem position is input as the *argument value* and the normalized valve area is input as the *function value*.

A9.17.12 Servo Valve Data and Initial Conditions, CCC0301-0399

These cards are required for a servo valve. The following words may be placed on one or more cards, and the card numbers need not be consecutive. The servo valve behaves as described for a motor valve except that the valve flow area or stem position is calculated by a control system. Input for control systems is discussed in Section A15. Input specifying the hydrodynamic losses for servo valves is also identical to that for motor valves.

- W1(I) Control variable number. The value of the indicated control variable is either the normalized valve area or the normalized stem position, depending on whether Word 2 is entered. The control variable is also the search argument for the CSUBV table if it is entered.
- W2(I) Valve table number. If this word is not entered, the control variable value is the normalized flow area. If it is entered, the control variable value is the normalized stem position, and the general table indicated by this word contains a table of normalized area versus normalized stem position. Input for the general table is identical to that for a motor valve.

A9.17.13 Motor, End, or Servo Valve CSUBV Table Factors, CCC0400

The CSUBV table may be input only for motor, end, and servo valves. If the CSUBV table is input, the smooth area change model must be specified on the valve junction geometry cards (cards CCC0101-0109). If the smooth area change model is specified, a CSUBV table must be input.

This card is optional for a motor, end, or servo valve. The factors apply to the flow area or the stem position and the flow coefficient entries in the CSUBV table.

W1(R) Normalized flow area or normalized stem position (See Section A9.17.10 for motor valve, Section A9.17.11 for end valve, and Section A9.17.12 for servo valve).

W2(R) Flow coefficient factor.

A9.17.14 Motor, End, or Servo Valve CSUBV Table Entries, CCC0401-0499

These cards are required if the motor, end, or servo valve requests a CSUBV table as part of its input. The CSUBV table contains forward and reverse flow coefficients as a function of normalized flow area or normalized stem position. These cards are required for a motor or servo valve.

The table is entered by using three-word sets. W1 is the flow area or stem position and must be normalized. The factor W1 on card CCC0400 can be used to normalize the flow area or stem position. In either case, the implication is that if the valve is fully closed, the normalized term is 0. If the valve is fully open, the normalized term is 1.0. Any value may be input that is between 0.0 and 1.0. The forward and reverse flow coefficients are W2 and W3, respectively. The code internally converts flow coefficients to energy loss coefficients by the formula $K = 2 \cdot A_j^2 / (\rho \cdot \text{CSUBV}^2)$, where ρ is a reference density of 991.091 kg/m³ (62.3706 lb_m/ft³), A_j is the full open valve area, and CSUBV is the flow coefficient. On card CCC0400, W2 may be used to modify the definition of CSUBV. A smooth area change must be specified in W6 on card CCC0101 to use the CSUBV table. CSUBV is entered in British units only.

W1(R) Normalized flow area or normalized stem position (See Section A9.17.10 for motor valve, Section A9.17.11 for end valve, and Section A9.17.12 for servo valve).

W2(R) Forward CSUBV $\{(\text{gal}/\text{min})/[(\text{lb}_f/\text{in}^2)^{0.5}]\}$. The CSUBV is input in British units only and is converted automatically by the code to SI units $[\frac{\text{m}^3/\text{s}}{\sqrt{\text{Pa}}} (\text{m}^7/\text{kg})^{0.5}]$ using a multiplier of 7.598055E-7 as the conversion factor.

W3(R) Reverse CSUBV $\{(\text{gal}/\text{min})/[(\text{lb}_f/\text{in}^2)^{0.5}]\}$.

The user is cautioned that when CSUBV = 0.0, the valve acts like a check valve and no flow occurs through the valve even if the normalized stem position is 1.0 or open. The user is referred to Volume 5, Section 4.6.7.6 for a complete description of CSUBV table input development.

A9.18 Pump Component, PUMP

A pump component is indicated by PUMP for Word 2 on card CCC0000. A pump consists of one volume and two junctions, one attached to each end of the volume. For major edits, minor edits, and plot variables, the volume in the pump component is numbered as CCC010000. The pump junctions are numbered CCC010000 for the inlet junction and CCC020000 for the outlet junction.

A9.18.1 Pump Volume Geometry, CCC0101-0107

This card (or cards) is required for a pump component. The seven words can be entered on one or more cards, and the card numbers need not be consecutive.

W1(R)	Area (m ² , ft ²).
W2(R)	Length (m, ft).
W3(R)	Volume (m ³ , ft ³). The program requires that the volume equals the area times the length (W3 = W1•W2). At least two of the three quantities, W1, W2, W3, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the words are 0, the volume must equal the area times the length within a relative error of 0.000001.
W4(R)	Azimuthal angle (degrees). The absolute value of this angle must be ≤ 360 degrees. This quantity is not used in the calculation but is specified for possible automated drawing of nodalization diagrams.
W5(R)	Inclination angle (degrees). The absolute value of this angle must be ≤ 90 degrees. The angle 0 degrees is horizontal, and positive angles have an upward direction, i.e., the outlet is at a higher elevation than the inlet. This angle is used in the interphase drag calculation.
W6(R)	Elevation change (m, ft). A positive value is an increase in elevation. The absolute value of this quantity must be equal to or less than the length. If the vertical angle orientation is 0, this quantity must be 0. If the vertical angle is nonzero, this quantity must also be nonzero and have the same sign. See Section 2.4.1 of Volume II of the manual for further discussion. For this component, this Word 6 is not compared to the length (Word 2) to decide if the horizontal or vertical flow regime is used. Rather, the pump flow regime map is used.
W7(I)	Volume control flags. This word has the format <u>tlpvbfe</u> . It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags. Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f-flag is the only coordinate direction oriented flag. This word enters the scalar oriented flags and the x-coordinate flag. The pump component forces all volume flags except for the e digit, and y- and z-coordinate flags are not read. The effective format is <u>000000e</u> .
e-flag	The equilibrium digit <u>e</u> specifies if nonequilibrium or equilibrium is to be used; <u>e</u> = 0 specifies that a nonequilibrium (unequal temperature) calculation is to be used, and <u>e</u> = 1 specifies that an equilibrium (equal temperature) calculation is to be used. Equilibrium

volumes should not be connected to nonequilibrium volumes. The equilibrium option is provided only for comparison to other codes.

A9.18.2 Pump Inlet Junction, CCC0108

This card is required for a pump component.

W1(I) Volume code of connecting volume on inlet side. This refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction (see Section 2.1 of Volume II). The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, only the number N equal to 1 or 2 is allowed.

W2(R) Area (m^2 , ft^2). If 0, the area is set to the minimum area of adjacent volumes. If an abrupt area change, the area must be equal to or less than the minimum of the adjacent volume areas. If a smooth area change, no restrictions exist.

W3(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. A variable loss coefficient may be specified (see Section A9.18.6). The interpretation and use of this loss coefficient vary based upon the "a" junction control flag selected in W5. If the smooth area change option is selected ($a = 0$), the entry here is the loss coefficient applied to the junction flow area resulting from the entry made in W3. If the abrupt area change option is selected ($\underline{a} = 1$ or 2), the loss coefficient entered here is adjusted by the square of the ratio of the junction area and the minimum flow area of the adjacent connected volumes. This adjusted loss coefficient is applied to the junction flow area. If the $\underline{a} = 1$ option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

Example: Two volumes with volume average flow areas of 0.3 m^2 and 0.4 m^2 are connected with a junction ($A_{\text{jun}} = 0.075 \text{ m}^2$). The user supplied $K_{\text{loss_user}}$ is 10.0. For the smooth area change option ($\underline{a} = 0$), the $K_{\text{loss_user}}$ of 10.0 is applied directly to the junction. For the abrupt area change option ($\underline{a} = 1$ or 2), the K_{loss} applied at the junction is $K_{\text{loss_jun}} = K_{\text{loss_user}} * (A_{\text{jun}}/A_{\text{min}})^2$. For this example, $K_{\text{loss_jun}} = 10.0 * (0.075/0.3)^2 = 0.625$.

W4(R) Reynolds number independent reverse flow energy loss coefficient, A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that

phase is negative. A variable loss coefficient may be specified (see Section A9.18.6). See information for the previous word regarding interpretation and use of this loss coefficient.

W5(I) Junction control flags. This word has the format jefvcahs. It is not necessary to input leading zeros. For the PUMP inlet junction, the junction control flag is restricted to the format 00f0cah0.

f-flag The digit f specifies CCFL options. f = 0 means that the CCFL model will not be applied, and f = 1 means that the CCFL model will be applied.

c-flag The digit c specifies choking options. c = 0 means that the choking model will be applied, and c = 1 means that the choking model will not be applied.

The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske or Moody (option 54) critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active.

Unlike junctions for certain other hydrodynamic components, only the default values of discharge coefficients and factors used in the critical flow models are allowed for junctions in the PUMP component. Problem renodalization should be used to circumvent this limitation should it prove to be significant.

For the Henry-Fauske or Moody critical flow model, the default value for the discharge coefficient is 1.0 and the default value for the thermal nonequilibrium constant is 0.14. For the original RELAP5 critical flow model, the default values for the subcooled, two-phase and superheated discharge coefficients each are 1.0.

a-flag The digit a specifies area change options. a = 0 means a smooth area change and only the user supplied K_{loss} is applied. a = 1 means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. a = 2 means a partial abrupt area change model and it is the same as a = 1 except no code calculated forward and reverse contraction/expansion K_{loss} is applied. See discussion for W3(R) regarding the K_{loss} flow area reference, which varies based on the a-flag option selected here.

h-flag The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two-velocity momentum equations) option; h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit printout will show a 1.

A9.18.3 Pump Outlet Junction, CCC0109

This card is required for a pump component. The format for this card is identical to card CCC0108 except data are for the outlet junction.

This card is required for a pump component.

W1(I) Volume code of connecting volume on outlet side. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction (see Section 2.1 of Volume II). The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, N is restricted to 1 or 2.

W2(R) Area (m^2 , ft^2). If 0, the area is set to the minimum area of adjacent volumes. If an abrupt area change, the area must be equal to or less than the minimum of the adjacent volume areas. If a smooth area change, no restrictions exist.

W3(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. A variable loss coefficient may be specified (see Section A9.18.6). The interpretation and use of this loss coefficient vary based upon the "a" junction control flag selected in W5. If the smooth area change option is selected ($\underline{a} = 0$), the entry here is the loss coefficient applied to the junction flow area resulting from the entry made in W3. If abrupt area change option is selected ($\underline{a} = 1$ or 2), the loss coefficient entered here is adjusted by the square of the ratio of the junction area and the minimum flow area of the adjacent connected volumes. This adjusted loss coefficient is applied to the junction flow area. If the $\underline{a} = 1$ option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

Example: Two volumes with volume average flow areas of 0.3 m^2 and 0.4 m^2 are connected with a junction ($A_{\text{jun}} = 0.075 \text{ m}^2$). The user supplied $K_{\text{loss_user}}$ is 10.0. For the smooth area change option ($\underline{a} = 0$), the $K_{\text{loss_user}}$ of 10.0 is applied directly to the junction. For the abrupt area change option ($\underline{a} = 1$ or 2), the K_{loss} applied at the junction is $K_{\text{loss_jun}} = K_{\text{loss_user}} * (A_{\text{jun}}/A_{\text{min}})^2$. For this example, $K_{\text{loss_jun}} = 10.0 * (0.075/0.3)^2 = 0.625$.

W4(R) Reynolds number independent reverse flow energy loss coefficient, A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that

phase is negative. A variable loss coefficient may be specified (see Section A9.18.6). See information for the previous word regarding interpretation and use of this loss coefficient.

W5(I) Junction control flags. This word has the format jefvcahs. It is not necessary to input leading zeros. For the PUMP outlet junction, the junction control flag is restricted to the format 00f0cah0.

f-flag The digit f specifies CCFL options. f = 0 means that the CCFL model will not be applied, and f = 1 means that the CCFL model will be applied.

c-flag The digit c specifies choking options. c = 0 means that the choking model will be applied, and c = 1 means that the choking model will not be applied.

The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske or Moody (option 54) critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active.

Unlike junctions for certain other hydrodynamic components, only the default values of discharge coefficients and factors used in the critical flow models are allowed for junctions in the PUMP component. Problem renodalization should be used to circumvent this limitation should it prove to be significant.

For the Henry-Fauske or Moody critical flow model, the default value for the discharge coefficient is 1.0 and the default value for the thermal nonequilibrium constant is 0.14. For the original RELAP5 critical flow model, the default values for the subcooled, two-phase and superheated discharge coefficients each are 1.0.

a-flag The digit a specifies area change options. a = 0 means a smooth area change and only the user supplied K_{loss} is applied. a = 1 means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. a = 2 means a partial abrupt area change model and it is the same as a = 1 except no code calculated forward and reverse contraction/expansion K_{loss} is applied. See discussion for W3(R) regarding the K_{loss} flow area reference, which varies based on the a-flag option selected here.

h-flag The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two-velocity momentum equations) option; h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit printout will show a 1.

A9.18.4 Pump Inlet Junction Diameter and CCFL Data, CCC0110

This card is optional. The defaults indicated for each word are used if the card is not entered. If this card is being used to specify only the junction hydraulic diameter for the interphase drag calculation (i.e., $\underline{f} = 0$ in Word 5 of card CCC0108), then the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (will not be used). If the card is being used for the CCFL model (i.e., $\underline{f} = 1$ in Word 5 of card CCC0108), then enter all four words for the appropriate CCFL model if values different from the default values are desired.

- W1(R) Inlet junction hydraulic diameter, D_j (m, ft). This is the junction hydraulic diameter used in the CCFL correlation equation and interphase drag and must be ≥ 0 . This number should be computed from $4.0 \cdot \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If a 0 is entered or the default is used, the junction diameter is computed from $2.0 \cdot \left(\frac{\text{area}}{\pi} \right)^{0.5}$. See Word 2 of card CCC0108 for the junction area.
- W2(R) Flooding correlation form, β . If 0, the Wallis CCFL form is used. If 1.0, the Kutateladze CCFL form is used. If between 0.0 and 1.0, Bankoff weighting between the Wallis and Kutateladze CCFL forms is used. This number must be ≥ 0 and ≤ 1 . The default value is 0 (Wallis form).
- W3(R) Gas intercept, c . This is the gas intercept used in the CCFL correlation (when $H_f^{1/2} = 0$) and must be > 0 . The default value is 1.
- W4(R) Slope, m . This is the slope used in the CCFL correlation and must be > 0 . The default value is 1.

A9.18.5 Pump Outlet Junction Diameter and CCFL Data, CCC0111

This card is optional. The defaults indicated for each word are used if the card is not entered. If this card is being used to just specify the junction hydraulic diameter for the interphase drag calculation (i.e., $\underline{f} = 0$ in Word 5 of card CCC0109), then the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (will not be used). If the card is being used for the CCFL model (i.e., $\underline{f} = 1$ in Word 5 of card CCC0109), then enter all four words for the appropriate CCFL model if values different from the default values are desired. The format for this card is identical to card CCC0110 except that data are for the outlet junction.

- W1(R) Outlet junction hydraulic diameter, D_j (m, ft). This is the junction hydraulic diameter used in the CCFL correlation equation and interphase drag and must be ≥ 0 . This number should be computed from $4.0 \cdot \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If a 0 is entered or the default is

used, the junction diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi} \right)^{0.5}$. See Word 2 of card CCC0108 for the junction area.

- W2(R) Flooding correlation form, β . If 0, the Wallis CCFL form is used. If 1.0, the Kutateladze CCFL form is used. If between 0.0 and 1.0, Bankoff weighting between the Wallis and Kutateladze CCFL forms is used. This number must be ≥ 0 and ≤ 1 . The default value is 0 (Wallis form).
- W3(R) Gas intercept, c . This is the gas intercept used in the CCFL correlation (when $H_f^{1/2} = 0$) and must be > 0 . The default value is 1.
- W4(R) Slope, m . This is the slope used in the CCFL correlation and must be > 0 . The default value is 1.

A9.18.6 Pump Inlet Junction Form Loss Data, CCC0112

This card is optional. The user-specified form loss is given in Words 3 and 4 of card CCC0108 if this card is not entered. If this card is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F \text{Re}^{-C_F}$$

$$K_R = A_R + B_R \text{Re}^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 3 and 4 of card CCC0108. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all four words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R .

- W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.
- W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.
- W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.
- W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

A9.18.7 Pump Outlet Junction Form Loss Data, CCC0113

This card is optional. The user-specified form loss is given in Words 3 and 4 of card CCC0109 if this card is not entered. If this card is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F \text{Re}^{-C_F}$$

$$K_R = A_R + B_R \text{Re}^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 3 and 4 of card CCC0109. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all four words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R . The format of this card is identical to card CCC0112 except data are for the outlet junction.

W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.

W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.

W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.

W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

A9.18.8 Pump Volume Initial Conditions, CCC0200

This card is required for a pump component.

W1(I) Control word. This word has the format $\underline{\epsilon}\underline{b}\underline{t}$. It is not necessary to input leading zeros.

ϵ -flag The digit $\underline{\epsilon}$ specifies the fluid, where $\underline{\epsilon} = 0$ is the default fluid. The value for $\underline{\epsilon} > 0$ corresponds to the position number of the fluid type indicated on the 120-129 cards (i.e., $\underline{\epsilon} = 1$ specifies H_2O , $\underline{\epsilon} = 2$ specifies D_2O , etc.). The default fluid is that set for the hydrodynamic system by cards 120-129 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-129 or these control words must be consistent (i.e., not specify different fluids). If cards 120-129 are not entered and all control words use the default $\underline{\epsilon} = 0$, then H_2O is assumed to be the fluid. It is recommended that the user use the 120-129 cards to set the fluid type in the systems.

b-flag The digit $\underline{b}$ specifies whether boron is present. $\underline{b} = 0$ specifies that the volume liquid does not contain boron; $\underline{b} = 1$ specifies that a boron concentration in mass of boron per mass of liquid (which may be 0) is being entered after the other required thermodynamic information.

t-flag The digit $\underline{t}$ specifies how the following words are to be used to determine the initial thermodynamic state. $\underline{t} = 0-3$ and 7 specifies one component (steam/water). Entering $\underline{t} = 4-6$ allows the specification of two components (steam/water and noncondensable gas).

With options $\underline{t}$ equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.

One Component (Steam/Water), Equilibrium or Non-equilibrium

[P,U_f,U_g, α_g] If $\underline{t} = 0$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. Enter boron in W6 if needed.

One Component (Steam/Water), Equilibrium

[T,x_s] If $\underline{t} = 1$, the next two words are interpreted as temperature (K, °F) and static quality in equilibrium condition. Enter boron in W4 if needed.

[P,x_s] If $\underline{t} = 2$, the next two words are interpreted as pressure (Pa, lb_f/in²) and static quality in equilibrium condition. Enter boron in W4 if needed.

[P,T] If $\underline{t} = 3$, the next two words are interpreted as pressure (Pa, lb_f/in²) and temperature (K, °F) in equilibrium condition. Enter boron in W4 if needed.

[P,T_f,T_g, α_g] If $\underline{t} = 7$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid temperature (K, F), vapor temperature (K, F), and vapor void fraction. These quantities will be interpreted as equilibrium conditions. The liquid temperature has to be equal to or less than the saturation temperature, and the vapor temperature has to be equal to or greater than the saturation temperature. Enter boron in W6 if needed. This input option was added to allow easier comparisons with TRACE calculations.

Two Components (Steam/Water/Air), Equilibrium or Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0\text{E-}9 \leq x_s \leq 0.99999999$,

Single phase if $x_s < 1.0 \text{ E-}9$ or $x_s > 0.99999999$

The static quality is given by $x_s = M_g/(M_g + M_f)$, where $M_g = M_s + M_n$. Section 3.2 of Volume I of the manual discusses this further.

[P,T,x_s] If $t = 4$, the next three words are interpreted as pressure (Pa, lb_f/in.²), temperature (K, °F), and static quality in equilibrium condition. Using this input option with static quality > 0.0 and ≤ 1.0 , saturated noncondensables will result. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. If the pressure is greater than the critical pressure, $P > P_{crit}$, then the static quality, x_s (mass steam + air)/(mass steam + air + water) has to be large enough to give a partial steam pressure that is less than the critical pressure, $P_s < P_{crit}$. Otherwise an input error will occur. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensables (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option. Enter boron in W5 if needed.

[T,x_s,x_n] If $t = 5$, the next three words are interpreted as steam saturation temperature (K, °F), static quality, and noncondensable quality in equilibrium condition. Enter boron in W5 if needed. Both the static and noncondensable qualities are restricted to be between 1.0 E-9 and 0.99999999. The liquid will be subcooled. The input temperature determines the steam partial pressure. However, the total pressure may be greater than the critical pressure, which will cause an input error.

[P,U_f,U_g,α_g,x_n] If $t = 6$, the next five words are interpreted as pressure (Pa, lb_f/in.²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing ($t = 0$). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor specific internal energy. Enter boron in W7 if needed.

W2-W7(R) Quantities as described under word 1. Depending on the control word, 2-6 quantities may be required. Enter only the minimum number required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions.

A9.18.9 Pump Inlet Junction Initial Conditions, CCC0201

This card is required for a pump component.

- W1(I) Control word. If 0, the next two words are velocities; if 1, the next two words are mass flow rates.
- W2(R) Initial liquid velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s).
- W3(R) Initial vapor velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s).
- W4(R) Initial interface velocity (m/s, ft/s). Enter 0.

A9.18.10 Pump Outlet Junction Initial Conditions, CCC0202

This card is required for a pump component. This card is similar to card CCC0201 except that data are for the outlet junction.

- W1(I) Control word. If 0, the next two words are velocities; if 1, the next two words are mass flow rates.
- W2(R) Initial liquid velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s).
- W3(R) Initial vapor velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s).
- W4(R) Initial interface velocity (m/s, ft/s). Enter 0.

A9.18.11 Pump Index and Option, CCC0301

This card is required for a pump component.

- W1(I) Pump table data indicator. Enter 0 if the single-phase homologous tables are entered with this component. Enter another pump component number (positive number) if the single-phase tables are to be obtained from another pump component. Enter -1 if the built-in Bingham pump single-phase homologous tables are to be used. Enter -2 if the built-in Westinghouse pump single-phase homologous tables are to be used.
- W2(I) Two-phase multiplier index. Enter -1 if the two-phase multiplier option is not used. Enter 0 if the two-phase multiplier option is used, and the two-phase multiplier tables are entered as part of this component. Enter another pump component number (positive

number) if the two-phase option is used and the two-phase multiplier table data are to be obtained from another pump component. There are no built-in Bingham or Westinghouse pump data for the two-phase multiplier table. See Table 9.18-1 for Word 2 and Word 3 combinations.

W3(I) Two-phase difference table index. Enter -3 if the two-phase difference table is not needed, i.e., if W2 is -1. Enter 0 if a two-phase difference table is entered as part of this pump component. Enter another pump component number (positive value) if the table is to be obtained from this other pump component. Enter -1 if the built-in two-phase difference table data for the Bingham pump is to be used. Enter -2 if the built-in two-phase difference table data for the Westinghouse pump is to be used. See Table 9.18-1 for permissible Word 2 and Word 3 combinations.

Table 9.18-1 Pump Index Permissible Combinations

W2(I) Two-phase Multiplier Index	W3(I) Two-phase Difference Table Index
-1	-3
0	0, -1, -2, Pump Component Number (Positive Value)
Pump Component Number (Positive Value)	0, -1, -2, Pump Component Number (Positive Value)

Head formulation: $H = H_{1\phi} - M_H(\alpha_g)(H_{1\phi} - H_{2\phi})$

Torque formulation: $\tau = \tau - M_\tau(\alpha_g)(\tau_{1\phi} - \tau_{2\phi})$

The quantity $(H_{1\phi} - H_{2\phi})$ is stored in the difference table.

W4(I) Pump motor torque table index. If -1, no table is used. If 0, a table is entered for this component. If nonzero, use the table from the component with this number.

W5(I) Time-dependent pump velocity index. If -1, no time-dependent pump rotational velocity table is used and the pump velocity is always determined by the torque-inertia equation. If 0, a table is entered with this component. If nonzero, the table from the pump component with this number is used. A pump velocity table cannot be used when the pump is connected to a shaft control component.

W6(I) Pump motor trip number. When the pump motor trip is off, electrical power is supplied to the pump motor; when the pump motor trip is on, electrical power is disconnected from

the pump motor. The pump velocity depends on the pump velocity table and associated pump velocity table trip (Card CCC6100), the pump motor torque data, and this pump motor trip. If the pump velocity table is being used (either the trip on Card CCC6100 is true or no trip is given on that card), the pump velocity is always computed from that table. If the pump velocity table is not being used (either the table is not entered or the trip on Card CCC6100 is false), the pump velocity depends on the pump motor torque data and this pump motor trip. If the pump motor trip is off and no pump motor torque data are present, the pump velocity is the same as for the previous time step. This will be the initial pump velocity if the pump motor trip has never been off. Usually the pump motor trip is a latched trip, but that is not necessary. If the pump motor trip is off and a pump motor torque table is present, the pump velocity is given by the torque-inertia equation where the net torque is given by the pump motor torque data and the homologous torque data. If the pump motor trip is on, the torque-inertia equation is used and the pump motor torque is set to 0. If the pump motor trip number is 0, no trip is tested, and the pump motor trip is assumed to always be off.

W7(I) Reverse indicator. If 0, no reverse is allowed; if 1, reverse is allowed.

A9.18.12 Pump Description, CCC0302-0304

This card (or cards) is required for a pump component.

W1(R)	Rated pump velocity (rad/s, rev/min).
W2(R)	Ratio of initial pump velocity to rated pump velocity. Used for calculating initial pump velocity.
W3(R)	Rated flow (m^3/s , gal/min).
W4(R)	Rated head (m, ft).
W5(R)	Rated torque ($\text{N}\cdot\text{m}$, $\text{lb}_f\cdot\text{ft}$).
W6(R)	Moment of inertia ($\text{kg}\cdot\text{m}^2$, $\text{lb}\cdot\text{ft}^2$). This includes all direct coupled rotating components, including the master for a motor driven pump.
W7(R)	Rated density (kg/m^3 , lb/ft^3). If 0, initial density is used. This is the density used to generate homologous data.
W8(R)	Rated pump motor torque ($\text{N}\cdot\text{m}$, $\text{lb}_f\cdot\text{ft}$). If this word is 0, the rated pump motor torque is computed from the initial pump velocity and the pump torque that is computed from the

initial pump velocity, initial volume conditions, and the homologous curves. This quantity must be nonzero if the relative pump motor torque table is entered.

- W9(R) TF2 friction torque coefficient (N·m, lb_f·ft). This parameter multiplies the speed ratio (absolute pump speed/rated speed) to the second power. The friction torque factors are summed together.
- W10(R) TF0, friction torque coefficient (N·m, lb_f·ft). This is constant frictional torque.
- W11(R) TF1, friction torque coefficient (N·m, lb_f·ft). This multiplies the speed ratio to the first power.
- W12(R) TF3, friction torque coefficient. (N·m, lb_f·ft). This multiplies the speed ratio to the third power.

A9.18.13 Pump Variable Inertia, CCC0308

This card is optional, and if this card is not entered, the pump inertia is given by Word 6 of card CCC0302. If this card is entered, pump inertia is computed from

$$I = I_3 S^3 + I_2 S^2 + I_1 S^1 + I_0$$

where S is the relative pump speed defined as the absolute value of the pump rotational velocity divided by the rated rotational velocity.

- W1(R) Relative speed at which to use the cubic expression for inertia. When the relative speed is less than this quantity, the inertia from Word 6 of card CCC0302 is used.
- W2-W5(R) I_3, I_2, I_1, I_0 (kg/m², lb·ft²).

A9.18.14 Pump-Shaft Connection, CCC0309

This card is optional, and if this card is entered, the pump is connected to a SHAFT component. The pump may still be driven by a pump motor that can be described in this component, by a turbine also connected to the SHAFT component, or from torque computed by the control system and applied to the SHAFT component. The pump speed table may not be entered if this card is entered.

- W1(I) Control component number of the shaft component.
- W2(I) Pump disconnect trip. If this quantity is omitted or 0, the pump is always connected to the SHAFT. If nonzero, the pump is connected to the shaft when the trip is false and disconnected when the trip is true.

A9.18.15 Pump Stop Data, CCC0310

If this card is optional, and if this card is not entered, the pump will not be stopped by the program.

- W1(R) Elapsed problem time for pump stop (s).
- W2(R) Maximum forward velocity for pump stop (rad/s, rev/min).
- W3(R) Maximum reverse velocity for pump stop (rad/s, rev/min). Reverse velocity is a negative number.

A9.18.16 Pump Single-Phase Homologous Curves, CCC1100-2699

These cards are required only if W1 of card CCC0301 is 0. There are sixteen possible sets of homologous curve data to completely describe the single-phase pump operation, that is, a curve for each head and torque for each of the eight possible curve types or regimes of operation. Entering all sixteen curves is not necessary, but an error will occur from an attempt to reference one that has not been entered.

Card numbering is CCC1100-1199 for the first curve, CCC1200-1299 for the second curve, and CCC2600-2699 for the sixteenth curve. Data for each individual curve are input on up to 99 cards, which need not be numbered consecutively.

- W1(I) Curve type. Enter 1 for a head curve; enter 2 for a torque curve.
- W2(I) Curve regime. See **Table 2.4-2** of this manual for definitions. The possible integer numbers and the corresponding homologous curve octants are: 1 (HAN or BAN), 2 (HVN or BVN), 3 (HAD or BAD), 4 (HVD or BVD), 5 (HAT or BAT), 6 (HVT or BVT), 7 (HAR or BAR), and 8 (HVR or BVR).
- W3(R) Independent variable. Values for each curve range from -1.0 to 0.0 or from 0.0 to 1.0 inclusive. The variable is v/a for W2(I) = 1, 3, 5, or 7 and a/v for W2(I) = 2, 4, 6, or 8. If the tabular data does not span the entire range of the independent variable, end point values are used for data outside the table. This usually leads to incorrect pump performance data. Thus, entering data to cover the complete range is recommended.
- W4(R) Dependent variable. The variable is h/a^2 or b/a^2 for W2(I) = 1, 3, 5, or 7 and h/v^2 or b/v^2 for W2(I) = 2, 4, 6, or 8. The variables are $h = \frac{H}{H_{\text{rated}}}$, $a = \frac{\omega}{\omega_{\text{rated}}}$, $b = \frac{\tau}{\tau_{\text{rated}}}$, and $v = \frac{Q}{Q_{\text{rated}}}$.

Additional pairs as needed are entered on this or following cards, up to a limit of 100 pairs.

A9.18.17 Pump Two-Phase Multiplier Tables, CCC3000-3199

These cards are required only if W2 of card CCC0301 is 0; cards CCC3201-3299 are for the pump head multiplier table and cards CCC3100-3199 are for the pump torque multiplier table.

W1(I) Extrapolation indicator. This is not used, enter 0.

W2(R) Void fraction.

W3(R) Head or torque difference multiplier depending on table type.

Additional pairs of data as needed are entered on this or additional cards as needed, up to a limit of 100 pairs. Void fractions must be in increasing order.

A9.18.18 Pump Initial Noncondensable Mass Fractions, CCC3201

This card is optional for a pump volume and is related to card 110. If card CCC3201 is not supplied, then the values on the card 115 are used for the noncondensable mass fractions. The number of words on card CCC3201 must equal the number of words on card 110. The mass fractions must sum to 1.0 to within a tolerance of $1.0\text{E}-10$.

W1-WN(R) Initial mass fraction for each noncondensable gas type as specified on card 110 where N is the number of gas types on card 110.

A9.18.19 Pump Two-Phase Difference Tables, CCC4100-5699

These cards are required only if W3 of card CCC0301 is 0. The two-phase difference tables are homologous curves entered in a similar manner to the single-phase homologous data. Card numbering is CCC4100-4199 for the first curve, CCC4200-4299 for the second curve, ... CCC5600-5699 for the sixteenth curve. Data are the same as the data for the single-phase data except that the dependent variable is the difference between single-phase and fully degraded two-phase data.

A9.18.20 Relative Pump Motor Torque Data, CCC6001-6099

These cards are required only if W4 of card CCC0301 is 0. If the pump velocity table is not being used and these cards are present, the torque-inertia equation is used. When the electrical power is supplied to the pump motor (the pump trip is off), the net torque is computed from the rated pump motor torque times the relative pump motor torque from this table and the torque from the homologous data. If the electrical power is disconnected from the pump (the pump trip is on), the pump motor torque is 0.

W1(R) Pump velocity (rad/s, rev/min).

W2(R) Relative pump motor torque.

Additional pairs as needed are added on this or additional cards, up to a maximum of 100 pairs.

A9.18.21 Time-Dependent Pump Velocity Control, CCC6100

This card is required only if W5 of card CCC0301 is 0. The pump velocity table, if present, has priority in setting the pump velocity over the pump motor trip, the pump motor torque data, and the torque-inertia equation.

W1(I) Pump velocity table trip number. If the pump velocity, i.e., pump speed, table trip number is 0, the pump velocity is always computed from this table. If the pump velocity table trip number is nonzero, this trip determines which table is to be used. If the pump velocity table trip is off, the pump velocity is set from the pump motor trip, the pump motor torque data, and the torque-inertia equation as if this pump velocity table had not been entered. If the pump velocity table trip is on, the pump velocity is computed from this pump velocity table.

If Words 2 and 3 are missing and the Word 1 is nonzero, the search variable in the pump velocity table is time and the search argument is time minus the trip time. If Words 2 and 3 are missing and Word 1 is 0, the search variable in the pump velocity table is time and the search argument is time.

W2(A) Alphanumeric part of variable request code. This quantity is optional. If present, this word and the next are a variable request code that specifies the search argument for the table lookup and interpolation. TIME can be selected and the search argument depends on the value entered for Word 1. If Word 1 is 0, the search argument is time, and if Word 1 is nonzero, the search argument is time minus the trip time.

W3(I) Numeric part of variable request code. This is assumed to be 0 if missing.

A9.18.22 Time-Dependent Pump Velocity, CCC6101-6199

These cards are required only if W5 of card CCC0301 is 0.

W1(R) Search variable. Units depend on the quantity selected for the search variable.

W2(R) Pump velocity (rad/s, rev/min).

Additional pairs as needed are added on this or additional cards, up to a maximum of 100 pairs. Time values must be in increasing order.

A9.19 Multiple Junction Component, MTPLJUN

A multiple junction component is indicated by MTPLJUN for Word 2 on card CCC0000.

The one or more junctions specified by this component can connect volumes in the same manner as several single-junction components except that all the volumes connected by the junctions in the

component must be in the same hydrodynamic system. If this restriction is violated, corrective action is to merge the hydrodynamic systems. For major edits, minor edits, and plot variables, the junctions in the multiple junction component are numbered CCCJJKK00, where KK is the set number and JJ is the junction number within the set. The quantity KK may be 01-99; JJ is 01 for the first junction described in a set and incremented by 1 for each additional junction ($01 \leq JJ \leq 99$).

A9.19.1 Multiple Junction Information, CCC0001

This card is required.

- W1(I) Number of junctions, **nj**. This number must between 1 and 99.
- W2(I) Initial condition control. This word is optional and, if missing, is assumed to be 0. If 0 is entered, the initial conditions on cards CCC1KKM are velocities; if 1 is entered, the initial conditions are mass flows.

A9.19.2 Multiple Junction Geometry, CCC0KKM

These cards are required. Junctions are described by one or more sets of data, KK being the set number and M being the card number within a set. The junctions are numbered as CCCJJKK00, where JJ is 01 for the first junction described in a set and increments by 1 for each additional junction. The quantity KK may be 01-99, and M may be 1-9. cards are processed by increasing set number KK, and cards within a set by increasing M. Neither KK or M need be strictly consecutive.

- W1(I) From connection code to a component. This refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction (see Section 2.1). The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, N is restricted to 1 or 2.
- W2(I) To connection code to a component. This refers to the component at which the junction coordinate direction ends. See the description for W1 above.
- W3(R) Area (m^2 , ft^2). If 0, the area is set to the minimum area of the adjoining volumes. For abrupt area changes, the area must be equal to or smaller than the minimum of the adjoining volume areas. For smooth area changes, there are no restrictions.
- W4(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. A variable loss coefficient may be specified (see Section A9.19.5). The interpretation and use of this loss coefficient vary based upon the “a” junction control

flag selected in W6. If the smooth area change option is selected ($\underline{a} = 0$), the entry here is the loss coefficient applied to the junction flow area resulting from the entry made in W3. If the abrupt area change option is selected ($\underline{a} = 1$ or 2), the loss coefficient entered here is adjusted by the square of the ratio of the junction area and the minimum flow area of the adjacent connected volumes. This adjusted loss coefficient is applied to the junction flow area. If the $\underline{a} = 1$ option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

Example: Two volumes with volume average flow areas of 0.3 m^2 and 0.4 m^2 are connected with a junction ($A_{\text{jun}} = 0.075 \text{ m}^2$). The user supplied $K_{\text{loss}_{\text{user}}}$ is 10.0. For the smooth area change option ($\underline{a} = 0$), the $K_{\text{loss}_{\text{user}}}$ of 10.0 is applied directly to the junction. For the abrupt area change option ($\underline{a} = 1$ or 2), the K_{loss} applied at the junction is $K_{\text{loss}_{\text{jun}}} = K_{\text{loss}_{\text{user}}} * (A_{\text{jun}}/A_{\text{min}})^2$. For this example, $K_{\text{loss}_{\text{jun}}} = 10.0 * (0.075/0.3)^2 = 0.625$.

- W5(R) Reynolds number independent reverse flow energy loss coefficient, A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. A variable loss coefficient may be specified (see Section A9.19.5). See information for the previous word regarding interpretation and use of this loss coefficient.
- W6(I) Junction control flags. This word has the format jefvcahs. It is not necessary to input leading zeros. For the multiple junction, the junction control flag is limited to the format 0ef0cahs.
- e-flag The digit e specifies the modified PV term in the energy equations. $\underline{e} = 0$ means that the modified PV term will not be applied, and $\underline{e} = 1$ means that it will be applied.
- f-flag The digit f specifies CCFL options. $\underline{f} = 0$ means that the CCFL model will not be applied, and $\underline{f} = 1$ means that the CCFL model will be applied.
- c-flag The digit c specifies choking options. $\underline{c} = 0$ means that the choking model will be applied, and $\underline{c} = 1$ means that the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske or Moody (option 54) critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active.
- a-flag The digit a specifies area change options. $\underline{a} = 0$ means a smooth area change and only the user supplied K_{loss} is applied. $\underline{a} = 1$ means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. $\underline{a} = 2$ means a partial abrupt area change model and it is the same as

a = 1 except no code calculated forward and reverse contraction/expansion K_{loss} is applied. See discussion for W4(R) regarding the K_{loss} flow area reference, which varies based on the a-flag option selected here.

h-flag The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two-velocity momentum equations) option; h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit printout will show h = 1.

s-flag The digit s specifies momentum flux options. s = 0 uses momentum flux in both the to and from volume. s = 1 uses momentum flux in the from volume, but not in the to volume. s = 2 uses momentum flux in the to volume, but not in the from volume. s = 3 does not use momentum flux in either the to volume or the from volume.

W7(R) The meaning of this word is different, depending on the critical flow model which has been selected (see W6).

If the Henry-Fauske or Moody (option 54) critical flow model is active, the discharge coefficient is entered. The recommended default value for this coefficient is 1.0.

If the original RELAP5 critical flow model is active, the subcooled discharge coefficient is entered. This entry applies only to subcooled liquid choked flow calculations. The entry must be > 0.0 and < 2.0. The recommended default value for this coefficient is 1.0.

W8(R) The meaning of this word is different, depending on the critical flow model which has been selected (see W6).

If the Henry-Fauske or Moody critical flow model is active, the thermal nonequilibrium constant is entered. The recommended default value for this coefficient is 0.14. If the entry is < 0.01, the equilibrium option is used and the entry is reset to 0.0. If the entry is > 1000.0, the frozen option is used and the entry is reset to 100.0.

If the original RELAP5 critical flow model is active, the two-phase discharge coefficient is entered. This entry applies only to two-phase choked flow calculations. The entry must be > 0.0 and < 2.0. The recommended default value for this coefficient is 1.0.

W9(R) The meaning of this word is different, depending on the critical flow model which has been selected (see W6).

If the Henry-Fauske or Moody critical flow model is active, enter 0.0 (this word is not used).

If the original RELAP5 critical flow model is active, the superheated discharge coefficient is entered. This entry applies only to superheated vapor choked flow calculations. The entry must be > 0.0 and < 2.0 . The recommended default value for this coefficient is 1.0.

- W10(I) From volume increment. Words 1 and 2 contain the from and to connection codes respectively for the first junction defined by the set. If the set defines more than one junction, connection codes for the following junctions are given by the connection code of the previous junction plus the increments in Words 10 and 11. The increments may be positive, negative, or 0. Junctions are defined up to the limit in Word 13. Words 3-9 apply to all junctions defined by the set. If additional sets are entered, Words 1 and 2 apply to the next junction, and increments are applied as with the first set. Word 13 for the second and following sets must be greater than Word 13 of the preceding set, and Word 13 of the last set must equal n_j . A new set is used whenever a new increment is needed, Words 3-9 need to be changed, or a change in junction numbering is desired.
- W11(I) To volume increment. See description for Word 10.
- W12(I) Enter 0. This is reserved for future capability.
- W13(I) Junction limit. Described above.

A9.19.3 Multiple Junction Initial Condition, CCC1KKM

These cards are required. Initial velocities are entered using one or more sets of data. The processing of sets of data is identical to that described in Section A9.19.2 except that there need be no relationship in the division of junctions within sets between these cards (CCC1KKM) and the multiple junction geometry cards (CCC0KKM). Likewise, these cards do not affect the numbering of the junctions.

- W1(R) Initial liquid velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s), depending on control Word 2 of card CCC0001.
- W2(R) Initial vapor velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s), depending on control Word 2 of card CCC0001.
- W3(I) Junction limit number.

A9.19.4 Multiple Junction Diameter and CCFL Data, CCC2KKM

These cards are optional. The defaults indicated for each word are used if the card is not entered. If the card is being used to specify only the junction hydraulic diameter for the interphase drag calculation (i.e., $\underline{f} = 0$ in Word 6 of cards CCC0KKM), then the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (will not be used). If this card is being used for the CCFL model (i.e., $\underline{f} = 1$ in Word 6 of cards CCC0KKM), then enter all four words for the appropriate CCFL model if values different from the default values are desired. The processing of sets of data is identical to

that described in Section A9.19.2 except that there need be no relationship in the division of junctions within sets between these cards (CCC1KKM) and the multiple junction geometry cards (CCC0KKM). Likewise, these cards do not affect the numbering of the junctions.

- W1(R) Junction hydraulic diameter, D_j (m, ft). This is the junction hydraulic diameter used in the CCFL correlation equation and interphase drag and must be ≥ 0 . This number should be computed from $4.0 \cdot \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If a 0 is entered or if the default is used, the junction diameter is computed from $2.0 \cdot \left(\frac{\text{area}}{\pi} \right)^{0.5}$. See Word 3 of card CCC0NNM for area.
- W2(R) Flooding correlation form, β . If 0, the Wallis CCFL form is used. If 1.0, the Kutateladze CCFL form is used. If between 0.0 and 1.0, Bankoff weighting between the Wallis and Kutateladze CCFL forms is used. This number must be ≥ 0 and ≤ 1 . The default value is 0 (Wallis form).
- W3(R) Gas intercept, c . This is the gas intercept used in the CCFL correlation (when $H_f^{1/2} = 0$) and must be > 0 . The default value is 1.
- W4(R) Slope, m . This is the slope used in the CCFL correlation and must be > 0 . The default value is 1.
- W5(I) Junction limit number.

A9.19.5 Multiple Junction Form Loss Data, CCC3KKM

These cards are optional. The user-specified form loss is given in Words 4 and 5 of card CCC0KKM if this card is not entered. If this card is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F \text{Re}^{-C_F}$$

$$K_R = A_R + B_R \text{Re}^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 4 and 5 of card CCC0KKM. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all five words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R .

- W1(R) B_F (≥ 0). This quantity must be greater than or equal to 0.

W2(R)	$C_F (\geq 0)$. This quantity must be greater than or equal to 0.
W3(R)	$B_R (\geq 0)$. This quantity must be greater than or equal to 0.
W4(R)	$C_R (\geq 0)$. This quantity must be greater than or equal to 0.
W5(I)	Junction limit number.

A9.20 Accumulator Component, ACCUM

An accumulator component is indicated by ACCUM for Word 2 on card CCC0000. For major edits, minor edits, and plot variables, the volume in the accumulator component is numbered CCC010000, and the junction in the accumulator component is numbered CCC010000.

An accumulator is a lumped parameter component treated by special numerical techniques that model both the tank and surpline until the accumulator is emptied of liquid. When the last of the liquid leaves the accumulator, the code automatically resets the accumulator to an equivalent single-volume with an outlet junction and proceeds with calculations using the normal hydrodynamic solution algorithm.

In the following input requirements, it is assumed that the component is an accumulator in which liquid completely fills the surge line but may or may not occupy the tank. It is further assumed that the accumulator is not initially in the injection mode. Hence, the initial pressure must be input lower than the injection point pressure, including elevation head effects. The junction initial conditions may not be input i.e., initial liquid and gas velocities are set to 0 in the code. The noncondensable gas in the accumulator is nitrogen and that the gas and liquid are initially in equilibrium. No other junctions should be connected to the accumulator volume. The geometry of the tank may be cylindrical or spherical. The standpipe/surge line inlet refers to the end of the pipe inside the tank itself (see Section 2.4.13 of Volume II).

A9.20.1 Accumulator Volume Geometry, CCC0101-0109

This card is required.

W1(R)	Area (m^2 , ft^2). This is the flow area of a cylindrical tank, or the maximum flow area of a spherical tank. In the case of a spherical tank, the flow area and the tank radius are related by the formula $A = \pi R^2$.
W2(R)	Length (m, ft). This is the length of the tank above the standpipe/surgeline inlet, where this inlet refers to the end of the pipe inside the tank itself.
W3(R)	Volume (m^3 , ft^3). This is the volume of the tank above the standpipe/surgeline inlet, where this inlet refers to the end of the pipe inside the tank itself. The code requires that the volume, area, and length are consistent. For a cylindrical tank, $W3 = W1 \bullet W2$, and at least two of the three quantities, W1, W2 or W3, must be nonzero. If one of the quantities is 0, it will be computed from the other two. For a spherical tank, W1 and W2 must be nonzero.

If W3 is 0, it will be computed from the other two. If none of the words are 0, they must satisfy the consistency condition within a relative error ± 0.000001 .

- W4(R) Azimuthal angle (degrees). The absolute value of this angle must be ≤ 360 degrees. This quantity is not used in the calculation but is specified for possible automated drawing of nodalization diagrams.
- W5(R) Inclination angle (degrees). Only +90 or -90 degrees is allowed. The accumulator is assumed to be a vertical tank with the standpipe/surgeline inlet (where this inlet refers to the end of the pipe inside the tank itself) at the bottom. This angle is used in the interphase drag calculation.
- W6(R) Elevation change (m, ft). This is the elevation change from the standpipe/surgeline inlet (where this inlet refers to the end of the pipe inside the tank itself) to the top of the tank. A positive value is an increase in elevation. The absolute value of this quantity must be nonzero, less than or equal to the length, and have the same sign as the angle for vertical orientation. See Section 2.4.1 of Volume II for further discussion. As with other components, this Word 6 is compared to the length (Word 2) to decide if the horizontal or vertical flow regime map is used. This is not important for this component, since the correlations that depend on the flow regime map are not needed for this component. The volume conditions are determined from the accumulator's special model.
- W7(R) Wall roughness (m, ft).
- W8(R) Hydraulic diameter (m, ft). This should be computed from $4.0 \bullet \left(\frac{\text{area}}{\text{wetted perimeter}} \right)$. If 0, the hydraulic diameter of the tank is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi} \right)^{0.5}$. A check is made that the pipe roughness is less than half the hydraulic diameter of the tank. See Word 1 for the area.
- W9(I) Volume control flags. This word has the format tlpvbfe. It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags. Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f-flag is the only coordinate direction oriented flag. This word enters the scalar oriented flags and the x-coordinate flag. The accumulator component forces all volume flags except for the x-coordinate f digit, and y- and z-coordinate flags are not read. For the accumulator volume control flag, the format is restricted to 00110f0.

- f-flag The digit $\underline{f}$ specifies whether wall friction is to be computed. $\underline{f} = 0$ specifies the wall friction effects are to be computed along the x-coordinate; and $\underline{f} = 1$ specifies the friction effects are not to be computed along the x-coordinate.
- W10(I) Geometry flag (optional). To specify a cylindrical tank, set the flag equal to 0 (default). To specify a spherical tank, set the flag equal to 1.

A9.20.2 Accumulator Additional Laminar Wall Friction, CCC0131

This card is optional. and if this card is not entered, the default values are 1.0 for the laminar shape factor and 0.0 for the viscosity ratio exponent.

- W1(R) Shape factor for x-coordinate.
- W2(R) Viscosity ratio exponent for x-coordinate.

A9.20.3 Accumulator Alternate Wall Friction Data, CCC0141

This card is optional. The input on this card, which does not distinguish between forward and reverse loss coefficients, has been superseded by the CCC1101-1102 cards. If this card is entered, the form loss coefficient is calculated from

$$K = A + BRe^{-C}$$

where K is the form loss coefficients. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all three words.

- W1(R) A (≥ 0). This quantity must be greater than or equal to 0.
- W2(R) B (≥ 0). This quantity must be greater than or equal to 0.
- W3(R) C (≥ 0). This quantity must be greater than or equal to 0.

A9.20.4 Accumulator Tank Initial Thermodynamics Conditions, CCC0200

This card is required.

- W1(R) Pressure (Pa, lb_f/in^2).
- W2(R) Temperature (K, $^{\circ}\text{F}$).
- W3(R) Boron concentration (mass of boron per mass of liquid). This word is optional.

A9.20.5 Accumulator Junction Geometry, CCC1101

This card is required.

- W1(I) To connection code to a component. The from connection is not entered, since it is always from the accumulator. The to connection code refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. A nonzero N specifies the expanded format. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction (see Section 2.1 of Volume II). The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, N is restricted to 1 or 2.
- W2(R) Area (m^2 , ft). This is the average area of the surpline and standpipe.
- W3(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. Because the smooth area change model (control flag a = 0, see W5) must be used for the junction in the ACCUM component, the loss coefficient entered here should be consistent with the junction flow area entered in W2.
- W4(R) Reynolds number independent reverse flow energy loss coefficient, A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. See information for the previous word regarding interpretation and use of this loss coefficient.
- W5(I) Junction control flags. This word has the format jefvcahs. It is not necessary to input leading zeros. For the accumulator, the junction control flag is restricted to the format 0000c0hs.
- The accumulator model automatically disables the following flags as long as liquid remains in the accumulator. However, when the accumulator empties of liquid, the model is automatically converted to an active normal volume. The following flags are then enabled and used as defined.
- c-flag The digit c specifies choking options. c = 0 means that the choking model will be applied, and c = 1 means the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske or Moody (option 54) critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active. Unlike junctions for certain other hydrodynamic components, only the default values of discharge coefficients and

factors used in the critical flow models are allowed for the junction in the ACCUM component. Problem renodalization should be used to circumvent this limitation should it prove to be significant. For the original RELAP5 critical flow model, the default values for the subcooled, two-phase and superheated discharge coefficients each are 1.0. For the Henry-Fauske or Moody critical flow model, the default value for the discharge coefficient is 1.0 and the default value for the thermal nonequilibrium constant is 0.14.

- h-flag The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two-velocity momentum equations) option; h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit will show h = 1.
- s-flag The digit s specifies momentum flux options. s = 0 uses momentum flux in both the to volume and the from volume. s = 1 uses momentum flux in the from volume, but not in the to volume. s = 2 or 3 is not allowed for an accumulator.

A9.20.6 Accumulator Junction Form Loss Data, CCC1102

This card is optional. The user-specified form loss is given in Words 3 and 4 of card CCC1101 if this card is not entered. If this card is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F \text{Re}^{-C_F}$$

$$K_R = A_R + B_R \text{Re}^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficients. A_F and A_R are the Words 3 and 4 of card CCC1101. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all four words for the appropriate expression.

W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.

W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.

W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.

W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

A9.20.7 Accumulator Miscellaneous Parameters, CCC2200

This card is required.

W1(R)	Liquid volume in tank (m^3 , ft^3). This is the volume of water contained in the tank above the standpipe/surgeline inlet (where this inlet refers to the end of the pipe inside the tank).
W2(R)	Liquid level in tank (m, ft). This is the liquid level of water contained in the tank above the standpipe/surgeline inlet (where this inlet refers to the end of the pipe inside the tank) entrance. For a cylindrical tank, either W1 or W2 must be specified as nonzero. For a spherical tank, W2 must be specified as nonzero. If one of the words is 0, it is computed from the other word.
W3(R)	Length of surgeline and standpipe (m, ft). If input as 0, then the surgeline and standpipe are not modeled.
W4(R)	Elevation drop of surgeline and standpipe (m, ft). This is the elevation drop from the standpipe/surgeline inlet (where this inlet refers to the end of the pipe inside the tank) entrance to the injection point. A positive number denotes a decrease in elevation.
W5(R)	Tank wall thickness (m, ft). This is not allowed to be 0.
W6(I)	Heat transfer flag. If 0, heat transfer will be calculated. If 1, no heat transfer will be calculated.
W7(R)	Tank density (kg/m^3 , lb/ft^3). If 0, the density will default to that for carbon steel.
W8(R)	Tank volumetric heat capacity ($\text{J}/\text{kg}\cdot\text{K}$, $\text{Btu}/\text{lb}\cdot^\circ\text{F}$). If 0, the heat capacity will default to that for carbon steel.
W9(I)	Trip number. If 0 or if no number is input, then no trip test is performed. If nonzero then this must be a valid trip number, the operations performed are similar to those performed for a trip valve. If the trip is false, then the accumulator is isolated and no flow through the junction can occur. If the trip is true, then the accumulator is not isolated and flow through the junction will occur in the normal manner for an accumulator.

A10 HEAT STRUCTURES

These are used in NEW and RESTART type problems and are required only if heat structures are described. The heat structure card numbers are of the form 1CCCGXNN where the fields are defined as follows:

- CCC is a heat structure number. The heat structure numbers need not be consecutive. We suggest, but the system does not require, that if heat structures and hydrodynamic volumes are related, they be given the same number.
- G is a geometry number. The combination CCCG is a heat structure geometry combination referenced in the heat structure input data. The G digit is provided to differentiate between different types of heat structures (such as fuel pins and core barrel) that might be associated with the same hydrodynamic volume.
- X is the card type.
- NN is the card number within a card type.

A10.1 General Heat Structure Data, 1CCCG000

This card is required for heat structures. Use 8 words for new data input or 1 word for deleting a heat structure.

- | | |
|-------|---|
| W1(I) | Number of axial heat structures with this geometry, nh. This number must be > 0 and < 100. |
| W2(I) | Number of radial mesh points for this geometry, np. This number must be < 100. Enter > 1 if no reflood is specified, and > 2 if reflood or metal water reaction is specified. |
| W3(I) | Geometry type. Enter 1 for rectangular, 2 for cylindrical, and 3 for spherical. Spherical geometry is not allowed if reflood is specified. Cylindrical geometry must be specified when the gap conductance model is used. |
| W4(I) | Steady-state initialization flag. Use 0 if the desired initial condition temperatures are entered on input cards 1CCCG401-G499; use 1 if the steady-state initial condition temperatures are to be calculated by the code. If option 1 is chosen, the user is still required to enter temperatures on cards 1CCCG402-G499. In this case, the temperatures are used as starting points for the steady-state solutions. The user should therefore enter temperatures either below or above the minimum film boiling point to assure the respective pre-DNB or post-DNB steady-state condition is calculated. This is because the boiling curve is multi-valued. |
| W5(R) | Left boundary coordinate (m, ft). |
| W6(I) | Reflood condition flag. This quantity is optional if no reflood calculation is to be performed. This quantity may be 0, 1, 2, or a trip number. |

If 0, no reflood calculation is to be performed.

If nonzero, all the heat structures in this heat structure/geometry are assumed to form a two-dimensional representation of a fuel pin. The radial mesh is defined on card 1CCCG1NN. Each heat structure represents an axial level of the fuel pin, with the first heat structure being the bottom level. Each heat structure should be connected to a hydrodynamic volume representing the same axial section of the coolant channel. The length of the axial mesh in the fuel pin is given by the height of the connected hydrodynamic volume. If the heat structure is fuel pins or heat exchanger tubes, the length factor (Word 5 on cards 1CCCG501-G509) is the product of the hydrodynamic volume length and the number of pins or tubes (see Section 3.2). The heat structures represent the temperatures at the midpoint of the axial mesh. Once the reflood calculation is initiated, additional mesh lines are introduced at each end of the fuel pin and between the heat structures. Once the reflood calculation is initiated, it remains activated, and the two-dimensional heat conduction calculation uses a minimum of $2nh + 1$ axial mesh nodes. Additional mesh lines are introduced and later eliminated as needed to follow the quench front.

If 1 is entered, the reflood calculation is initiated in this heat structure geometry when the average pressure in the connected hydrodynamic volumes is less than 1.2×10^6 Pa, and the average void fraction in the interconnected hydrodynamic volumes is greater than 0.9 (i.e., nearly empty).

If 2 is entered, the reflood calculation is initiated in this heat structure geometry when the average pressure in the connected hydrodynamic volumes is less than 1.2×10^6 Pa and the average void fraction in the interconnected hydrodynamic volumes is greater than 0.1 (i.e., dryout begins).

If a trip number is entered, the reflood calculation is initiated when the trip is set true. When using the expanded trip number format, 1 and 2 are possible trip numbers. A 1 or 2 entered in this word is not treated as a trip number.

W7(I) Boundary volume indicator. Enter 0 or 1 to indicate that reflood heat transfer applies to the left or right boundary, respectively.

W8(I) Maximum number of axial intervals. Enter 2, 4, 8, 16, 32, 64, or 128 to indicate the maximum number of axial subdivisions a heat structure can have. Storage is allocated for the number indicated, even though a transient may not require that level of subdivision.

A10.2 Heat Structure Deletion, 1CCCG000

This card can only be entered for RESTART problems. If entered, all heat structures associated with the heat structure geometry number CCCG are deleted.

W1(A) Enter DELETE.

A10.3 Gap Data, 1CCCG001

This card is optional and is needed only if the gap conductance model is to be used. If the card is entered, Word 1 of card 1CCCG100 must be 0, cards 1CCCG011-G099, and cards 1CCCG201-G299 are required. Word 2 of card 201MMM00 must be 3, and a table of the gas component name and mole fraction must be specified in the gap material data (cards 201MMM01-49).

- W1(R) Initial gap internal pressure (Pa, lb_f/in^2).
- W2(I) Gap conductance reference volume. This word is required. The pressure of the gas in a fuel pin for the gap conductance model is given by $P(t) = (P(0)T(t))/(T(0))$, where $P(t)$ is the pressure in the fuel pin and $T(t)$ is the temperature in the reference volume. $P(0)$ is Word 1 above, and $T(0)$ is the initial value, if the volume is also being defined with these input data or the value from the restart block. The reference volume is usually the hydrodynamic volume (i.e., the nine-digit number CCCNN0000) most closely associated with the nonfuel region in a fuel pin at the top of a stack of fuel pellets.

A10.4 Metal-Water Reaction Control, 1CCCG003

This card is optional, and if this card is not present, no metal-water reaction will be calculated. The initial oxide thickness is assumed to be 0 on the inner surface. It remains 0 unless cladding rupture occurs. CCCG is a heat structure geometry number.

- W1(R) Initial oxide thickness on cladding's outer surface (m, ft).
- W2(I) Hydrogen addition control word. This word is optional and defaults to 0 if not entered. If 0, hydrogen from metal-water reaction will not be added to the noncondensables in the attached volume. If 1, hydrogen from metal-water reaction will be added to the noncondensables in the attached volume.
- W3(R) Coefficient k_{ocp} in oxide thickness equation. This word is optional and defaults to $2.252\text{e-}6 \text{ m}^2/\text{s}$ if not entered. See Section 4.13 in Volume 1. The Baker-Just value for k_{ocp} , $7.9848\text{e-}5 \text{ m}^2/\text{s}$, can be obtained by using the CANDU option, Card 1, Option 81 (m^2/s , ft^2/s).
- W4(R) Coefficient a_{ocp} in oxide thickness equation. This word is optional and defaults to 18062.0 K if not entered. See Section 4.13 in Volume 1. The Baker-Just value for a_{ocp} , 22900.0 K, can be obtained by using the CANDU option, Card 1, Option 81 (K, R).

A10.5 Fuel Cladding Deformation Model Control, 1CCCG004

If this card is optional, and if it is not present, no deformation calculations will be done. If this card is present, then card 1CCCG001 must also be present. CCCG is a heat structure geometry number.

W1(I) Form loss factor flag. Enter 0 if no additional form loss factors are to be calculated after a rod ruptures. Enter 1 if additional form loss factors are to be calculated. Either a 0 or a 1 must be entered.

A10.6 Gap Deformation Data, 1CCCG011-G099

These cards are required for the gap conductance model only. The card format is sequential format, five or six words per set, describing nh heat structures. The user also needs to enter a table containing gas component name and mole fraction on the 201MMM01-49 cards. These cards can be entered in either 5-word or 6-word format as shown below.

A10.6.1 5-Word Format

W1(R) Fuel surface roughness (m, ft). This number must be ≥ 0 . An appropriate value is 10^{-6} m. A negative entry is reset to 10^{-6} m with no errors.

W2(R) Cladding surface roughness (m, ft). This number must be either positive or 0. An appropriate value is 2×10^{-6} m. A negative entry is reset to 2×10^{-6} m with no errors.

W3(R) Radial displacement due to fission gas-induced fuel swelling and densification (m, ft). This number must be ≥ 0 . A negative entry is reset to 0. An appropriate value can be obtained from calculations using FRAPCON-2 or FRAP-T6.

W4(R) Radial displacement due to cladding creepdown (m, ft). The value is normally negative. A positive entry is reset to 0. An appropriate value can be obtained from calculations using FRAPCON-2 or FRAP-T6.

W5(I) Heat structure number.

A10.6.2 6-Word Format

W1(R) Fuel surface roughness (m, ft). This number must be ≥ 0 . An appropriate value is 10^{-6} m. A negative entry is reset to 10^{-6} m with no errors.

W2(R) Cladding surface roughness (m, ft). This number must be either positive or 0. An appropriate value is 2×10^{-6} m. A negative entry is reset to 2×10^{-6} m with no errors.

W3(R) Radial displacement due to fission gas-induced fuel swelling and densification (m, ft). This number must be ≥ 0 . A negative entry is reset to 0. An appropriate value can be obtained from calculations using FRAPCON-2 or FRAP-T6.

- W4(R) Radial displacement due to cladding creepdown (m, ft). The value is normally negative. A positive entry is reset to 0. An appropriate value can be obtained from calculations using FRAPCON-2 or FRAP-T6.
- W5(R) Fuel-cladding centerline shift ratio ($0 \leq r \leq 1$). This variable is used in the modified gap conductance model for the new method of computing gap width only when Card 1, Option 28 is turned on.
- W6(I) Heat structure number.

A10.7 Heat Structure Mesh Flags, 1CCCG100

This card is required for heat structure input.

- W1(I) Mesh location flag. If 0, geometry data, including mesh interval data, composition data, and source distribution data, are entered with this heat structure input. If nonzero, that information is taken from the geometry data from the heat structure geometry (CCCG) number in this word. If this word is nonzero, the remaining geometry information described in Section A10.7-Section A10.10 is not entered.
- W2(I) Mesh format flag. This word is needed only if Word 1 is 0, though no error occurs if it is present when Word 1 is nonzero. The mesh interval data are given as a sequence of pairs of numbers in one of two formats to be used in cards 1CCCG101-G199. If this word is 1 (format 1 on cards 1CCCG101-G199), the pairs of numbers contain the number of intervals in this region and the right boundary coordinate. For the first pair, the left coordinate of the region is the left boundary coordinate previously entered in Word 5 of card 1CCCG000; for succeeding pairs, the left coordinate is the right coordinate of the previous pair. If this word is 2 (format 2 on cards 1CCCG101-G199), the format is a sequential expansion of mesh intervals; i.e., the distance in Word 1 on cards 1CCCG101-G199 is used for each interval starting from the leftmost, as yet unspecified, interval to and including the interval number specified in Word 2.

A10.8 Format 1 Number of Mesh Intervals, 1CCCG101-G199

These cards are required if Word 1 of card 1CCCG100 is 0. The sum of the numbers of intervals must be np-1. The card numbers need not be sequential.

- W1(I) Number of intervals. Enter the number of intervals, not the interval number.
- W2(R) Right coordinate (m, ft).

A10.9 Format 2 Mesh Intervals, 1CCCG101-G199

These cards are required if Word 1 of card 1CCCG100 is 0. The sequential expansion must be for np-1 intervals. The card numbers need not be sequential.

W1(R) Mesh interval (m, ft.)

W2(I) Interval number.

A10.10 Heat Structure Composition Data, 1CCCG201-G299

These cards are required if Word 1 of card 1CCCG100 is 0 and must not be entered otherwise. The card format is two numbers per set in sequential expansion format for np-1 intervals. The card numbers need not be in sequential order.

W1(I) Composition number. The absolute value of this quantity is the composition number, and it must be identical to the subfield MMM used in Heat Structure Thermal Property Data, Section A12. The sign indicates whether the region over which this composition is applied is to be included or excluded from the volume averaged temperature computation. If positive, the region is included; if negative, the region is not included. The option to exclude regions from the volume averaged temperature integration is to limit the integration to fuel regions only for use in reactivity feedback calculations. Gap and cladding regions should not be included in this case. If the gap conductance model is used, only one interval can be used for the gap model.

W2(I) Interval number.

A10.11 Fission Product Decay Heat Flag, 1CCCG300

This card is required if fission product decay heat is used on this heat structure. This card sets the fission product decay heat flag. The code will then treat card 1CCCG301 as a gamma attenuation coefficient card. This card is not needed if fission product decay heat is not used on this heat structure.

W1(A) DKHEAT.

A10.12 Heat Structure Source Distribution Data, 1CCCG301-G399

These cards are required if Word 1 of card 1CCCG100 is 0 and must not be entered otherwise. The card format is two numbers per set in sequential expansion format for np-1 intervals. The card numbers need not be in sequential order. Radial power peaking factors are entered here.

W1(R) Source value. These are relative values only and can be scaled by any factor without changing the results. By entering different values for the various mesh intervals, a characteristic shape of a power curve can be described.

W2(I) Mesh interval number.

If card 1CCCG300 is entered, then card 1CCCG301 is treated as a gamma attenuation coefficient card.

W1(R) Gamma attenuation coefficient. These are values dependent on the heat structure material. A value of 50 is recommended for stainless steel.

W2(I) Mesh interval number.

A10.13 Initial Temperature Flag, 1CCCG400

This card is optional; if missing, Word 1 is assumed to be 0.

W1(I) Initial temperature flag. If this word is 0 or -1, initial temperatures are entered with the input data for this heat structure geometry. If greater than 0, initial temperatures for this heat structure geometry are taken from the heat structure geometry number in this word, and the initial temperature distribution from Section A10.12 is not needed.

A10.14 Format 1 Initial Temperature Data, 1CCCG401-99

These cards are required if Word 1 of card 1CCCG400 is 0. One temperature distribution is entered; and the same distribution is applied to all of the nh heat structures. The card format is two numbers per set in sequential expansion format for np mesh points

W1(R) Temperature (K, °F).

W2(I) Mesh point number.

A10.15 Format 2 Initial Temperature Data, 1CCCG401-499

These cards are required if Word 1 of card 1CCCG400 is -1. A separate temperature distribution must be entered for each of the nh heat structures. The distribution for the first heat structure is entered on card 1CCCG401, the distribution for the second heat structure is entered on card 1CCCG402, and the remaining distributions are entered on consecutive card numbers. Continuation cards can be used if the data do not fit on one card.

W1-WNP(R) Temperature (K, °F). Enter the np mesh point temperatures in order from left to right.

A10.16 Left Boundary Condition, 1CCCG501-599

These cards are required. The boundary condition data for the heat structures with this geometry are entered in a slightly modified form of sequential expansion using six quantities per set for the number of heat structures with this geometry (nh sets). The modification deals with Words 1 and 2.

W1(I) Boundary volume number or general table number. Enter a boundary volume number as a positive number and a general table number as a negative number.

If positive, this word specifies the hydrodynamic volume number (of the form CCCNN000F) or general table associated with the left surface of this heat structure. These are used to specify the sink temperature. If 0, no volume or general table is associated with the left surface of this heat structure, and a symmetry or insulated boundary condition is used (i.e., a 0 temperature gradient at the boundary), or a temperature of 0 is used for a surface temperature or a sink temperature in boundary conditions. A boundary volume number is entered as a positive number. If F is 0 or 4, the volume coordinate associated values such as average volume velocity are taken from the x-coordinate; if F is 2 or 1, volume coordinate associated values are taken from the y- or z-axes, respectively. Specifying a volume coordinate not in use is an input error.

If negative (-1 through -999), the negative of this word specifies a general table number.

W2(I) Increment. This word and Word 1 of this card are treated differently from the standard sequential expansion. Word 1 of the first set applies to the first heat structure of the heat structure geometry set. The increment (normally 10000) is added to Word 1, which results in the hydraulic cell number associated with heat structure 2; etc. The increment is applied up to the limit in Word 6 of a set. Word 1 of the next set applies to the next heat structure, and increments are applied as for the first set. The increment may be 0 or nonzero, positive or negative. If Word 1 is 0, this word should be 0. Additional examples are shown in Section 4 of Volume V.

W3(I) Boundary condition type.

If 0, a symmetry or insulated boundary condition is used (i.e., a 0 temperature gradient is used at the boundary). The boundary volume must be 0.

If 1 or 1nn, a convective boundary condition where the heat transfer coefficient obtained from Heat Transfer Package 1 is used. The allowed values of nn are shown in **Table 10.16-1**. The sink temperature is the temperature of the boundary volume. Word 1 must specify a boundary volume with this boundary condition type. The boundary volume cannot be a time-dependent volume.

There are several numbers allowed for Word 3 to activate convective boundary conditions for nonstandard geometries. A 1, 100, or 101 give the default values. The numbers 1, 100, and 101 use the same correlations. The number 101 is recommended; the numbers 1 and 100 are allowed so that the code is backwards compatible with previous input decks. The default convection and boiling correlations were derived mainly based on data from internal vertical pipe flow. Other possible input values are shown in **Table 10.16-1**. When modeling a vertical bundle, the rod or tube pitch-to-diameter ratio should be input on the

801 card. This has the effect of increasing the convective part of heat transfer such that users can input the true hydraulic diameter and get reasonable predictions.

Table 10.16-1 Card 501 Word 3 Convection Boundary Type

Word 3	Geometry Type
1, 100, 101	Default
102	Parallel plates (ORNL, ANS reactor; set gap and span on CCC3101-3199 hydro cards for pipes and CCC0111 hydro card for single volumes and branches, set b = 2 in volume control flag on CCC1001-1099 hydro cards for pipes and CCC0101-0109 hydro cards for single volumes and branches).
110	Vertical bundle without crossflow (set P/D on 801 card)
111	Vertical bundle with crossflow (set P/D on 801 card)
115	Parallel plates Large gap - uses Petukhov-Gnielinski heat transfer correlation with Swanson-Catton multiplier for mixed convection
124	CANDU fuel bundle element heat transfer
130	Flat plate above fluid
134	Horizontal bundle

If 115, a time constant (in seconds) for relaxing the heat transfer coefficient can be added in the front of the 115, e.g., 60115 will be parsed as a 60 second relaxation time constant for geometry type 115. The relaxation is of the type $1 - \exp\left(-\frac{t}{\tau}\right)$ where t is time and τ is the time constant entered as part of word 3.

If 1000, the temperature of the boundary volume or the temperature from the general table (as specified in Word 1) is used as the left surface temperature. If Word 1 is 0, the surface temperature is set to 0.

If 1xxx, the temperature in general Table xxx is used as the left surface temperature.

If 2xxx, the heat flux from Table xxx is used as the left boundary condition.

If 3xxx, a convective boundary condition is used where the heat transfer coefficient as a function of time is obtained from general Table xxx. The sink temperature is the temperature of the boundary volume or from the table specified in Word 1. If Word 1 is 0, the sink temperature is set to 0.

If 4xxx, a convective boundary condition is used where the heat transfer coefficient as a function of surface temperature is obtained from general Table xxx. The sink temperature is the temperature of the boundary volume or from the table specified in Word 1. If Word 1 is 0, the sink temperature is set to 0.

If reflood is specified, the left boundary condition type must be same for all nh heat structures and, similarly, for the right boundary condition type. The left and right boundary types need not be the same, but neither can be 1000 or 1xxx.

W4(I) Surface area code. If 0, Word 5 is the left surface area. If 1, Word 5 is (a) the surface area in rectangular geometry, (b) the cylinder height or equivalent in cylindrical geometry, or (c) the fraction of a sphere (0.5 is a hemisphere) in spherical geometry.

W5(R) Surface area or factor. As indicated in Word 4, this word contains the surface area (m^2 , ft^2) or a geometry dependent multiplier (m^2 , ft^2) for rectangular; (m, ft) for cylindrical; or dimensionless for spherical geometries). If the symmetry boundary condition is specified (Word 3 = 0), this word must still be entered nonzero.

W6(I) Heat structure number.

A10.17 Right Boundary Condition, 1CCCG601-G699

These cards are required. The left and right surface areas must be compatible with the geometry. The boundary condition data for the heat structures with this geometry are entered in a slightly modified form of sequential expansion using six quantities per set for the number of heat structures with this geometry (nh sets). The modification deals with Words 1 and 2.

W1(I) Boundary volume number or general table. This word specifies the hydrodynamic volume number (of the form CCCNN000F) or general table associated with the right surface of this heat structure. These are used to specify the sink temperature. If 0, no volume or general table is associated with the right surface of this heat structure, and a symmetry or insulated boundary condition is used (i.e., a 0 temperature gradient at the boundary), or a temperature of 0 is used for a surface temperature or a sink temperature in boundary conditions. A boundary volume number is entered as a positive number. If F is 0 or 4, the volume coordinate associated values such as average volume velocity are taken from the x-coordinate; if F is 2 or 1, volume coordinate associated values are taken from the y- or z-axes, respectively. Specifying a volume coordinate not in use is an input error. A general table is entered as a negative number (-1 through -999).

W2(I) Increment. This word and Word 1 of this card are treated differently from the standard sequential expansion. Word 1 of the first set applies to the first heat structure of the heat structure geometry set. The increment (normally 10000) is added to Word 1, which results in the hydraulic cell number associated with heat structure 2; etc. The increment is applied

up to the limit in Word 6 of a set. Word 1 of the next set applies to the next heat structure, and increments are applied as for the first set. The increment may be 0 or nonzero, positive or negative. If Word 1 is 0, this word should be 0. Additional examples are shown in Section 4 of Volume V.

W3(I) Boundary condition type.

If 0, a symmetry or insulated boundary condition is used (i.e., a 0 temperature gradient is used at the boundary). The boundary volume must be 0.

If 1 or 1nn, a convective boundary condition where the heat transfer coefficient obtained from Heat Transfer Package 1 is used. The allowed values of nn are shown in **Table 10.17-1**. The sink temperature is the temperature of the boundary volume. Word 1 must specify a boundary volume with this boundary condition type. The boundary volume cannot be a time-dependent volume.

There are several numbers allowed for Word 3 to activate convective boundary conditions for nonstandard geometries. A 1, 100, or 101 give the default values. The numbers 1, 100, and 101 use the same correlations. The number 101 is recommended; the numbers 1 and 100 are allowed so that the code is backwards compatible with previous input decks. The default convection and boiling correlations were derived mainly based on data from internal vertical pipe flow. Other possible input values are shown in **Table 10.17-1**. When modeling a vertical bundle, the rod or tube pitch-to-diameter ratio should be input on the 901 card. This has the effect of increasing the convective part of heat transfer such that users can input the true hydraulic diameter and get reasonable predictions.

Table 10.17-1 Card 601 Word 3 Convection Boundary Type

Word 3	Geometry Type
1, 100, 101	Default
102	Parallel plates (ORNL, ANS reactor; set gap and span on CCC3101-3199 hydro cards for pipes and CCC0111 hydro card for single volumes and branches, set b = 2 in volume control flag on CCC1001-1099 hydro cards for pipes and CCC0101-0109 hydro cards for single volumes and branches).
110	Vertical bundle without crossflow (set P/D on 901 card)
111	Vertical bundle with crossflow (set P/D on 901 card)
115	Parallel plates Large gaps - uses Petukhov-Gnielinski heat transfer correlation with Swanson-Catton multiplier for mixed convection
124	CANDU fuel bundle element heat transfer

Table 10.17-1 Card 601 Word 3 Convection Boundary Type

Word 3	Geometry Type
130	Flat plate above fluid
134	Horizontal bundle

If 115, a time constant (in seconds) for relaxing the heat transfer coefficient can be added in the front of the 115, e.g., 60115 will be parsed as a 60 second relaxation time constant for geometry type 115. The relaxation is of the type $1 - \exp\left(-\frac{t}{\tau}\right)$ where t is time and τ is the time constant entered as part of word 3.

If 1000, the temperature of the boundary volume or the temperature from the general table (as specified in Word 1) is used as the right surface temperature. If Word 1 is 0, the surface temperature is set to 0.

If 1xxx, the temperature in general Table xxx is used as the right surface temperature.

If 2xxx, the heat flux from Table xxx is used as the right boundary condition.

If 3xxx, a convective boundary condition is used where the heat transfer coefficient as a function of time is obtained from general Table xxx. The sink temperature is the temperature of the boundary volume or from the table specified in Word 1. If Word 1 is 0, the sink temperature is set to 0.

If 4xxx, a convective boundary condition is used where the heat transfer coefficient as a function of surface temperature is obtained from general Table xxx. The sink temperature is the temperature of the boundary volume or from the table specified in Word 1. If Word 1 is 0, the sink temperature is set to 0.

If reflood is specified, the right boundary condition type must be same for all nh heat structures and, similarly, for the left boundary condition type. The left and right boundary types need not be the same, but neither can be 1000 or 1xxx.

W4(I) Surface area code. If 0, Word 5 is the right surface area. If 1, Word 5 is (a) the surface area in rectangular geometry, (b) the cylinder height or equivalent in cylindrical geometry, or (c) the fraction of a sphere (0.5 is a hemisphere) in spherical geometry.

W5(R) Surface area or factor. As indicated in Word 4, this word contains the surface area (m^2 , ft^2) or a geometry dependent multiplier (m^2 , ft^2) for rectangular; (m, ft) for cylindrical; or dimensionless for spherical geometries). If the symmetry boundary condition is specified (Word 3 = 0), this word must still be entered nonzero.

W6(I) Heat structure number.

A10.18 Source Data, 1CCCG701-G799

These cards are required for heat structure data. The card format is sequential expansion format, five words per set, describing nh heat structures.

W1(I) Source type. If 0, no source is used. If a positive number is less than 1000, power from the general table with this number is used as the source. If 1000-1004, the number has the form 100t, and the source is taken from a point kinetics calculation. The field t = 0 specifies total reactor power, t = 1 specifies total decay power, t = 2 specifies fission power, t = 3 specifies fission product decay power, and t = 4 specifies actinide decay power. If 10001-19999, the source is the control variable whose number is this quantity minus 10000.

Note that in British-units problems, source powers in general tables are entered in MW and are internally converted to W by RELAP5. Reactor kinetics and control variable source powers are specified in W for both British-units and SI-units problems.

To run the RELAP5/PARCS 3-D kinetics code, the user must enter Option 88 on Card 1. During the heat structure initialization, PARCS is not called, so it is up to RELAP5 to supply the initial power and heat structure temperatures. Depending on how the user models the initial heat structure temperature, the user may need to input an initial power. If the steady-state initialization flag W4 on card 1CCCG000 is a 0, the initial temperatures are entered on cards 1CCCG401-G499, and it is not necessary to enter a power source. If the steady-state initialization flag W4 on card 1CCCG000 is a 1, the steady-state initial temperatures are calculated by the code, and the user needs to enter an initial power. The initial power can be entered using any of the methods discussed in the previous paragraph. After the RELAP5 initial power calculation, PARCS will supply the power, and the RELAP5 power calculations is skipped.

It is acceptable to restart a RELAP5/PARCS 3-D kinetics coupled code calculation from a RELAP5 stand-alone restart file in which the user can use any of the heat source types. However, the reverse case, restarting a RELAP5 stand-alone calculation from a RELAP5/PARCS coupled code restart file, is not allowed.

W2(R) Internal source multiplier. Axial peaking factors may be entered here. These values are multiplied by the power in the general table number in Word 1 to obtain the total power generated in this heat structure. These factors are not relative factors.

W3(R) Direct moderator heating multiplier for left boundary volume (see Section 3.3 of Volume II).

W4(R) Direct moderator heating multiplier for right boundary volume.

W5(I) Heat structure number.

A10.19 Additional Left Boundary Option, 1CCCG800

If this card is not entered or if this word is 0, the nine-word format is used on cards 1CCCG801-G899.

W1(I) If this word is 0, the nine-word format is used on cards 1CCCG801-G899.

If this word is 1, the twelve-word format is used on the cards 1CCCG801-G899.

If this word is 2, the thirteen-word format is used on the cards 1CCCG801-G899 (needed for PG-CHF correlation).

A10.20 9-Word Format Additional Left Boundary, 1CCCG801-G899

These cards are required whenever the left boundary communicates energy with the left hand fluid volume, and Word 1 = 0 on card 1CCCG800. The cards are in sequential expansion format, nine words per set, describing nh heat structures. Sequential expansion would only be used where the critical heat flux value was not of importance, since the length to all heat structures in the expansion would be the same. Words 2-8 are used for the CHF correlation.

W1(R) Heat transfer hydraulic diameter (i.e., heated equivalent diameter) (m, ft). This is $4\left(\frac{\text{flow area}}{\text{heated perimeter}}\right)$ and is recommended to be greater than or equal to the volume hydraulic diameter since $(\text{heated perimeter}) \leq (\text{wetted perimeter})$. It is possible to input this diameter to be less than the volume hydraulic diameter. If Word 1 equals 0.0, the volume hydraulic diameter is used. See Section 3.5 of Volume II of the manual for further guidelines.

W2(R) Heated length forward (m, ft). Distance is from the heated inlet to the center of this slab. This quantity will be used when the liquid volume velocity is positive or 0. This is used to get the hydraulic entrance length effect. This is used only for the CHF correlation. It must be > 0 . To ignore the length effect, put in a large number (i.e., ≥ 10.0).

W3(R) Heated length reverse (m, ft). Distance is from the heated outlet to the center of this slab. This quantity will be used when the liquid volume velocity is negative. This is used to get the hydraulic entrance length effect. This is used only for the CHF correlation. It must be > 0 . To ignore the length effect, put in a large number (i.e., ≥ 10.0).

W4(R) Grid spacer length forward (m, ft). Distance is from the center of this slab to the nearest grid or obstruction upstream. This quantity will be used when the liquid volume velocity is

positive or 0. This is used to get the boundary layer disturbance and atomization effect of a grid spacer in rod bundles. This is used only for the CHF correlation. If the grid K loss (Word 6) is 0, Word 4 is not used.

- W5(R) Grid spacer length reverse (m, ft). Distance is from the center of the slab to the nearest grid or obstruction downstream. This quantity will be used when the liquid volume velocity is negative. This is used to get the boundary layer disturbance and atomization affect of a grid space in rod bundles. This is used only for the CHF correlation. If the grid K loss (Word 7) is 0, Word 5 is not used.
- W6(R) Grid loss coefficient forward. Used for forward flow in rod bundles. This quantity is used when the liquid volume velocity is positive or 0. This is used only for CHF calculation.
- W7(R) Grid loss coefficient reverse. Used for reverse flow in rod bundles. This quantity is used when the liquid volume velocity is negative. This is used only for the CHF correlation.
- W8(R) Local boiling factor. Enter 1.0 if there is no power source in the heat structure or if the local equilibrium quality is negative (i.e., liquid is subcooled and void is 0). This is the local heat flux/average heat flux from start of boiling. If the power profile is not flat, a steady-state run may help determine this number. This number must be greater than 0.0.
- W9(I) Heat structure number.

A10.21 12-Word Format Additional Left Boundary, 1CCCG801-G899

These cards are required whenever the left boundary communicates energy with the left hand fluid volume, and Word 1 = 1 on card 1CCCG800. The cards are in sequential expansion format, twelve words per set, describing nh heat structures.

- W1(R) Heat transfer hydraulic diameter (i.e., heated equivalent diameter) (m, ft). This is $4\left(\frac{\text{flow area}}{\text{heated perimeter}}\right)$ and is recommended to be greater than or equal to the volume hydraulic diameter since $(\text{heated perimeter}) \leq (\text{wetted perimeter})$. It is possible to input this diameter to be less than the volume hydraulic diameter. If Word 1 equals 0.0, the volume hydraulic diameter is used. See Section 3.5 of Volume II of the manual for further guidelines.
- W2(R) Heated length forward (m, ft). Distance is from the heated inlet to the center of this slab. This quantity will be used when the liquid volume velocity is positive or 0. This is used to get the hydraulic entrance length effect. This is used only for the CHF correlation. It must be > 0 . To ignore the length effect, put in a large number (i.e., ≥ 10.0).

- W3(R) Heated length reverse (m, ft). Distance is from the heated outlet to the center of this slab. This quantity will be used when the liquid volume velocity is negative. This is used to get the hydraulic entrance length effect. This is used only for the CHF correlation. It must be > 0 . To ignore the length effect, put in a large number (i.e., ≥ 10.0).
- W4(R) Grid spacer length forward (m, ft). Distance is from the center of this slab to the nearest grid or obstruction upstream. This quantity will be used when the liquid volume velocity is positive or 0. This is used to get the boundary layer disturbance and atomization effect of a grid spacer in rod bundles. This is used only for the CHF correlation. If the grid K loss (Word 6) is 0, Word 4 is not used.
- W5(R) Grid spacer length reverse (m, ft). Distance is from the center of the slab to the nearest grid or obstruction downstream. This quantity will be used when the liquid volume velocity is negative. This is used to get the boundary layer disturbance and atomization affect of a grid space in rod bundles. This is used only for the CHF correlation. If the grid K loss (Word 7) is 0, Word 5 is not used.
- W6(R) Grid loss coefficient forward. Used for forward flow in rod bundles. This quantity is used when the liquid volume velocity is positive or 0. This is used only for CHF calculation.
- W7(R) Grid loss coefficient reverse. Used for reverse flow in rod bundles. This quantity is used when the liquid volume velocity is negative. This is used only for the CHF correlation.
- W8(R) Local boiling factor. Enter 1.0 if there is no power source in the heat structure or if the local equilibrium quality is negative (i.e., liquid is subcooled and void is 0). This is the local heat flux/average heat flux from start of boiling. If the power profile is not flat, a steady-state run may help determine this number. This number must be greater than 0.0.
- W9(R) Natural circulation length (m, ft). This should be the height of a hydraulic natural convection cell. For a heated vertical plate, this is the total height of the plate. For inside a horizontal tube, this should be the inside tube diameter. For the outer surface of vertical or horizontal bundles, it is suggested to use the heated bundle height in the vertical direction. When using the nine word format, this quantity is set to Word 1, the heat transfer hydraulic diameter.
- W10(R) Rod or tube pitch-to-diameter ratio. The default is 1.1. The maximum is 1.6 ft. It is not used unless Word 3 on the 501 card is 110, 111, 114, or 135. If CANDU geometry (124 option on Card 1CCCG5XX) was selected, this value is the relative height of the fuel element, and its value should be within ± 1.0 .
- W11(R) Fouling factor. This factor is applied to the heat transfer correlations and may be used to represent fouling or to run sensitivity studies. This quantity must be a positive nonzero number. When using the nine-word format, this quantity is set to 1.0.

W12(I) Heat structure number.

A10.22 13-Word Format Additional Left Boundary, 1CCCG801-G899

These cards are required whenever the left boundary communicates energy with the left hand fluid volume, and Word 1 = 2 on card 1CCCG800. The cards are in sequential expansion format, thirteen words per set, describing nh heat structures.

W1(R) Not used, enter 0.0.

W2(R) Reduced heated length forward (m, ft). This is the product $(y \bullet T_a)$. The first term is the distance from the heated channel inlet to the point of the predicted CHFR when the liquid volume velocity is positive or 0. The second term is the ratio of average heat flux from the heated channel inlet to the axial coordinate y (m, ft), i.e., at the point of the predicted CHFR, to local heat flux q at y . Word 2 should be determined as follows:

$$y \bullet T_a = \frac{1}{q(y)} \int_0^y q(z) dz.$$

W3(R) Reduced heated length reverse (m, ft). This is the product $(y \bullet T_a)$. The first term is the distance from the heated channel outlet to the point of the predicted CHFR when the liquid volume velocity is negative. The second term is the ratio of average heat flux from the heated channel outlet to the axial coordinate y (m, ft), i.e., at the point of the predicted CHFR, to local heat flux q at y . Word 3 should be determined as follows:

$$y \bullet T_a = \frac{1}{q(y)} \int_0^y q(z) dz.$$

W4(R) Grid spacer factor forward. This should be input as follows:

If Word 12 = 11, 12, 21, 22, 31, 32, 41, or 42, i.e., CHFR for the tube or the internally heated annulus, then Word 4 must be input as $W4 = 1.0$.

If Word 12 = 13, 23, 33, or 43, i.e., CHFR for the rod bundle with vaneless grid spacers, then Word 4 should be input either as $W4 = 1.0 / \bar{R}$, if the statistical evaluation data for the rod bundles are available ($\bar{R}$ is the mean of variable R . R is the statistical random variable representing CHFR, i.e., predicted CHF to measured CHF ratio), or as $W4 = 1.0$, if the statistical evaluation data for the rod bundle are not available.

If Word 12 = 14, 24, 34, or 44, i.e., CHFR for the rod bundle with vane grid spacers, then Word 4 should be input as: $W4$ could be determined from statistical evaluation data of specific fuel design.

If Word 12 = 15, then W4 should be input as W4 = 1.0.

W5(R) Grid spacer factor reverse. This should be input as follows:

If Word 12 = 11, 12, 21, 22, 31, 32, 41, or 42, i.e., CHFR for the tube or the internally heated annulus, then Word 5 must be input as W5 = 1.0.

If Word 12 = 13, 23, 33, or 43, i.e., CHFR for the rod bundle with vaneless grid spacers, then Word 5 should be input either as $W5 = 1.0 / \bar{R}$, if the statistical evaluation data for the rod bundles are available ($\bar{R}$ is the mean of variable R. R is the statistical random variable representing CHFR, i.e., predicted CHF to measured CHF ratio), or as W5 = 1.0, if the statistical evaluation data for the rod bundle are not available.

If Word 12 = 14, 24, 34, or 44, i.e., CHFR for the rod bundle with vane grid spacers, then Word 5 should be input as: W5 could be determined from statistical evaluation data of specific fuel design.

If Word 12 = 15, then W5 should be input as W5 = 1.0.

W6(R) Factor of the radial heat flux distribution. This should be input as:

$$T_r = q \frac{\sum_i r_i}{\sum_i r_i q_i}$$

This is the ratio of local heat flux on referred perimeter to average heat flux on perimeters pertaining to the subchannel (or the annulus).

W7(I) Heated channel upstream hydrodynamic volume number. This refers to the hydrodynamic components which represents the inlet for the heated channel. This is to get the heated channel inlet quality in the case of forward flow direction.

W8(I) Heated channel downstream hydrodynamic volume number. This refers to the hydrodynamic components which represents the outlet for the heated channel. It applies when the flow reverses.

W9(R) Natural circulation length (m, ft). This should be the height of a hydraulic natural convection cell. For a heated vertical plate, this is the total height of the plate. For inside a horizontal tube, this should be the inside tube diameter. For the outer surface of vertical or horizontal bundles, it is suggested to use the heated bundle height in the vertical direction. When using the nine word format, this quantity is set to Word 1, the heat transfer hydraulic diameter.

- W10(R) Rod or tube pitch-to-diameter ratio. The default is 1.1. The maximum is 1.6 ft. It is not used unless Word 3 on the 501 card is 110, 111, 114, or 135. If CANDU geometry (124 option on Card 1CCCG5XX) was selected, this value is the relative height of the fuel element, and its value should be within ± 1.0 .
- W11(R) Fouling factor. This factor is applied to the heat transfer correlations and may be used to represent fouling or to run sensitivity studies. This quantity must be a positive nonzero number. When using the nine-word format, this quantity is set to 1.0.
- W12(I) CHFR correlation option. This is input in mn format. The first digit specifies the CHFR correlation form.
- If $m = 1$, then the basic form of PG CHFR correlation is used.
- If $m = 2$, then the flux form of the PG CHFR correlation is used.
- If $m = 3$, then the geometry form of PG CHFR correlation is used.
- If $m = 4$, then the power form of PG CHFR correlation is used.
- The second digit specifies the geometry of heated structure. If this is the rod bundle, it specifies if and how the statistical evaluation data are applied for the grid spacer factor (see Word 4 and Word 5).
- If $n = 1$, then this is the tube.
- If $n = 2$, then this is the internally heated annulus.
- If $n = 3$, then this is the rod bundle. The use of an isolated subchannel model is recommended. This is used if the applicable PG CHFR correlation statistical evaluation data are not available.
- If $n = 4$, then this is the rod bundle. The use of an isolated subchannel model is recommended. An extended use of the PG CHFR statistical evaluation data is enabled.
- If $n = 5$, then this is the rod bundle. This is only used in combination with $m = 1$. Applicable for a subchannel code respecting lateral mixing.
- W13(I) Heat structure number.

A10.23 Additional Right Boundary Option, 1CCCG900

This card is optional, and if it is not entered, the nine-word format is used on cards 1CCCG901-G999.

W1(I) If this word is 0, the nine-word format is used on cards 1CCCG901-G999.

If this word is 1, the twelve-word format is used on the cards 1CCCG901-G999.

If this word is 2, the thirteen-word format is used on the cards 1CCCG901-G999 (needed for PG-CHF correlation).

A10.24 9-Word Format Additional Right Boundary1CCCG901-G999

These cards are required whenever the right boundary communicates energy with the right hand fluid volume, and Word 1 = 0 on card 1CCCG900. The cards are in sequential expansion format, nine words per set, describing the heat structures. Sequential expansion would only be used where the critical heat flux value was not of importance, since the length to all heat structures in the expansion would be the same. Words 2-8 are used for the CHF correlation.

W1(R) Heat transfer hydraulic diameter (i.e., heated equivalent diameter) (m, ft). This is $4\left(\frac{\text{flow area}}{\text{heated perimeter}}\right)$ and is recommended to be greater than or equal to the volume hydraulic diameter since $(\text{heated perimeter}) \leq (\text{wetted perimeter})$. It is possible to input this diameter to be less than the volume hydraulic diameter. If Word 1 equals 0.0, the volume hydraulic diameter is used. See Section 3.5 of Volume II of the manual for further guidelines.

W2(R) Heated length forward (m, ft). Distance is from the heated inlet to the center of this slab. This quantity will be used when the liquid volume velocity is positive or 0. This is used to get the hydraulic entrance length effect. This is used only for the CHF correlation. It must be > 0 . To ignore the length effect, put in a large number (i.e., ≥ 10.0).

W3(R) Heated length reverse (m, ft). Distance is from the heated outlet to the center of this slab. This quantity will be used when the liquid volume velocity is negative. This is used to get the hydraulic entrance length effect. This is used only for the CHF correlation. It must be > 0 . To ignore the length effect, put in a large number (i.e., ≥ 10.0).

W4(R) Grid spacer length forward (m, ft). Distance is from the center of this slab to the nearest grid or obstruction upstream. This quantity will be used when the liquid volume velocity is positive or 0. This is used to get the boundary layer disturbance and atomization effect of a grid spacer in rod bundles. This is used only for the CHF correlation. If the grid K loss (Word 6) is 0, Word 4 is not used.

W5(R) Grid spacer length reverse (m, ft). Distance is from the center of the slab to the nearest grid or obstruction downstream. This quantity will be used when the liquid volume velocity is negative. This is used to get the boundary layer disturbance and atomization effect of a grid space in rod bundles. This is used only for the CHF correlation. If the grid K loss (Word 7) is 0, Word 5 is not used.

- W6(R) Grid loss coefficient forward. Used for forward flow in rod bundles. This quantity is used when the liquid volume velocity is positive or 0. This is used only for CHF calculation.
- W7(R) Grid loss coefficient reverse. Used for reverse flow in rod bundles. This quantity is used when the liquid volume velocity is negative. This is used only for the CHF correlation.
- W8(R) Local boiling factor. Enter 1.0 if there is no power source in the heat structure or if the local equilibrium quality is negative (i.e., liquid is subcooled and void is 0). This is the local heat flux/average heat flux from start of boiling. If the power profile is not flat, a steady-state run may help determine this number. This number must be greater than 0.0.
- W9(I) Heat structure number.

A10.25 12-Word Format Additional Right Boundary, 1CCCG901-G999

These cards are required whenever the right boundary communicates energy with the right hand fluid volume, and Word 1 = 1 on card 1CCCG900. The cards are in sequential expansion format, twelve words per set, describing nh heat structures.

- W1(R) Heat transfer hydraulic diameter (i.e., heated equivalent diameter) (m, ft). This is $4\left(\frac{\text{flow area}}{\text{heated perimeter}}\right)$ and is recommended to be greater than or equal to the volume hydraulic diameter since $(\text{heated perimeter}) \leq (\text{wetted perimeter})$. It is possible to input this diameter to be less than the volume hydraulic diameter. If Word 1 equals 0.0, the volume hydraulic diameter is used. See Section 3.5 of Volume II of the manual for further guidelines.
- W2(R) Heated length forward (m, ft). Distance is from the heated inlet to the center of this slab. This quantity will be used when the liquid volume velocity is positive or 0. This is used to get the hydraulic entrance length effect. This is used only for the CHF correlation. It must be > 0 . To ignore the length effect, put in a large number (i.e., ≥ 10.0).
- W3(R) Heated length reverse (m, ft). Distance is from the heated outlet to the center of this slab. This quantity will be used when the liquid volume velocity is negative. This is used to get the hydraulic entrance length effect. This is used only for the CHF correlation. It must be > 0 . To ignore the length effect, put in a large number (i.e., ≥ 10.0).
- W4(R) Grid spacer length forward (m, ft). Distance is from the center of this slab to the nearest grid or obstruction upstream. This quantity will be used when the liquid volume velocity is positive or 0. This is used to get the boundary layer disturbance and atomization effect of a grid spacer in rod bundles. This is used only for the CHF correlation. If the grid K loss (Word 6) is 0, Word 4 is not used.

- W5(R) Grid spacer length reverse (m, ft). Distance is from the center of the slab to the nearest grid or obstruction downstream. This quantity will be used when the liquid volume velocity is negative. This is used to get the boundary layer disturbance and atomization affect of a grid space in rod bundles. This is used only for the CHF correlation. If the grid K loss (Word 7) is 0, Word 5 is not used.
- W6(R) Grid loss coefficient forward. Used for forward flow in rod bundles. This quantity is used when the liquid volume velocity is positive or 0. This is used only for CHF calculation.
- W7(R) Grid loss coefficient reverse. Used for reverse flow in rod bundles. This quantity is used when the liquid volume velocity is negative. This is used only for the CHF correlation.
- W8(R) Local boiling factor. Enter 1.0 if there is no power source in the heat structure or if the local equilibrium quality is negative (i.e., liquid is subcooled and void is 0). This is the local heat flux/average heat flux from start of boiling. If the power profile is not flat, a steady-state run may help determine this number. This number must be greater than 0.0.
- W9(R) Natural circulation length (m, ft). This should be the height of a hydraulic natural convection cell. For a heated vertical plate, this is the total height of the plate. For inside a horizontal tube, this should be the inside tube diameter. For the outer surface of vertical or horizontal bundles, it is suggested to use the heated bundle height in the vertical direction. When using the nine word format, this quantity is set to Word 1, the heat transfer hydraulic diameter.
- W10(R) Rod or tube pitch-to-diameter ratio. The default is 1.1. The maximum is 1.6 ft. It is not used unless Word 3 on the 601 card is 110, 111, 114, or 135. If CANDU geometry (124 option on Card 1CCCG6XX) was selected, this value is the relative height of the fuel element, and its value should be within ± 1.0 .
- W11(R) Fouling factor. This factor is applied to the heat transfer correlations and may be used to represent fouling or to run sensitivity studies. This quantity must be a positive nonzero number. When using the nine-word format, this quantity is set to 1.0.
- W12(I) Heat structure number.

A10.26 13-Word Format Additional Right Boundary, 1CCCG901-G999

These cards are required whenever the right boundary communicates energy with the right hand fluid volume, and Word 1 = 2 on card 1CCCG900. The cards are in sequential expansion format, thirteen words per set, describing nh heat structures.

- W1(R) Not used, enter 0.0.

W2(R) Reduced heated length forward (m, ft). This is the product $(y \bullet T_a)$. The first term is the distance from the heated channel inlet to the point of the predicted CHFR when the liquid volume velocity is positive or 0. The second term is the ratio of average heat flux from the heated channel inlet to the axial coordinate y (m, ft), i.e., at the point of the predicted CHFR, to local heat flux q at y . Word 2 should be determined as follows:

$$y \bullet T_a = \frac{1}{q(y)} \int_0^y q(z) dz.$$

W3(R) Reduced heated length reverse (m, ft). This is the product $(y \bullet T_a)$. The first term is the distance from the heated channel outlet to the point of the predicted CHFR when the liquid volume velocity is negative. The second term is the ratio of average heat flux from the heated channel outlet to the axial coordinate y (m, ft), i.e., at the point of the predicted CHFR, to local heat flux q at y . Word 3 should be determined as follows:

$$y \bullet T_a = \frac{1}{q(y)} \int_0^y q(z) dz.$$

W4(R) Grid spacer factor forward. This should be input as follows:

If Word 12 = 11, 12, 21, 22, 31, 32, 41, or 42, i.e., CHFR for the tube or the internally heated annulus, then Word 4 must be input as $W4 = 1.0$.

If Word 12 = 13, 23, 33, or 43, i.e., CHFR for the rod bundle with vaneless grid spacers, then Word 4 should be input either as $W4 = 1.0 / \bar{R}$, if the statistical evaluation data for the rod bundles are available ($\bar{R}$ is the mean of variable R . R is the statistical random variable representing CHFR, i.e., predicted CHF to measured CHF ratio), or as $W4 = 1.0$, if the statistical evaluation data for the rod bundle are not available.

If Word 12 = 14, 24, 34, or 44, i.e., CHFR for the rod bundle with vane grid spacers, then Word 4 should be input as: $W4$ could be determined from statistical evaluation data of specific fuel design.

If Word 12 = 15, then $W4$ should be input as $W4 = 1.0$.

W5(R) Grid spacer factor reverse. This should be input as follows:

If Word 12 = 11, 12, 21, 22, 31, 32, 41, or 42, i.e., CHFR for the tube or the internally heated annulus, then Word 5 must be input as $W5 = 1.0$.

If Word 12 = 13, 23, 33, or 43, i.e., CHFR for the rod bundle with vaneless grid spacers, then Word 5 should be input either as $W5 = 1.0 / \bar{R}$, if the statistical evaluation data for the

rod bundles are available ($\bar{R}$ is the mean of variable R. R is the statistical random variable representing CHFR, i.e., predicted CHF to measured CHF ratio), or as $W5 = 1.0$, if the statistical evaluation data for the rod bundle are not available.

If Word 12 = 14, 24, 34, or 44, i.e., CHFR for the rod bundle with vane grid spacers, then Word 5 should be input as: W5 could be determined from statistical evaluation data of specific fuel design.

If Word 12 = 15, then W5 should be input as $W5 = 1.0$.

W6(R) Factor of the radial heat flux distribution. This should be input as:

$$T_r = q \frac{\sum_i r_i}{\sum_i r_i q_i}$$

This is the ratio of local heat flux on referred perimeter to average heat flux on perimeters pertaining to the subchannel (or the annulus).

W7(I) Heated channel upstream hydrodynamic volume number. This refers to the hydrodynamic components which represents the inlet for the heated channel. This is to get the heated channel inlet quality in the case of forward flow direction.

W8(I) Heated channel downstream hydrodynamic volume number. This refers to the hydrodynamic components which represents the outlet for the heated channel. It applies when the flow reverses.

W9(R) Natural circulation length (m, ft). This should be the height of a hydraulic natural convection cell. For a heated vertical plate, this is the total height of the plate. For inside a horizontal tube, this should be the inside tube diameter. For the outer surface of vertical or horizontal bundles, it is suggested to use the heated bundle height in the vertical direction. When using the nine word format, this quantity is set to Word 1, the heat transfer hydraulic diameter.

W10(R) Rod or tube pitch-to-diameter ratio. The default is 1.1. The maximum is 1.6 ft. It is not used unless Word 3 on the 601 card is 110, 111, 114, or 135. If CANDU geometry (124 option on Card 1CCCG6XX) was selected, this value is the relative height of the fuel element, and its value should be within ± 1.0 .

- W11(R) Fouling factor. This factor is applied to the heat transfer correlations and may be used to represent fouling or to run sensitivity studies. This quantity must be a positive nonzero number. When using the nine-word format, this quantity is set to 1.0.
- W12(I) CHFR correlation option. This is input in mn format. The first digit specifies the CHFR correlation form.
- If $m = 1$, then the basic form of PG CHFR correlation is used.
- If $m = 2$, then the flux form of the PG CHFR correlation is used.
- If $m = 3$, then the geometry form of PG CHFR correlation is used.
- If $m = 4$, then the power form of PG CHFR correlation is used.
- The second digit specifies the geometry of heated structure. If this is the rod bundle, it specifies if and how the statistical evaluation data are applied for the grid spacer factor (see Word 4 and Word 5).
- If $n = 1$, then this is the tube.
- If $n = 2$, then this is the internally heated annulus.
- If $n = 3$, then this is the rod bundle. The use of an isolated subchannel model is recommended. This is used if the applicable PG CHFR correlation statistical evaluation data are not available.
- If $n = 4$, then this is the rod bundle. The use of an isolated subchannel model is recommended. An extended use of the PG CHFR statistical evaluation data is enabled.
- If $n = 5$, then this is the rod bundle. This is only used in combination with $m = 1$. Applicable for a subchannel code respecting lateral mixing.
- W13(I) Heat structure number.

A11 RADIATION MODEL

Any heat structure may radiate to any other heat structure or set of heat structures in a user-defined enclosure.

A11.1 Number of Radiation Heat Slab Sets, 60000000

An enclosure is a set of heat structures that communicate via thermal radiation. The calculation ignores fluid in the enclosure.

W1(I) Number of sets of radiation enclosures, nset. This word must be less than 100 and greater than 0 for radiation to be on.

A11.2 Radiation Heat Slab Sets, 6SS00000

SS is the set number. One of these cards must be input for each radiating set. The first word is required; the other three are optional.

W1(I) Number of radiating heat slabs, nrh. This word is the number of radiating heat slabs (surfaces) that participate in radiation heat transfer in set SS. The maximum is 99.

W2(R) Minimum temperature, trmin (K, °F). This word is the minimum temperature of all surfaces in a radiation enclosure below which radiation will no longer be calculated. The default value is 900 K.

W3(R) Minimum void fraction, voidmn. This word is the minimum void fraction below which radiation will no longer be calculated. The default value is 0.75. Each volume connected to any of the radiating surfaces in the set is checked and if any have a void fraction greater than Word 3, radiation stays on in the enclosure set.

W4(I) View factor set, refset. If input, this is the number of the set (SS) from which view factors are to be obtained. If no number is found here, the view factors must be input for this set.

A11.3 Radiation Heat Structure Data, 6SSNN001-NN099

For these cards, SS must take on every value from 1 to nset (Word 1 in card 60000000), and NN must take on every value from 1 to nrh (Word 1 on card 6SS00000) for each SS. Data are entered for each conductor surface that participates in radiation heat transfer.

W1(I) Heat structure geometry level, jrh. This word is CCCG0ZZ, where CCCG is the heat structure geometry combination of nh, and ZZ is the axial level number participating in radiation.

W2(I) Surface flag, jlr. For this word, 0 = left surface, and 1 = right surface of NN.

W3(R) Emissivity of surface NN.

A11.4 Radiation View Factors, 6SSNN101-NN199

There are $nrh \bullet nrh$ values in each set. SS is the set number (from 1 to nset). NN is the surface number (from 1 to nrh). For a given NN, the sum of the view factors must be 1.0, and the view factor times the surface area must equal the view factor times the area of the receiving surface.

W1(R) View factor, v_{fij} . View factor from surface NN to surface W2(I).

W2(I) Radiation surface number to which NN radiates. Repeat the above two words until view factors to all nhr surfaces from all surfaces are entered. Sequential expansion is used.

A12 HEAT STRUCTURE THERMAL PROPERTIES

These cards are used in NEW or RESTART problems. These cards are required if cards 1CCCGXNN, Heat Structure Input cards, Section A10 are entered. These data, if present, are processed and stored even if no cards 1CCCGXNN are entered.

The subfield MMM is the composition number, and the cards with this subfield describe the thermal properties of composition MMM. The composition numbers entered on cards 1CCCG201-1CCCG299 correspond to this subfield. A set of cards 201MMMNN must be entered for each composition number used, but MMM need not be consecutive. During RESTART, thermal property may be deleted, new compositions may be added, or data may be modified by entering new data for an existing composition.

A12.1 Composition Type and Data Format, 201MMM00

This card is required.

W1(A) Material type. Thermal properties for four materials are stored within the program: carbon steel (C-STEEL), stainless steel (S-STEEL), uranium dioxide (UO₂), zirconium (ZR), and fuel rod gap (GAP). These properties are selected by entering the name in parentheses for this word. At present, the data are primarily to demonstrate capability. The user should check whether the data are satisfactory.

If a user-supplied table or function is to be used, enter TBL/FCTN for this word.

The word DELETE may be entered in RESTART problems to delete a composition.

The next two words are required only if TBL/FCTN is entered for W1.

W2(I) Thermal conductivity format flag or gap mole fraction flag.

Enter 1 if a table containing temperature and thermal conductivity is to be entered.

Enter 2 if functions are to be entered.

Enter 3 if the gap conductance model is used, and thus a table containing gas component names and mole fractions is to be entered.

W3(I) Volumetric heat capacity flag.

Enter -1 if a table containing only volumetric heat capacities is to be entered and the temperature values are identical to the thermal conductivity table.

Enter 1 if a table containing temperature and volumetric heat capacity is to be entered.

Enter 2 if functions are to be entered.

A12.2 Constant Thermal Conductivity, 201MMM01-49

One of these cards is required if W1 of card 201MMM00 contains TBL/FCTN, W2 contains a 1, and the thermal conductivity is constant for all temperatures.

W1(R) Constant thermal conductivity (W/m•K, Btu/s•ft•°F).

A12.3 Temperature-Thermal Conductivity, 201MMM01-49

These cards are required if W1 of card 201MMM00 contains TBL/FCTN and W2 of contains a 1. In this case, the thermal conductivity varies with temperature, so the user has to enter pairs of temperatures and thermal conductivities. More than one pair can be entered on a single card. The total number of pairs is limited to 100. The temperatures must be in increasing order. The first and last temperatures must bracket the expected temperatures during the calculation. A code failure will occur if the calculated material temperature is outside the bracketed range.

W1(R) Temperature (K, °F).

W2(R) Thermal conductivity (W/m•K, Btu/s•ft•°F).

A12.4 Thermal Conductivity Function, 201MMM01-49

These cards are required if W1 of card 201MMM00 contains TBL/FCTN and W2 contains a 2. Sets of nine quantities are entered, each set containing one function and its range of application. The function is

$$k = A_0 + A_1 T_x + A_2 T_x^2 + A_3 T_x^3 + A_4 T_x^4 + A_5 / T_x$$

where $T_x = T - C$, T is the temperature argument, and C is a constant. Each function has a lower and upper limit of application. The first function entered must be for the lowest temperature range. The lower limit of each following function must equal the upper bound of the previous function.

W1(R) Lower limit temperature (K, °F).

W2(R) Upper limit temperature (K, °F).

W3(R) A_0 (W/m•K, Btu/s•ft•°F).

W4(R) A_1 (W/m•K², Btu/s•ft•°F²).

W5(R) A_2 (W/m•K³, Btu/s•ft•°F³).

W6(R) A_3 (W/m•K⁴, Btu/s•ft•°F⁴).

W7(R) A_4 (W/m•K⁵, Btu/s•ft•°F⁵).

W8(R) A_5 (W/m, Btu/s•ft).

W9(R) C (K, °F).

A12.5 Gap Mole Fraction, 201MMM01-49

These cards are required if W1 of card 201MMM00 contains TBL/FCTN and W2 contains a 3, Enter pairs of gas component names and mole fractions. More than one pair can be entered on a single card. One to 7 pairs of gas names and their mole fractions can be entered. The gas component names that may be entered are HELIUM, ARGON, KRYPTON, XENON, NITROGEN, HYDROGEN, and OXYGEN. No particular order of the pairs is required. Do not enter any gas component with a zero mole fraction. Normalization of the total mole fraction to one is performed if the sum of the mole fractions entered is not one. The table of gas composition data is applicable to any gap and is required if card 1CCCG001 is present.

The volumetric heat capacity cards also need to be added since only the conductivity values are build into the code.

W1(R) Gas name. Possible names include HELIUM, ARGON, KRYPTON, XENON, NITROGEN, HYDROGEN, and OXYGEN.

W2(R) Mole fraction.

A12.6 Volumetric Heat Capacity, 201MMM51-99

These cards are required if W1 of card 201MMM00 contains TBL/FCTN and W3 contains a -1. The card numbers need not be consecutive. More than one value can be entered on a single card. Since this format assumes the temperatures are identical to the values in the temperature-thermal conductivity table, only the volumetric heat capacities need be entered. The number of entries must be the same as the number of pairs in the temperature-thermal conductivity table, which is limited to 100.

W1(R) Volumetric heat capacity (J/m³K, Btu/ft³ °F). This is ρC_p , where ρ is density (kg/m³, lb/ft³) and C_p is specific heat capacity (J/kg•K, Btu/lb•°F).

A12.7 Constant Volumetric Heat Capacity, 201MMM51-99

One of these cards is required if W1 of card 201MMM00 contains TBL/FCTN, W3 contains a 1, and the volumetric heat capacity is constant for all temperatures.

W1(R) Constant volumetric heat capacity ($\text{J/m}^3\text{K}$, $\text{Btu/ft}^3\text{ }^\circ\text{F}$). This is ρC_p , where ρ is density (kg/m^3 , lb/ft^3) and C_p is specific heat capacity ($\text{J/kg}\cdot\text{K}$, $\text{Btu/lb}\cdot\text{ }^\circ\text{F}$).

A12.8 Temperature-Volumetric Heat Capacity, 201MMM51-99

These cards are required if W1 of card 201MMM00 contains TBL/FCTN and W3 contains a 1. The card numbers need not be consecutive. More than one pair can be entered on a single card. Since the volumetric heat capacity varies with temperature, enter pairs of temperatures and volumetric heat capacities. More than one pair can be entered on a single card. The total number of pairs is limited to 100. The temperatures must be in increasing order. The first and last temperatures must bracket the expected temperatures during the calculation. A code failure will occur if the calculated material temperature is outside the bracketed range.

W1(R) Temperature (K, $^\circ\text{F}$).

W2(R) Volumetric heat capacity ($\text{J/m}^3\text{K}$, $\text{Btu/ft}^3\text{ }^\circ\text{F}$). This is ρC_p , where ρ is density (kg/m^3 , lb/ft^3) and C_p is specific heat capacity ($\text{J/kg}\cdot\text{K}$, $\text{Btu/lb}\cdot\text{ }^\circ\text{F}$).

A12.9 Volumetric Heat Capacity Function, 201MMM51-99

These cards are required if W1 of card 201MMM00 contains TBL/FCTN and W3 contains a 2. The card numbers need not be consecutive. Sets of nine quantities are entered, each set containing one function and its range of application. More than one set can be entered on a single card. The function is

$$\rho C_p = A_0 + A_1 T_x + A_2 T_x^2 + A_3 T_x^3 + A_4 T_x^4 + A_5 / T_x$$

where $T_x = T - C$, T is the temperature argument, and C is a constant. Each function has a lower and upper limit of application. The first function entered must be for the lowest temperature range. The lower limit of each following function must equal the upper bound of the previous function.

W1(R) Lower limit temperature (K, $^\circ\text{F}$).

W2(R) Upper limit temperature (K, $^\circ\text{F}$).

W3(R) A_0 ($\text{J/m}^3\text{ K}$, $\text{Btu/ft}^3\text{ }^\circ\text{F}$).

W4(R) A_1 ($\text{J/m}^3\text{ K}^2$, $\text{Btu/ft}^3\text{ }^\circ\text{F}^2$).

W5(R) A_2 ($\text{J/m}^3\text{ K}^3$, $\text{Btu/ft}^3\text{ }^\circ\text{F}^3$).

W6(R) A_3 ($\text{J/m}^3 \text{K}^4$, $\text{Btu/ft}^3 {}^\circ\text{F}^4$).

W7(R) A_4 ($\text{J/m}^3 \text{K}^5$, $\text{Btu/ft}^3 {}^\circ\text{F}^5$).

W8(R) A_5 (J/m^3 , Btu/ft^3).

W9(R) C (K, ${}^\circ\text{F}$).

A13 GENERAL TABLE DATA

These cards are used only in NEW or RESTART type problems and are required only if any input references general tables. TTT is the table number, and table references such as for power, heat transfer coefficients, and temperatures refer to this number. Data must be entered for each table that is referenced, but TTT need not be consecutive. Tables entered but not referenced are stored, and this is not considered an error. During RESTART, general tables may be added, existing tables may be deleted, or existing tables may be modified by entering new data.

A13.1 Table Type and Multiplier Data, 202TTT00

W1(A) Table type. In RESTART problems, DELETE can be entered to delete general table TTT. For non restart problems, the names in the following table can be used.

Table 13.1-1 Types of General Tables

Table Type	Quantity in Table
POWER	Power versus time
HTRNRATE	Heat flux versus time
HTC-T	Heat transfer coefficient versus time
HTC-TEMP	Heat transfer coefficient versus temperature
TEMP	Temperature versus time
REAC-T	Reactivity versus time or Control system variable versus time (prevents undesirable unit conversion)
NORMAREA	Normalized area versus normalized stem position

The following two, three, or four words are optional and allow trips and factors or units changes to be applied to the table entries. If the factors are omitted, the data are used as entered. One multiplier is used for time, power, heat flux, heat transfer coefficient, reactivity, normalized stem position, and normalized area; a multiplier and additive constant are used for temperature as $T = MT_x + C$, where M is the multiplier, C is the additive constant, and T_x is the temperature entered. The first one or two factors apply to the argument variable, time or temperature; one factor is applied if the argument is time, and two factors are used if the argument is temperature. The remaining one or two factors are used for the function, two factors being used if temperature is the function. Note that the temperature can be either the argument (HTC-TEMP table type) or the function (TEMP table type). When factors are used for a table involving temperature as the argument, the first two factors apply to the temperature. The second and third factors apply to the temperature when temperature is the function.

W2(I) Table trip number. This number is optional unless factors are entered. If missing or 0, no trip is used, and the time argument in the following table is the time supplied to the table

for interpolation. If nonzero, the number is the trip number, and the time argument in the following table is -1.0 if the trip is false and the time supplied to the table minus the trip time if the trip is true. This field may be omitted if no factors are entered. This number must be 0 or blank for tables that are not a function of time.

W3-W5(R) Factors. As described above, enter factors such that when applied to the table values entered, the resultant values have the appropriate units. For the NORMAREA table, the resultant values for both the normalized length and area must be ≥ 0 and ≤ 1.0 .

A13.2 General Table Data, 202TTT01-99

The card numbers need not be consecutive. The units given are the units required after the factors on card 202TTT00 have been applied. Pairs of numbers are entered; the limit on the number of pairs is 99.

W1(R) Argument value (s, if time; K, °F, if temperature; dimensionless, if normalized stem position).

W2(R) Function value (W, MW, if power; K, °F, if temperature; W/m^2 , $Btu/s \cdot ft^2$, if heat flux; $W/m^2 \cdot K$, $Btu/s \cdot ft^2 \cdot °F$, if heat transfer coefficient; dollars, if reactivity; dimensionless, if normalized area).

The tables use linear interpolation for segments between table search argument values. For search arguments beyond the range of entered data, the end-point values are used.

Note that table entries in British-units problems are converted internally in RELAP5 to the equivalent SI units prior to use.

A14 PLOT REQUESTS, 20300000-203499999

The plotting capability that was built into RELAP5 is not currently active. This capability was not converted from a machine-dependent form that was in the original CDC-7600 version to be used on other computers. It also required a proprietary plotting package.

Users can use the strip option to write an ASCII coded file containing data to be plotted and interface this file to plotting routines available within their organizations or they can use the XMGR5 or ACGRACE interactive plotting program that is distributed with RELAP5.

A15 CONTROL SYSTEM

These cards are used in NEW and RESTART problems if a control system is desired. They are also used to define the generic control components employed with the self-initialization option. Input can also be used to compute additional quantities from the normally computed quantities. These additional quantities can then be output in major and minor edits and plots.

Two different card types are available for entering control system data, but only one type can be used in a problem. The digits CCC or CCCC form the control variable number (i.e., control component number). The card format 205CCCNN allows 999 control variables, where CCC ranges from 001-999. The card format 205CCCCN allows 9999 control variables, where CCCC ranges from 1-9999.

If the self-initialization option is selected, the data cards described in Section A15.3.19, Section A15.3.20, and Section A15.3.21 must be included. If loop flow control is to be included, the data cards described in Section A15.3.19 must also be included.

The names of the various types of components and other information needed for a control component should be entered as lower case in the input deck. In the following descriptions, the names are written in upper case, but the code will compare the names on the input cards to lower case names in the code.

A15.1 Control Variable Card Type, 20500000

If this card is omitted, card type 205CCCNN is used. If this card is entered, either card format can be selected. This card cannot be entered on RESTART problems if control components exist from the restart problem, in which case the card format from the restart problem must be used.

W1(I) Enter 999 to select the 205CCCNN format or 9999 (4095 also allowed) to select the 205CCCCN format.

A15.2 Control Component Type, 205CCC00 or 205CCCC0

One card must be entered for each of the generic control components when using the self-initialization option.

W1(A) Alphanumeric name. Enter a name descriptive of the component. This name will appear in the printed output along with the component number. A limit of 10 characters is allowed for CDC 7600 computers, and a limit of 8 characters is allowed for most other computers.

W2(A) Control component type. Enter one of the component names, SUM, MULT, DIV, DIFFRENI, DIFFREND, INTEGRAL, FUNCTION, STDFNCTN, DELAY, TRIPUNIT, TRIPDLAY, POWERI, POWERR, POWERX, PROP-INT, LAG, LEAD-LAG, CONSTANT, SHAFT, PUMPCTL, STEAMCTL, FEEDCTL DIGITAL, or the command, DELETE. If DELETE is entered, enter any alphanumeric word in Word 1 and zeros in Words 3-8. No other control system cards are needed when DELETE is entered.

- W3(R) Scaling factor. For a CONSTANT component, this quantity is the constant value. No additional words are entered on this card, and cards 205CCC01-CCC09 or 205CCCC1-CCCC9 are not entered. For the PUMPCTL, STEAMCTL, or FEEDCTL components, this is the gain multiplier (G) for the output signal.
- W4(R) Initial value of Y.
- W5(I) Initial value flag.
- 0 means no initial condition calculation, and W4 is used as the initial value Y_0 .
- 1 means compute initial value of Y_0 . This implies that any time integral from zero to time t that is part of the control component be set to zero, since no input is provided for the initial value of the integral.
- W6(I) Limiter control. Enter 0, or omit this and the following words if no limits on the control variable are to be imposed. Enter 1 if only a minimum limit is to be imposed, 2 if only a maximum limit is to be imposed, and enter 3 if both minimum and maximum limits are to be imposed.
- W7(R) Minimum or maximum value. This word is the minimum or maximum value if only one limit is to be imposed or is the minimum value if both limits are to be imposed.
- W8(R) Maximum value. This word is used if both limits are to be imposed.

A15.3 Control Component Data, 205CCC01-CCC98 or 205CCCC1-CCCC8

The format of these cards depends on the control component type. An equation is used to describe the processing by each component. The symbol Y represents the control variable defined by the component. The symbols A_j , $j = 1, 2, \dots, J$, represent constants defined by the control component input data. The variables V_j , $j = 1, 2, \dots, J$, represent any of the variables listed in the minor edit input description. Besides hydrodynamic component data, heat structure data, reactor kinetic data, etc., any of the control variables including the variable being defined may be specified. The symbol S is the scale factor (or G, the gain multiplier, for self-initialization control components) on card 205CCC00 or 205CCCC0. The variables V_j use the code's internal units (SI). To use British units, the user must convert from SI to British using the scale factor S (or the gain multiplier G) and the constants A_j .

See Section 6 of Volume I for a detailed description of these models.

A15.3.1 Sum-Difference Component

This component is indicated by SUM in Word 2 of card 205CCC00 or 205CCCC0. The sum-difference component is defined by

$$Y = S(A_0 + A_1V_1 + A_2V_2 + \dots + A_jV_j)$$

W1(R) Constant A_0 .

W2(R) Constant A_1 .

W3(A) Alphanumeric name of variable request code for V_1 .

W4(I) Integer name of the variable request code for V_1 . At least four words that define a constant and one product term must be entered. Additional sets of three words corresponding to Words 2-4 can be entered for additional product terms up to twenty product terms. One or more cards may be used as desired. Card numbers need not be strictly consecutive. The sign of A_j determines addition or subtraction of the product terms.

A15.3.2 Multiplier Component

This component is indicated by MULT in Word 2 of card 205CCC00 or 205CCCC0. The multiplier component is defined by

$$Y = SV_1V_2\dots V_j$$

W1(A) Alphanumeric name of the variable request code for V_1 .

W2(I) Integer name of the variable request code for V_1 . At least two words must be entered. Additional pairs of words can be entered on this or additional cards to define additional factors. Card numbers need not be strictly consecutive.

A15.3.3 Divide Component

This component is indicated by DIV in Word 2 of card 205CCC00 or 205CCCC0. The divide component is defined by

$$Y = \frac{S}{V_1} \quad \text{or} \quad Y = \frac{SV_2}{V_1}$$

Specifying two words on the card indicates the first form, and specifying four words on the card indicates the second form. Execution will terminate if a divide by 0 is attempted.

W1(A) Alphanumeric name of the variable request code for V_1 .

W2(I) Integer name of the variable request code for V_1 .

W3(A) Alphanumeric name of the variable request code for V_2 .

W4(I) Integer name of the variable request code for V_2 .

A15.3.4 Differentiating Components

These components are indicated by DIFFRENI or DIFFREND in Word 2 of card 205CCC00 or 205CCCC0. The differentiating component is defined by

$$Y = S \frac{dV_1}{dt}$$

This is evaluated by

$$Y = \frac{2S}{\Delta t}(V_1 - V_{10}) - Y_0 \quad (\text{DIFFRENI})$$

$$Y = S \frac{(V_1 - V_{10})}{\Delta t} \quad (\text{DIFFREND})$$

where Δt is the time step, and V_{10} and Y_0 are values at the beginning of the time step. The numerical approximations for the DIFFRENI and INTEGRAL components are exact inverses of each other. However, an exact initial value is required to use the DIFFRENI component, and erroneous results are obtained if an exact initial value is not furnished. The DIFFREND component uses a simple difference approximation that is less accurate and is not consistent with the integration approximation, but does not require an initial value. For these reasons, use of DIFFRENI is not recommended.

Since differentiation, especially numerical differentiation, can introduce noise into the calculation, it should be avoided if possible. When using control components to solve differential equations, the equations can be arranged such that INTEGRAL components can handle all indicated derivatives except possibly those involving noncontrol variables.

W1(A) Alphanumeric name of variable request code for V_1 .

W2(I) Integer name of variable request code for V_1 .

A15.3.5 Integrating Component

This component is indicated by INTEGRAL in Word 2 of card 205CCC00 or 205CCCC0. The integrating component is defined by

$$Y = Y(t_1) + S \int_{t_1}^t V_1 dt$$

or, in Laplace notation,

$$Y(s) = \frac{SV_1(s)}{s}$$

This is evaluated by

$$Y = Y_o + S \bullet (V_1 + V_{1o}) \frac{\Delta t}{2}$$

where Δt is the time step, Y_o and V_{1o} are values at the beginning of the time step.

If the initialization flag is 0 or 1, Word 5 = 0 or 1 on card 205CCC00 or 205CCCC0, $Y_o = \text{Word 4}$ for the first time step. Thus, the initial value of the integral term is 0.

W1(A) Alphanumeric name of the variable request code for V_1 .

W2(I) Integer name of the variable request code for V_1 .

A15.3.6 Functional Component

This component is indicated by FUNCTION in Word 2 of card 205CCC00 or 205CCCC0. The component is defined by

$$Y = S[\text{FUNCTION}(V_1)]$$

where FUNCTION is defined by a general table. This allows the use of any function that is conveniently defined by a table lookup and linear interpolation procedure. The function component can also be used to set limiting values.

W1(A) Alphanumeric name of the variable request code for V_1 .

W2(I) Integer name of the variable request code for V_1 .

W3(I) General table number of the function.

A15.3.7 Standard Function Component

This component is indicated by STDFNCTN in Word 2 of card 205CCC00 or 205CCCC0. The component is defined by

$$Y = S[\text{FNCTN}(V_1, V_2, \dots)]$$

where FNCTN is ABS (absolute value), SQRT (square root), EXP (e raised to power), LOG (natural logarithm), SIN (sine), COS (cosine), TAN (tangent), CTN (cotangent), MIN (minimum value), or MAX (maximum value). All function types except MIN and MAX must have only one argument; MIN and MAX function types must have at least two arguments and may have up to twenty arguments. If the control variable being defined also appears in the argument list of MIN or MAX, the old time value is used in the comparison.

W1(A) FNCTN.

W2(A) Alphanumeric name of the variable request code for V_1 .

W3(I) Integer name of the variable request code for V_1 .

A15.3.8 Delay Component

This component is indicated by DELAY in Word 2 of card 205CCC00 or 205CCCC0. The component is defined by

$$Y = SV_1(t - t_d)$$

where t is time and t_d is the delay time.

W1(A) Alphanumeric name of the variable request code for V_1 .

W2(I) Integer name of the variable request code for V_1 .

W3(R) Delay time, t_d (s).

W4(I) Number of hold positions. This quantity, h , must be > 0 and ≤ 100 . This quantity determines the length of the table used to store past values of the quantity V_1 . The maximum number of time-function pairs that can be stored is $h + 2$. The delay table time increment, d_{TM} , is $d_{TM} = \frac{t_d}{h}$. The delayed function is obtained by linear interpolation for $V_1(t - t_d)$ using the stored past history. As the problem is advanced in time, new time values are added to the table. Once the table is filled, new values replace values that are

older than the delay time. There are no restrictions on t_d or d_{TM} relative to the time steps on cards 201-299. When a change in advancement time is made at a restart, the time values in this table are changed to have time values as if the problem in the restart had run to the new advancement time.

A15.3.9 Unit Trip Component

This component is indicated by TRIPUNIT in Word 2 of card 205CCC00 or 205CCCC0. The component is defined by

$$Y = S \bullet U(\pm T_1)$$

where U is 0.0 if the trip, T_1 , is false and is 1.0 if the trip is true. If the complement of T_1 is specified, U is 1.0 if the trip is false and 0.0 if the trip is true.

W1(I) Trip number. A minus sign may prefix the trip number to indicate that the complement of the trip is to be used.

A15.3.10 Trip Delay Component

This component is indicated by TRIPDLAY in Word 2 of card 205CCC00 or 205CCCC0. The component is defined by

$$Y = ST_{rptim}(T_1)$$

where T_{rptim} is the time the trip last turned true. If the trip is false, the value is -1.0; if the trip is true, the value is 0 or a positive number.

W1(I) Trip number, T_1 .

A15.3.11 Integer Power Component

This component is indicated by POWERI in Word2 of card 205CCC00 or 205CCCC0. The component is defined by

$$Y = SV_1^I$$

W1(A) Alphanumeric name of the variable request code for V_1 .

W2(I) Integer name of the variable request code for V_1 .

W3(I) I .

A15.3.12 Real Power Component

This component is indicated by POWERR in Word 2 of card 205CCC00 or 205CCCC0. The component is defined by

$$Y = S V_1^R$$

W1(A) Alphanumeric name of the variable request code for V_1 .

W2(I) Integer name of the variable request code for V_1 .

W3(R) R.

A15.3.13 Variable Power Component

This component is indicated by POWERX in Word 2 of card 205CCC00 or 205CCCC0. The component is defined by

$$Y = S V_1^{V_2}$$

W1(A) Alphanumeric name of the variable request code for V_1 .

W2(I) Integer name of the variable request code for V_1 .

W3(A) Alphanumeric name of the variable request code for V_2 .

W4(I) Integer name of the variable request code for V_2 .

A15.3.14 Proportional-Integral Component

This component is indicated by PROP-INT in Word 2 of card 205CCC00 or 205CCCC0. The component is defined by

$$Y = S \left[A_1 V_1 + A_2 \int_0^t V_1 dt + A_2 I_o \right]$$

where I_o is the initial value of the integral.

In Laplace transform notation,

$$Y(s) = S \left[A_1 + \frac{A_2}{s} \right] V_1(s)$$

If the initialization flag is 0, Word 5 = 0 on card 205CCC00 or 205CCCC0, $Y_o = \text{Word 4}$ and the initial value of the integral is $I_o = \frac{Y_o - SA_1 V_{1o}}{A_2 S}$.

If the initialization flag is 1, Word 5 = 1 on card 205CCC00 or 205CCCC0, the initial value of the integral is $I_o = 0$ and the initial value of Y is $Y_o = SA_1 V_{1o}$.

If it is desired that the output quantity Y remain constant as long as the input quantity remains constant, V_1 must initially be 0 regardless of the initialization flag because the integration of V_1 will otherwise continue to make a contribution as time increases.

W1(R) A_1 .

W2(R) A_2 .

W3(A) Alphanumeric name of the variable request code for V_1 .

W4(I) Integer name of the variable request code for V_1 .

A15.3.15 Lag Component

This component is indicated by LAG in Word 2 of card 205CCC00 or 205CCCC0. This component is defined by

$$Y = \int_0^t \left(\frac{SV_1 - Y}{A_1} \right) dt$$

or, in Laplace transform notation,

$$Y(s) = \frac{S}{1 + A_1 S} V_1(s)$$

If the initialization flag is 0, Word 5 = 0 on card 205CCC00 or 205CCCC0, $Y_o = \text{Word 4}$.

If the initialization flag is 1, Word 5 = 1 on card 205CCC00 or 205CCCC0, the initial value of Y is set to $Y_o = SV_{1o}$ so that there is no contribution from the integral initially. There is no input quantity that can be used to set a non-zero initial value for the integral. Y remains constant as long as V_1 retains its initial value.

W1(R) Lag time, A_1 (s).

W2(A) Alphanumeric name of the variable request code for V_1 .

W3(I) Integer name of the variable request code for V_1 .

A15.3.16 Lead-Lag Component

This component is indicated by LEAD-LAG in Word 2 of card 205CCC00 or 205CCCC0. The component is defined by

$$Y = \frac{A_1 S V_1}{A_2} + \int_0^t \left(\frac{S V_1 - Y}{A_2} \right) dt + I_o$$

where I_o is the initial value of the integral.

In Laplace transform notation,

$$Y(s) = S \left[\frac{1 + A_1 s}{1 + A_2 s} \right] V_1(s)$$

If the initialization flag is 0, Word 5 = 0 on card 205CCC00 or 205CCCC0, $Y_o = \text{Word 4}$ and the initial value of the integral is $I_o = Y_o - \frac{A_1 S V_{1o}}{A_2}$.

If the initialization flag is 1, Word 5 = 1 on card 205CCC00 or 205CCCC0, the initial value of Y is set to $Y_o = S V_{1o}$ so that there is no contribution from the integral initially. There is no input quantity that can be used to set a non-zero initial value for the integral term. Y remains constant as long as V_1 retains its initial value.

W1(R) Lead time, A_1 (s).

W2(R) Lag time, A_2 (s).

W3(A) Alphanumeric name of the variable request code for V_1 .

W4(I) Integer name of the variable request code for V_1 .

A15.3.17 Constant Component

Cards 205CCC01-CCC09 or 205CCCC1-CCCC9 are not entered. The quantity in Word 3 of card 205CCC00 or 205CCCC0 is the constant value used for this component.

A15.3.18 Shaft Component

This component is indicated by SHAFT in Word 2 of card 205CCC00 or 205CCCC0. A GENERATR component may optionally be associated with a SHAFT component. The SHAFT component advances the rotational velocity equation

$$\sum_i I_i \frac{d\omega}{dt} = \sum_i \tau_i - \sum_i f_i \omega + \tau_c$$

where I_i is the moment of inertia of component i , ω is rotational velocity, τ_i is torque of component i , f_i is the friction factor of component i , and τ_c is an optional torque from a control component. The summations include the shaft as well as the pump, turbine, and generator components that are connected to the shaft.

The SHAFT control component differs somewhat from other control components. The scale factor on card 205CCC00 or 205CCCC0 must be 1.0. The initial value and optional minimum and maximum values have units (rad/s, rev/min), and British-SI units conversion are applied to these quantities. The output of the SHAFT in minor and major edits is in the requested units. Card number ranges are restricted so that both data to complete the SHAFT component description and optional data to describe a generator can be entered. Units conversion is applied to the following cards.

A15.3.18.1 Shaft Description, 205CCC01-CCC05 or 205CCCC1-CCCC5.

- | | |
|-------|--|
| W1(I) | Torque control variable number. If 0, there is no contribution to torque from the control system. If nonzero, the control variable with this number is assumed to be a torque and is added to the torques from the other components attached to the shaft. The torque must be in SI units. |
| W2(R) | Shaft moment of inertia, I_i ($\text{kg}\cdot\text{m}^2$, $\text{lb}\cdot\text{ft}^2$). |
| W3(R) | Friction factor for the shaft, f_i ($\text{N}\cdot\text{m}\cdot\text{s}$, $\text{lb}_f\cdot\text{ft}\cdot\text{s}$). |
| W4(A) | Type of attached component. Enter either TURBINE, PUMP, or GENERATR. |
| W5(I) | Component number. This is the hydrodynamic component number for a TURBINE or PUMP, or the control variable number for this SHAFT component if GENERATR. |

Additional two-word pairs may be entered to attach additional components to the shaft, up to a total of ten components. Only one generator, the one which is defined as part of this SHAFT component, may be attached.

A15.3.18.2 Generator Description, 205CCC06 or 205CCCC6. Each SHAFT component may optionally define an associated GENERATR component.

W1(R)	Initial rotational velocity (rad/s, rev/min).
W2(R)	Synchronous rotational velocity (rad/s, rev/min).
W3(R)	Moment of inertia, I_i ($\text{kg}\cdot\text{m}^2$, $\text{lb}\cdot\text{ft}^2$).
W4(R)	Friction factor, f_i ($\text{N}\cdot\text{m}\cdot\text{s}$, $\text{lb}_f\cdot\text{ft}\cdot\text{s}$).
W5(I)	Generator trip number. When the trip is false, the generator is connected to an electrical distribution system and rotational velocity is forced to the synchronous speed. When the trip is true, the generator is not connected to an electrical system and the generator and shaft rotational velocity is computed from the rotational velocity equation.
W6(I)	Generator disconnect trip number. If 0, the generator is always connected to the shaft. If nonzero, the generator is connected to the shaft when the trip is false and disconnected when the trip is true.

A15.3.19 PUMPCTL Component

This component is specified when using the self-initialization option and loop flow control is desired, but it is not limited to that use. For each PUMPCTL component enter:

W1(A)	Alphanumeric name of setpoint variable.
W2(I)	Parameter part of setpoint variable.
W3(A)	Alphanumeric name of sensed variable.
W4(I)	Parameter part of sensed variable.
W5(R)	Scale factor(s) applied to sensed and setpoint values, S_i . Must be nonzero.
W6(R)	Integral name time constant, T_2 (s).
W7(R)	Proportional part time constant, T_1 (s).

Standard use of PUMPCTL controller require the following interpretation of the input data. W1 and W2 contain CNTRLVAR and CCC (or CCCC), respectively, where CCC (or CCCC) is a CONSTANT type control element containing the desired (setpoint) flow rate. W3 is MFLOWJ, and W4 is the junction number at which the flow is to be sensed and compared to the setpoint. W5 is the S_i value used to divide

the difference between the desired (setpoint) and sensed flow rate to produce the error signal E_1 . E_1 must be initially 0 if it is intended to have the controller output remain constant as long as the input quantities remain constant. W6 and W7 are the T_2 and T_1 values, respectively. All variables having units must be in SI units.

A15.3.20 STEAMCTL Component

This component is specified when using the self-initialization option to control steam flow from one or more steam generators, but it is not limited to that use. For each STEAMCTL component enter:

W1(A)	Alphanumeric name of setpoint variable.
W2(I)	Parameter part of setpoint variable.
W3(A)	Alphanumeric name of sensed variable.
W4(I)	Parameter part of sensed variable.
W5(R)	Scale factor(s) applied to sensed and setpoint values, S_j . Must be nonzero.
W6(R)	Integral name time constant, T_4 (s).
W7(R)	Proportional part time constant, T_3 (s).

Standard use of the STEAMCTL controller requires the following interpretation of the input data. W1 and W2 would contain CNTRLVAR and CCC (or CCCC), respectively, where CCC (or CCCC) is a CONSTANT type control element. This constant would be the desired (setpoint) cold leg temperature (for suboptions A and B) or secondary pressure (suboptions C and D). W3 would be TEMPF (for suboptions A and B) or P (for suboptions C and D), and W4 would be the volume number where the temperature (suboptions A and B) or pressure (suboptions C and D) is sensed. W5 is the S_j value used to divide the difference between the desired (setpoint) and sensed temperature (suboptions A and B) or pressure (suboptions C and D) to produce the error signal E_2 . E_2 must be initially 0 if it is intended to have the controller output remain constant as long as the input quantities remain constant. W6 and W7 are the T_4 and T_3 values respectively. All variables having units must be in SI units.

A15.3.21 FEEDCTL Component

This component is specified when using the self-initialization option to control feedwater flow to a steam generator, but it is not limited to that use. For each FEEDCTL component enter:

W1(A)	Alphanumeric name of first setpoint variable.
W2(I)	Parameter part of first setpoint variable.

W3(A)	Alphanumeric name of sensed variable to be compared with first setpoint.
W4(I)	Parameter part of sensed variable to be compared with first setpoint.
W5(R)	Scale factor applied to sensed and setpoint values (first setpoint), S_k . Must be nonzero.
W6(A)	Alphanumeric name of second setpoint variable.
W7(I)	Parameter part of second setpoint variable.
W8(A)	Alphanumeric name of sensed variable to be compared with second setpoint.
W9(I)	Parameter part of sensed variable to be compared with second setpoint.
W10(R)	Scale factor applied to sensed and setpoint values (second setpoint), S_m . Must be nonzero.
W11(R)	Integral name time constant, T_6 (s).
W12(R)	Proportional part time constant, T_5 (s).

Standard use of the FEEDCTL controller requires the following interpretation of the input data. W1 and W2 contain CNTRLVAR and CCC (or CCCC), respectively, where CCC (or CCCC) is a CONSTANT type control element. This constant would be the desired (setpoint) steam generator secondary side water level. The latter may be expressed alternatively as a desired secondary coolant mass or as a differential pressure measured between two locations in the steam generator downcomer. W3 and W4 would contain CNTRLVAR and CCC (or CCCC), respectively, where CCC (or CCCC) is the number of the control component that describes the summing algorithm to compute the sensed variable (e.g., collapsed water level may be computed by summing the product of VOIDF and volume length over the control volumes in the riser section). W5 is the S_k value used to divide the difference between the desired (setpoint) and sensed water level to produce the first portion of the error signal E_3 . W6 is MFLOWJ, and W7 is the junction number of the steam exit junction from the steam generator. W8 is MFLOWJ, and W9 is the junction number of the feedwater inlet junction. W10 is the S_m value used to divide the difference between the sensed steam flow and sensed feedwater flow to produce the second portion of the error signal E_3 . E_3 must be initially 0 if it is intended to have the controller output remain constant as long as the input quantities remain constant. W11 and W12 are the T_6 and T_5 values, respectively. All variables having units must be in SI units.

A15.3.22 DIGITAL Component

This component is indicated by DIGITAL in Word 2 of Card 205CCC00 or 205CCCC0. The component is defined by

$$Y = SV_1 f\{t_s\}(t - t_d)$$

where t is time and t_d is the delay time.

W1(A) Alphanumeric name of the variable request code for V_1 .

W2(I) Integer name of the variable request code for V_1 .

W3(R) Sampling time, t_s (s).

W4(R) Delay time, t_d (s).

A16 REACTOR KINETICS

These cards are required if a space-independent (point) reactor kinetics or a nodal reactor kinetics calculation is desired. These cards may be entered in a new problem or on a restart. If no reactor kinetics data are present in a restart problem, the data will be added; if reactor kinetics data are already present, the data are deleted and replaced by the new data. A complete set of reactor kinetics data must always be entered. Initial conditions are computed the same for new or restart problems; the initial conditions can be obtained from assuming infinite operating time at the input power or from an input power history.

A16.1 Reactor Kinetics Type, 30000000

This card is required. Read in by rrkin.

- W1(A) Kinetics type. Enter POINT or DELETE. Enter POINT for the point reactor kinetics option. Enter DELETE in a restart problem if reactor kinetics is to be deleted. No other data are needed if reactor kinetics is being deleted.
- W2(A) Feedback type. Enter SEPARABL, TABLE3, TABLE4, TABLE3A, or TABLE4A. If Word 2 is not entered, a default value is assumed. If the kinetics type is POINT, the default is SEPARABL. If SEPARABL is entered, reactor kinetics feedback due to moderator density, void fraction weighted moderator temperature, and fuel temperature is assumed to be separable, and feedback data are entered on cards 30000501-0899. If TABLE3, TABLE4, TABLE3A, or TABLE4A is entered, reactivity is obtained from a table defining reactivity as a function of three or four variables using cards 30001001-2999. If TABLE3 or TABLE4 are entered, the variables are moderator density, void fraction weighted moderator temperature, fuel temperature, and boron density. If TABLE3A or TABLE4A is entered, the variables are void fraction, liquid moderator temperature, volume average fuel temperature, and boron concentration. If TABLE3 or TABLE3A is entered, the first three of the variables in one of the sets defined above are used, and if TABLE4 or TABLE4A is entered, all four variables are used.

A16.2 Reactor Kinetics Information, 30000001

Read in by rrkin.

- W1(A) Fission product decay type. Enter NO-GAMMA for no fission product decay calculations, GAMMA for standard fission product decay calculations, or GAMMA-AC for fission product decay plus actinide decay calculations.
- W2(R) Total reactor power (W). This is the sum of fission power, fission product decay power, and actinide decay power. Watts are used for both SI and British units. This must be > 0.0.

- W3(R) Initial reactivity (dollars). This quantity must be less than or equal to 0.0. It is recommended that this quantity be less than or equal to $-1.0\text{E-}60$; otherwise, the CPU time may increase dramatically.
- W4(R) Delayed neutron fraction over prompt neutron generation time (s^{-1}).
- W5(R) Fission product yield factor. This is usually 1.0 for best-estimate problems, and 1.2 has been used with ANS73 data for conservative mode problems. The factor 1.0 is assumed if this word is not entered.
- W6(R) ^{239}U yield factor. This is the number of ^{239}U atoms produced per fission times any conservative factor desired. The factor 1.0 is assumed if this word is not entered.
- W7(R) Fissions per initial fissile atom, ψ . Used in factor $G(t) [= 1.0 + (3.24 \cdot 10^{-6} + 5.23 \cdot 10^{-10}t) T^{0.4}\psi]$ to account for neutron capture in fission products when using ANS79-1 or ANS79-3 option. Entering this quantity as a nonzero includes the G factor as part of the decay heat. The factor is not included if this quantity is not entered or is entered as 0. Entering this word as a positive quantity indicates that the equation is to be used for shutdown time up to 10^4 seconds, and the table is to be used from that time on. Entering this word as a negative number indicates that the table is to be used for all shutdown times. Note that there is a discontinuity in $G(t)$ when switching between and equation and the table. The standard indicates that the table can be used for all shutdown times and that would result in a higher neutron absorption capture effect. The magnitude of this quantity if nonzero must be greater than or equal to 1.0 and less than or equal to 3.0.
- W8(R) Reactor operating time T. This quantity is the T in the expression given in W7 above. The unit for this quantity is given in the next word. If not entered or entered as 0, this quantity defaults to 52 wk. This quantity is used only if the power history data in Section A16.7 are not entered. When the power history data are entered, the reactor operating time is obtained from that data. When the power history data are not entered, an infinite operating time is assumed in initializing the decay heat variables, and if the equation form of $G(T)$ is being used, the quantity in this word is used with the shutdown period t set to 0 to determine the G factor at the start of the simulation. The limit for this quantity is $1.2614 \cdot 10^8$ seconds.
- W9(A) Units for W8 above. Must be sec, min, hr, day, wk.
- W10(R) Differential boron worth in $\$/\text{ppm}$. This value is normally given in core design reports in units of pcm/ppm . The abbreviation pcm stands for per cent mille or 10^{-5} of $(\Delta k)/k$, while dollars are measured in units of the delayed neutron fraction, $\$ = (\Delta k)/(k\beta)$.

W11(I) Boron feedback treatment flag. This word is optional, and a value of 0 is the default which indicates that the original method of multiplying the boron worth times the liquid density will be used to obtain the boron density. A value of 1 indicates that the method of multiplying the boron worth times the mixture density will be used to obtain the boron density.

A16.3 Fission Product Decay Information, 30000002

This card is optionally entered for POINT problems if W1 of card 30000001 contains GAMMA or GAMMA-AC. If this card is not entered, the Proposed 1973 ANS Standard fission product data are used if default data are used. Read in by rkin.

W1(A) Fission product type. Enter ANS73, ANS79-1, ANS79-3, or ANS79-4. If default fission product data are used, ANS73 specifies the Proposed 1973 ANS Standard data. If fission product data are entered, ANS73 and ANS79-1 specify only one isotope. ANS79-3 specifies three isotopes and ANS94-4 specifies four isotopes.

ANS79-1 specifies the 1979 Standard data for ^{235}U .

ANS79-3 specifies the 1979 ANS Standard data for the three isotopes, ^{235}U , ^{238}U , and ^{239}Pu . This option requires that power fractions for each isotope must be entered. The entry of the number of decay heat groups for each isotope is optional and defaults to 23. The sum of the power fractions must add to 1.0. The number of groups for each isotope must be less than or equal to 50.

ANS94-4 specifies the 1994 ANS Standard data for the four isotopes, ^{235}U , ^{238}U , ^{239}Pu , and ^{241}Pu . To use this option, the user enters all six words for the ANS79-3 option (three words for the power fractions and three words for the number of decay heat groups), and then adds two additional words for the power fraction of ^{241}Pu and the number of decay heat groups for ^{241}Pu . The sum of the power fractions must add to 1.0. The number of groups for each isotope must be less than or equal to 50.

W2(R) Energy release per fission (MeV/fission). If not entered or 0, the default value of 200 MeV/fission is used.

W3(R) Fraction of power generated in ^{235}U . This word is required for ANS79-3 and ANS94-4.

W4(R) Fraction of power generated in ^{238}U . This word is required for ANS79-3 and ANS94-4.

W5(R) Fraction of power generated in ^{239}Pu . This word is required for ANS79-3 and ANS94-4.

- W6(I) Number of decay groups for ^{235}U . This word is optional for ANS79-3 (defaults to 23) and required for ANS94-4. The number of groups for each isotope must be less than or equal to 50.
- W7(I) Number of decay groups for ^{238}U . This word is optional for ANS79-3 (defaults to 23) and required for ANS94-4. The number of groups for each isotope must be less than or equal to 50.
- W8(I) Number of decay groups for ^{239}Pu . This word is optional for ANS79-3 (defaults to 23) and required for ANS94-4. The number of groups for each isotope must be less than or equal to 50.
- W9(R) Fraction of power generated in ^{241}Pu . This word is required for ANS94-4.
- W10(I) Number of decay groups for ^{241}Pu . This word is required for ANS94-4. The number of groups for each isotope must be less than or equal to 50.

A16.4 Delayed Neutron Constants, 30000101-0199

If these cards are missing, constants for the six generally accepted delayed neutron groups are supplied. Otherwise, two numbers for each delay group are entered, one or more pairs per card. Card numbers need not be consecutive. The number of pairs on these cards defines the number of delay groups. Up to 50 delay groups may be entered. Read in by rkin.

- W1(R) Delayed neutron precursor yield ratio.
- W2(R) Delayed neutron decay constant (s^{-1}).

A16.5 Fission Product Decay Constants, 30000201-0299

These cards are not needed if W1 of card 30000001 is NO-GAMMA. If this word is GAMMA or GAMMA-AC, data from these cards or default data are used to define fission product decay. If the cards are missing, data as defined in W1 of card 30000002 are supplied. Up to 50 fission product groups may be entered. Data are entered on cards similarly to cards 30000101-0199. The factor in W5 of card 30000001 is applied to the yield fractions. Read in by rkin.

- W1(R) Fission product yield fraction.
- W2(R) Fission product decay constant (s^{-1}).

A16.6 Actinide Decay Constants, 30000301-0399

These cards are not needed unless W1 of card 30000001 is GAMMA-AC. If GAMMA-AC is entered, data from these cards or default data are used to define actinide decay. If the cards are missing, default data are supplied. Read in by rkin.

W1(R) Energy yield from ^{239}U decay (Mev).

W2(R) Decay constant of ^{239}U (s^{-1}).

W3(R) Energy yield from ^{239}Np (Mev).

W4(R) Decay constant of ^{239}Np (s^{-1}).

A16.7 Power History Data, 30000401-0499

If these cards are not present, initial conditions for fission product and actinide groups are for steady-state operation at the power given in W2 of card 30000001. This is equivalent to operation at that power for an infinite time. If these cards are present, the power history consisting of power and time duration is used to determine the fission product and actinide initial conditions. The power from gamma and actinide decay is assumed to be 0 at the beginning of the first time duration. Data are entered in three-word, six-word, or seven-word sets; one or more sets per card. Card numbers need not be consecutive. Read in by rkin.

W1(R) Reactor power (W). This quantity is the total reactor power, that is, the sum of fission power and decay power, and must be ≥ 0 . If a decay power obtained from the power history exceeds this quantity, the fission power is assumed to be 0.

W2(R) Time duration. Units are as given in next word. This quantity must be ≥ 0 .

W3(A) Time duration units. Must be sec, min, hr, day, or wk.

W4(R) Fraction of power generated in ^{235}U . This word is required if ANS79-3 or ANS94-4 is entered in W1 of Card 30000002.

W4(R) Fraction of power generated in ^{235}U . This word is required if ANS79-3 or ANS94-4 is entered in W1 of Card 30000002. The sum of the power fractions must add to 1.0.

W5(R) Fraction of power generated in ^{238}U . This word is required if ANS79-3 or ANS94-4 is entered in W1 of Card 30000002. The sum of the power fractions must add to 1.0.

W6(R) Fraction of power generated in ^{239}Pu . This word is required if ANS79-3 or ANS94-4 is entered in W1 of Card 30000002. The sum of the power fractions must add to 1.0.

W7(R) Fraction of power generated in ^{241}Pu . This word is required if ANS94-4 is entered in W1 of Card 30000002. The sum of the power fractions must add to 1.0.

A16.8 Feedback Input

Feedback information for point kinetics information are entered on the following cards. For steady-state computations in which constant power is desired, these cards can be omitted and the feedback reactivity will be 0.

A16.8.1 Reactivity Curve or Control Variable Numbers, 30000011-0020

Reactivity (or scram) curves from the general tables (cards 202TTTNN) or control variables that contribute to reactivity feedback are specified on these cards. These cards are not used if there are no references to reactivity contributions from general tables or control variables. Tables and control variables referenced must be defined. No error is indicated if reactivity curves are defined but not referenced on this card, but memory space is wasted. Curve numbers, which are the TTT of the general table card number or control variable number code, are entered one or more per card. Card numbers need not be consecutive. Read in by rkin.

W1(I) Table or control variable number. Up to 20 numbers may be entered. Numbers from 1-999 indicate general table numbers. Numbers greater than 10000 indicate the control variable whose number is the entered number minus 10000.

A16.8.2 Density Reactivity Table, 30000501-0599

This table is required if the SEPARABL option is being used and if cards 30000701-0709 are entered. One or more pairs of numbers are entered to define reactivity as a function of moderator density. Data are entered one or more pairs per card, and card numbers need not be consecutive. Up to 100 pairs may be entered. The table uses linear interpolation for segments between table search argument values. For search arguments beyond the range of entered data, the end-point values are used. Read in by rkinp.

W1(R) Moderator density (kg/m^3 , lb/ft^3).

W2(R) Reactivity (dollars).

A16.8.3 Doppler Reactivity Table, 30000601-0699

This table is required if the SEPARABL option is being used and if cards 30000801-0899 are entered. One or more pairs of numbers are entered to define Doppler reactivity as a function of volume-averaged fuel temperature. Heat structure composition data cards 1CCCG201-9 need to exclude the gap and the cladding for the volume-average fuel temperatures. Data are entered one or more pairs per card, and card numbers need not be consecutive. Up to 100 pairs may be entered. The table uses linear interpolation for segments between table search argument values. For search arguments beyond the range of entered data, the end-point values are used. Read in by rkinp.

W1(R) Temperature (K, °F).

W2(R) Reactivity (dollars).

A16.8.4 Volume Weighting Factors, 30000701-0799

These cards are used only if the SEPARABL option is being used and are omitted if no reactor kinetics feedback from hydrodynamics is present. Each card contains the input for reactivity feedback due to conditions in one or more hydrodynamic volumes. Words 1 and 2 are a volume number and an increment. Words 3 and 4 are the reactivity data for the volume defined by Word 1; Words 5 and 6 are the reactivity data for the volume defined by Word 1 plus Word 2; Words 7 and 8 contain data for the volume defined by Word 1 plus two times Word 2; etc. Each card must contain at least four words. Volumes must be defined by hydrodynamic component data cards, and any volume reactivity data must be defined only once on these cards. Card numbers need not be consecutive.

W1(I) Hydrodynamic volume number.

W2(I) Increment.

W3(R) Weighting factor for density feedback, $W_{\rho i}$. See Volume 1 of this manual for a discussion of the symbols.

W4(R) Water temperature coefficient, a_{Wi} (dollars/K, dollars/°F). As defined in Volume 1, the weighting factor in Word 3 is not applied to this quantity.

A16.8.5 Heat Structure Weighting Factors, 30000801-0899

These cards are used only if the SEPARABL option is being used and are omitted if no reactor kinetics feedback from heat structures are present. Each card contains the input for reactivity feedback due to conditions in one or more heat structures representing fueled portions of the reactor. Data are entered in a manner similar to cards 30000701-0799. Read in by rkinp.

For each heat structure specified on these cards, input on the heat structure data cards 1CCCG2NN must define the fueled region as the region over which the volume-average temperature is computed.

Usually, either Word 3 or 4 is 0.

W1(I) Heat structure number.

W2(I) Increment.

W3(R) Weighting factor for doppler feedback, W_{Fi} . See Volume I of this manual for a discussion of the symbols.

W4(R) Fuel temperature coefficient, a_{Fi} (dollars/K, dollars/ $^{\circ}$ F). As defined in Volume 1, the weighting factor in Word 3 is not applied to this quantity.

A16.8.6 Volume-Weighting Factors, 30001701-1799

These cards are used only if the TABLE3, TABLE3A, TABLE4, or TABLE4A option is being used. Each card contains the weighting factor for reactivity feedback due to moderator density (void fraction), void weighted moderator temperature (liquid moderator temperature), and spatial boron density (boron concentration) in one or more hydrodynamic volumes. The quantities preceding the quantities within parentheses are used if TABLE3 or TABLE4 has been entered; the quantities within parentheses are used if TABLE3A or TABLE4A has been entered. The same factor is assumed to apply to all three effects, so only one factor is entered for each value. At least three quantities must be entered on each card. The use of the increment field is similar to that in Section A16.8.4. Read in by rkinp.

W1(I) Hydrodynamic volume number.

W2(I) Increment.

W3(R) Weight factor, $W_{\rho i}$.

A16.8.7 Heat Structure Weighting Factors, 30001801-1899

These cards are used only if the TABLE3, TABLE 3A, TABLE4, or TABLE4A option is being used. Each card contains the weighting factor for reactivity feedback due to temperature in one or more heat structures. At least three quantities must be entered on each card. The use of the increment field is similar to that in Section A16.8.4. Read in by rkinp.

W1(I) Heat structure number.

W2(I) Increment.

W3(R) Weight factor, W_{Fi} .

A16.8.8 Feedback Table Coordinate Data, 300019C1-19C9

If the TABLE3 or TABLE3A option is being used, the feedback table is a function of three variables: moderator density or void fraction ($C = 1$), void weighted moderator temperature or liquid moderator temperature ($C = 2$), and fuel temperature ($C = 3$). If the TABLE4 or TABLE4A option is being used, the feedback table is a function of four variables: the three above and spatial boron density or boron concentration ($C = 4$). Which variables are used depend on the feedback option used (see W2, card 30000000 in Section A16.1). These cards define the coordinates of the table, and table values are entered (on another card set) for each point defined by all combinations of the coordinate values. The table size is the product of the number of coordinate values entered for each variable. At least two coordinate points must be entered, and up to twenty points may be entered for each variable. The table uses multi-dimensional interpolation for values between table coordinate values. For values beyond the range of the

entered coordinate values, the end-point coordinate values are used. Coordinate values are entered in increasing magnitude, one or more per card on one or more cards as desired. Card numbers need not be consecutive. The C in the parentheses above defines the C to be used in the card number. Read in by rrkinp.

W1(R) Coordinate value (kg/m^3 , lb/ft^3 for moderator and spatial boron densities; K, $^{\circ}\text{F}$ for moderator and heat structure temperatures; void fractions and boron concentrations are dimensionless).

A16.8.9 Feedback Table Data, 30002001-2999

Values defining the table are entered in pairs. The first is a coded number defining the position of the table entry. The second number is the table entry. One or more pairs may be entered on one or more cards as needed. Card numbers need not be consecutive. There is no required ordering for the coded number, but a coded number may be entered only once. Read in by rrkinp.

W1(I) Coded number. The coded number has the form ddmmffbb, where the letter pairs represent coordinate numbers of the independent variables of the table. The dd pair refers to moderator density or void fraction, mm refers to void weighted moderator temperature or liquid moderator temperature, ff refers to heat structure temperature, and bb refers to spatial boron density or boron concentration. The paired numbers range from 00 to one less than the number of coordinate values for that variable. The 00 pair refers to the first coordinate value. If boron dependence is not included, bb is always 00. All table values must be entered. (A future version may allow gaps that are filled in by interpolation.)

W2(R) Table value.

Feedback Input

A17 INSTALLATION AND OPERATING PROCEDURES

RELAP5/MOD3.3 Patch04 has been installed on Windows, Linux, Macintosh (PowerPC), and Macintosh (Intel) computers. In addition, the following compilers have been tested: g77, g95, gfortran, Intel Fortran Compiler (IFC), Lahey/Fujitsu F95, and Compaq Visual Fortran (CVF). The following table shows the combinations of compilers and platforms that have been tested.

Table 17.0-1 Matrix of tested platforms and compilers.

	WindowsXP (SP3), 32-bit,	Linux, RH9, 2.4.28	Linux, RHEL5, 2.6.18	Mac (Intel), OS X 10.6.4	Mac (PowerPC), OS X 10.5.2
Intel Fortran Compiler	v11.1	v9.0	v11.0	v11.1	
Compaq Visual Fortran	v6.6B				
Lahey/ Fujitsu F95		v6.2c	v8.1b		
gfortran	gcc 3.4.4		gcc 4.5.0	gcc 4.6.0	gcc 4.4.1
g95			v0.92		v0.91
g77	gcc 3.4.4	gcc 3.2.2			gcc 3.4.6

A description of the installation procedure is provided below in Section A17.1, which includes a description of installing from pre-built executables along with a description of building the code from source on Unix, Linux, Windows, and Macintosh (Intel and PowerPC) platforms. In Section A17.2 is a description of the RELAP5 execution procedure, which includes a description of running RELAP5 from within SNAP and also running from the command line.

A17.1 RELAP5 Installation

There are two basic methods of installing RELAP5. The first is to utilize the pre-built executables provided on the DVD. The second is to build the RELAP5 executable from the source files provided on the DVD. The following sections describe both methods.

A17.1.1 Installing RELAP5 Using Pre-Built Executables

Pre-built executables of RELAP5/MOD3.3 Patch04 (both optimized and debug) are available for each of the platform/OS/compiler combinations listed above in **Table 17.0-1**. These executables, along with the thermodynamic property files (tpf*), can be found on the distribution DVD for Patch04 under the folder titled "Executables". In this folder are two subfolders, one for big endian machines (e.g. MAC PPC) and one for little endian machines (e.g. Intel-based machines). All of the executables available can be found in these two subfolders, along with ReadMe files which describe the OS and compiler version and the compiler options used to build each executable.

To use the pre-built executables, simply copy the appropriate executable for the user's platform/OS/compiler configuration to a desired location on the computer. The sample input files can be copied from the SampleInput folder on the DVD as well; this folder also contains scripts provided to run the cases and a ReadMe explaining how to run the cases.

To run all the sample input decks contained in the SampleInput folder, first copy the SampleInput folder from the DVD to a location on the user's computer. Copy one of the desired executables from the Executables folder on the DVD into the local SampleInput folder on the user's computer, along with the tpf* files. Rename the executable to "relap5.x" (even if it's a Windows executable with a *.exe extension). Then, open a BASH shell (requires Cygwin to be installed if on Windows), cd to the location where the local SampleInput folder resides, and perform the following:

```
> ./runx
```

Note that this script calls the "runit" script, which also resides in this folder. Note also that the "runit" script is expecting to find "relap5.x", along with the tpf* files (tpfh2o, tpfd2o, tpfh2onew), in this folder. So, as discussed above, these files will need to be copied in from the Executables folder on the DVD.

To run a single sample input deck in this folder, open a BASH shell, cd to the location where the local SampleInput folder resides, and perform the following:

```
> ./runit [input.i]
```

where input.i should be replaced with the actual name of the input file.

After running any or all of the sample cases, a utility is provided called "runcheck" which scans the output file(s) for any errors. To run this script, simply type:

```
> ./runcheck
```


A17.1.2 Building RELAP5 From Source

The following sections describe how to build RELAP5/MOD3.3 Patch04 from source on UNIX, Linux, Macintosh, and Windows platforms. These build procedures were all tested from a BASH shell.

A17.1.2.1 UNIX, Linux, and Macintosh Installation

On the installation CD-ROM, a ReadMe file is included that contains detailed instructions on the installation procedure. Briefly, the installation procedure consists of typing the following commands:

```
> ./configure
```

```
> ./make
```

The “configure” script looks at the CPU configuration and builds the “.config” file. The “.config” file can be altered later if needed to modify the installation parameters, e.g., add the debug option. After the “configure” script is run, the installer types “make”, which makes all programs needed to install RELAP5/MOD3.3 Patch04 on the computer. The following text inset provides a description of the options available to the “configure” script:

List of “configure” Options

configure [-m {mach}] [-c {code}] [-G {gui}] [-b[m]] [-eghr] [-map] [-nan] [-trap]
[-endian (big, little)]

-m {mach} Set machine type manually. *The current available options are:*

cray	sgi64o	hp3	win32cvf
cray2	stardent	hp90	win32ifc
decrisc	solaris	vax	win32g77
decrisc2	solaris2	linuxg77	win32gft
decalpha	solaris3	linuxg95	macppg77
decalp90	solarisg	linuxgft	macppg95
ibmrisc	sun-os	linuxlf	macppgft
sgi	hp	linuxnw	macifc
sgi64	hp2	linuxifc	macgft

The following are the generally recommended options:

win32cvf	win32gft	linuxgft	macgft
win32ifc	linuxifc	macifc	macppgft

The following describes the different Linux options:

linuxg77:	linux with g77
linuxg95:	linux with g95
linuxgft:	linux with gfortran
linuxlf:	linux with Lahey/Fujitsu f95
linuxnw:	linux with NAGWare f95
linuxifc:	linux with Intel Fortran Compiler (IFC)

-b[m] Build the code with a bigger container for running large problems, where 'm' can be equal to 0 or 1. If -b0 is specified, the code will be built with a container size of 6,600,000 words (this is the default). If -b1 is specified, the code will be built with a container size of 10,000,000 words. Note: -b0 is the default, and -b is equivalent to -b1.

-c {code version} Set code version, e.g., scdap or relap (relap is default).

-e Turn on extra variables.

-g Compile the code with debug information.

-h Print this help information.

-G {gui type} Set GUI type, e.g., npa, snap, or nonpa (nonpa is default).

-map Compile the code with surplot capability in phantv, phantj, and htcond.

-nan Compile the code to set scratch variables to NaN after use (for use on hp).

-r Turn on RDMR for 3D kinetics.

-trap Compile the code with strict run-time checking (for use on linuxlf only).

-endian Set the byte order to big or little endian. The default is set automatically for the most common configurations for a given architecture. (e.g. for HP, endian='big'; for winCVF, endian='little').

If no options are specified, the script will execute with the following options:

configure -c relap -G nonpa

where the machine type will be determined by this script with some input from the user.

A17.1.2.2 Windows Installation

Installation on Windows machines involves the same scripts, and essentially the same procedure with a few additions and requirements. NOTE: The Windows installation is highly sensitive to different operating systems and user environment configurations. No guarantees are provided that the installation will proceed without problems. A strong understanding of Windows environments and the CYGWIN software is recommended before continuing with the installation process. The following is a list of software requirements for building RELAP5 on Windows:

Required:

- (1) One of the following Fortran compilers:
 - Compaq Visual Fortran
 - Intel Visual Fortran
 - gcc / gfortran
 - gcc / g77
- (2) Cygwin Tools (see “CYGWIN Installation” below for download and installation instructions)

Optional:

- (1) PVM (only for RDMR)
- (2) Microsoft Visual C++ Compiler (only for building SNAP GUI hooks)

A17.1.2.2.1 CYGWIN Installation

Starting with RELAP5 MOD3.3/Patch.01, the installation on Windows machines requires that the user have a working CYGWIN installation. CYGWIN is a UNIX-emulation package that provides UNIX functions and capabilities on Windows machines. The following provides a brief description of how to install the CYGWIN tools.

- (1) Point your web browser to: www.cygwin.com
- (2) Click on the “Install or update now!” icon, and follow the prompts described in the text inset below:
- (3) After Cygwin is installed, you will need to add the Cygwin root-bin folder (e.g. C:\cygwin\bin) to your Windows environment path. Consult your Windows User Manual or the Help menu from the Start Menu for information on how to change your path.

CYGWIN Installation Prompts

Cygwin Setup

- Click "Next"

Cygwin Setup - Choose Installation Type

- The simplest option is "Install from Internet".
- Click "Next"

Cygwin Setup - Choose Installation Directory

- Enter the directory path where you want Cygwin installed in the box titled "Root Directory".
- Choose "All Users" under "Install For".
- Click "Next"

Cygwin Setup - Select Local Package Directory

- Enter the directory path where you want the Cygwin installation files to be downloaded.
- Click "Next"

Cygwin Setup - Select Connection Type

- Generally choose "Direct Connection".
- Click "Next"

Cygwin Setup - Choose Download Site(s)

- Choose any FTP or HTTP site, preferably one that is close to you
- Click "Next"

Cygwin Setup - Select Packages

- Select the radio button for "Curr".
 - Click the "View" button until you get the "Full" view.
 - Select whichever packages you wish to install, but select these as a minimum:
 ash, bash, binutils, cygutils, cygwin, diffutils, gawk, gettext, grep, less,
 make, ncurses, sed, tcsh, which
 These packages ARE REQUIRED to compile RELAP5 on Windows. To select a particular package, simply click on the item under the "New" column until you see a version number (most of the items will originally be listed as "Skip").
 NOTE: If you are using the g77 or gfortran compiler instead of the CVF or IFC compiler, you will need to also select the "gcc-g77" or "gcc4-fortran" package as well.
 - Click "Next"
-

A17.1.2.2.2 RELAP5 Build on Windows

To install RELAP5/MOD3.3 Patch04, open an MS-DOS command prompt window and type “**bash**” to open a BASH shell. Change the directory to the RELAP5 root directory (where all the installation files reside), and perform the following commands:

```
> source ./winconf
> ./configure
> make
```

The “**source ./winconf**” command will check that the necessary CYGWIN tools are available, will modify the user’s path to ensure that the path to CVF and CYGWIN are set correctly. It is necessary to run “winconf” each time a new BASH shell is started prior to executing any commands related to RELAP5 installation and execution.

Note that sourcing of the "winconf" script is only needed when using the CVF compiler. For "win32ifc" installations with the IFC compiler, the user needs to open the "Intel Fortran Build Environment" command shell (located in Program Files for the IFC compiler) and then start BASH. For "win32g77" or "win32gft" installations, simply open a BASH shell and then run "configure" and "make". The "winconf" script does not need to be sourced for "win32ifc", "win32g77", or "win32gft".

A17.2 RELAP5 Execution

The following sections describe how to execute RELAP5 from either the SNAP GUI or a command shell.

A17.2.1 Execution Within SNAP

The SNAP GUI is a convenient tool for creating/editing RELAP5 models and also for submitting RELAP5 calculations. For information on downloading, installing and running SNAP, visit the following website:

<http://www.nrcsnap.com/>

A17.2.1.1 Configuring the SNAP Calculation Server

The first step in utilizing SNAP to run RELAP5 calculations is to configure and start the calculation server. First, open the SNAP Configuration Tool from the Start Menu: "All Programs" -- "SNAP" -- "Configuration Tool". A window should appear as shown in **Figure 17.2-1**.

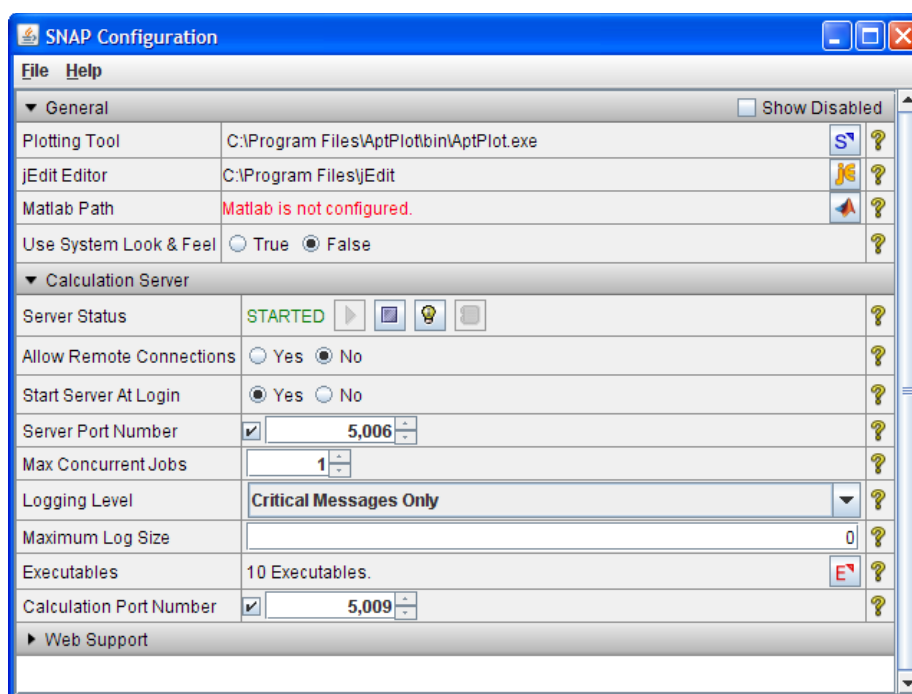


Figure 17.2-1 Main SNAP Configuration Window

Click the expand button for "Executables", which will open a window similar to that shown in **Figure 17.2-2**. To add a new RELAP5 executable, click the "Add" button. Change the Plug-in type to "RELAP", provide a name for the executable in the "Exec ID" box, and provide a description for the executable in the "Description" box. Then in the "Executable" box, click the select button, and browse to the location of the RELAP5 executable on the computer. For "Executable Arguments", none are necessary, but it is advised

that the user create a central location on his/her computer to store the TPF files relative to the RELAP5 version, and then provide the "-Z" and "-d" arguments specifying the full path to the TPF files. If the RELAP5 executable was compiled with the SNAP GUI option (configure -G snap), then select "Yes" next to "Supports Interactive Execution"; otherwise, select "No".

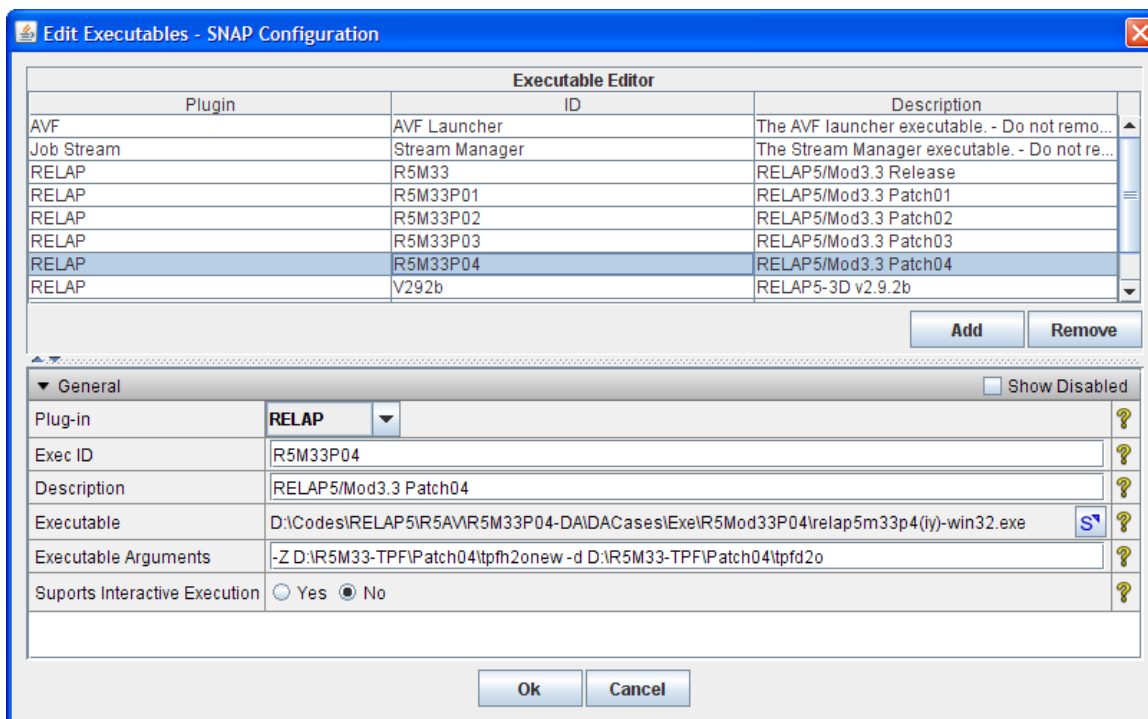


Figure 17.2-2 SNAP Configuration Window - Edit Executables

Once the desired executables have been added, close the window by selecting "Ok". If the calculation server is already started, stop the server and restart it from the main SNAP configuration window (**Figure 17.2-1**). Once the calculation server has been restarted, close the configuration window.

A17.2.1.2 Importing RELAP5 Input Decks Into SNAP

To import RELAP5 ASCII input decks into SNAP, begin by opening the SNAP Model Editor. This can be found from the Start Menu: "All Programs" -- "SNAP" -- "Model Editor". An introduction screen will open as shown in **Figure 17.2-3**. Under "Import a New Model", click "Continue >>". This will open the import file window shown in **Figure 17.2-4**. Select "RELAP ASCII" and click "OK". A window will appear allowing the user to browse to the location of the RELAP input deck, as seen in **Figure 17.2-5**. Browse to the desired location, select the RELAP input deck, choose the "RELAP5/MOD3.3" radio button, and then click "Open". SNAP will then import the model and set up the hydraulic and control system layouts (if chosen).

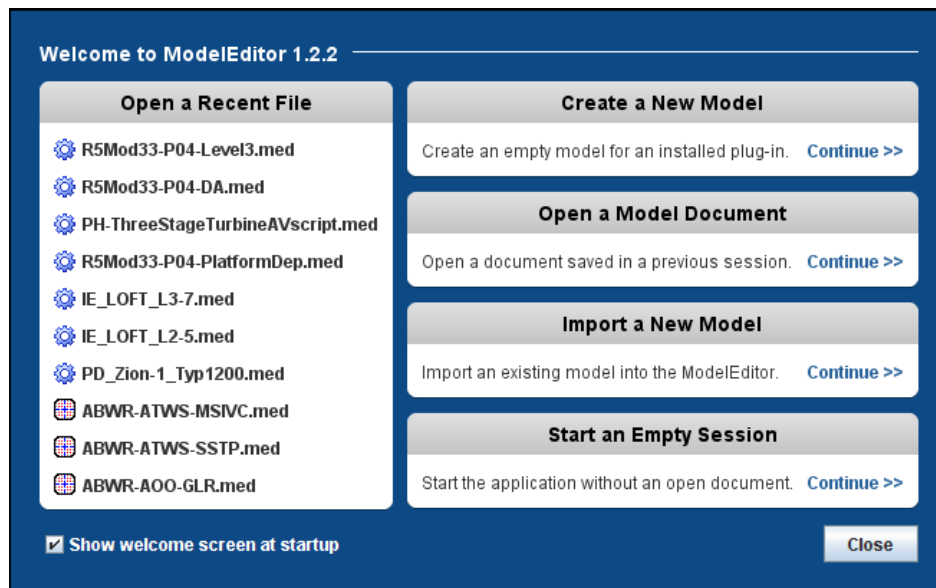


Figure 17.2-3 SNAP Model Editor Welcome Window

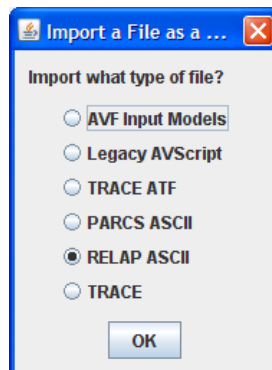


Figure 17.2-4 SNAP Import File Window

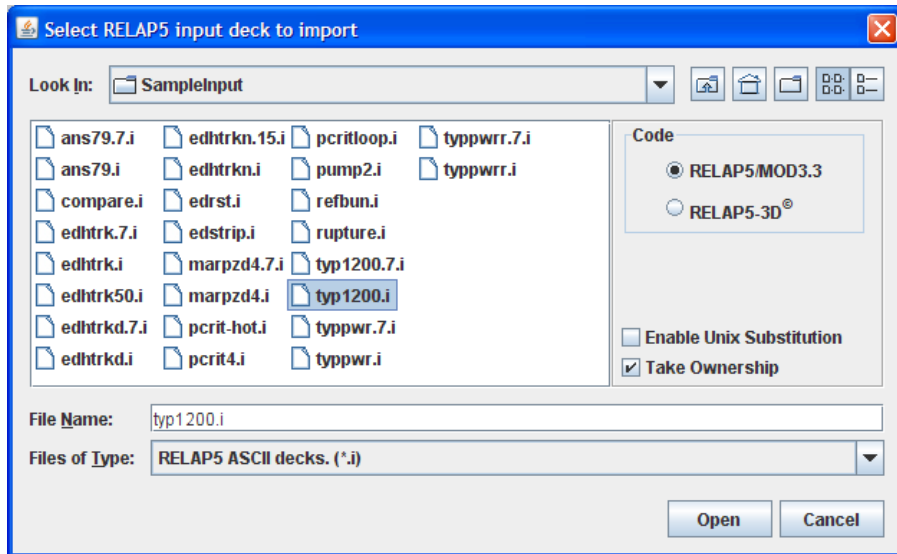


Figure 17.2-5 SNAP Import RELAP5 Input Deck Window

Once the model is imported, an initial system layout similar to that shown in **Figure 17.2-6** will be displayed. At this point, the user can rearrange the layout as desired and can modify the model as needed. See the SNAP user guide for more details.

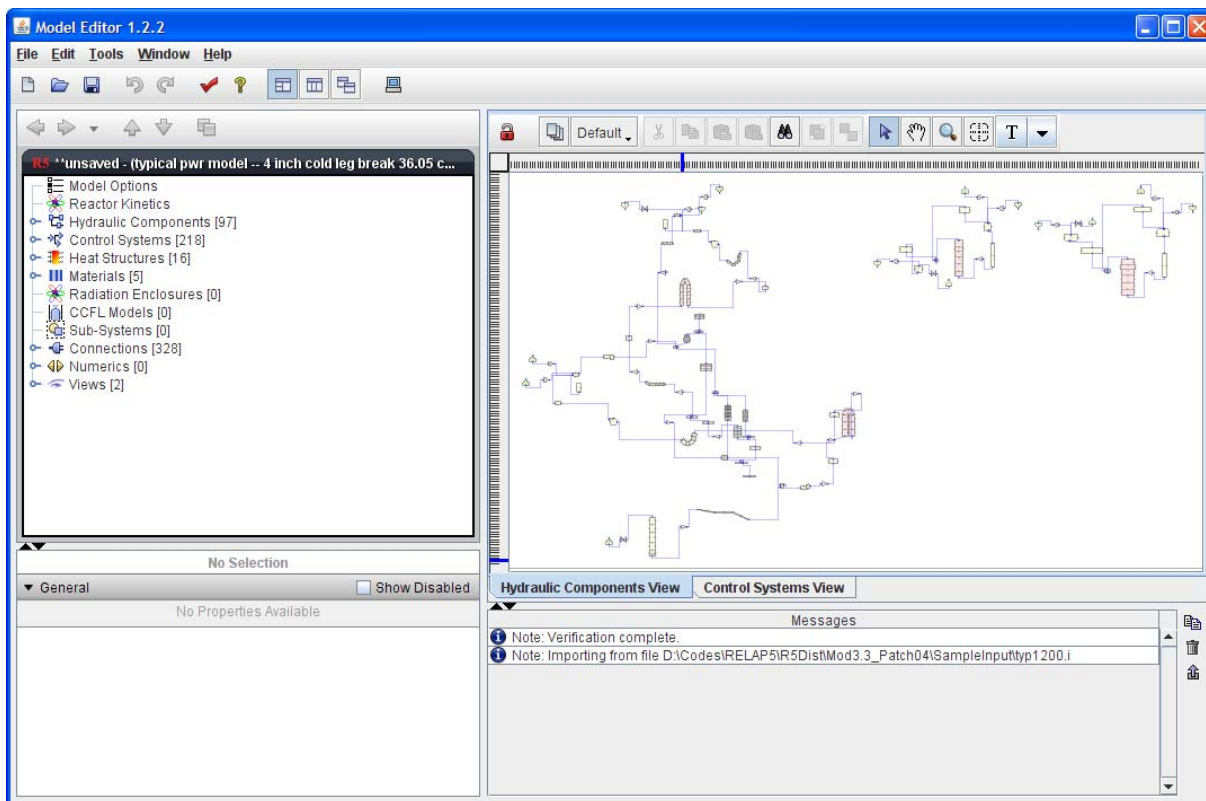


Figure 17.2-6 Main SNAP Model Editor Window

A17.2.1.3 Submitting RELAP5 Job From SNAP

When a model is ready to be submitted to the calculation server, the user can click on "Tools" at the top of the Model Editor. Select "Submit Job...". This will open a "Submit Calculation" window similar to that shown in **Figure 17.2-7**. For "Executable:", select the desired RELAP5 executable that was configured earlier in Section A17.2.1.1. For the "Target Folder", click the browse button. This will open a window similar to that shown in **Figure 17.2-8**. Browse to a folder in which to place the calculation results. If no folders exist under "localhost", then a new folder will need to be mounted.

To mount a new folder, select "localhost", then right-click and select "Mount Root Folder". This will open a window similar to that shown in **Figure 17.2-9**. Browse to a desired location and click "Mount". The user will be prompted to input the mounted name of the root folder, as shown in **Figure 17.2-10**. Type in a mount point name and click "OK". This will add the selected folder to the list under "localhost" in the "Select Target Folder" window (**Figure 17.2-8**). Select the mounted folder, or a subfolder within the newly mounted root folder, and click "OK".

Submit Calculation

Server: localhost

Executable: R5M33P04

Target Folder: /R5_SNAP_Runs/

Run Options

Name: ☒ Typ1200

Overwrite: ☒ Yes ☐ No ☐ Prompt

Priority: 5

Interactive: ☐ True ☒ False

Start Paused: ☐ True ☒ False

Parametric

☐ Enabled

First: < none >

Second: < none >

Restart

☒ No Restart ☐ Selected Restart File ☐ Selected Restart Job

☒ Submit Into Separate Folder

Animation

☐ Connect Animation After Submit

C:\Documents and Settings\Doug Barber\animation.med

☒ View Console Output

Submit Cancel

Figure 17.2-7 SNAP/RELAP5 "Submit Calculation" Window

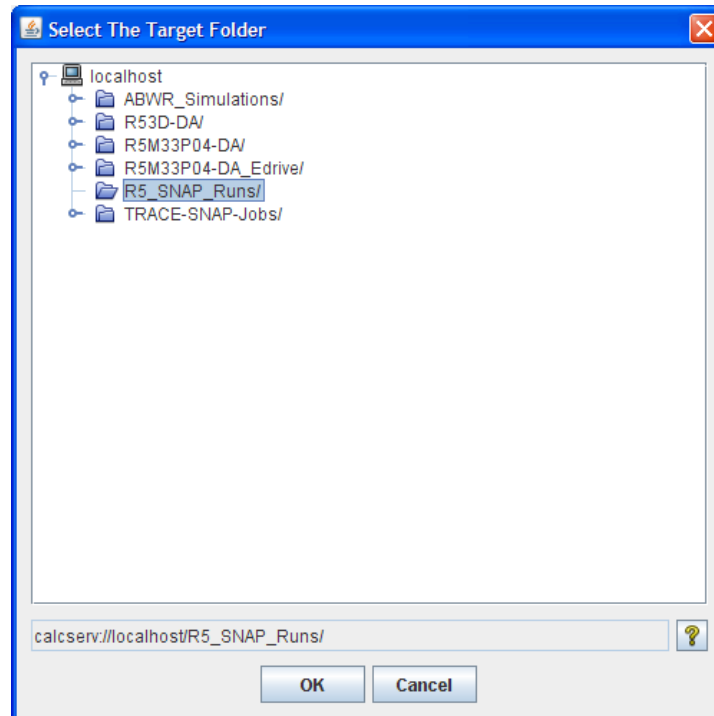


Figure 17.2-8 SNAP "Select The Target Folder" Window

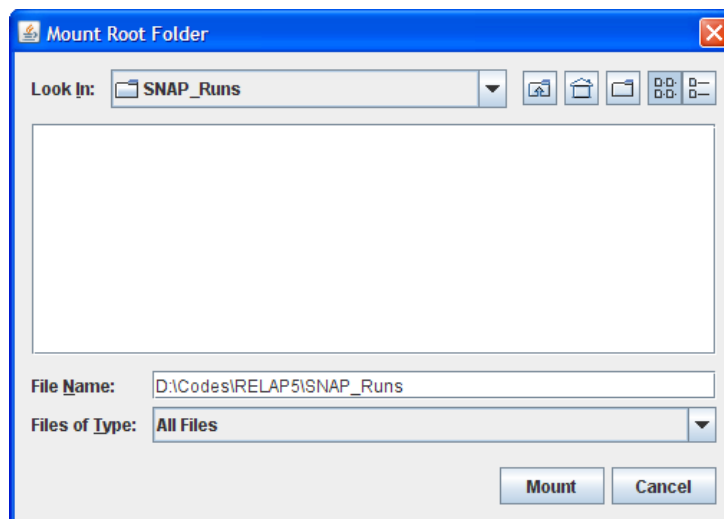


Figure 17.2-9 SNAP "Mount Root Folder" Window

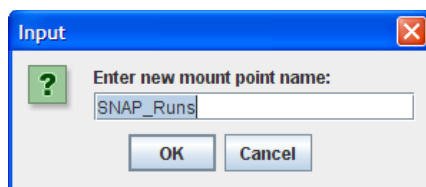


Figure 17.2-10 SNAP Select Mount Point Name Window

Once the target folder has been selected, specify the "Run Options" to be used for this case (see **Figure 17.2-8**). The first option is "Name". Check the box and specify a name for this submitted calculation. Click whether or not to overwrite existing calculations (if present), and select a priority for this job (jobs submit with a default priority of 5). If this calculation is a restarted case, select "Selected Restart File" or "Selected Restart Job" under the "Restart" box, and select either the name of the restart file or the name of a previously submitted job from which to restart the current run. If the user wishes to output results of this job under a separate folder under the target folder, check the "Submit Into Separate Folder" box. The user can also connect animation models if one exists for the model being submitted. Lastly, check the box for "View Console Output" in order to have SNAP present a job status window. If this option is chosen a window similar to that shown in **Figure 17.2-11** will appear displaying the screen output from the submitted RELAP5 calculation.

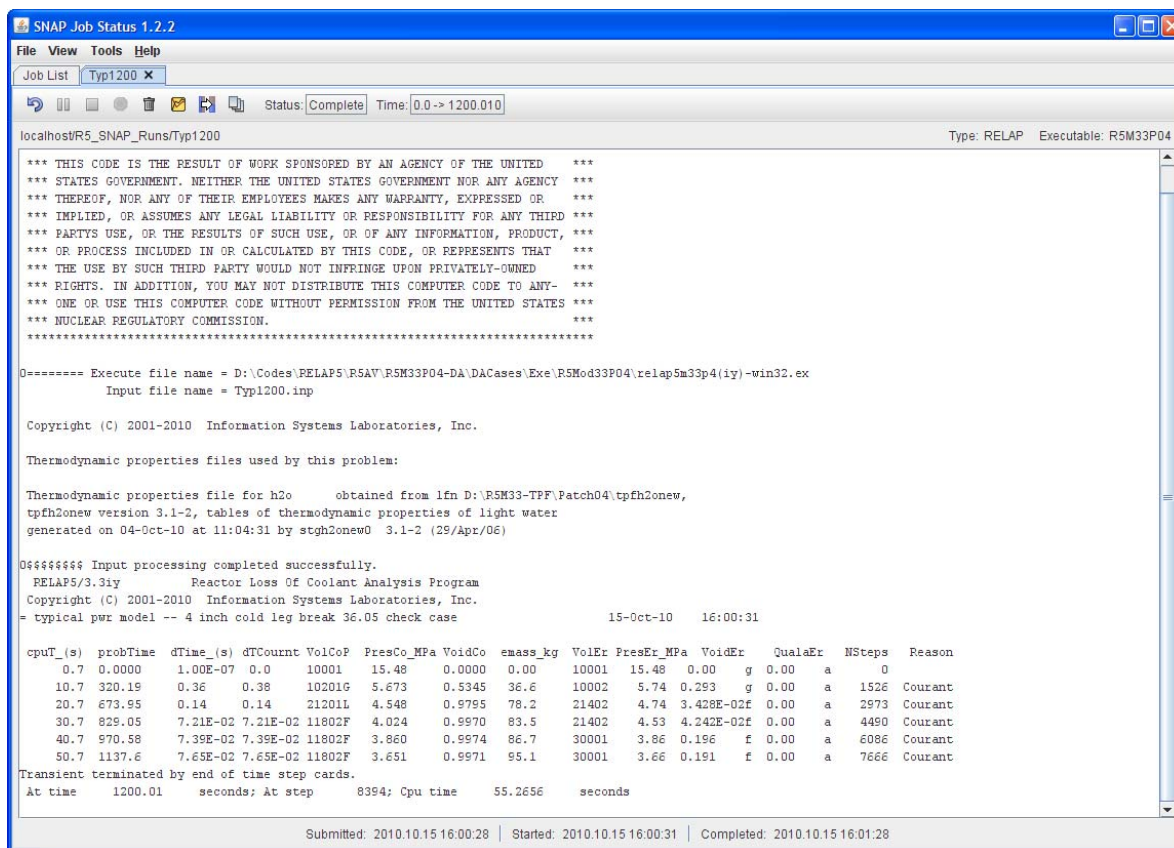


Figure 17.2-11 SNAP Job Status Window

A17.2.2 Command Line Execution

When operating on UNIX, Linux, and Windows systems, the RELAP5-based program can interpret a UNIX-style command line. The command line below is written with all of the allowed options (prefixed by a minus sign), and each option is followed by its default value:

```
relap5.x -f ftb1 -i indta -o outdta -r rstplt -s stripf -j jbinfo \
        -n null -c cdffile -a coupfl -X -C 0 -A dumpfil1 -B dumpfil2 \
        -w tpfh2o -d tpfd2o -Z tpfh2onew
```

If an option and its parameter are not entered, the default is used. With the exception of the X option, an option character must always be followed with a file name, and an option may not be repeated. The f option specifies a scratch word addressable file used only on CRAY versions. This file is a word addressable file, and the I/O is done by CRAY library subroutines that have successfully resisted external open and close statements. This file is small and can fit in most directories. This file should be removed after execution, but no error occurs in subsequent calculations if it is not removed. The file indta contains input data, outdta contains printed output, rstplt is the restart-plot file, and stripf is the strip file. The files dumpfil1 and dumpfil2 are files used to dump common and dynamic blocks for debugging purposes. The files beginning with tpf (e.g., tpfh2o) are thermodynamic property files where the characters following the tpf are chemical names of the material. The RELAP5 code uses three materials: new light water (H₂O), old light water (H₂OOLD), and heavy water (D₂O). The default is the new light water material. The table below shows each of the fluid materials along with its corresponding tpf filename and Card 120 name.

Table 17.2-1 RELAP5 Thermodynamic Property Files

Fluid Material	TPF Filename	Name Used on Card 120
Old Light Water	tpfh2o	h2oold
Heavy Water	tpfd2o	d2o
New Light Water	tpfh2onew	h2o

The file jbinfo is an optional file created by the user who wishes to place some additional information on his output file (such as a listing of the control cards used to run the job). The indta file must be available and the outdta file must not exist, or else a diagnostic message followed by immediate termination will follow. The rstplt file must not exist for a NEW problem and must exist for other type problems. The dumpfil1 and dumpfil2 files must not exist for the run that creates them and must exist for the CMPCOMS run. Most of the other options are for operation with the Nuclear Plant Analyzer (NPA), which is not included on the distribution CD-ROM.

The command line capability eliminates the need to have all files needed for execution in the same directory or to copy/rename files to match the default names. For example, the command:

```
relap5.x -i myprob.i -o /usr/tmp/rjw/myprob.o \
        -r /usr/tmp/rjw/myprob.r -w /u2/rjw/relap5/tpfh2o
```

takes the executable file and input file from the current directory, uses a temporary disk for the output and restart-plot files, and uses a water property file from a different directory.

A programmer convenience feature that was added to this version of RELAP5 is the use of capital -O and capital -R instead of -o and -r for designating the output and restart filenames on the command line. When capital letters are used, the requirement that the output and restart files have to be new files is removed so that existing files can be overwritten. This convenience feature allows the program developer to eliminate the removal of the output and restart files between runs. It is very useful during successive debugging runs in which the input, output, and restart filenames are not changed between runs.

A17.3 Plotting RELAP5 Results

AptPlot is the recommended tool to be used for plotting information from the RELAP5 restart/plot file (*.r, *.rst) or from demultiplexed restart/plot files (*.dmx). Supplied with AptPlot is also the utility for demultiplexing restart/plot files. Information on obtaining AptPlot can be found at the following website:

<http://www.aptpplot.com/>

18 REVISION HISTORY

Manual Revision History

Date	Notes
2006-06-28	Corrections to the lag and lead-lag control system components, Section A15
2007-03-09	Addition of total noncondensable mass (TMASSN) minor edit and plot variable, Section A4
2008-04-11	Addition of Card 1, Option 28, to increase hif when TEMPF approaches the liquid spinodal temperature. Added liquid spinodal temperature (TSPINF) and vapor spinodal temperature (TSPING) to minor edit list and plot variables (Patch03 Release)
2008-07-03	Revised description for Section A4.10, Expanded Plot Variables, 2080xxxx. Added statement about what happens in a restart and when components are deleted.
2008-07-08	Cleaned up the section on the DELETE option for the CCC0000 card to explain how the 2080XXXX cards are handled in a restart.
2008-07-23	Added liquid and vapor mass in volume, volmasf and volmasg. Added liquid and vapor volume mass errors, volmerf and volmerg. Added total liquid and vapor mass in volume, tmassvf and tmassvg. These can be used in minor edits and plotted.
2008-10-02	Added liquid, vapor, and noncondensable system mass (systmf, systmg, systmn). These can be used in minor edits and plotted using 2028xxxx cards.
2008-10-21	Added continuity equation system masses for total, liquid, vapor, noncondensable, and boron masses, systmfc, systmgc, systmnc, systmbc. Added state equation system masses for boron, systmb. Added system mass errors for liquid, vapor, noncondensable, and boron: sysmerf, sysmerg, sysmern, sysmerb. Added state equation volume masses for boron, tmassbv. Changed the name of state equation noncondensable volume mass from tmassn to tmassnv. Added state equation volume mass for boron, tmassbv, and continuity equation boron volume mass, volmasb. Added volume mass error for boron, volmerb, and noncondensables, volmern. Added boronj and mflowbj, mflownj.
2009-03-26	Added information about the Moody model being a subset of the Henry-Fauske choking model.
2010-04-01	Added end valve input to the valve section A 9.17. Increased number of card 1 options from 90 to 190. Added option 91 to output three tabular files for the volumes, junctions, and heat structures in a problem.
2010-04-02	Added a paragraph at the front of the Control System section to remind users to use lower case names for component types, etc.
2010-08-23	Added HS gas gap variables to the minor edits and plot variables: hgap, kgap, pgap, tgap, gapwd, clstrn, clhoop.
2010-09-21	Added minor corrections from Glen's input manual.

Manual Revision History

Date	Notes
2010-09-22	Added input for centerline shift ratio used in gap conductance model. Also added Card 1, Option 28 to use the new coding to compute gap width.
2010-10-18	Updated Section 17 for installing and executing RELAP5/Mod3.3 Patch04.